A Course in Differential Privacy

Manuel V. Loureiro

Version 2026-10-28

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Preface

This book is a course in differential privacy. It starts from the failures that motivate the subject, builds the definition and its basic algorithms, and goes on to answering many queries, private statistics and machine learning, lower bounds, and deployments.

Each chapter takes one topic. The text is short, and the mathematics is complete. Definitions, theorems and proofs are set apart and numbered, and every proof is given in full or, where it is long and standard, cited with what it needs. The prose between them says why each result comes next and what it means.

The reader should know discrete probability, including expectation, variance, independence and the inequalities of Markov, Chebyshev and Hoeffding, and basic linear algebra and calculus. Results from probability that the course needs beyond these are stated and proved where they are first used.

Notation. A dataset is $D = (x_1, \dots, x_n) \in \mathcal{X}^n$, a list of n records from a universe $\mathcal{X}$. Two datasets are neighbours, $D \sim D'$, when they differ in one record, which is the replace-one relation. A mechanism M maps a dataset to a random output in a set $\mathcal{Y}$. The privacy parameters are $\varepsilon \geq 0$ and $\delta \in [0, 1]$. The natural logarithm is written ln, and the logarithm to base 2 is written $\log_2$.

Part I.

Privacy Failures

Privacy is easy to promise and hard to deliver, so this part shows what goes wrong without a definition before the course gives one. The first chapter looks at real releases that their owners protected with pseudonyms, removed names, summary statistics or a trained model. An attacker defeated each one by combining the release with information from another source. The second chapter turns these cases into theorems: answers to counting queries that are too accurate let an analyst rebuild almost every secret bit. Together the chapters set the requirements for a definition of privacy. It must hold against any side information and apply to any kind of output. It must degrade gracefully as more data comes out, and it must allow the error that the attacks show to be necessary.

Table 1.: The chapters of this part.

Chapter	Question it answers
Attempts at Data Privacy	Why do pseudonyms, removing identifiers, releasing models or statistics, and k-anonymity fail to protect people?
Reconstruction Attacks	How much error must answers to counting queries have before an analyst cannot rebuild the data?

1. Attempts at Data Privacy

This chapter looks at five releases of sensitive data whose owners believed them safe. An attacker undid each one by combining the release with information from another source. Each protection came from intuition, and nobody could say what it guaranteed or against whom. The last section shows that k-anonymity, the first formal answer to these attacks, fails in the same way.

1.1. Pseudonyms: the taxi trips

In 2014, a public-records request obtained the record of every New York City taxi trip of 2013. Each record holds the times and places of the pickup and drop-off, the fare, the cab's medallion number and the driver's licence number. The two numbers would show where each driver worked and how much they earned, so the regulator replaced each one by a string of 32 hexadecimal digits. This step is called pseudonymization, and the attack below uses its one essential property: the same identifier always gets the same code.

Definition 1.1 (Pseudonymization)**.** Let $\mathcal{U}$ be a set of identifiers, such as licence numbers. A **pseudonymization** replaces each identifier $u \in \mathcal{U}$ in a table by $h(u)$, where h is a fixed injective function, so that the same person gets the same pseudonym in every row.

An analyst saw that one licence pseudonym earned far more per day than any driver could. It was the MD5 hash of the string `0`, a placeholder for a missing licence, so the function h was MD5 with no secret key. A hash function is hard to invert when its inputs come from a huge set. But the identifier formats allowed only about 22 million strings: about 3 million licence numbers and 19 million medallion numbers. With so few inputs, anyone can hash all of them and look up each pseudonym.

Proposition 1.1 (Dictionary attack)**.** *Let the set of identifiers $\mathcal{U}$ be finite, and let the pseudonymization function h be injective and computable by anyone. Then $|\mathcal{U}|$ evaluations of h recover the identifier behind every pseudonym in the table.*

Proof.

1. Compute $h(u)$ for every $u \in \mathcal{U}$ and store the pairs $(h(u), u)$ in a table sorted by $h(u)$. This takes $|\mathcal{U}|$ evaluations of h.
2. For each pseudonym v in the released table, look up v. Since h is injective, exactly one stored pair has first entry v, and its second entry is the identifier.

□

A licence number is not yet a name, so the attacker joined the decoded table with a public list of licensed drivers. That list gives the name for each licence number, so every driver's trips and income became public. This join with a second table that names people turns every attack in this chapter into a disclosure.

Definition 1.2 (Linkage attack)**.** Let T be a released table and T' an auxiliary table that names people, and let Q be a set of attributes that both tables contain. A **linkage attack** joins T and T' on Q. A row of T is **re-identified** when its values on Q match exactly one row of T' and no other row of T.

Random pseudonyms would stop the dictionary attack, but not linkage. A passenger knows the time and place of their own pickup and can read the driver's licence number in the cab, so that one row gives the driver's pseudonym, which then points to every other trip by the same driver. Passengers are exposed too, because the time and place of a pickup identify the trip, and its drop-off point is often a home.

1.2. Rich records: the Netflix Prize

In 2006, Netflix released about 100 million film ratings from about 480,000 subscribers for a contest. Each rating was a user pseudonym, a film, a score and a date. Such histories are sensitive, because they can hint at a person's politics or sexuality.

Narayanan and Shmatikov showed in 2008 that the pseudonyms protected little, because many subscribers also post reviews on the Internet Movie Database (IMDb) under their name or a stable handle. A linkage attack in the sense of Definition 1.2, with Q the films, scores and dates, matches the two sources. The match need not be exact. The authors scored each candidate pair of users by how many films both rated with similar scores on nearby dates, and they gave rarely rated films more weight. They reported that 8 ratings, 2 of which may be wrong, with dates known only to within 14 days, single out 99% of the records.

So in a sparse table with many columns, almost every record is unique, and no attribute is harmless. A handful of a person's ratings is enough to identify them.

1.3. Releasing a model: memorization

A trained model seems safer to release than a table. Carlini and coauthors tested this in 2019 on language models. Such a model predicts each word from the words before it, with a vector of parameters θ learned from a corpus of text. To test whether a model has memorized a string, the authors measured how surprised the model is by it, in bits, so a string the model finds likely gets a small score.

Definition 1.3 (Log-perplexity)**.** Write $\Pr_\theta$ for the probability that the model with parameters θ assigns to a sequence of words, or to the next word given the words before it. The **log-perplexity** of a sequence $x_1, \dots, x_\ell$ of ℓ words under this model is

$$P_\theta(x_1, \dots, x_\ell) = -\log_2 \Pr_\theta[x_1, \dots, x_\ell] = \sum_{i=1}^{\ell} -\log_2 \Pr_\theta\left[x_i \mid x_1, \dots, x_{i-1}\right]. \tag{1.1}$$

The second form follows from the chain rule of probability. A model of English gives "the cat sat on the mat" a low score and a string of unrelated words a high one. The question is whether a rare string in the training text, such as a password or an identity number, gets a much lower score than it would otherwise. If it does, the model memorized it, and anyone who can query the model may find it.

To measure this, the authors put a **canary** into the training text: a sentence of a fixed format with a random secret, such as "my identity number is 314-15-9265". They then compared its score with those of all other sentences of that format. Exposure is large when the model ranks the true canary above most of them.

Definition 1.4 (Exposure)**.** Let $\mathcal{R}$ be the finite set of candidate secrets, all sentences of the canary's format, and let $N = |\mathcal{R}|$. Sort $\mathcal{R}$ by increasing log-perplexity under the model, and let $\operatorname{rank}(r) \in \{1, \dots, N\}$ be the position of r, so that rank 1 is the most likely candidate. The **exposure** of r is

$$\operatorname{exposure}(r) = \log_2 N - \log_2 \operatorname{rank}(r). \tag{1.2}$$

Exposure goes from 0, for the least likely candidate, to $\log_2 N$, for the most likely one, and an attacker can read a canary at the top off the model by trying the candidates in order. To know what a large exposure is, you need the exposure of a canary that the model never saw, which the next result bounds.

Proposition 1.2 (Exposure without memorization)**.** *Suppose the canary r is drawn uniformly from $\mathcal{R}$, independently of the model, and the ranking breaks ties in a fixed way. Then for every $t \geq 0$,*

$$\Pr[\operatorname{exposure}(r) \geq t] \leq 2^{-t}, \qquad \mathbb{E}[\operatorname{exposure}(r)] < \frac{1}{\ln 2} \approx 1.443. \tag{1.3}$$

Proof.

1. The ranking depends only on the model, and r is uniform and independent of it. So $\operatorname{rank}(r)$ is uniform on $\{1, \dots, N\}$.
2. The exposure is at least t exactly when $\operatorname{rank}(r) \leq N2^{-t}$. By step 1 this has probability $\lfloor N2^{-t} \rfloor / N \leq 2^{-t}$.
3. By step 1, the average exposure is $\log_2 N - \frac{1}{N}\sum_{k=1}^{N} \log_2 k = \frac{1}{N \ln 2}\left(N \ln N - \ln N!\right)$.
4. Since ln is increasing, $\ln k$ is at least the integral of $\ln x$ over $[k-1, k]$ for $k \geq 2$. Summing over k gives

$$\ln N! \geq \int_1^N \ln x \, \mathrm{d}x = N \ln N - N + 1. \tag{1.4}$$

5. Steps 3 and 4 give $\mathbb{E}[\text{exposure}(r)] \le \frac{N-1}{N \ln 2} < \frac{1}{\ln 2}$.

□

So if the model never saw the canary, an exposure of t bits is as surprising as an event of probability 2^{-t}. With nine-digit secrets, $N = 10^9$, and the largest possible exposure is $\log_2 10^9 \approx 29.9$.

The experiments found both outcomes. A translation model trained on about 100,000 pairs of sentences got a canary with a nine-digit identity number, and after four insertions the canary reached the largest possible exposure. Gmail's Smart Compose, trained on billions of word sequences, got canaries with N of order 10^{12}, and even after 10,000 insertions their exposure reached only about 10 of a possible 39.9. Standard defences against overfitting, such as dropout and regularization, did not reduce exposure. Training with differential privacy, the subject of the next part, was the only method the authors found that did.

1.4. Releasing statistics: genomic studies

Medical studies must publish their findings, and aggregate statistics seemed a safe middle ground. But suppose a study enrols only people with a stigmatized disease. Then the fact that a person took part reveals their diagnosis, and a test that decides membership from the published statistics is an attack.

Homer and coauthors showed in 2008 that such a test exists for genetic data. It compares a person's genotype with the published allele frequencies of the study group and of a reference population. Under conditions on the number of genetic markers and the size of the group, it tells whether the person's DNA was in the group. In practice the test makes more false positives than this analysis suggests, because its threshold depends on how well the reference population matches the study group. Even so, the National Institutes of Health in the United States moved aggregate genomic statistics behind an approval process.

1.5. Removing identifiers: k-anonymity

In the mid-1990s, the Group Insurance Commission of Massachusetts gave researchers the hospital records of every state employee, with names and social security numbers removed. Each record still held the patient's ZIP code, date of birth, sex and diagnosis. Latanya Sweeney bought the voter roll of Cambridge, Massachusetts, which lists every voter's name, ZIP code, date of birth and sex. A linkage attack (Definition 1.2) on these three attributes found the governor's hospital records. The attack works at scale, because Sweeney later estimated that 87% of the population of the United States is the only person with their five-digit ZIP code, sex and date of birth.

Her answer, with Samarati, sorts the attributes of a table into three kinds. The release removes **identifiers**, such as names, and keeps **sensitive attributes**, such as the diagnosis, because they are what the release is for. In between are **quasi-identifiers**, such as ZIP code, age and sex, which are harmless alone but identify people in combination. So the release coarsens them, for example a ZIP code to three digits. It stops when each person hides among at least k rows that look the same.

Definition 1.5 (k-anonymity)**.** Let $k \geq 1$ be an integer. A table is **k-anonymous** with respect to a set of quasi-identifiers if every combination of quasi-identifier values that appears in the table appears in at least k rows. The rows that share a combination form an **equivalence class**.

A k-anonymous table stops the attack on the governor, because a linkage on the quasi-identifiers matches every person to at least k rows. But the definition says nothing about the sensitive values in a class, or about other releases. Two examples show what goes wrong, with tables from two hospitals in the same town.

Table 1.1.: Hospital A's release, 3-anonymous with respect to ZIP code and age.

ZIP code	Age	Diagnosis
021**	20–29	influenza
021**	20–29	asthma
021**	20–29	diabetes
021**	30–39	hepatitis
021**	30–39	hepatitis
021**	30–39	hepatitis
024**	40–59	influenza
024**	40–59	migraine
024**	40–59	asthma

Table 1.2.: Hospital B's release, 3-anonymous with respect to ZIP code and age.

ZIP code	Age	Diagnosis
02***	18–24	influenza
02***	18–24	asthma
02***	18–24	fracture
02***	25–34	diabetes
02***	25–34	migraine
02***	25–34	fracture

The first example shows that a class whose rows all share one sensitive value hides nothing.

Example 1.1 (Homogeneity)**.** A neighbour aged 35, living in ZIP code 02139, was a patient of hospital A. In Table 1.1, the neighbour must be in the class with ZIP code 021** and age 30–39. All three of its rows say hepatitis, so the attacker learns the diagnosis without knowing which row is the neighbour's.

The second example has varied diagnoses in each class, but combines the two releases.

Example 1.2 (Intersection)**.** A colleague aged 27, living in ZIP code 02139, was treated at both hospitals for the same condition. In Table 1.1 the colleague's class allows influenza, asthma or diabetes. In Table 1.2 the class with age 25–34 allows diabetes, migraine or fracture. Only diabetes is in both lists, so the two releases together reveal the diagnosis, although each is 3-anonymous.

The second example is a failure of composition: each hospital checked its own table and could not know what the other released. Ganta, Kasiviswanathan and Smith showed in 2008 that such attacks succeed in practice, also against refinements of k-anonymity.

The five cases give three requirements for a definition of privacy. It must hold whatever side information an attacker has, because every attack above used some. It must apply to any output, such as a table, a model or a statistic. And it must degrade gracefully when several releases are combined. Differential privacy, defined in the chapter on randomized response, meets all three.

1.6. Summary

Table 1.3.: The chapter's results.

Result	Statement
Definition 1.1	replace each identifier u by $h(u)$ for a fixed injective h
Proposition 1.1	a public h on a set of N identifiers is inverted by N evaluations
Definition 1.2	join a release with named auxiliary data on shared attributes
Definition 1.4	$\log_2 N - \log_2 \operatorname{rank}(r)$ for a canary among N candidates
Proposition 1.2	without memorization, exposure $\geq t$ has probability at most 2^{-t} and mean below $1/\ln 2$
Definition 1.5	every quasi-identifier combination appears in at least k rows
Example 1.1	a class with one sensitive value reveals it
Example 1.2	two k-anonymous releases together can reveal a value

1.7. References

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2. Reconstruction Attacks

This chapter asks when answers to counting questions about a dataset, each only slightly wrong, let an attacker rebuild almost the whole dataset. It does not yet define privacy. Instead it defines a failure so flagrant that nobody would call it private, and shows that a wide class of algorithms fails in this way. If a curator answers every possible counting query, it must make errors of order n, the number of people, to stop the attack. If it answers only about n random queries, errors of order $\sqrt{n}$ are needed. ## Reconstruction from exact statistics {#sec-rec-census}

Exact statistics give the data away. A census collects **microdata**, one record per person with fields such as age, sex and race. It publishes tables of counts, medians and means for each small area, such as a city block. It suppresses cells about fewer than three people, but suppression alone protects nobody, because tables overlap. If three of a group's four residents are women, a published median age for its men is the age of its one man.

Every published number is a constraint on the unknown microdata, and enough constraints leave only one possibility. The next example counts the age combinations that one line of a census table leaves for three people.

Example 2.1 (One line of a census table)**.** A block's table says that it has three men, with median age 35 and mean age 47. Call the youngest man's age u_1, the middle one's u_2 and the oldest one's u_3, so that $u_1 \le u_2 \le u_3$. Ages are whole numbers between 0 and 125.

1. Without the statistics, the number of such triples is the number of multisets of size 3 from 126 values, $\binom{128}{3} = 341{,}376$.
2. The median of three numbers is the middle one, so $u_2 = 35$. Then u_1 has 36 values and u_3 has 91, which leaves $36 \times 91 = 3{,}276$ triples.
3. The mean gives $u_1 + u_2 + u_3 = 141$, so $u_3 = 106 - u_1$. Each u_1 from 0 to 35 gives a u_3 between 71 and 106, and every such u_3 lies between u_2 and 125. So exactly 36 triples remain, from $(0, 35, 106)$ to $(35, 35, 71)$.

So one line of the table cuts the possibilities by a factor of almost 10,000. The other lines group the same people differently and cut them further. A reconstruction attack searches for microdata that satisfies every constraint. This search is an integer program, which is hard in the worst case, but solvers handle such instances quickly. For a fictional block of seven people, Garfinkel, Abowd and Martindale found that the constraints of its full table have exactly one solution, which a solver finds in a fraction of a second.

The attack scales. The United States Census Bureau ran it on its 2010 tables, and the rebuilt records matched the confidential ones exactly for 46% of the population. They matched for 71% if ages could be off by one year. A link of these records with commercial databases then gave the names of more than 50 million people. So the Bureau adopted differential privacy for the 2020 census.

2.1. Subset queries and blatant non-privacy

To analyse such attacks, strip the setting to its core. There are n people. Each has public attributes that identify them, such as a name, and one secret bit, such as whether they have a disease. The secret bits form the vector $d = (d_1, \dots, d_n) \in \{0,1\}^n$. An analyst asks how many people who satisfy a condition on the identifiers have secret bit 1. A condition such as "the name is Alice, Bob or Eve" can single out any set of people, so a query is a subset $S \subseteq [n] = \{1, \dots, n\}$. We identify it with its indicator vector $S \in \{0,1\}^n$, whose entry i is 1 when person i is in the set and 0 otherwise.

A curator holds the secret bits and answers the queries. The definition assumes only a bound E on how far each answer may be from the truth.

Definition 2.1 (Subset queries and a curator with error E)**.** Let $d \in \{0,1\}^n$ be the vector of secret bits. The true answer to a **subset query** $S \in \{0,1\}^n$ is the number of people in S whose bit is 1,

$$a(S) = \sum_{i \in S} d_i = \langle S, d \rangle, \tag{2.1}$$

where $\langle S, d \rangle = \sum_i S_i d_i$ is the inner product of the indicator vector S with d.

A **curator with error** E answers each query S with a response $r(S)$, a number within E of the truth, $|r(S) - a(S)| \le E$. Within that bound, the curator may choose its answers in any way, depending on d and on all the queries.

The error need not be random. The curator may pick the most misleading answer in the allowed range, so every attack below also works against random noise. Some error is necessary, because the exact answer to the query $S = \{i\}$ is the secret bit of person i.

To measure how much an attacker learns, count the bits it gets wrong. The attacker outputs a guess $c \in \{0,1\}^n$ of the secret vector d, and the **Hamming distance** $\mathrm{dist}(c, d) = |\{i : c_i \neq d_i\}|$ is the number of positions where the guess is wrong. The next definition names the clearest failure of privacy: for every secret vector, some attacker gets all but a vanishing fraction of the bits right, except with a vanishing probability.

Definition 2.2 (Blatant non-privacy)**.** Consider a family of curators, one for each number of people n. The family is **blatantly non-private** if three things exist. The first is an attacker that interacts with the curator and outputs a guess $c \in \{0,1\}^n$. The second is a bound $g(n)$ on the number of wrong bits, with $g(n)/n \to 0$. The third is a failure probability $\beta(n)$, with $\beta(n) \to 0$. They must satisfy, for every n and every secret vector $d \in \{0,1\}^n$,

$$\Pr\left[\mathrm{dist}(c, d) \le g(n)\right] \ge 1 - \beta(n). \tag{2.2}$$

Such an attacker, with $o(n)$ wrong bits (a vanishing fraction of n), learns the secret of almost everyone. Whatever privacy means, this violates it.

2.2. Asking every query

The first attack ignores cost. It asks all 2^n subset queries and outputs any candidate that agrees with every answer.

Theorem 2.1 (Reconstruction from all subset queries)**.** *Let $E \geq 0$, and let a curator with error E answer all 2^n subset queries, giving the answer $r(S)$ to the query S. An attacker that outputs any $c \in \{0,1\}^n$ satisfying*

$$|\langle S, c\rangle - r(S)| \leq E \quad \text{for every } S \subseteq [n] \tag{2.3}$$

finds such a c, and every such c differs from the secret vector d in at most $4E$ positions, $\mathrm{dist}(c, d) \leq 4E$.

Proof.

1. The true vector d satisfies Equation 2.3 by Definition 2.1, so a candidate exists.
2. Let $I_0 = \{i : d_i = 0\}$. The attacker asked the query I_0, so Equation 2.3 gives $\langle I_0, c\rangle \leq r(I_0) + E$. The curator's bound gives $r(I_0) \leq \langle I_0, d\rangle + E = E$. Hence $\langle I_0, c\rangle \leq 2E$. This sum counts the positions of I_0 where $c_i = 1 \neq d_i$, so c and d differ in at most $2E$ positions of I_0.
3. Let $I_1 = \{i : d_i = 1\}$. In the same way, $\langle I_1, c\rangle \geq r(I_1) - E \geq \langle I_1, d\rangle - 2E = |I_1| - 2E$. The number of positions of I_1 where $c_i = 0$ is $|I_1| - \langle I_1, c\rangle \leq 2E$.
4. The sets I_0 and I_1 partition $[n]$, so $\mathrm{dist}(c, d) \leq 4E$.

□

The attacker does not know I_0 and I_1, which is why it asks every query. With errors that are a vanishing fraction of n, the bound meets Definition 2.2.

Corollary 2.1. *If $E = n/401$, the attacker of Theorem 2.1 recovers more than 99% of the bits. If $E = E(n)$ with $E(n)/n \to 0$, the curator is blatantly non-private, with $g(n) = 4E(n)$ and $\beta(n) = 0$.*

Proof. For the first claim, $4n/401 < n/100$, since $4 \times 100 < 401$. The second claim is Theorem 2.1 with Definition 2.2, because the attacker never fails. □

Errors of order n are the most a counting query can bear, since every answer lies between 0 and n. The next result shows that error E always hides about $E/2$ bits, however many queries the attacker asks.

Proposition 2.1 (Error E hides $E/2$ bits)**.** *Let $E \geq 1$ and $k = \min(\lfloor E \rfloor, n)$. For every attacker, which may ask any queries in any adaptive order, there are a vector $d \in \{0,1\}^n$ and a curator with error E for which* $\Pr[\mathrm{dist}(c, d) \geq k/2] \geq 1/2$.

Proof.

1. Let d be all zeros, and let d' have ones in its first k positions and zeros elsewhere, so $\mathrm{dist}(d, d') = k$.
2. On the data d', let the curator answer every query exactly, $r(S) = \langle S, d' \rangle$. On the data d, let it give the same answers. Their error is $|\langle S, d' - d \rangle| \le k \le E$, so both are curators with error E.
3. The answers to every query are the same on d and on d'. So the attacker's output c has the same distribution in the two cases.
4. For every c, the triangle inequality gives $\mathrm{dist}(c, d) + \mathrm{dist}(c, d') \ge k$, so at least one of the two distances is at least $k/2$. Hence $\Pr[\mathrm{dist}(c, d) \ge k/2] + \Pr[\mathrm{dist}(c, d') \ge k/2] \ge 1$, and one of the two probabilities is at least $1/2$.

□

So for the attacker who asks everything, safety needs error of order n, and Theorem 2.1 is tight up to a factor of 8. But an attack with 2^n queries is not a practical threat once n is more than a few dozen.

2.3. Random queries

The second attack asks about n queries, each a uniformly random subset of the people. It defeats errors of order $\sqrt{n}$, much smaller than the first attack can handle but still large next to the error a useful statistic can tolerate.

The idea is to fix a wrong candidate c and show that a random query is likely to expose it. If c differs from d in k positions, the answers $\langle S, c \rangle$ and $\langle S, d \rangle$ differ by a sum of about $k/2$ random signs, which is typically of size $\sqrt{k}$. A curator with error E can hide a difference of at most $2E$, so when $\sqrt{k}$ is much larger than E, a random query rules c out with constant probability. To make "typically" precise, the proof needs a lower bound on the size of a random sum, the opposite of a concentration inequality.

Lemma 2.1 (A random subset sees a fixed vector)**.** *Let $w \in \mathbb{R}^n$, with Euclidean length $\|w\|_2 = (\sum_i w_i^2)^{1/2}$. Let S be a uniformly random subset of $[n]$, so that each i is in S independently with probability $1/2$. Then the sum $\langle S, w \rangle$ of the entries of w in S satisfies*

$$\Pr\left[\, |\langle S, w \rangle| \ge \tfrac{1}{4}\|w\|_2 \,\right] \ge \frac{3}{32}. \tag{2.4}$$

Proof.

1. If $w = 0$ the event always holds, so let $w \ne 0$. Write $S_i = (1 + \xi_i)/2$, where the signs $\xi_i \in \{-1, +1\}$ are independent and uniform. Then $\langle S, w \rangle = \frac{1}{2}(a + Z)$, with the constant $a = \sum_i w_i$ and the random sum $Z = \sum_i \xi_i w_i$.
2. *Moments of Z.* Expand the powers and use independence. A product of the ξ_i has mean 0 unless each index appears an even number of times. So $\mathbb{E}[Z^2] = \sum_i w_i^2 = \|w\|_2^2$, and

$$\mathbb{E}[Z^4] = \sum_i w_i^4 + 3\sum_{i \ne j} w_i^2 w_j^2 = 3\|w\|_2^4 - 2\sum_i w_i^4 \le 3\|w\|_2^4. \tag{2.5}$$

3. *Paley–Zygmund inequality.* Let $Y \geq 0$ have $\mathbb{E}[Y] > 0$, and let $s \in [0,1]$. Split $\mathbb{E}[Y]$ according to whether $Y > s\,\mathbb{E}[Y]$, and apply the Cauchy–Schwarz inequality to the second part:

$$\mathbb{E}[Y] \leq s\,\mathbb{E}[Y] + \mathbb{E}\left[Y\mathbf{1}[Y > s\,\mathbb{E}[Y]]\right] \leq s\,\mathbb{E}[Y] + \sqrt{\mathbb{E}[Y^2]\,\Pr[Y > s\,\mathbb{E}[Y]]}. \tag{2.6}$$

Rearranging gives $\Pr[Y > s\,\mathbb{E}[Y]] \geq (1-s)^2\,\mathbb{E}[Y]^2/\mathbb{E}[Y^2]$.

4. Apply step 3 to $Y = Z^2$ with $s = 1/4$, using step 2. The event $|Z| > \|w\|_2/2$ has probability at least $(3/4)^2/3 = 3/16$.
5. *Removing the constant a.* For every z, $|a+z|+|a-z| \geq 2|z|$, so $|a+Z| \geq |Z|$ or $|a-Z| \geq |Z|$. Flipping all signs maps Z to $-Z$ and preserves the distribution and the event of step 4. Hence the events $\{|Z| > \|w\|_2/2,\ |a+Z| \geq |Z|\}$ and $\{|Z| > \|w\|_2/2,\ |a-Z| \geq |Z|\}$ have the same probability. Together they cover the event of step 4, so each has probability at least $3/32$.
6. On the first of these events, $|\langle S, w\rangle| = \frac{1}{2}|a+Z| \geq \frac{1}{2}|Z| > \frac{1}{4}\|w\|_2$.

□

The lemma rules out one wrong candidate with constant probability per query. A union bound over the 2^n candidates in $\{0,1\}^n$ then shows that about n queries rule out all the wrong ones at once.

Theorem 2.2 (Reconstruction from random queries)**.** *Let $E \geq 0$ and $\beta \in (0,1)$, and let*

$$m \geq \frac{32}{3}\left(n \ln 2 + \ln \frac{1}{\beta}\right). \tag{2.7}$$

The attacker draws m independent, uniformly random subsets $S_1, \ldots, S_m$ of $[n]$ before seeing any answer. A curator with error E answers them with $r_1, \ldots, r_m$. The attacker outputs any $c \in \{0,1\}^n$ with $|\langle S_j, c\rangle - r_j| \leq E$ for all j. Then such a c exists and, with probability at least $1-\beta$ over the queries, every such c has

$$\operatorname{dist}(c,d) \leq 64E^2. \tag{2.8}$$

Proof.

1. The true vector d satisfies all m constraints by Definition 2.1, so a candidate exists.
2. *One far candidate.* Fix c with $k = \operatorname{dist}(c,d) > 64E^2$, and let $w = c - d$. Its entries are in $\{-1,0,1\}$ and exactly k are nonzero, so $\|w\|_2 = \sqrt{k} > 8E$. By Lemma 2.1, each query has $|\langle S_j, w\rangle| \geq \sqrt{k}/4 > 2E$ with probability at least $3/32$. The queries are independent, so the probability that $|\langle S_j, w\rangle| \leq 2E$ for every j is at most $(1-3/32)^m \leq e^{-3m/32}$.
3. *All far candidates.* There are at most 2^n candidates. By the union bound, the event B that some candidate with $\operatorname{dist}(c,d) > 64E^2$ has $|\langle S_j, c-d\rangle| \leq 2E$ for every j has probability at most $2^n e^{-3m/32}$, which is at most β by the choice of m.

4. *Consistent candidates are close.* Suppose B does not occur. If c is consistent with the answers, the triangle inequality gives $|\langle S_j, c-d\rangle| \le |\langle S_j, c\rangle - r_j| + |r_j - \langle S_j, d\rangle| \le 2E$ for every j. So c is not far, which is Equation 2.8. The event B depends only on the queries, so this holds whatever answers the curator chose.

□

The number of queries is about $7.4n$: for $n = 1{,}000$ and $\beta = 0.01$ the theorem asks for $m = 7{,}443$. Setting $\beta = 2^{-n}$ gives the asymptotic statement.

Corollary 2.2. *If $E = E(n)$ with $E(n)/\sqrt{n} \to 0$, the curator is blatantly non-private against $15n$ random queries, with $g(n) = 64E(n)^2$ and $\beta(n) = 2^{-n}$. If $E = \alpha\sqrt{n}$, the same attacker gets at most $64\alpha^2 n$ bits wrong with probability at least $1 - 2^{-n}$.*

Proof. With $\beta = 2^{-n}$, the bound on m in Theorem 2.2 is $\frac{32}{3} \cdot 2n \ln 2 \approx 14.79\, n \le 15n$. Then $g(n) = 64E(n)^2 = 64n\,(E(n)/\sqrt{n})^2$, and $g(n)/n \to 0$. □

2.3.1. An efficient attack

The attacker of Theorem 2.2 asks few queries but searches 2^n candidates. A search over fractional vectors is a linear program, so this attack is called linear programming (LP) decoding. It looks for $x \in [0,1]^n$ with $|\langle S_j, x\rangle - r_j| \le E$ for all j, which takes time polynomial in n and m, and rounds x to the nearest bit vector. The analysis is the same as before with one extra step: a fractional candidate is not one of 2^n points, so a fine grid replaces the set of candidates.

Theorem 2.3 (Reconstruction by linear programming)**.** *Let $E > 0$, $\beta \in (0,1)$, $K = \lceil n/(2E) \rceil$, and*

$$m \ge \frac{32}{3}\left(n \ln(2K+1) + \ln\frac{1}{\beta}\right). \tag{2.9}$$

The attacker draws m random subsets and receives answers from a curator with error E, as in Theorem 2.2. It finds any $x \in [0,1]^n$ with $|\langle S_j, x\rangle - r_j| \le E$ for all j, and it outputs $c_i = 1$ if $x_i \ge 1/2$ and $c_i = 0$ otherwise. With probability at least $1 - \beta$ over the queries,

$$\operatorname{dist}(c, d) \le 676E^2. \tag{2.10}$$

Proof.

1. The vector d is feasible, so the linear program has a solution x. Let $z = x - d \in [-1,1]^n$. As in step 4 of Theorem 2.2, $|\langle S_j, z\rangle| \le 2E$ for every j.
2. *Rounding errors.* If $c_i \ne d_i$, then either $d_i = 0$ and $x_i \ge 1/2$, or $d_i = 1$ and $x_i < 1/2$. In both cases $|z_i| \ge 1/2$, so $(2z_i)^2 \ge 1$. Summing over i gives $\operatorname{dist}(c, d) \le 4\|z\|_2^2$.
3. *The grid.* Let $h = 1/K \le 2E/n$, and let G be the set of vectors whose entries are multiples of h in $[-1,1]$. It has $(2K+1)^n$ points. Round each entry of z to the nearest multiple of h to get $z' \in G$, with $|z_i - z'_i| \le h/2 \le E/n$. Then $|\langle S_j, z - z'\rangle| \le \sum_i |z_i - z'_i| \le E$, so $|\langle S_j, z'\rangle| \le 3E$ for every j. Also $\|z - z'\|_2 \le \sqrt{n} \cdot E/n \le E$.

4. *Far grid points are ruled out.* Fix $g \in G$ with $\|g\|_2 > 12E$. By Lemma 2.1, each query has $|\langle S_j, g\rangle| \geq \|g\|_2/4 > 3E$ with probability at least $3/32$. As in Theorem 2.2, the probability that $|\langle S_j, g\rangle| \leq 3E$ for every j is at most $e^{-3m/32}$. A union bound over the $(2K+1)^n$ grid points bounds the probability that this happens for some far grid point by $(2K+1)^n e^{-3m/32} \leq \beta$.
5. Outside that event, step 3 shows that z' is not far, so $\|z'\|_2 \leq 12E$ and $\|z\|_2 \leq 13E$. Step 2 gives $\mathrm{dist}(c, d) \leq 4 \cdot 169E^2 = 676E^2$.

□

For $1 \leq E \leq n$, the factor $\ln(2K+1)$ is at most $\ln(n+3)$, so the attack runs in polynomial time with $O(n \log n)$ queries and gets at most $676E^2$ bits wrong. So error $o(\sqrt{n})$ again lets an attacker recover all but a vanishing fraction of the bits. The constants 64 and 676 are far from the truth, as Figure 2.1 shows: on simulated data the attack recovers almost every bit at errors the theorem cannot handle.

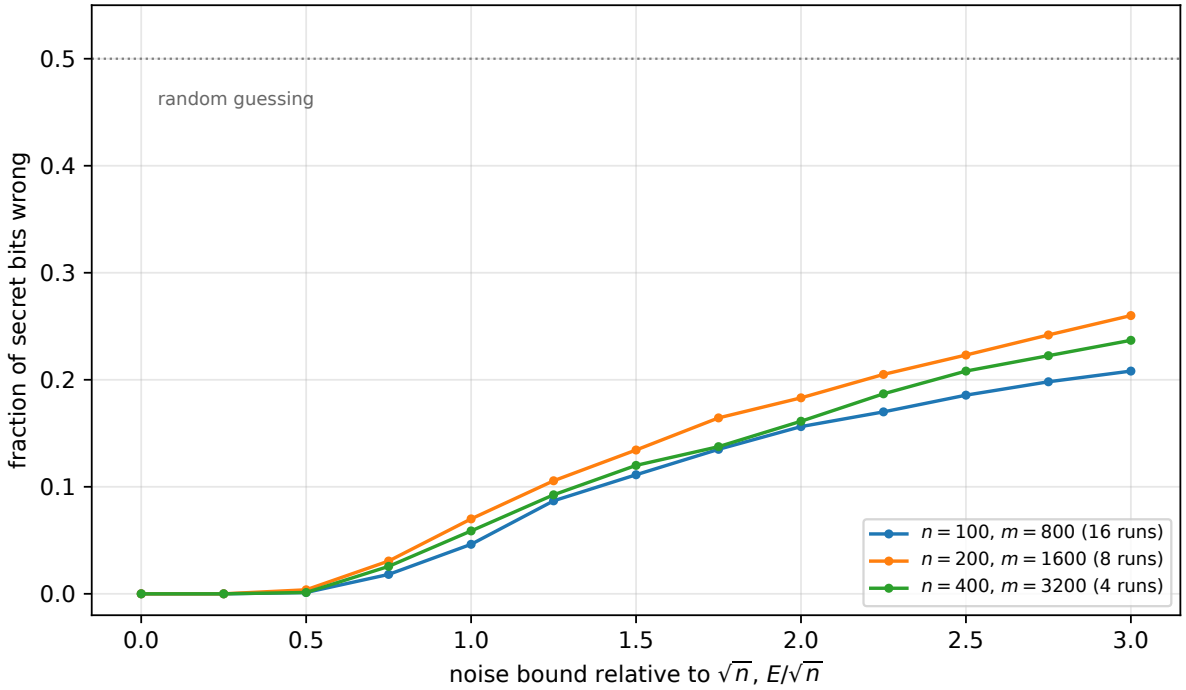


Figure 2.1.: The linear-programming attack on simulated data, with $m = 8n$ random subset queries and an error drawn uniformly from $[-E, E]$ for each answer. The attacker finds a point of the feasible set of Theorem 2.3, up to a violation of at most 0.01, by projected gradient descent and rounds it. Each point averages $1600/n$ runs. The curves for different n nearly coincide, so the error the attack tolerates grows like $\sqrt{n}$. For $n = 400$, a single run gets none of the bits wrong at $E = \sqrt{n}/2$ and 17% of them at $E = 2\sqrt{n}$.

Bounds on random matrices, which this course does not develop, show that $O(n)$ queries suffice for the linear program (Dwork and Roth, 2014, Section 8.1). They also show that a linear program

that minimizes the total error tolerates a constant fraction of arbitrary answers (Dwork, McSherry and Talwar, 2007). The argument above gives an inefficient version of that tolerance.

Proposition 2.2 (Tolerating some arbitrary answers)**.** *Let $E \geq 0$, let $\beta \in (0,1)$ be a failure probability, let $\rho \in [0, 3/128)$ be the fraction of answers that may be arbitrary, and let*

$$m \geq \frac{2048}{9}\left(n \ln 2 + \ln \frac{1}{\beta}\right). \tag{2.11}$$

The attacker draws m random subsets as in Theorem 2.2. The curator answers at least $(1-\rho)m$ of them with error at most E and the others arbitrarily, choosing which after seeing the queries. The attacker outputs any $c \in \{0,1\}^n$ with $|\langle S_j, c\rangle - r_j| \leq E$ for at least $(1-\rho)m$ of the queries. With probability at least $1-\beta$, every such c has $\operatorname{dist}(c,d) \leq 64E^2$.

Proof.

1. The true vector d meets the condition, so a candidate exists.
2. Fix c with $\operatorname{dist}(c,d) > 64E^2$ and let $V_j = \mathbf{1}[|\langle S_j, c-d\rangle| > 2E]$ indicate that query j exposes c. As in step 2 of Theorem 2.2, the V_j are independent with mean at least $3/32$. Hoeffding's inequality gives $\Pr\left[\sum_j V_j < 3m/64\right] \leq \exp(-2m(3/64)^2) = e^{-9m/2048}$.
3. A union bound over at most 2^n candidates shows that, with probability at least $1 - 2^n e^{-9m/2048} \geq 1-\beta$, every far candidate has $\sum_j V_j \geq 3m/64$.
4. Suppose this holds, and let c be an output. The queries where c is consistent and those answered with error at most E each number at least $(1-\rho)m$, so at least $(1-2\rho)m$ queries are of both kinds. On each of them, the triangle inequality gives $|\langle S_j, c-d\rangle| \leq 2E$, so $V_j = 0$. Hence $\sum_j V_j \leq 2\rho m < 3m/64$, and c is not far.

□

These attacks are nearly tight. The chapters on approximate differential privacy and advanced composition show that a curator can answer m subset queries with error of order $\sqrt{m}$ and still meet the definition of differential privacy, up to logarithmic factors and factors that depend on the privacy parameters. So for n queries, the line between blatant non-privacy and a formal guarantee runs at error $\sqrt{n}$.

2.4. Reconstruction in practice

The random-query attack seems theoretical, because it asks about arbitrary sets of people, but it broke a deployed system. The company Aircloak built Diffix, an interface that let analysts ask an unlimited number of counting queries on sensitive data. It promised privacy with less error than differential privacy adds.

Diffix had defences against the attacks above. Each condition in a query added its own Gaussian noise with standard deviation 1, so the error grew with the number of conditions. The interface also rejected conditions joined by OR, suppressed small counts and altered extreme values. A

random subset of n people needs about $n/2$ conditions joined by OR to describe person by person, so both defences block that route.

Cohen and Nissim reversed the order of the attack. Instead of a random set that they then had to describe, they drew a random single condition from a family whose sets look random. Each condition tests a digit of a power of each person's numeric identifier. One query they sent kept a row when

$$\left\lfloor 100\,(2 \cdot \text{identifier})^{0.7} + 0.5 \right\rfloor = \left\lfloor 100\,(2 \cdot \text{identifier})^{0.7} \right\rfloor, \tag{2.12}$$

which holds exactly when the third decimal digit of $(2 \cdot \text{identifier})^{0.7}$ is at most 4. Of the 1,001 identifiers from 2000 to 3000, it keeps 501, about half, as a random subset would.

Each such query has few conditions, so its error does not grow with the number of rows. Diffix added noise of standard deviation 4 to it. Cohen and Nissim targeted four ranges of identifiers with 73 to 142 rows each and asked 3,500 queries about each range. Gaussian noise is not bounded, so they used the linear program that minimizes the total error, which tolerates a few large errors as Proposition 2.2 does. It recovered the secret attribute of every row in all four ranges.

The theorems of this chapter do not apply to these queries as stated, because a fixed formula is not a random subset. But the proofs use the randomness of the queries only through Lemma 2.1, by which each query exposes a fixed wrong candidate with constant probability. The digit tests behave that way on real identifiers, as an empirical fact and not a theorem.

2.5. Summary

Table 2.1.: The chapter's results.

Result	Statement
Example 2.1	a median and a mean cut 341,376 possible ages to 36
Definition 2.2	an attacker recovers all but $o(n)$ secret bits with probability tending to 1
Theorem 2.1	all 2^n queries with error E reveal all but $4E$ bits
Proposition 2.1	error E can hide $E/2$ bits from any attacker
Lemma 2.1	a random subset has $\lvert\langle S, w\rangle\rvert \geq \lVert w\rVert_2/4$ with probability at least 3/32
Theorem 2.2	$\frac{32}{3}(n \ln 2 + \ln \frac{1}{\beta})$ random queries with error E reveal all but $64E^2$ bits
Theorem 2.3	a linear program does the same in polynomial time with $O(n \log n)$ queries, all but $676E^2$ bits
Proposition 2.2	a fraction below 3/128 of arbitrary answers does not stop the attack

2.6. References

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- Cynthia Dwork, Frank McSherry and Kunal Talwar, “The Price of Privacy and the Limits of LP Decoding”, 2007. Gives the LP attack with $O(n)$ queries that tolerates some arbitrary answers.
- Cynthia Dwork and Aaron Roth, *The Algorithmic Foundations of Differential Privacy*, 2014. Presents both attacks in Section 8.1.
- Raymond Paley and Antoni Zygmund, “A Note on Analytic Functions in the Unit Circle”, 1932. Gives the inequality in Lemma 2.1.
- Aloni Cohen and Kobbi Nissim, “Linear Program Reconstruction in Practice”, 2020. Breaks Diffix with digit-test queries.

Part II.

The Basics of Differential Privacy

The first part showed that statistics about people can expose them. This part gives the answer that the rest of the course builds on, the definition of differential privacy (DP). Under DP, a change to one person's data changes the probability of any outcome by at most a factor e^{ε}, for a small privacy parameter ε. The part then builds the first private algorithms and the rules that combine them. Every later part uses these results, so each chapter gives exact constants and full proofs.

Table 2.2.: The chapters of this part and the questions they answer.

Chapter	Question it answers
Randomized response and the definition	How can one collect a sensitive bit from each person, and what does it mean for an algorithm to be private?
The Laplace mechanism	How much noise makes a numeric statistic private, and how do private algorithms combine?
Approximate differential privacy	What does a small additive slack δ allow, how small must it be, and what does the Gaussian mechanism gain?
Advanced composition	Why does the privacy cost of k analyses grow only like $\sqrt{k}$?
The exponential mechanism	How can one privately choose the best object from a set?

3. Randomized Response and the Definition

A group of people each hold one sensitive bit, and an analyst wants the fraction of ones without learning anyone's bit. This chapter solves the problem with a biased coin, the oldest private algorithm. It then turns the idea behind the coin, plausible deniability, into the definition of differential privacy (DP), on which the rest of the course builds.

3.1. Randomized response

Write n for the number of people. Person number i holds a bit $x_i \in \{0, 1\}$, for instance whether they cheated on an exam, and the analyst wants the fraction of ones,

$$p = \frac{1}{n} \sum_{i=1}^{n} x_i. \tag{3.1}$$

Each person sends the analyst one message Y_i, which may depend on x_i and on coins that only person i sees. At one extreme, if $Y_i = x_i$, the analyst computes p exactly but sees every bit. At the other, if Y_i is a fair coin independent of x_i, the analyst learns nothing about anyone. But the analyst also learns nothing about p, because the average of the reports has the distribution $\frac{1}{n}\mathrm{Bin}(n, \frac{1}{2})$ whatever the data. Randomized response goes between the two.

Definition 3.1 (Randomized response)**.** For a bias $\kappa \in [0, 1/2]$, which sets the probability $\frac{1}{2} + \kappa$ of reporting the truth, person i reports

$$Y_i = \begin{cases} x_i & \text{with probability } \frac{1}{2} + \kappa, \\ 1 - x_i & \text{with probability } \frac{1}{2} - \kappa, \end{cases} \tag{3.2}$$

with fresh, independent coins for each person.

The value $\kappa = 1/2$ gives the honest protocol and $\kappa = 0$ the fair coin. In between, say at $\kappa = 1/4$, a person who reports $Y_i = 1$ can claim that their bit was 0 and the coin flipped it, which happens with probability $1/4$. That plausible deniability is the privacy, and it grows as κ gets smaller.

The reports lean towards the truth, so their average is not p. But the analyst knows the bias and can undo it, because the average of one report,

$$\mathbb{E}[Y_i] = \left(\tfrac{1}{2} + \kappa\right) x_i + \left(\tfrac{1}{2} - \kappa\right)(1 - x_i) = 2\kappa x_i + \tfrac{1}{2} - \kappa, \tag{3.3}$$

is linear in x_i. Solving for x_i gives an unbiased estimate of each bit, and the average of those estimates is the estimator of p.

Definition 3.2 (Debiased estimator)**.** For $\kappa > 0$, the analyst outputs

$$\tilde{p} = \frac{1}{n} \sum_{i=1}^{n} \frac{Y_i - \frac{1}{2} + \kappa}{2\kappa}. \tag{3.4}$$

The next theorem checks that the estimator is right on average and measures, through its variance, how far it typically strays.

Theorem 3.1. *For every $\kappa \in (0, 1/2]$ and every input $D \in \{0,1\}^n$, the estimator is unbiased, $\mathbb{E}[\tilde{p}] = p$, and its variance is*

$$\mathrm{Var}(\tilde{p}) = \frac{1 - 4\kappa^2}{16\kappa^2 n} \leq \frac{1}{16\kappa^2 n}. \tag{3.5}$$

Proof.

1. Each term $Z_i = (Y_i - \frac{1}{2} + \kappa)/(2\kappa)$ has mean x_i, by the formula for $\mathbb{E}[Y_i]$ above. Averaging over i and using linearity of expectation gives $\mathbb{E}[\tilde{p}] = p$.
2. The constants $\frac{1}{2} - \kappa$ and 2κ do not change the variance except through the scaling, so $\mathrm{Var}(Z_i) = \mathrm{Var}(Y_i)/(4\kappa^2)$.
3. The report Y_i is a Bernoulli variable that equals 1 with probability $\frac{1}{2} + \kappa$ or $\frac{1}{2} - \kappa$, depending on x_i. In both cases its variance is $(\frac{1}{2} + \kappa)(\frac{1}{2} - \kappa) = \frac{1}{4} - \kappa^2$.
4. The reports are independent because each person uses their own coins. So the variance of the sum is the sum of the variances, and

$$\mathrm{Var}(\tilde{p}) = \frac{1}{n^2} \sum_{i=1}^{n} \frac{\mathrm{Var}(Y_i)}{4\kappa^2} = \frac{\frac{1}{4} - \kappa^2}{4\kappa^2 n} = \frac{1 - 4\kappa^2}{16\kappa^2 n}. \tag{3.6}$$

□

By Chebyshev's inequality, the error is at most two standard deviations, $|\tilde{p} - p| \leq 1/(2\kappa\sqrt{n})$, with probability at least $3/4$. For a small failure probability β, Hoeffding's inequality does better, because each Z_i takes only two values.

Theorem 3.2. *For every $\kappa \in (0, 1/2]$ and $\beta \in (0, 1)$, with probability at least $1 - \beta$ over the coins,*

$$|\tilde{p} - p| \leq \frac{1}{2\kappa} \sqrt{\frac{\ln(2/\beta)}{2n}}. \tag{3.7}$$

Proof.

1. The term Z_i takes the values $(\frac{1}{2} + \kappa)/(2\kappa)$ and $-(\frac{1}{2} - \kappa)/(2\kappa)$, so it lies in an interval of length $1/(2\kappa)$.
2. The Z_i are independent with mean x_i. Hoeffding's inequality for the average of n independent variables, each in an interval of length ℓ, gives

$$\Pr\left[\,|\tilde{p}-p|\geq t\,\right]\leq 2\exp\!\left(-\frac{2nt^2}{\ell^2}\right)=2\exp(-8\kappa^2 nt^2)\,. \tag{3.8}$$

3. Setting the right-hand side equal to β and solving for t gives Equation 3.7.

□

So the error is of order $1/(\kappa\sqrt{n})$ and goes to 0 as n grows. Read the other way, an error of at most α with probability $1-\beta$ needs

$$n\geq\frac{\ln(2/\beta)}{8\kappa^2\alpha^2} \tag{3.9}$$

people. A smaller κ hides each bit better and needs more people for the same accuracy. To say how much better it hides a bit, the next section gives a definition.

3.2. Differential privacy

The setting is the **central model**. Each of n people sends their data point to a **trusted curator**, and they trust no one else. The curator runs an algorithm M on the dataset $D=(x_1,\ldots,x_n)\in\mathcal{X}^n$ and publishes the output, and anyone who sees the output may be an adversary. The guarantee is a property of the algorithm, called a **mechanism** in this field: no single person's data has a large effect on the distribution of the output. To make "one person's data" precise, compare two datasets that differ only in one person's record.

Definition 3.3 (Neighbouring datasets)**.** Two datasets $D, D'\in\mathcal{X}^n$ are **neighbours**, written $D\sim D'$, if they differ in exactly one record. This is the **replace-one** relation.

The definition asks that the distribution of the output barely change between neighbours. On every set of outputs, the probability may grow by at most a factor e^{ε}, where the privacy parameter ε sets how much change is allowed.

Definition 3.4 (Differential privacy)**.** Let $\varepsilon\geq 0$. A randomized algorithm $M\colon\mathcal{X}^n\to\mathcal{Y}$, which maps a dataset to an output in a set $\mathcal{Y}$, is **ε-differentially private** (ε-DP) if for all neighbours $D\sim D'$ and all events $S\subseteq\mathcal{Y}$,

$$\Pr[M(D)\in S]\leq e^{\varepsilon}\,\Pr[M(D')\in S]. \tag{3.10}$$

The probability is over the coins of M only. The datasets are fixed.

Its other name is **pure** differential privacy, which sets it apart from the relaxation of a later chapter. Four of its features matter later.

First, the definition is quantitative. At $\varepsilon = 0$ the output distribution is the same on every dataset, which is perfect privacy and allows no accuracy. An algorithm that is not randomized is ∞-DP, which is no guarantee at all. Typical values of ε lie between about 0.1 and 5, and smaller is more private. Because the bound is e^{ε}, privacy weakens quickly once ε is above 1.

Second, it is a worst-case guarantee: it holds for every pair of neighbours, however unlikely the data, with no distribution over datasets.

Third, it is symmetric. Swapping D and D' in Equation 3.10 gives the lower bound $\Pr[M(D) \in S] \geq e^{-\varepsilon} \Pr[M(D') \in S]$.

Fourth, it bounds a ratio, not a difference. For small ε the factor e^{ε} is close to $1 + \varepsilon$. The exponential form makes factors multiply when guarantees are chained, $e^{\varepsilon_1} e^{\varepsilon_2} = e^{\varepsilon_1 + \varepsilon_2}$, which the next chapter uses for group privacy. An additive notion of closeness, such as a small total variation distance, sounds natural but gives no meaningful guarantee.

DP with these neighbours, which replace one record, has the name **bounded** DP. The **add/remove** relation, or **unbounded** DP, instead adds or removes one record. A replacement is a removal followed by an addition. So, by the chaining argument of the next chapter's group privacy, ε-DP under add/remove neighbours gives 2ε-DP under replace-one neighbours. The converse can fail, because under add/remove neighbours the size n is itself private. The course uses replacement, which is more convenient mathematically, and you can compare two mechanisms only under the same relation.

For discrete outputs it is enough to check single outcomes, because the probability of an event is the sum of the probabilities of its outcomes.

Lemma 3.1. *Let $\mathcal{Y}$ be countable. If $\Pr[M(D) = y] \leq e^{\varepsilon} \Pr[M(D') = y]$ for all neighbours $D \sim D'$ and all outputs $y \in \mathcal{Y}$, then M is ε-DP.*

Proof. For an event S, sum the pointwise inequality over the outputs $y \in S$:

$$\Pr[M(D) \in S] = \sum_{y \in S} \Pr[M(D) = y] \leq e^{\varepsilon} \sum_{y \in S} \Pr[M(D') = y] = e^{\varepsilon} \Pr[M(D') \in S]. \tag{3.11}$$

□

A first consequence of the definition is that privacy needs randomness.

Proposition 3.1. *For $\varepsilon < \infty$, a deterministic ε-DP algorithm is constant.*

Proof.

1. Let M be deterministic and $D \sim D'$. Take the event $S = \{M(D)\}$. The definition gives $1 = \Pr[M(D) \in S] \leq e^{\varepsilon} \Pr[M(D') \in S]$, so $\Pr[M(D') = M(D)] > 0$. Since $M(D')$ is a single value, $M(D') = M(D)$.
2. Any two datasets in $\mathcal{X}^n$ are joined by a path of at most n neighbours, changing one record at a time. By step 1, M takes the same value at every step of the path, so it is constant.

□

3.2.1. Hypothesis testing

The definition has an operational meaning from the adversary's side. Suppose an adversary knows that the dataset is one of two neighbours, D or D', and sees the output of M. It must decide between the **null hypothesis** and the **alternative**,

$$H_0 \colon \text{the dataset is } D, \qquad H_1 \colon \text{the dataset is } D'. \tag{3.12}$$

A test is a function $\phi \colon \mathcal{Y} \to [0,1]$, the probability of answering H_1 given the output, so a test can be random. The test's **false positive rate** FPR is the chance of answering H_1 when H_0 holds, and its **false negative rate** FNR is the chance of answering H_0 when H_1 holds:

$$\mathrm{FPR} = \mathbb{E}[\phi(M(D))], \qquad \mathrm{FNR} = 1 - \mathbb{E}[\phi(M(D'))]. \tag{3.13}$$

A test takes values in $[0,1]$, while the definition bounds only probabilities of events. The proof of the next theorem needs the bound for such functions, which the following lemma gives.

Lemma 3.2. *Let M be ε-DP and let $h \colon \mathcal{Y} \to [0,1]$. For all neighbours $D \sim D'$,*

$$\mathbb{E}[h(M(D))] \le e^{\varepsilon}\, \mathbb{E}[h(M(D'))]. \tag{3.14}$$

Proof.

1. A variable with values in $[0,1]$ has mean equal to the integral of its tail, so $\mathbb{E}[h(M(D))] = \int_0^1 \Pr[h(M(D)) > t]\, \mathrm{d}t$.
2. Each tail is the probability of the event $\{y : h(y) > t\}$, which Equation 3.10 bounds by $e^{\varepsilon} \Pr[h(M(D')) > t]$.
3. Integrating over $t \in [0,1]$ gives $e^{\varepsilon}\, \mathbb{E}[h(M(D'))]$.

□

An adversary wants both error rates small. The next theorem says that differential privacy is exactly a limit on how small they can be together, for every test.

Theorem 3.3. *A mechanism M is ε-DP if and only if, for every pair of neighbours and every test, the false positive rate* FPR *and the false negative rate* FNR *satisfy*

$$\mathrm{FPR} + e^{\varepsilon}\mathrm{FNR} \ge 1 \qquad \textit{and} \qquad e^{\varepsilon}\mathrm{FPR} + \mathrm{FNR} \ge 1. \tag{3.15}$$

Proof.

1. *DP extends from events to tests.* Let M be ε-DP and let ϕ take values in $[0,1]$. Then $\mathbb{E}[\phi(M(D))] \le e^{\varepsilon}\,\mathbb{E}[\phi(M(D'))]$, and the same holds with D and D' swapped. This is Lemma 3.2, applied to ϕ in each direction.
2. *DP implies the inequalities.* Apply step 1 to ϕ, from D' to D: $1-\mathrm{FNR} = \mathbb{E}[\phi(M(D'))] \le e^{\varepsilon}\,\mathbb{E}[\phi(M(D))] = e^{\varepsilon}\mathrm{FPR}$. This is the second inequality. Apply step 1 to $1-\phi$, from D to D': $1-\mathrm{FPR} = \mathbb{E}[1-\phi(M(D))] \le e^{\varepsilon}\,\mathbb{E}[1-\phi(M(D'))] = e^{\varepsilon}\mathrm{FNR}$. This is the first.
3. *The inequalities imply DP.* Take an event S and the test $\phi(y) = \mathbf{1}[y \in S]$, which answers H_1 exactly when the output lands in S. Its error rates are $\mathrm{FPR} = \Pr[M(D) \in S]$ and $\mathrm{FNR} = 1 - \Pr[M(D') \in S]$. The second inequality says $e^{\varepsilon}\Pr[M(D) \in S] + 1 - \Pr[M(D') \in S] \ge 1$, that is, $\Pr[M(D') \in S] \le e^{\varepsilon}\Pr[M(D) \in S]$. The pair of neighbours was arbitrary, so the same holds with the roles of D and D' swapped, which is Equation 3.10.

□

At $\varepsilon = 0$ the two inequalities force $\mathrm{FPR} + \mathrm{FNR} \ge 1$, which a test that ignores the output already achieves, such as a random guess ($\mathrm{FPR} = \mathrm{FNR} = 1/2$). As ε grows, the region of possible error pairs gets larger, and the adversary can beat a blind guess by more.

3.2.2. What the definition promises

In words, the probability of any event is about the same whether a given person's data is in the dataset or is replaced by someone else's. So a harm that comes from the output is not much more likely because the person took part. A linkage attack, for example, succeeds about as often without their record, whatever auxiliary data exists now or later. At $\varepsilon = 0.1$, the chance of any bad event grows by a factor of at most $e^{0.1} \approx 1.105$. The guarantee holds against an adversary with unlimited computation and side information, so it is information theoretic, unlike most of cryptography.

The definition does not prevent inference from aggregate patterns. Suppose a study with differential privacy finds that smoking causes cancer, and an insurer then raises the premium of a smoker who took part. The study's conclusion would be the same without that person's record, so the harm comes from a fact about the population, not from their participation. Learning such facts is the purpose of statistics.

3.3. Randomized response revisited

Randomized response now gets a privacy guarantee. The theorem treats the whole vector of reports as the output, and the estimator Equation 3.4, a function of that vector, inherits the guarantee by post-processing, which the next chapter proves.

Theorem 3.4. *Randomized response with bias $\kappa \in [0, 1/2)$, releasing $(Y_1, \ldots, Y_n)$, is ε-DP for*

$$\varepsilon = \ln \frac{\frac{1}{2} + \kappa}{\frac{1}{2} - \kappa}. \tag{3.16}$$

Equivalently, to obtain ε-DP, report the truth with probability $\frac{1}{2} + \kappa = e^{\varepsilon}/(1 + e^{\varepsilon})$.

Proof.

1. Fix an output $y \in \{0,1\}^n$. The coins are independent across people, so $\Pr[M(D) = y] = \prod_{i=1}^{n} \Pr[Y_i = y_i]$, where each factor depends only on x_i.
2. Let D and D' differ only in record j, and write Y_j' for the report of person j on D'. All factors with $i \neq j$ are equal for the two datasets and cancel in the ratio:

$$\frac{\Pr[M(D) = y]}{\Pr[M(D') = y]} = \frac{\Pr[Y_j = y_j]}{\Pr[Y_j' = y_j]}. \tag{3.17}$$

3. Each probability on the right is $\frac{1}{2} + \kappa$ or $\frac{1}{2} - \kappa$, so the ratio is at most $(\frac{1}{2} + \kappa)/(\frac{1}{2} - \kappa) = e^{\varepsilon}$.
4. Lemma 3.1 turns this pointwise bound into the definition.
5. Solving $e^{\varepsilon} = (\frac{1}{2} + \kappa)/(\frac{1}{2} - \kappa)$ for $\frac{1}{2} + \kappa$ gives $e^{\varepsilon}/(1 + e^{\varepsilon})$.

□

For small κ the privacy parameter is $\varepsilon = 4\kappa + O(\kappa^3)$, so a bias of order ε gives ε-DP. Exactly, the bias that gives ε-DP is

$$2\kappa = \frac{e^{\varepsilon} - 1}{e^{\varepsilon} + 1} = \tanh \frac{\varepsilon}{2}, \tag{3.18}$$

so Theorem 3.2 and Theorem 3.1 become, with probability at least $1 - \beta$,

$$|\tilde{p} - p| \leq \coth\Big(\frac{\varepsilon}{2}\Big) \sqrt{\frac{\ln(2/\beta)}{2n}}, \qquad \mathrm{Var}(\tilde{p}) = \frac{e^{\varepsilon}}{(e^{\varepsilon} - 1)^2\, n}. \tag{3.19}$$

Here p is again the fraction of ones and $\tilde{p}$ its estimate. The factor $\coth(\varepsilon/2)$ lies between $2/\varepsilon$ and $2/\varepsilon + \varepsilon/6$ for every $\varepsilon > 0$, so the error is $O(1/(\varepsilon\sqrt{n}))$ for $\varepsilon \leq 1$. Figure 3.1 compares this bound with simulation, and Example 3.1 gives the sizes at a typical privacy level.

Example 3.1. With $\varepsilon = 1$, each person tells the truth with probability $e/(1 + e) = 0.7311$. With $n = 10{,}000$ and $\beta = 0.05$, the bound Equation 3.19 is $\coth(0.5)\sqrt{\ln 40/20{,}000} = 2.1640 \times 0.01358 = 0.0294$. The standard deviation from Equation 3.19 is $\sqrt{e/((e-1)^2 \cdot 10{,}000)} = 0.0096$.

Randomized response gives more than the definition asks for. Each person randomizes their own bit before sending it, so no one ever sees a true bit and the curator needs no trust. This is **local differential privacy**, the subject of a later chapter, and it implies the central guarantee proved here. The price is accuracy, because the $\sqrt{n}$ in the error is not necessary under central privacy: the next chapter's Laplace mechanism estimates p with error $O(1/(\varepsilon n))$.

3.4. Summary

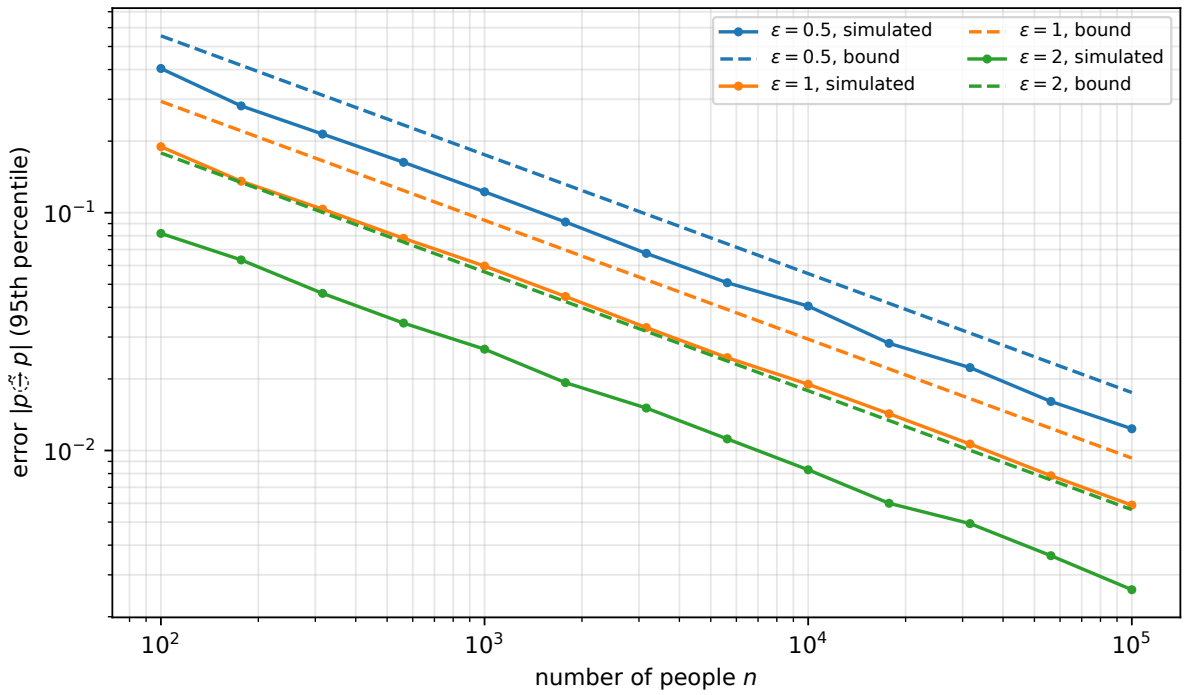


Figure 3.1.: Error of randomized response. For each ε, the solid line is the 95th percentile of $|\tilde{p} - p|$ over 2,000 simulated runs with $p = 0.3$, and the dashed line is the bound of Equation 3.19 with $\beta = 0.05$. Both fall as $1/\sqrt{n}$. The bound is 1.4 to 2.2 times the simulated error, and it is loosest for large ε, because Hoeffding's inequality uses only the range of each term and not its smaller variance.

Table 3.1.: The chapter's results.

Result	Statement
Theorem 3.2	randomized response with bias κ has error $\frac{1}{2\kappa}\sqrt{\ln(2/\beta)/(2n)}$
Definition 3.4	$\Pr[M(D) \in S] \leq e^{\varepsilon} \Pr[M(D') \in S]$ for all neighbours and events
Lemma 3.1	for discrete outputs, single outcomes suffice
Proposition 3.1	a deterministic private algorithm is constant
Lemma 3.2	the bound extends from events to functions with values in $[0, 1]$
Theorem 3.3	ε-DP holds exactly when every test has $\mathrm{FPR} + e^{\varepsilon}\mathrm{FNR} \geq 1$ and $e^{\varepsilon}\mathrm{FPR} + \mathrm{FNR} \geq 1$
Theorem 3.4	truth with probability $e^{\varepsilon}/(1 + e^{\varepsilon})$ gives ε-DP
Equation 3.19	error $\coth(\varepsilon/2)\sqrt{\ln(2/\beta)/(2n)} = O(1/(\varepsilon\sqrt{n}))$

3.5. References

- Stanley L. Warner, "Randomized Response: A Survey Technique for Eliminating Evasive Answer Bias", 1965. Gives the original protocol, for surveys on sensitive questions.
- Cynthia Dwork, Frank McSherry, Kobbi Nissim and Adam Smith, "Calibrating Noise to Sensitivity in Private Data Analysis", 2006. Defines differential privacy.
- Larry Wasserman and Shuheng Zhou, "A Statistical Framework for Differential Privacy", 2010. Gives the hypothesis-testing reading of the definition.
- Cynthia Dwork and Aaron Roth, *The Algorithmic Foundations of Differential Privacy*, 2014. Explains the definition and its meaning, with the smoking example.
- Salil Vadhan, "The Complexity of Differential Privacy", 2017. Shows in Section 1.6 why total variation distance gives no meaningful guarantee.
- Thomas Steinke and Jonathan Ullman, "The Pitfalls of Average-Case Differential Privacy", 2020, a post on the DifferentialPrivacy.org blog. Explains why the worst case over datasets matters.

4. The Laplace Mechanism

Randomized response privatizes one bit per person. But most questions an analyst asks are numbers computed from the whole dataset, such as a count or a histogram. This chapter answers them with the Laplace mechanism, which computes the true answer and adds noise scaled to how much one person can change it. On the fraction of ones with privacy parameter ε, its error is of order $1/(\varepsilon n)$ rather than the $1/(\varepsilon\sqrt{n})$ of randomized response. The chapter then proves three properties of differential privacy (DP) that let you build private algorithms from private parts: post-processing, group privacy and basic composition.

4.1. Sensitivity

The output must hide the contribution of any one person, so the noise must be at least as large as the change that one person can cause in the answer. The sensitivity measures the largest such change. Recall from the randomized response chapter that two datasets are neighbours, written $D \sim D'$, when they differ in exactly one record.

Definition 4.1 ($\ell 1$ sensitivity)**.** Let $f\colon \mathcal{X}^n \to \mathbb{R}^k$ map a dataset of n records to a vector of d numbers. The ℓ_1 **sensitivity** of f is the largest change in f, measured by the sum of the absolute changes of its coordinates, between neighbours $D \sim D'$:

$$\Delta_1 f = \sup_{D \sim D'} \|f(D) - f(D')\|_1, \tag{4.1}$$

where $\|v\|_1 = \sum_{j=1}^{d} |v_j|$ for a vector $v \in \mathbb{R}^d$.

The ℓ_1 norm is what makes the privacy proof of Theorem 4.1 work. Other mechanisms use other norms, such as the ℓ_2 norm for the Gaussian mechanism of the next chapter. For $d = 1$ all these norms are the absolute value, and for $d > 1$ the ℓ_1 and ℓ_2 sensitivities can differ by a factor of up to $\sqrt{d}$.

The running example of this chapter is the fraction of ones that randomized response estimates. Its sensitivity is small, because one person moves the fraction only a little.

Example 4.1. Let $\mathcal{X} = \{0, 1\}$ and let $f(D) = \frac{1}{n}\sum_{i=1}^{n} x_i$ be the fraction of ones. Changing one bit changes the sum by at most 1, so $\Delta_1 f = 1/n$.

4.2. The Laplace distribution

The noise must hide a shift of the true answer by up to the sensitivity. A density that falls exponentially in the distance from zero does this, because a shift by a fixed amount changes its value by at most a fixed factor. The Laplace distribution is the simplest such density.

Definition 4.2 (Laplace distribution)**.** The **Laplace distribution** $\mathrm{Lap}(b)$ with scale $b > 0$ has density

$$\frac{1}{2b}\exp\left(-\frac{|x|}{b}\right), \qquad x \in \mathbb{R}. \tag{4.2}$$

It is a two-sided exponential distribution: the exponential density, proportional to $e^{-x/b}$ on $[0, \infty)$, reflected to the negative half-line. Its tails decay like $e^{-|x|/b}$, more slowly than a Gaussian's but still exponentially. The accuracy of the mechanism rests on three facts about this distribution: its mean, its variance and the exact size of its tails.

Lemma 4.1. *Let $Z \sim \mathrm{Lap}(b)$. Then $\mathbb{E}[Z] = 0$, $\mathrm{Var}(Z) = 2b^2$, and for every $t \geq 0$,*

$$\Pr[|Z| \geq tb] = e^{-t}. \tag{4.3}$$

Proof.

1. The density is symmetric about 0, so the mean is 0.
2. The absolute value $|Z|$ has density $\frac{1}{b}e^{-u/b}$ on $[0, \infty)$, the exponential distribution with mean b. Its second moment is $\int_0^\infty u^2 \frac{1}{b} e^{-u/b}\,\mathrm{d}u = 2b^2$, by integrating by parts twice. Since the mean of Z is 0, $\mathrm{Var}(Z) = \mathbb{E}[Z^2] = 2b^2$.
3. The tail of $|Z|$ is $\int_{tb}^\infty \frac{1}{b} e^{-u/b}\,\mathrm{d}u = e^{-t}$.

□

4.3. The Laplace mechanism

The mechanism adds independent Laplace noise to each coordinate, with scale equal to the sensitivity divided by ε.

Definition 4.3 (Laplace mechanism)**.** For $f\colon \mathcal{X}^n \to \mathbb{R}^d$ with $0 < \Delta_1 f < \infty$ and for $\varepsilon > 0$, the Laplace mechanism outputs

$$M(D) = f(D) + (Z_1, \ldots, Z_d), \tag{4.4}$$

where $Z_1, \ldots, Z_d$ are independent with distribution $\mathrm{Lap}(\Delta_1 f/\varepsilon)$.

Before the privacy proof, consider the accuracy of the mechanism on the running example. The mechanism is unbiased, and Lemma 4.1 gives its error.

Example 4.2. For $f(D) = p$, the fraction of ones, Example 4.1 gives $\Delta_1 f = 1/n$. The mechanism outputs $\tilde{p} = p + Z$ with $Z \sim \mathrm{Lap}(1/(\varepsilon n))$. So $\mathbb{E}[\tilde{p}] = p$, and $\mathrm{Var}(\tilde{p}) = 2/(\varepsilon n)^2$. By Equation 4.3 with $t = \ln(1/\beta)$, with probability at least $1 - \beta$,

$$|\tilde{p} - p| \leq \frac{\ln(1/\beta)}{\varepsilon n}. \tag{4.5}$$

With $\varepsilon = 1$, $n = 10{,}000$ and $\beta = 0.05$, this is $\ln(20)/10{,}000 = 3.0 \times 10^{-4}$. Randomized response with the same ε, n and β has the bound 0.0294, which is 98 times larger.

The gap grows with n, because randomized response has error of order $1/(\varepsilon\sqrt{n})$ and the Laplace mechanism of order $1/(\varepsilon n)$. The price of the smaller error is a trusted curator who sees the true data, which randomized response does not need. It remains to prove that the mechanism is private.

Theorem 4.1 (Privacy of the Laplace mechanism)**.** *For every $f\colon \mathcal{X}^n \to \mathbb{R}^d$ with $0 < \Delta_1 f < \infty$ and every $\varepsilon > 0$, the Laplace mechanism is ε-DP.*

Proof.

1. Fix neighbours $D \sim D'$, and write p and q for the densities of $M(D)$ and $M(D')$ on $\mathbb{R}^d$. In this proof p is a density, not the fraction of ones. Let $b = \Delta_1 f/\varepsilon$. The coordinates are independent, so at a point $y \in \mathbb{R}^d$,

$$\frac{p(y)}{q(y)} = \prod_{j=1}^{d} \frac{\exp(-|f(D)_j - y_j|/b)}{\exp(-|f(D')_j - y_j|/b)} = \exp\left(\frac{\varepsilon}{\Delta_1 f} \sum_{j=1}^{d} \left(|f(D')_j - y_j| - |f(D)_j - y_j| \right) \right). \tag{4.6}$$

2. By the triangle inequality, $|f(D')_j - y_j| - |f(D)_j - y_j| \leq |f(D')_j - f(D)_j|$ for each j. Summing over j gives at most $\|f(D) - f(D')\|_1$, which is at most $\Delta_1 f$ by Definition 4.1. So $p(y) \leq e^{\varepsilon} q(y)$ for every y.
3. For an event $S \subseteq \mathbb{R}^d$, integrate the pointwise bound over S: $\Pr[M(D) \in S] = \int_S p \leq e^{\varepsilon} \int_S q = e^{\varepsilon} \Pr[M(D') \in S]$.

□

The proof shows why the Laplace distribution fits the definition. Its density falls exponentially in the distance from the true value, and the definition bounds a ratio by an exponential. So a shift of the centre by the sensitivity changes the density by a factor of at most e^{ε}. Figure 4.1 shows the two densities for a one-dimensional statistic.

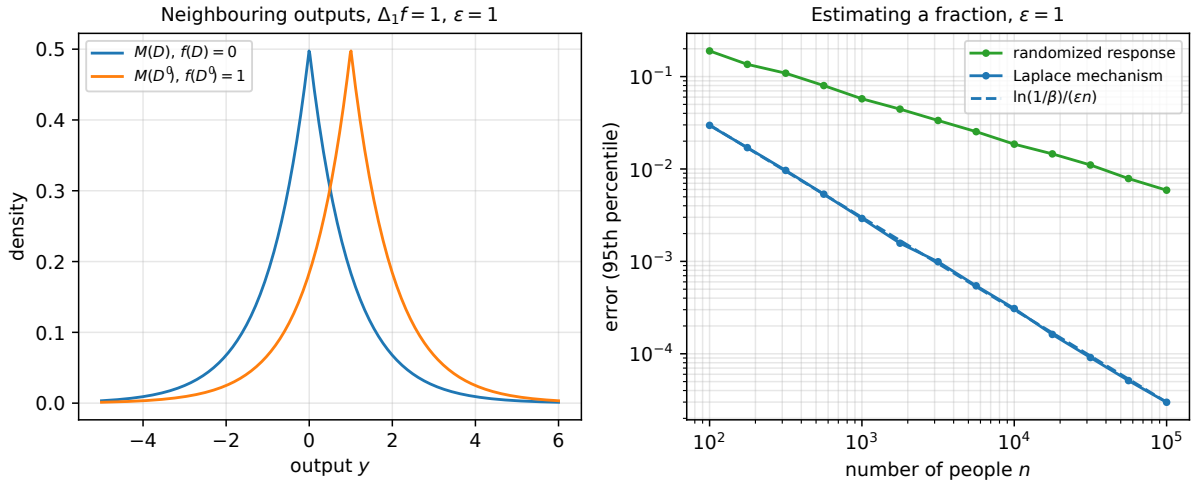


Figure 4.1.: Left, the densities of the Laplace mechanism at $\varepsilon = 1$ on two neighbours whose values differ by the sensitivity $\Delta_1 f = 1$. Their ratio stays between e^{-1} and e. Right, the 95th percentile of the error in estimating a fraction of ones, over 2,000 simulated runs with $p = 0.3$. Randomized response falls as $1/\sqrt{n}$ and the Laplace mechanism as $1/n$. The dashed line is the exact 95th percentile $\ln(20)/(\varepsilon n)$ from Equation 4.3.

4.4. Counting queries

A **counting query** asks how many people in the dataset have some property. It is given by a predicate $\phi\colon \mathcal{X} \to \{0,1\}$ and has the value $f(D) = \sum_{i=1}^{n} \phi(x_i)$. One record changes the count by at most 1, so $\Delta_1 f = 1$, and the Laplace mechanism adds $\mathrm{Lap}(1/\varepsilon)$. By Equation 4.3, its error is at most $\ln(1/\beta)/\varepsilon$ with probability $1 - \beta$, whatever the size of the dataset.

Often an analyst asks many counting queries $f = (f_1, \dots, f_k)$, with predicates $\phi_1, \dots, \phi_k$, on the same data, such as whether a person smokes and whether their income is above a threshold. One person can then change all of the answers at once.

Proposition 4.1. *The vector of k counting queries has ℓ_1 sensitivity at most k, with equality when some record satisfies every predicate and another satisfies none. For every $\beta \in (0,1)$, with probability at least $1 - \beta$, the Laplace mechanism answers all k queries with error at most*

$$\frac{k \ln(k/\beta)}{\varepsilon}. \tag{4.7}$$

Proof.

1. Let D and D' differ in one record, x in D and x' in D'. In each query the other terms cancel, so $\|f(D) - f(D')\|_1 = \sum_{j=1}^{k} |\phi_j(x) - \phi_j(x')| \le k$. The sum equals k when $\phi_j(x) = 1$ and $\phi_j(x') = 0$ for all j.
2. The noise in each answer is $\mathrm{Lap}(k/\varepsilon)$. By Equation 4.3 with $t = \ln(k/\beta)$, each noise is at least $k \ln(k/\beta)/\varepsilon$ in absolute value with probability β/k.
3. The union bound over the k answers gives a failure probability of at most β.

□

This analysis needs all k queries to be fixed in advance. The composition theorem of Section 4.6 gives the same privacy when each query is chosen after seeing the earlier answers.

Compare this error of order k/ε per query with the reconstruction chapter, where about n counting queries answered with error $o(\sqrt{n})$ let an attacker recover almost the whole dataset. Here, n counting queries get error of order n/ε, up to a logarithm. The wide gap between $\sqrt{n}$ and n is real slack: advanced composition, under a relaxation of the definition, lowers the error to about $\sqrt{k}/\varepsilon$ per query, up to logarithmic factors.

4.5. Histograms

When the queries have structure, a sharper sensitivity bound is possible. In a **histogram**, each record falls into exactly one of k bins, such as an age in whole years, and the query counts the records in each bin: $f_j(D) = |\{i : x_i \text{ is in bin } j\}|$.

Proposition 4.2. *A histogram with $k \geq 2$ bins has ℓ_1 sensitivity 2. For every $\beta \in (0, 1)$, with probability at least $1 - \beta$, the Laplace mechanism, which adds $\mathrm{Lap}(2/\varepsilon)$ to each bin, answers every bin with error at most*

$$\frac{2\ln(k/\beta)}{\varepsilon}. \tag{4.8}$$

Proof.

1. Let D and D' differ in one record, which lies in bin a in D and in bin a' in D'. If $a = a'$ the histograms are equal. Otherwise count a drops by 1 and count a' rises by 1, so $f(D) - f(D') = e_a - e_{a'}$, where e_j is the j-th unit vector, and $\|e_a - e_{a'}\|_1 = 2$.
2. By Equation 4.3 with $t = \ln(k/\beta)$, each of the k noises is at least $2\ln(k/\beta)/\varepsilon$ in absolute value with probability β/k. The union bound over the bins finishes the proof.

□

The sensitivity is 2 because a replacement moves a record from one bin to another. A single bin has error at most $2\ln(1/\beta)/\varepsilon$ with probability $1 - \beta$, and the proposition pays only $\ln k$ more to control all bins at once. So private histograms over very large domains are practical. For $k = 100$ bins, $\varepsilon = 1$ and $\beta = 0.05$, the bound is $2\ln(2000) = 15.20$, and in a simulation of 20,000 runs the 95th percentile of the largest error is 15.17. The same 100 bins treated as arbitrary counting queries would get the bound $100\ln(2000) = 760.1$ from Proposition 4.1.

4.6. Properties of differential privacy

Three properties let you reason about a private algorithm piece by piece.

4.6.1. Post-processing

No computation on a private output can undo its privacy, as long as the computation does not look at the data again. The randomized response chapter used this to turn the private reports into an estimate.

Theorem 4.2 (Post-processing)**.** *Let $M\colon \mathcal{X}^n \to \mathcal{Y}$ be ε-DP, and let $F\colon \mathcal{Y} \to \mathcal{Z}$ be a randomized map whose coins are independent of those of M. Then $F \circ M$ is ε-DP.*

Proof.

1. A randomized map is a random choice of a deterministic one. Write $F(y) = g_R(y)$, where R is the coins of F and each $g_r\colon \mathcal{Y} \to \mathcal{Z}$ is deterministic.
2. Fix neighbours $D \sim D'$ and an event $S \subseteq \mathcal{Z}$. For fixed coins, $g_R(M(D)) \in S$ exactly when $M(D) \in g_R^{-1}(S)$, so

$$\Pr[F(M(D)) \in S] = \mathbb{E}_R\Big[\Pr\left[M(D) \in g_R^{-1}(S)\right]\Big]. \tag{4.9}$$

3. Inside the expectation, apply the definition of ε-DP to the event $g_R^{-1}(S)$, then undo step 2 for D':

$$\mathbb{E}_R\Big[\Pr\left[M(D) \in g_R^{-1}(S)\right]\Big] \le e^{\varepsilon}\,\mathbb{E}_R\Big[\Pr\left[M(D') \in g_R^{-1}(S)\right]\Big] = e^{\varepsilon}\Pr[F(M(D')) \in S]. \tag{4.10}$$

□

4.6.2. Group privacy

The definition protects one record. Datasets that differ in several records, such as the records of one family, are still protected, with a guarantee that degrades in proportion to their number.

Theorem 4.3 (Group privacy)**.** *Let $M\colon \mathcal{X}^n \to \mathcal{Y}$ be ε-DP, and let $D, D' \in \mathcal{X}^n$ differ in k records. For every event $S \subseteq \mathcal{Y}$,*

$$\Pr[M(D) \in S] \le e^{k\varepsilon}\Pr[M(D') \in S]. \tag{4.11}$$

Proof.

1. The proof is a **hybrid argument**. Change the k differing records one at a time, from their values in D to their values in D'. This gives datasets $D = D^{(0)}, D^{(1)}, \ldots, D^{(k)} = D'$ in which each consecutive pair are neighbours.
2. Apply the definition to each consecutive pair, from the first to the last:

$$\Pr[M(D^{(0)}) \in S] \le e^{\varepsilon} \Pr[M(D^{(1)}) \in S] \le e^{2\varepsilon} \Pr[M(D^{(2)}) \in S] \le \dots \le e^{k\varepsilon} \Pr[M(D^{(k)}) \in S]. \tag{4.12}$$

□

So a group of k people has the protection of $k\varepsilon$-DP, which means something only while $k\varepsilon$ is small. The theorem also relates the two kinds of neighbours of the randomized response chapter. A replacement is a removal followed by an addition, so ε-DP under add/remove neighbours gives 2ε-DP under replace-one neighbours.

4.6.3. Basic composition

An analyst rarely runs one mechanism. Usually each mechanism is chosen in the light of the outputs of the earlier ones and runs on the same dataset.

Definition 4.4 (Adaptive composition)**.** Let $M_1, \dots, M_k$ be mechanisms. For each i and each sequence of earlier outputs $y_{<i} = (y_1, \dots, y_{i-1})$, the mechanism $M_i(\cdot\,; y_{<i})$ takes the dataset D and returns y_i. Their **adaptive composition** M runs them in order on the same D, each with fresh coins, and releases $(y_1, \dots, y_k)$.

Basic composition says that the privacy parameters add up, even though each mechanism may depend on the earlier outputs.

Theorem 4.4 (Basic composition)**.** *If each $M_i(\cdot\,; y_{<i})$ is ε_i-DP for every $y_{<i}$, then the adaptive composition of $M_1, \dots, M_k$ is $(\sum_{i=1}^{k} \varepsilon_i)$-DP. In particular, k mechanisms that are each ε-DP compose to a $k\varepsilon$-DP mechanism.*

The proof is simplest for countable outputs, where the probability of a sequence of outputs factors into conditional probabilities.

Proof. Countable outputs.

1. Fix neighbours $D \sim D'$ and outputs $y = (y_1, \dots, y_k)$. Each M_i uses fresh coins, so the chain rule gives $\Pr[M(D) = y] = \prod_{i=1}^{k} \Pr[M_i(D; y_{<i}) = y_i]$, and likewise for D'.
2. Each factor for D is at most e^{ε_i} times the matching factor for D', by the privacy of $M_i(\cdot\,; y_{<i})$ applied to the event $\{y_i\}$. Multiplying the k bounds gives

$$\Pr[M(D) = y] = \prod_{i=1}^{k} \Pr[M_i(D; y_{<i}) = y_i] \le \prod_{i=1}^{k} e^{\varepsilon_i} \Pr[M_i(D'; y_{<i}) = y_i] = \exp\Big(\sum_{i=1}^{k} \varepsilon_i\Big) \Pr[M(D') = y]. \tag{4.13}$$

3. The pointwise lemma of the randomized response chapter turns this bound on single outcomes into the definition.

□

For outputs that are not countable, such as the real numbers that the Laplace mechanism releases, the same idea works with averages in place of products. It uses one fact about functions h with values in $[0, 1]$: if M is ε-DP, then for all neighbours

$$\mathbb{E}[h(M(D))] \le e^{\varepsilon}\, \mathbb{E}[h(M(D'))]. \tag{4.14}$$

This is Lemma 3.2 of the randomized response chapter.

Proof. General outputs.

1. The composition of the first $k-1$ mechanisms is itself a mechanism, so by induction on k it suffices to treat $k = 2$.
2. Fix neighbours $D \sim D'$ and an event S of pairs. For an output y_1, write $S_{y_1} = \{y_2 : (y_1, y_2) \in S\}$ and $g_D(y_1) = \Pr[M_2(D; y_1) \in S_{y_1}]$, and define $g_{D'}$ in the same way. Conditioning on the first output gives $\Pr[M(D) \in S] = \mathbb{E}_{y_1 \sim M_1(D)}[g_D(y_1)]$, and likewise for D'.
3. By the privacy of $M_2(\cdot\,; y_1)$, $g_D(y_1) \le e^{\varepsilon_2} g_{D'}(y_1)$ for every y_1.
4. The function $g_{D'}$ takes values in $[0, 1]$, so the fact above, applied to M_1, gives

$$\Pr[M(D) \in S] \le e^{\varepsilon_2}\, \mathbb{E}_{y_1 \sim M_1(D)}[g_{D'}(y_1)] \le e^{\varepsilon_1 + \varepsilon_2}\, \mathbb{E}_{y_1 \sim M_1(D')}[g_{D'}(y_1)] = e^{\varepsilon_1 + \varepsilon_2} \Pr[M(D') \in S]. \tag{4.15}$$

□

Basic composition recovers the analysis of Proposition 4.1. Each counting query with $\mathrm{Lap}(k/\varepsilon)$ noise is (ε/k)-DP, and k of them compose to ε-DP, now even when the analyst chooses the queries adaptively. So the cost of k analyses grows linearly in k. It can grow like $\sqrt{k}$ instead, but only under the relaxation of the definition that the next chapter introduces.

4.7. Summary

Table 4.1.: The chapter's results.

Result	Statement
Definition 4.1	$\Delta_1 f$ is the largest ℓ_1 change in f between neighbours
Lemma 4.1	$\mathrm{Lap}(b)$ has variance $2b^2$ and $\Pr[\lvert Z\rvert \ge tb] = e^{-t}$
Theorem 4.1	adding $\mathrm{Lap}(\Delta_1 f/\varepsilon)$ to each coordinate is ε-DP
Example 4.2	a fraction of ones with error $\ln(1/\beta)/(\varepsilon n)$
Proposition 4.1	k counting queries: sensitivity k, error $k \ln(k/\beta)/\varepsilon$
Proposition 4.2	a histogram: sensitivity 2, error $2\ln(k/\beta)/\varepsilon$ over all bins

Result	Statement
Theorem 4.2	post-processing keeps ε-DP
Theorem 4.3	datasets differing in k records: factor $e^{k\varepsilon}$
Theorem 4.4	adaptive composition adds the ε_i

4.8. References

- Cynthia Dwork, Frank McSherry, Kobbi Nissim and Adam Smith, "Calibrating Noise to Sensitivity in Private Data Analysis", 2006. Defines sensitivity and the Laplace mechanism.
- Irit Dinur and Kobbi Nissim, "Revealing Information while Preserving Privacy", 2003. Gives the reconstruction attacks that set the scale for the error.
- Cynthia Dwork and Aaron Roth, *The Algorithmic Foundations of Differential Privacy*, 2014. Covers the mechanism and its uses in Section 3.3, and the properties in Sections 2.3 and 3.5.
- Cynthia Dwork, Guy Rothblum and Salil Vadhan, "Boosting and Differential Privacy", 2010. Proves advanced composition, which improves the linear growth of basic composition.

5. Approximate Differential Privacy

Pure differential privacy (DP) bounds the ratio of two output probabilities by e^{ε} on every event. This chapter adds a small additive slack δ to that bound. The relaxed definition, (ε, δ)-DP, lets a mechanism fail with probability about δ, and the examples of the chapter show that δ must be much smaller than $1/n$. In exchange, the relaxation admits the Gaussian mechanism, which adds less noise than the Laplace mechanism to statistics with many coordinates, and the advanced composition of the next chapter.

5.1. The definition

An additive bound alone is the wrong notion of privacy. Suppose we asked only that, for all neighbours and all events,

$$\Pr[M(D) \in S] \leq \Pr[M(D') \in S] + \delta. \tag{5.1}$$

This allows no useful accuracy when δ is small and allows harmful mechanisms when δ is large (Vadhan, 2017, Section 1.6), as Section 5.3 shows. But together with the multiplicative bound of pure DP, an additive term gives the right definition. Recall that two datasets in $\mathcal{X}^n$ are neighbours, written $D \sim D'$, when they differ in exactly one record (the replace-one relation).

Definition 5.1 (Approximate differential privacy)**.** Let $\varepsilon \geq 0$ and $\delta \in [0, 1]$. A randomized algorithm $M \colon \mathcal{X}^n \to \mathcal{Y}$ is (ε, δ)**-differentially private** if for all neighbours $D \sim D'$ and all events $S \subseteq \mathcal{Y}$,

$$\Pr[M(D) \in S] \leq e^{\varepsilon} \Pr[M(D') \in S] + \delta. \tag{5.2}$$

At $\delta = 0$ this is the ε-DP of the randomized response chapter, which from now on is called **pure DP**. A definition with $\delta > 0$ is called **approximate DP**.

5.2. The privacy loss random variable

Definition 5.1 bounds probabilities of events, but it is easier to understand one outcome at a time. Fix neighbours $D \sim D'$. An observer who sees an output y and asks whether the dataset was D or D' compares how likely y is under each. Write P and Q for the distributions of $M(D)$ and $M(D')$, and $p(y)$ and $q(y)$ for the likelihood of y under each: probabilities when $\mathcal{Y}$ is countable, and densities when both outputs have densities. (In general, p and q are densities with respect to the measure $P + Q$, which always exist.)

The privacy loss is the logarithm of the ratio $p(y)/q(y)$. An outcome that can occur under D but never under D', with $p(y) > 0$ and $q(y) = 0$, rules D' out entirely, so its loss is $+\infty$.

Definition 5.2 (Privacy loss)**.** Let $p(y)$ and $q(y)$ be the probabilities, or densities, of an outcome y under $M(D)$ and under $M(D')$. The **privacy loss** of an outcome y with $p(y) > 0$ is

$$L_{D,D'}(y) = \ln \frac{p(y)}{q(y)}, \tag{5.3}$$

where $L_{D,D'}(y) = +\infty$ if $q(y) = 0$. Once the pair is fixed we write it $L(y)$. The **privacy loss random variable** is $L = L(Y)$ with $Y = M(D)$: the privacy loss of the outcome the mechanism actually produces on D.

The outcome $M(D)$ has $p(M(D)) > 0$ with probability 1, so L is always defined. A large value of L means that the outcome is much more likely under D than under D', so it is evidence for D. A negative value is evidence for D', and an infinite value means that the outcome is impossible under D'.

The Laplace mechanism of the previous chapter gives a first example. For a function $f\colon \mathcal{X}^n \to \mathbb{R}$ whose value changes by at most $\Delta_1 f$ between neighbours, it releases $f(D) + Z$, where Z has the Laplace density $\frac{1}{2b}e^{-|z|/b}$ with scale $b = \Delta_1 f/\varepsilon$, and it is ε-DP.

Example 5.1. Take neighbours with $f(D) = a$ and $f(D') = a - \Delta_1 f$, the largest gap allowed. The densities give

$$L(y) = \frac{\varepsilon}{\Delta_1 f}(|y - a + \Delta_1 f| - |y - a|), \tag{5.4}$$

which equals ε for $y \geq a$, equals $-\varepsilon$ for $y \leq a - \Delta_1 f$, and is linear in between.

1. The event $y \geq a$ has probability $\frac{1}{2}$ under P, because the noise is symmetric. So L has an atom of mass $\frac{1}{2}$ at ε.
2. The event $y \leq a - \Delta_1 f$ needs noise below $-\Delta_1 f = -\varepsilon b$, which has probability $\frac{1}{2}e^{-\varepsilon}$. So L has an atom of mass $\frac{1}{2}e^{-\varepsilon}$ at $-\varepsilon$.
3. In between, write $y = a - s$ with $s \in (0, \Delta_1 f)$, so that $L = \varepsilon - 2\varepsilon s/(\Delta_1 f)$. The variable s has density $\frac{\varepsilon}{2\Delta_1 f}e^{-\varepsilon s/(\Delta_1 f)}$, and the change of variables gives L the density $u \mapsto \frac{1}{4}e^{(u-\varepsilon)/2}$ on $(-\varepsilon, \varepsilon)$, of total mass $\frac{1}{2}(1 - e^{-\varepsilon})$.

The three masses add up to 1. The privacy loss never leaves $[-\varepsilon, \varepsilon]$, as Figure 5.1 shows on the left.

The example is an instance of a general fact. Pure DP says that the privacy loss is at most ε with probability 1, and approximate DP says, roughly, that it is at most ε except with probability δ. Both follow from one identity for the smallest δ that works for a pair, which is the largest violation of the multiplicative bound over all events. The identity writes it as an average over the privacy loss, with the positive part $(z)_+ = \max(z, 0)$.

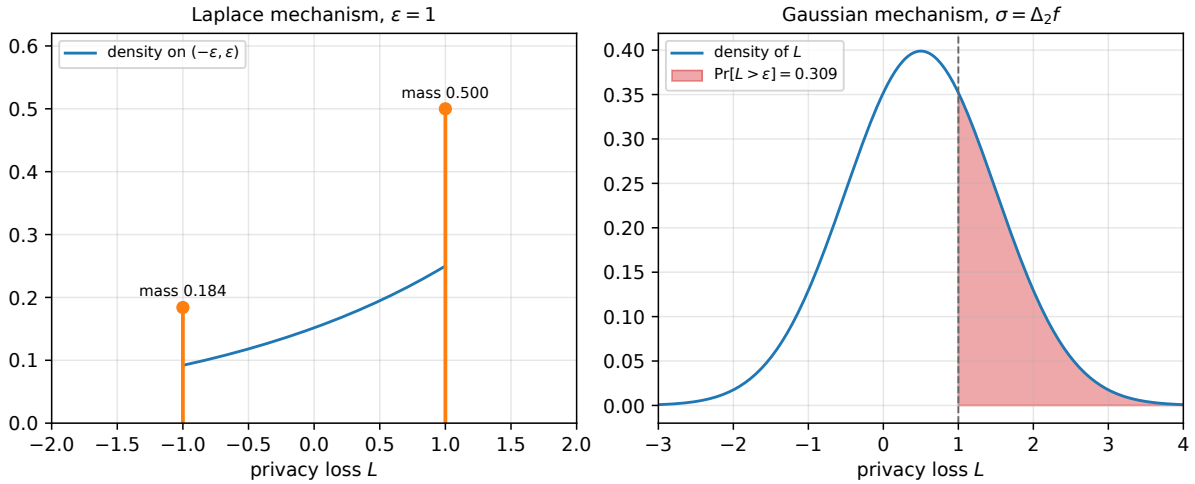


Figure 5.1.: Privacy loss for two mechanisms on neighbours whose values differ by the full sensitivity. Left, the Laplace mechanism with $\varepsilon = 1$, from Example 5.1: atoms of mass 0.5 at ε and 0.184 at $-\varepsilon$, and a density in between. Right, the Gaussian mechanism with noise standard deviation $\sigma = \Delta_2 f$, from Lemma 5.4: the loss is Gaussian with mean $1/2$ and variance 1, so it exceeds $\varepsilon = 1$ with probability 0.309.

Lemma 5.1. *For neighbours $D \sim D'$ and $\varepsilon \geq 0$,*

$$\sup_{S \subseteq \mathcal{Y}} \Big(\Pr[M(D) \in S] - e^{\varepsilon} \Pr[M(D') \in S] \Big) = \mathbb{E}\Big[\big(1 - e^{\varepsilon - L}\big)_+ \Big], \tag{5.5}$$

where L is the privacy loss random variable of Definition 5.2 for this pair, $(z)_+ = \max(z, 0)$ is the positive part of z, and $e^{\varepsilon - L}$ is read as 0 when $L = +\infty$. So M is (ε, δ)-DP if and only if the right-hand side is at most δ for every ordered pair of neighbours.

Proof.

1. For an event S, write the difference as an integral, $\Pr[M(D) \in S] - e^{\varepsilon} \Pr[M(D') \in S] = \int_S (p - e^{\varepsilon} q)$. The integrand is positive exactly on $B = \{y : p(y) > e^{\varepsilon} q(y)\}$, so the integral is largest for $S = B$, where it equals $\int (p - e^{\varepsilon} q)_+$.
2. Where $p(y) = 0$, the integrand $(p - e^{\varepsilon} q)_+$ is 0. Where $p(y) > 0$, it equals $p(y)(1 - e^{\varepsilon} q(y)/p(y))_+ = p(y)(1 - e^{\varepsilon - L(y)})_+$, which is also right when $q(y) = 0$ and $L(y) = \infty$.
3. Integrating $p(y)$ times a function of y is taking the expectation under P, so $\int (p - e^{\varepsilon} q)_+ = \mathbb{E}[(1 - e^{\varepsilon - L})_+]$.
4. Definition 5.1 asks that the left-hand side of Equation 5.5 be at most δ for every ordered pair of neighbours, which proves the last sentence.

□

The right-hand side of Equation 5.5 is called the **hockey-stick divergence** of P from Q, after the shape of the function $z \mapsto (1 - e^{\varepsilon - z})_+$. Its integrand is 0 when $L \leq \varepsilon$ and at most 1 otherwise.

So the lemma ties (ε, δ)-DP to how often L exceeds ε, and the next theorem makes that tie precise in both directions.

Theorem 5.1. *Let $M \colon \mathcal{X}^n \to \mathcal{Y}$ be a randomized algorithm, and let L be its privacy loss random variable for an ordered pair of neighbours $D \sim D'$.*

1. *M is ε-DP if and only if $L \le \varepsilon$ with probability 1 for every ordered pair of neighbours. In that case $|L| \le \varepsilon$ with probability 1.*
2. *If $\Pr[L > \varepsilon] \le \delta$ for every ordered pair of neighbours, then M is (ε, δ)-DP.*
3. *If M is (ε, δ)-DP, then for every $t > 0$ and every ordered pair of neighbours,*

$$\Pr[L > \varepsilon + t] \le \frac{\delta}{1 - e^{-t}}. \tag{5.6}$$

Proof.

1. With $\delta = 0$, Lemma 5.1 says that M is ε-DP exactly when $\mathbb{E}[(1 - e^{\varepsilon - L})_+] = 0$ for every pair. The integrand is nonnegative, so this holds exactly when it is 0 with probability 1, that is, when $L \le \varepsilon$ with probability 1. For the lower bound, apply the same fact to the pair (D', D). It gives $q \le e^{\varepsilon} p$ except on a set of Q-probability 0, and that set also has P-probability 0, because pure DP gives $\Pr[M(D) \in S] \le e^{\varepsilon} \Pr[M(D') \in S]$. So $L \ge -\varepsilon$ with probability 1 under P.
2. The integrand $(1 - e^{\varepsilon - L})_+$ is 0 when $L \le \varepsilon$ and at most 1 when $L > \varepsilon$. So its expectation is at most $\Pr[L > \varepsilon] \le \delta$, and Lemma 5.1 gives (ε, δ)-DP.
3. On the event $L > \varepsilon + t$, the integrand is at least $1 - e^{-t}$. So $(1 - e^{-t}) \Pr[L > \varepsilon + t] \le \mathbb{E}[(1 - e^{\varepsilon - L})_+] \le \delta$, by Lemma 5.1.

□

The converse of part 2 fails with the same parameters, which is why part 3 pays in both ε and δ. For example, the Gaussian mechanism on the right of Figure 5.1 has privacy loss above 1 with probability 0.309, yet Proposition 5.3 below shows that it is $(1, 0.127)$-DP. With $t = \varepsilon$, part 3 says that an (ε, δ)-DP mechanism has privacy loss above 2ε with probability at most $\delta/(1 - e^{-\varepsilon})$. So, up to these constants, (ε, δ)-DP says that the privacy loss is at most ε except with probability about δ, and it says nothing about what happens on that rare event.

5.3. How small δ must be

On the event of probability δ, the privacy loss may be infinite. The first example shows this in its crudest form.

Example 5.2. Let $\perp$ be a fixed symbol that carries no information about the data. The mechanism $M(D)$ outputs $\perp$ with probability $1 - \delta$ and outputs the whole dataset D with probability δ. For neighbours $D \neq D'$ and an event S,

$$\Pr[M(D) \in S] \le (1 - \delta)\, \mathbf{1}[\perp \in S] + \delta \le \Pr[M(D') \in S] + \delta, \tag{5.7}$$

so M is $(0, \delta)$-DP. With probability $1 - \delta$ the privacy loss is $\ln \frac{1-\delta}{1-\delta} = 0$, and with probability δ it is infinite, because the output D is impossible under D'.

This mechanism publishes everyone's data with probability δ, so δ must be small. The next example shows how small, because it publishes each record on its own.

Definition 5.3 (Name and shame)**.** For $\delta \in [0, 1]$, the mechanism NS_δ draws independent coins $c_1, \ldots, c_n$, each equal to 1 with probability δ, and outputs the set of pairs $\{(i, x_i) : c_i = 1\}$.

Each record, which might be a person's medical file, is published in the clear with probability δ. Even so, the mechanism meets the definition with $\varepsilon = 0$, because a person's record affects the output only when that person's coin comes up 1, which has probability δ.

Proposition 5.1. *For every $\delta \in [0, 1]$, NS_δ is $(0, \delta)$-DP.*

Proof.

1. Let D and D' differ only in record i, and run the mechanism on both datasets with the same coins. Let E be the event $c_i = 1$, so $\Pr[E] = \delta$.
2. On the complement $\bar{E}$, record i is not published, and the output is a function of the coins and of the records $j \neq i$, which the two datasets share. So on $\bar{E}$ the two outputs are equal.
3. For an event S, split on E and use step 2:

$$\Pr[\mathrm{NS}_\delta(D) \in S] = \Pr[\mathrm{NS}_\delta(D) \in S,\ \bar{E}] + \Pr[\mathrm{NS}_\delta(D) \in S,\ E] \leq \Pr[\mathrm{NS}_\delta(D') \in S,\ \bar{E}] + \Pr[E]. \tag{5.8}$$

4. The first term is at most $\Pr[\mathrm{NS}_\delta(D') \in S]$ and the second is δ, which is Equation 5.2 with $\varepsilon = 0$.

□

So a mechanism that plainly harms people meets $(0, \delta)$-DP. Its harm depends on the chance that it publishes at least one record, which the next proposition computes.

Proposition 5.2. *The probability that NS_δ publishes at least one record is $1 - (1 - \delta)^n$, and*

$$n\delta - \frac{(n\delta)^2}{2} \leq 1 - (1 - \delta)^n \leq n\delta. \tag{5.9}$$

Proof.

1. The coins are independent, so no record is published with probability $(1 - \delta)^n$, and the complement gives the formula.
2. The upper bound is the union bound over the n records, each published with probability δ.
3. For the lower bound, $1 - \delta \leq e^{-\delta}$ gives $(1 - \delta)^n \leq e^{-n\delta}$. The inequality $e^{-x} \leq 1 - x + x^2/2$ for $x \geq 0$ then gives $1 - (1 - \delta)^n \geq n\delta - (n\delta)^2/2$.

□

So the chance that someone is named is about $n\delta$ when $n\delta$ is small. At $\delta = 1/n$, step 3 of the proof shows that some record is in the clear with probability at least $1 - e^{-1} \approx 0.632$. A guarantee that allows this is not privacy, so δ must be well below $1/n$. Even $\delta = n^{-1.1}$ allows a chance of up to $n\delta = n^{-0.1}$, which is 0.25 for a million records. Common choices are $\delta \le 1/n^2$, or a **cryptographically small** δ, smaller than every inverse polynomial in n.

In both examples the privacy loss is 0 or infinite, so the rare event is a catastrophe. Without more information about a mechanism, you must assume this worst case. But many mechanisms degrade gracefully. Their privacy loss exceeds ε with probability about δ but rarely exceeds it by much, so they meet a family of guarantees at once, such as $(1, 10^{-3})$-DP and $(2, 10^{-6})$-DP. The Gaussian mechanism of the next section is the main example, as Example 5.5 shows.

The definition must be stated for events, not single outcomes. For pure DP the two are equivalent on countable outputs, by the pointwise lemma of the randomized response chapter, but for approximate DP the pointwise version is much weaker.

Example 5.3. Let $m = \lceil 1/\delta \rceil$ for some $\delta \in (0, 1)$. The mechanism $M(D) = (D, U)$ outputs the whole dataset together with a number U drawn uniformly from $\{1, \dots, m\}$. Every single outcome has probability at most $1/m \le \delta$, so for all neighbours and all outcomes y,

$$\Pr[M(D) = y] \le e^{\varepsilon} \Pr[M(D') = y] + \delta. \tag{5.10}$$

Yet the event $S = \{(D, u) : 1 \le u \le m\}$ has probability 1 under D and 0 under D', so M is not (ε, δ_0)-DP for any ε and any $\delta_0 < 1$.

The privacy loss of this mechanism is infinite with probability 1. So Theorem 5.1, which looks at the privacy loss, catches the failure that the pointwise inequality misses.

5.4. The Gaussian mechanism

The Laplace mechanism calibrates its noise to the ℓ_1 sensitivity. The Gaussian mechanism adds Gaussian noise instead, calibrated to the ℓ_2 sensitivity, which can be much smaller.

Definition 5.4 ($\ell 2$ sensitivity)**.** The ℓ_2 **sensitivity** of a function $f \colon \mathcal{X}^n \to \mathbb{R}^d$ is

$$\Delta_2 f = \sup_{D \sim D'} \|f(D) - f(D')\|_2. \tag{5.11}$$

The ℓ_1 sensitivity $\Delta_1 f$ of the previous chapter is the same supremum with the ℓ_1 norm. The two sensitivities are within a factor $\sqrt{d}$ of each other, because the two norms are.

Lemma 5.2. *For every $v \in \mathbb{R}^d$, $\|v\|_2 \le \|v\|_1 \le \sqrt{d}\,\|v\|_2$. Hence $\Delta_2 f \le \Delta_1 f \le \sqrt{d}\,\Delta_2 f$.*

Proof.

1. Expanding the square, $\|v\|_1^2 = \sum_i v_i^2 + \sum_{i \neq j} |v_i||v_j| \geq \|v\|_2^2$.
2. The Cauchy–Schwarz inequality applied to $(|v_1|, \dots, |v_d|)$ and the all-ones vector gives $\|v\|_1 = \sum_i |v_i| \cdot 1 \leq \|v\|_2 \sqrt{d}$.
3. Applying both inequalities to $f(D) - f(D')$ and taking the supremum over neighbours gives the bounds on the sensitivities.

□

The Gaussian distribution $\mathcal{N}(\mu, \sigma^2)$ with mean μ and variance σ^2 has density

$$\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right). \tag{5.12}$$

Its tails fall like $e^{-x^2/(2\sigma^2)}$, faster than the $e^{-|x|/b}$ of the Laplace distribution. Write $\mathcal{N}(0, \sigma^2 I_d)$ for the vector of d independent $\mathcal{N}(0, \sigma^2)$ coordinates. The Gaussian mechanism adds such a vector to the true answer.

Definition 5.5 (Gaussian mechanism)**.** For $f \colon \mathcal{X}^n \to \mathbb{R}^d$ and a noise level $\sigma > 0$, the Gaussian mechanism outputs

$$M(D) = f(D) + Z, \qquad Z \sim \mathcal{N}(0, \sigma^2 I_d). \tag{5.13}$$

The theorem below says how large the noise level σ must be for a given ε and δ.

Theorem 5.2 (Privacy of the Gaussian mechanism)**.** *Let f have $0 < \Delta_2 f < \infty$, and let $\varepsilon > 0$ and $\delta \in (0, 1)$. The Gaussian mechanism with*

$$\sigma = \frac{\Delta_2 f}{\varepsilon} \sqrt{2\ln(1/\delta) + \varepsilon} \tag{5.14}$$

is (ε, δ)-DP.

The proof is in Section 5.4.1. As $\delta \to 0$, the noise level is $\sigma = \frac{\Delta_2 f}{\varepsilon}\sqrt{2\ln(1/\delta)}\,(1 + o(1))$. The Laplace mechanism has noise of scale $\Delta_1 f/\varepsilon$ in each coordinate, so the Gaussian mechanism differs in two ways. It pays a factor $\sqrt{2\ln(1/\delta)}$ for only approximate DP, but it measures sensitivity in ℓ_2, which can save a factor $\sqrt{d}$. The next example shows that the saving can be the larger effect.

Example 5.4. Each of n people answers d yes-or-no questions, so the dataset is $D \in \{0,1\}^{n \times d}$ with rows $x_i \in \{0,1\}^d$. The analyst wants the vector of averages $f(D) = \frac{1}{n}\sum_i x_i$, with small error in ℓ_2 norm.

1. Replacing row x by row x' changes f by $(x - x')/n$. The largest change is from the all-ones row to the all-zeros row, so $\Delta_1 f = d/n$ and $\Delta_2 f = \sqrt{d}/n$.
2. The Laplace mechanism adds $\mathrm{Lap}(d/(\varepsilon n))$ to each coordinate, and each coordinate has variance $2d^2/(\varepsilon n)^2$. The root mean squared ℓ_2 error is

$$\sqrt{\mathbb{E}\|Z\|_2^2} = \sqrt{d \cdot \frac{2d^2}{\varepsilon^2 n^2}} = \frac{\sqrt{2}\, d^{3/2}}{\varepsilon n}. \tag{5.15}$$

3. The Gaussian mechanism with Equation 5.14 adds $\mathcal{N}(0, \sigma^2)$ to each coordinate with $\sigma = \frac{\sqrt{d}}{\varepsilon n}\sqrt{2\ln(1/\delta) + \varepsilon}$, so its root mean squared error is

$$\sqrt{d}\,\sigma = \frac{d\sqrt{2\ln(1/\delta) + \varepsilon}}{\varepsilon n}. \tag{5.16}$$

4. The ratio of the two errors is $\sqrt{2d/(2\ln(1/\delta) + \varepsilon)}$. So the Gaussian mechanism is more accurate as soon as $d > \ln(1/\delta) + \varepsilon/2$, and its advantage grows like $\sqrt{d}$.

With $d = 100$ questions, $n = 10{,}000$ people, $\varepsilon = 1$ and $\delta = 10^{-6}$, the Laplace error is $\sqrt{2} \cdot 1000/10{,}000 = 0.1414$ and the Gaussian error is $100\sqrt{27.63 + 1}/10{,}000 = 0.0535$, smaller by a factor 2.64. A simulation of 20,000 runs of each mechanism gives 0.1413 and 0.0535.

So the choice is a trade-off: the Laplace mechanism gives the stronger guarantee of pure DP, and the Gaussian mechanism a weaker guarantee with less noise in high dimension.

5.4.1. Proof of privacy

The proof has two steps. First, the privacy loss of the Gaussian mechanism is itself Gaussian. Second, a Gaussian tail bound shows that it exceeds ε with probability at most δ, and part 2 of Theorem 5.1 concludes. The lemma below collects four facts about Gaussians: the first two give the distribution of the privacy loss, and the last two bound its tail.

Lemma 5.3. *Let $W \sim \mathcal{N}(0, 1)$.*

1. *For constants μ and $a \neq 0$, $\mu + aW \sim \mathcal{N}(\mu, a^2)$.*
2. *If $V_1 \sim \mathcal{N}(0, a_1^2)$ and $V_2 \sim \mathcal{N}(0, a_2^2)$ are independent, then $V_1 + V_2 \sim \mathcal{N}(0, a_1^2 + a_2^2)$.*
3. *For every $u \geq 0$, $\Pr[W > u] \leq e^{-u^2/2}$.*
4. *For every $u > 0$, $\Pr[W > u] \leq e^{-u^2/2}/(u\sqrt{2\pi})$.*

Proof.

1. Completing the square in the density gives the moment generating function $\mathbb{E}[e^{\lambda W}] = e^{\lambda^2/2}$. A distribution is determined by its moment generating function when that function is finite near 0, as it is for all the variables here.
2. For part 1, $\mathbb{E}[e^{\lambda(\mu + aW)}] = e^{\lambda\mu + \lambda^2 a^2/2}$, which is the moment generating function of $\mathcal{N}(\mu, a^2)$.
3. For part 2, independence makes the moment generating function of the sum the product $e^{\lambda^2 a_1^2/2} e^{\lambda^2 a_2^2/2} = e^{\lambda^2(a_1^2 + a_2^2)/2}$.
4. For part 3, Markov's inequality applied to e^{uW} gives $\Pr[W > u] \leq e^{-u^2}\,\mathbb{E}[e^{uW}] = e^{-u^2/2}$.
5. For part 4, write $\varphi(z) = e^{-z^2/2}/\sqrt{2\pi}$ for the density, whose derivative is $-z\varphi(z)$. On $z \geq u$ the ratio z/u is at least 1, so $\Pr[W > u] \leq \int_u^\infty \frac{z}{u}\varphi(z)\,\mathrm{d}z = \frac{\varphi(u)}{u}$.

□

The first step of the plan computes the privacy loss, which is Gaussian, with a mean and a variance set by the distance between $f(D)$ and $f(D')$.

Lemma 5.4. *Let $D \sim D'$, let M be the Gaussian mechanism with noise level σ, and write $v = f(D) - f(D')$. The privacy loss random variable of $M(D)$ against $M(D')$ is distributed as*

$$L \sim \mathcal{N}(\rho,\ 2\rho)\,, \qquad \rho = \frac{\|v\|_2^2}{2\sigma^2}. \tag{5.17}$$

Proof.

1. The output is $f(D) + z$ with $z \sim \mathcal{N}(0, \sigma^2 I_d)$. Its density under D is proportional to $\exp(-\|z\|_2^2/(2\sigma^2))$. Under D', the same point is at distance $z + v$ from the mean $f(D')$, so its density is proportional to $\exp(-\|z + v\|_2^2/(2\sigma^2))$, with the same constant.
2. The constants cancel in the ratio, and the logarithm gives

$$L = \frac{\|z + v\|_2^2 - \|z\|_2^2}{2\sigma^2} = \frac{1}{2\sigma^2}\sum_{j=1}^{d}\left((z_j + v_j)^2 - z_j^2\right) = \frac{1}{2\sigma^2}\sum_{j=1}^{d}\left(2z_j v_j + v_j^2\right). \tag{5.18}$$

3. The constant part is $\sum_j v_j^2/(2\sigma^2) = \rho$, the claimed mean.
4. The random part is $\frac{1}{\sigma^2}\sum_j v_j z_j$, and the terms with $v_j = 0$ vanish. By part 1 of Lemma 5.3 each remaining term $v_j z_j$ is $\mathcal{N}(0, \sigma^2 v_j^2)$, and the terms are independent. By part 2 their sum is $\mathcal{N}(0, \sigma^2\|v\|_2^2)$. Dividing by σ^2 gives $\mathcal{N}(0, \|v\|_2^2/\sigma^2) = \mathcal{N}(0, 2\rho)$.

□

So the variance of the privacy loss is twice its mean. For a single coordinate, $d = 1$, the proof is just the identity $L = vz/\sigma^2 + v^2/(2\sigma^2)$. A Gaussian privacy loss is special to this mechanism: the loss of the Laplace mechanism, by Example 5.1, has atoms.

Proof. Proof of Theorem 5.2.

1. By part 2 of Theorem 5.1, it suffices to show $\Pr[L > \varepsilon] \le \delta$ for every ordered pair of neighbours. If $v = 0$ then $L = 0$, so assume $v \neq 0$.
2. By Lemma 5.4 and part 1 of Lemma 5.3, $L = \rho + \sqrt{2\rho}\,W$ with $W \sim \mathcal{N}(0, 1)$. So

$$\Pr[L > \varepsilon] = \Pr\left[W > g(\rho)\right], \qquad g(\rho) = \frac{\varepsilon - \rho}{\sqrt{2\rho}} = \frac{\varepsilon}{\sqrt{2\rho}} - \sqrt{\frac{\rho}{2}}. \tag{5.19}$$

3. The function g decreases in ρ, and $\rho \le (\Delta_2 f)^2/(2\sigma^2)$ by the definition of $\Delta_2 f$. Write $t = \sqrt{2\ln(1/\delta) + \varepsilon}$, so that Equation 5.14 reads $\sigma = t\,\Delta_2 f/\varepsilon$ and the largest value of ρ is $\varepsilon^2/(2t^2)$. Then

$$g(\rho) \ge g\left(\frac{\varepsilon^2}{2t^2}\right) = t - \frac{\varepsilon}{2t} =: u. \tag{5.20}$$

4. The value u is positive, because $t^2 \geq \varepsilon > \varepsilon/2$. Its square is $u^2 = t^2 - \varepsilon + \varepsilon^2/(4t^2) \geq t^2 - \varepsilon = 2\ln(1/\delta)$.
5. By part 3 of Lemma 5.3, $\Pr[L > \varepsilon] \leq \Pr[W > u] \leq e^{-u^2/2} \leq e^{-\ln(1/\delta)} = \delta$.

□

Only the upper tail of L matters in this proof, because part 2 of Theorem 5.1 bounds $\Pr[L > \varepsilon]$, not $\Pr[|L| > \varepsilon]$. For $\varepsilon \leq 1$ the noise level Equation 5.14 is at most $\frac{\Delta_2 f}{\varepsilon}\sqrt{2\ln(1.65/\delta)}$. The sharper tail bound of part 4 of Lemma 5.3 lowers the constant 1.65 to 1.25, the form that later chapters use.

Corollary 5.1 (Gaussian mechanism for $\varepsilon \leq 1$)**.** *Let f have $0 < \Delta_2 f < \infty$, and let $\varepsilon \in (0, 1]$ and $\delta \in (0, 1)$. The Gaussian mechanism with*

$$\sigma = \frac{\Delta_2 f}{\varepsilon}\sqrt{2\ln(1.25/\delta)} \tag{5.21}$$

is (ε, δ)-DP.

Proof.

1. Write $t = \sqrt{2\ln(1.25/\delta)}$, so that $\sigma = t\,\Delta_2 f/\varepsilon$ and $e^{-t^2/2} = \delta/1.25$. Steps 1 to 3 of the proof of Theorem 5.2 used only $\sigma = t\,\Delta_2 f/\varepsilon$, so they give $\Pr[L > \varepsilon] \leq \Pr[W > u]$ with $u = t - \varepsilon/(2t)$.
2. Since $\varepsilon \leq 1$, the value u is at least $t - 1/(2t)$, which increases in t. Also $u^2 = t^2 - \varepsilon + \varepsilon^2/(4t^2) \geq t^2 - 1$.
3. *Case $\delta \leq 0.74$.* Then $t \geq \sqrt{2\ln(1.25/0.74)} \geq 1.0239$, and by step 2, $u \geq 0.5356$. By part 4 of Lemma 5.3 and step 2,

$$\Pr[W > u] \leq \frac{e^{-u^2/2}}{u\sqrt{2\pi}} \leq \frac{e^{1/2}\,e^{-t^2/2}}{0.5356\sqrt{2\pi}} = \frac{e^{1/2}}{1.25 \cdot 0.5356\sqrt{2\pi}}\,\delta \leq 0.983\,\delta. \tag{5.22}$$

4. *Case $\delta > 0.74$.* Then $t > \sqrt{2\ln 1.25} \geq 0.6680$, and by step 2, $u > -0.0805$. The density of W is at most $1/\sqrt{2\pi}$, so $\Pr[W > u] \leq \frac{1}{2} + 0.0805/\sqrt{2\pi} \leq 0.533 < \delta$.
5. In both cases $\Pr[L > \varepsilon] \leq \delta$, and part 2 of Theorem 5.1 gives (ε, δ)-DP.

□

This is the bound of Dwork and Roth (2014, Theorem A.1). On a grid of ε from 0.01 to 1 and δ from 0.89 down to 10^{-20}, the exact δ of Proposition 5.3 at this noise level is at most 0.233 times the target. Both constants are loose, because the tail bounds of Lemma 5.3 are loose. The exact δ comes from the identity of Lemma 5.1 for the Gaussian privacy loss, evaluated instead of bounded.

Proposition 5.3. *Let Φ be the distribution function of $\mathcal{N}(0,1)$, let f have $0 < \Delta_2 f < \infty$, and let $s = \Delta_2 f/\sigma$. For every $\varepsilon \geq 0$, the Gaussian mechanism with noise level σ is $(\varepsilon, \delta(\varepsilon))$-DP for*

$$\delta(\varepsilon) = \Phi\Big(\frac{s}{2} - \frac{\varepsilon}{s}\Big) - e^{\varepsilon}\,\Phi\Big(-\frac{s}{2} - \frac{\varepsilon}{s}\Big), \tag{5.23}$$

and for no smaller δ when some pair of neighbours has $\|f(D) - f(D')\|_2 = \Delta_2 f$.

Proof.

1. Fix a pair of neighbours and let $r = \|v\|_2/\sigma \in (0, s]$, so that $\rho = r^2/2$ and $L = r^2/2 + rW$. By Lemma 5.1, the smallest δ for this pair is $\mathbb{E}[(1 - e^{\varepsilon - L})\mathbf{1}[L > \varepsilon]]$.
2. The event $L > \varepsilon$ is $W > c$ with $c = \varepsilon/r - r/2$, so it has probability $\Phi(r/2 - \varepsilon/r)$.
3. Write φ for the standard normal density. Completing the square gives $e^{-rz}\varphi(z) = e^{r^2/2}\varphi(z + r)$, so $\mathbb{E}[e^{-rW}\mathbf{1}[W > c]] = e^{r^2/2}\,\Phi(-c - r)$. Since $e^{\varepsilon - L} = e^{\varepsilon - r^2/2}e^{-rW}$ and $-c - r = -r/2 - \varepsilon/r$,

$$\mathbb{E}\big[e^{\varepsilon - L}\mathbf{1}[L > \varepsilon]\big] = e^{\varepsilon - r^2/2}\,e^{r^2/2}\,\Phi\Big(-\frac{r}{2} - \frac{\varepsilon}{r}\Big) = e^{\varepsilon}\,\Phi\Big(-\frac{r}{2} - \frac{\varepsilon}{r}\Big). \tag{5.24}$$

4. Subtracting step 3 from step 2 gives Equation 5.23 with r in place of s. Call that function $h(r)$.
5. Write $c_1 = r/2 - \varepsilon/r$ and $c_2 = -r/2 - \varepsilon/r$. Then $(c_1^2 - c_2^2)/2 = (c_1 - c_2)(c_1 + c_2)/2 = -\varepsilon$, so $e^{\varepsilon}\varphi(c_2) = \varphi(c_1)$. Differentiating, $h'(r) = \varphi(c_1)(\frac{1}{2} + \varepsilon/r^2) + e^{\varepsilon}\varphi(c_2)(\frac{1}{2} - \varepsilon/r^2) = \varphi(c_1) > 0$. 6. So h increases in r, the worst pair is the one with the largest r, and $h(r) \leq h(s) = \delta(\varepsilon)$, with equality when some pair attains $\Delta_2 f$. □

The exact formula shows the graceful degradation promised in Section 5.3: at one noise level, the Gaussian mechanism meets a whole family of guarantees.

Example 5.5. For $\varepsilon = 1$ and $\delta = 10^{-5}$, Equation 5.14 gives $\sigma = \sqrt{2\ln(10^5) + 1}\,\Delta_2 f = 4.9016\,\Delta_2 f$. At this noise level, Equation 5.23 gives the following guarantees, all at once:

- $(0.5,\ 6.1 \times 10^{-4})$-DP,
- $(1,\ 3.0 \times 10^{-8})$-DP,
- $(2,\ 3.0 \times 10^{-24})$-DP.

The guarantee at $\varepsilon = 1$ is far better than the $\delta = 10^{-5}$ that was asked for, and the smallest noise level that gives $(1, 10^{-5})$-DP is $\sigma = 3.7306\,\Delta_2 f$. As ε grows, δ falls faster than exponentially, so the privacy loss of this mechanism degrades gracefully.

5.5. Properties

The properties of pure DP proved in the Laplace mechanism chapter carry over to approximate DP. Post-processing and composition keep their proofs, with δ added to each bound, and group privacy gets a worse δ.

Theorem 5.3 (Post-processing)**.** *Let $M\colon \mathcal{X}^n \to \mathcal{Y}$ be (ε, δ)-DP, and let $F\colon \mathcal{Y} \to \mathcal{Z}$ be a randomized map whose coins are independent of those of M. Then $F \circ M$ is (ε, δ)-DP.*

Proof.

1. The proof of Theorem 4.2 writes F as a random choice g_R of deterministic functions and conditions on the coins R. Those steps do not use the definition of privacy, so they are unchanged.
2. Only its last step changes. For neighbours $D \sim D'$ and an event $S \subseteq \mathcal{Z}$, apply Equation 5.2, instead of pure DP, to the event $g_R^{-1}(S)$, so δ joins the bound inside the expectation:

$$\Pr[F(M(D)) \in S] = \mathbb{E}_R\Big[\Pr[M(D) \in g_R^{-1}(S)]\Big] \leq \mathbb{E}_R\Big[e^\varepsilon \Pr[M(D') \in g_R^{-1}(S)] + \delta\Big] = e^\varepsilon \Pr[F(M(D')) \in S] + \delta. \tag{5.25}$$

□

For group privacy, recall that an ε-DP mechanism meets the bound with $e^{k\varepsilon}$ on datasets that differ in k records (Theorem 4.3). With approximate DP, each step of the chain also multiplies the earlier δ terms by e^ε.

Theorem 5.4 (Group privacy)**.** *Let M be (ε, δ)-DP, and let $D, D' \in \mathcal{X}^n$ differ in $k \geq 1$ records. For every event S,*

$$\Pr[M(D) \in S] \leq e^{k\varepsilon} \Pr[M(D') \in S] + \delta \sum_{j=0}^{k-1} e^{j\varepsilon} \leq e^{k\varepsilon} \Pr[M(D') \in S] + k e^{(k-1)\varepsilon}\delta. \tag{5.26}$$

Proof.

1. Change the differing records one at a time to get a chain $D = D^{(0)} \sim D^{(1)} \sim \cdots \sim D^{(k)} = D'$ of neighbours. Write $P_j = \Pr[M(D^{(j)}) \in S]$.
2. We show $P_0 \leq e^{j\varepsilon} P_j + \delta \sum_{i=0}^{j-1} e^{i\varepsilon}$ by induction on j. The case $j = 1$ is Equation 5.2.
3. If the bound holds for j, apply Equation 5.2 to $D^{(j)} \sim D^{(j+1)}$, which gives $P_j \leq e^\varepsilon P_{j+1} + \delta$. Substituting,

$$P_0 \leq e^{j\varepsilon}\big(e^\varepsilon P_{j+1} + \delta\big) + \delta \sum_{i=0}^{j-1} e^{i\varepsilon} = e^{(j+1)\varepsilon} P_{j+1} + \delta \sum_{i=0}^{j} e^{i\varepsilon}. \tag{5.27}$$

4. At $j = k$ this is the first inequality in Equation 5.26. Each of the k terms of the sum is at most $e^{(k-1)\varepsilon}$, which gives the second.

□

So the additive term grows exponentially in the group size, and the bound says nothing once k is a little over $\ln(1/\delta)/\varepsilon$.

Composition needs one more tool, because its proof bounds averages of functions with values in $[0, 1]$, not only probabilities of events. For pure DP this tool is Lemma 3.2, and the lemma below adds δ to it.

Lemma 5.5. *Let M be (ε,δ)-DP and let $h\colon \mathcal{Y}\to[0,1]$. For all neighbours $D\sim D'$,*

$$\mathbb{E}[h(M(D))]\le e^{\varepsilon}\,\mathbb{E}[h(M(D'))]+\delta. \tag{5.28}$$

Proof.

1. The proof is that of Lemma 3.2 with $+\delta$. Its first step, which writes $\mathbb{E}[h(M(D))]=\int_0^1\Pr[h(M(D))>t]\,\mathrm{d}t$, does not use privacy and is unchanged.
2. Only the bound on each tail changes. Equation 5.2 bounds the probability of the event $\{y: h(y)>t\}$ by $e^{\varepsilon}\Pr[h(M(D'))>t]+\delta$, and integrating over $t\in[0,1]$ gives $e^{\varepsilon}\,\mathbb{E}[h(M(D'))]+\delta$.

□

Adaptive composition is as in Definition 4.4: each M_i may depend on the earlier outputs $y_{<i}$, and now each $M_i(\cdot\,;y_{<i})$ is (ε_i,δ_i)-DP.

Theorem 5.5 (Basic composition)**.** *If each $M_i(\cdot\,;y_{<i})$ is (ε_i,δ_i)-DP for every $y_{<i}$, where the parameters ε_i and δ_i are fixed in advance, then the adaptive composition of $M_1,\dots,M_k$ is $(\sum_i\varepsilon_i,\sum_i\delta_i)$-DP.*

Proof. The proof is that of Theorem 4.4 for general outputs, with δ added. Its steps 1 and 2 do not change. By induction it suffices to treat $k=2$. For an event S of pairs, with S_{y_1} and $g_D(y_1)=\Pr[M_2(D;y_1)\in S_{y_1}]$ as there, conditioning on the first output gives $\Pr[M(D)\in S]=\mathbb{E}_{y_1\sim M_1(D)}[g_D(y_1)]$. Steps 3 and 4 change, because the bound on g_D now has an additive term.

3. Since $M_2(\cdot\,;y_1)$ is (ε_2,δ_2)-DP and $g_D\le 1$, we have $g_D(y_1)\le\min\{1,e^{\varepsilon_2}g_{D'}(y_1)+\delta_2\}\le h(y_1)+\delta_2$, where $h=\min\{1,e^{\varepsilon_2}g_{D'}\}$ takes values in $[0,1]$.
4. By Lemma 5.5 applied to M_1 and h,

$$\Pr[M(D)\in S]\le\mathbb{E}_{M_1(D)}[h]+\delta_2\le e^{\varepsilon_1}\,\mathbb{E}_{M_1(D')}[h]+\delta_1+\delta_2. \tag{5.29}$$

5. Finally $h\le e^{\varepsilon_2}g_{D'}$, and $\mathbb{E}_{y_1\sim M_1(D')}[g_{D'}(y_1)]=\Pr[M(D')\in S]$ by the conditioning identity above, applied to D'. So $\Pr[M(D)\in S]\le e^{\varepsilon_1+\varepsilon_2}\Pr[M(D')\in S]+\delta_1+\delta_2$.

□

Fix the parameters in advance. If the analyst chooses ε_i from earlier outputs, the sum is no longer a valid bound without further work. With k mechanisms that are each (ε,δ)-DP, basic composition gives $(k\varepsilon,k\delta)$-DP. The next chapter proves that, for every $\delta'>0$, the same composition is also $(\varepsilon',k\delta+\delta')$-DP with

$$\varepsilon'=\varepsilon\sqrt{2k\ln(1/\delta')}+k\varepsilon\,(e^{\varepsilon}-1). \tag{5.30}$$

When ε is of order $1/\sqrt{k}$ or smaller, the second term is no larger than the first, so ε' grows like $\sqrt{k}$ rather than k. That allows quadratically more analyses for the same privacy, at the price of a $\delta'>0$ even when every mechanism is pure.

5.6. Summary

Table 5.1.: The chapter's results.

Result	Statement
Definition 5.1	$\Pr[M(D) \in S] \le e^{\varepsilon} \Pr[M(D') \in S] + \delta$ for all neighbours and events
Lemma 5.1	the largest violation over events is $\mathbb{E}[(1 - e^{\varepsilon - L})_+]$
Theorem 5.1	pure DP is $L \le \varepsilon$ almost surely; $\Pr[L > \varepsilon] \le \delta$ implies (ε, δ)-DP; the converse holds with $\varepsilon + t$ and $\delta/(1 - e^{-t})$
Proposition 5.1	publishing each record with probability δ is $(0, \delta)$-DP, so δ must be well below $1/n$
Example 5.3	for approximate DP, single outcomes do not suffice
Lemma 5.2	$\Delta_2 f \le \Delta_1 f \le \sqrt{d}\, \Delta_2 f$
Lemma 5.4	the Gaussian mechanism's privacy loss is $\mathcal{N}(\rho, 2\rho)$ with $\rho = \lVert f(D) - f(D') \rVert_2^2 / (2\sigma^2)$
Theorem 5.2	$\sigma = \Delta_2 f \sqrt{2 \ln(1/\delta) + \varepsilon}/\varepsilon$ gives (ε, δ)-DP
Corollary 5.1	for $\varepsilon \le 1$, $\sigma = \Delta_2 f \sqrt{2 \ln(1.25/\delta)}/\varepsilon$ suffices
Proposition 5.3	exact $\delta(\varepsilon) = \Phi(s/2 - \varepsilon/s) - e^{\varepsilon} \Phi(-s/2 - \varepsilon/s)$ with $s = \Delta_2 f / \sigma$
Theorem 5.3	post-processing keeps (ε, δ)
Theorem 5.4	groups of k: $(k\varepsilon,\ k e^{(k-1)\varepsilon} \delta)$
Theorem 5.5	adaptive composition adds the ε_i and the δ_i

5.7. References

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Equation 5.23.

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6. Advanced Composition

When you run k private analyses on the same data, basic composition says that their privacy costs add up. This chapter shows that the total cost grows much more slowly, about as $\sqrt{k}$, if you accept a small extra failure probability. The proof turns each step into one simple pair of outputs with four outcomes. Then it bounds k copies of that pair with Hoeffding's inequality.

6.1. The theorem

Write $D \sim D'$ for neighbouring datasets, which differ in one record (the replace-one relation). Recall from the approximate differential privacy (DP) chapter that a mechanism M is (ε, δ)**-DP** if, for all neighbours $D \sim D'$ and all events S,

$$\Pr[M(D) \in S] \le e^{\varepsilon} \Pr[M(D') \in S] + \delta. \tag{6.1}$$

The proofs below compare two output distributions at a time, so the next definition names this condition for a pair of random variables.

Definition 6.1 (Indistinguishable pair)**.** Let $\varepsilon, \delta \ge 0$. Two random variables A and B with values in the same space are (ε, δ)**-indistinguishable** if, for every event S,

$$\Pr[A \in S] \le e^{\varepsilon} \Pr[B \in S] + \delta \qquad \text{and} \qquad \Pr[B \in S] \le e^{\varepsilon} \Pr[A \in S] + \delta. \tag{6.2}$$

So M is (ε, δ)-DP exactly when $M(D)$ and $M(D')$ are (ε, δ)-indistinguishable for every pair of neighbours.

The theorem is about the adaptive composition of Definition 4.4: mechanisms $M_1, \dots, M_k$ run in order on the same D, where $M_j(D; y_{<j})$ may depend on the earlier outputs $y_{<j} = (y_1, \dots, y_{j-1})$, and the composition outputs the whole transcript $(y_1, \dots, y_k)$.

Basic composition, proved in the two previous chapters, says that if each $M_j(\,\cdot\,; y_{<j})$ is (ε, δ)-DP for every fixed $y_{<j}$, then M is $(k\varepsilon, k\delta)$-DP. The theorem of this chapter improves the first parameter. Throughout, every output space is countable, so that probabilities are sums over outcomes.

Theorem 6.1 (Advanced composition)**.** *Let $\varepsilon \ge 0$, $\delta \in [0, 1)$ and $\delta' \in (0, 1)$. Let M be the adaptive composition of k mechanisms such that $M_j(\,\cdot\,; y_{<j})$ is (ε, δ)-DP for every j and every $y_{<j}$. Then M is $(\tilde{\varepsilon}, \tilde{\delta})$-DP with*

$$\tilde{\varepsilon} = \varepsilon \sqrt{2k \ln(1/\delta')} + k\varepsilon \, \frac{e^{\varepsilon} - 1}{e^{\varepsilon} + 1}, \qquad \tilde{\delta} = k\delta + \delta'. \tag{6.3}$$

The first term of $\tilde{\varepsilon}$ is the important one, and it grows as $\varepsilon\sqrt{k}$ instead of $k\varepsilon$. The second term is small when ε is small. Its factor is $(e^{\varepsilon}-1)/(e^{\varepsilon}+1) = \tanh(\varepsilon/2)$, and $\tanh x \le x$ for $x \ge 0$, so the term is at most $k\varepsilon^2/2$. That is also at most $k\varepsilon(e^{\varepsilon}-1)$, because $e^{\varepsilon}-1 \ge \varepsilon$, so the theorem implies the weaker form that the previous chapter quoted. The extra slack δ' is a free parameter, and a larger δ' lowers $\tilde{\varepsilon}$, so the theorem trades a bigger δ for a smaller ε.

The proof is in Section 6.3. Before it, Example 6.1 and Figure 6.1 say how much the theorem gains.

Example 6.1. Take $\varepsilon = 0.1$ and $\delta' = 10^{-6}$, so that $\ln(1/\delta') = 13.8155$ and $\tanh(0.05) = 0.04996$. For $k = 100$ mechanisms, basic composition gives 10. Advanced composition gives $0.1\sqrt{200 \times 13.8155} + 100 \times 0.1 \times 0.04996 = 5.257 + 0.500 = 5.756$. For $k = 1{,}000$, basic composition gives 100 and advanced composition gives $16.623 + 4.996 = 21.618$. Advanced composition is the smaller of the two from $k = 31$ on.

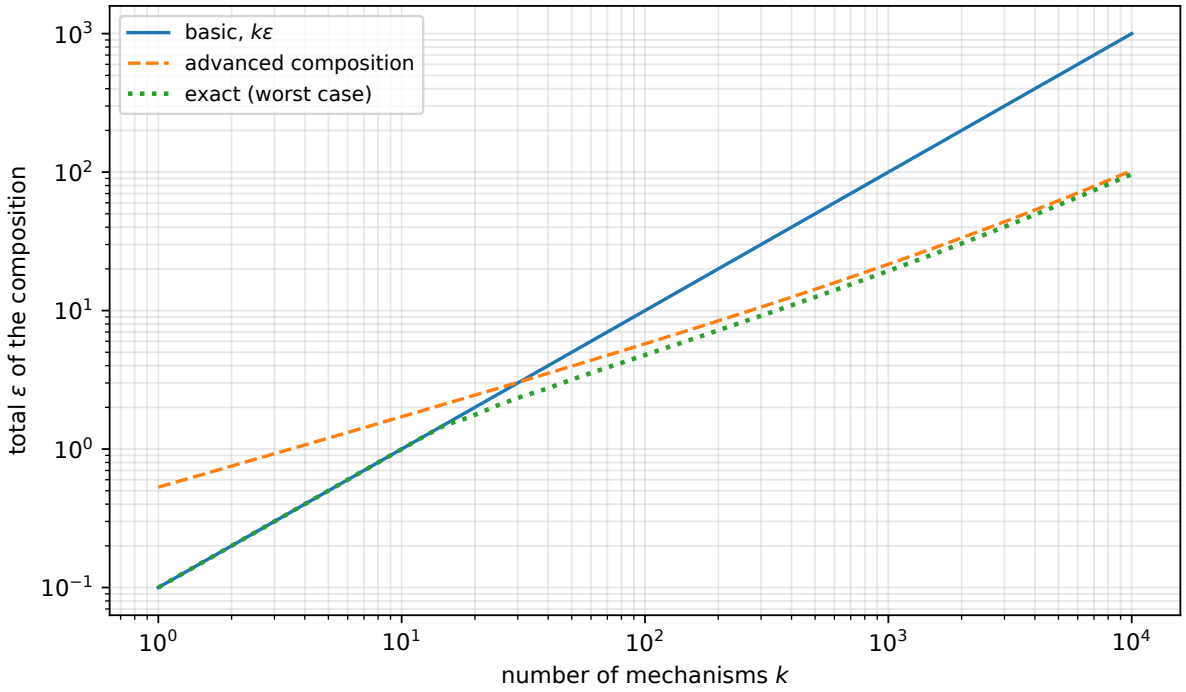


Figure 6.1.: The total privacy parameter of k mechanisms that are each 0.1-DP, at $\delta' = 10^{-6}$. The solid line is basic composition, the dashed line is Equation 6.3, and the dotted line is the exact smallest $\tilde{\varepsilon}$ for k independent randomized responses, which is the worst case for pure DP by Section 6.2. Advanced composition overtakes basic composition at $k = 31$ and is within 21% of the exact value for every k from 100 to 10,000.

The exact curve in Figure 6.1 shows that Theorem 6.1 is close to the truth, within 7% of it at $k = 10{,}000$. So growth as $\varepsilon\sqrt{k\ln(1/\delta')}$ is how privacy really degrades, not an artefact of the proof. Kairouz, Oh and Viswanath (2015) computed the exact optimal $\tilde{\varepsilon}$, which is tighter but harder to use.

With a total budget $\varepsilon_{\text{total}}$ for k analyses, you can invert Equation 6.3. The next corollary gives a simple choice that always works.

Corollary 6.1. *Let $\varepsilon_{\text{total}} \in (0, 1]$ and $\delta' \in (0, 1/2]$. If each of k adaptively composed mechanisms is (ε, δ)-DP with*

$$\varepsilon = \frac{\varepsilon_{\text{total}}}{2\sqrt{2k \ln(1/\delta')}}, \tag{6.4}$$

then the composition is $(\varepsilon_{\text{total}}, k\delta + \delta')$-DP.

Proof.

1. By Theorem 6.1, the composition is $(\tilde{\varepsilon}, k\delta + \delta')$-DP. Its first term is $\varepsilon\sqrt{2k \ln(1/\delta')} = \varepsilon_{\text{total}}/2$.
2. The second term is $k\varepsilon \tanh(\varepsilon/2) \le k\varepsilon^2/2$, because $\tanh x \le x$. Substituting ε gives $k\varepsilon^2/2 = \varepsilon_{\text{total}}^2/(16 \ln(1/\delta'))$.
3. Since $\delta' \le 1/2$, we have $\ln(1/\delta') \ge \ln 2 > 1/8$, and since $\varepsilon_{\text{total}} \le 1$, the second term is at most $\varepsilon_{\text{total}}^2/2 \le \varepsilon_{\text{total}}/2$.
4. Adding the two terms gives $\tilde{\varepsilon} \le \varepsilon_{\text{total}}$. A mechanism that is $(\tilde{\varepsilon}, \tilde{\delta})$-DP is also $(\varepsilon_{\text{total}}, \tilde{\delta})$-DP for every $\varepsilon_{\text{total}} \ge \tilde{\varepsilon}$.

□

The corollary is conservative by about a factor of 2. With $\varepsilon_{\text{total}} = 1$, $k = 1{,}000$ and $\delta' = 10^{-6}$, it allows $\varepsilon = 0.00301$ per mechanism, for which Equation 6.3 gives a total of only 0.5045. Solving Equation 6.3 numerically allows $\varepsilon = 0.00591$, almost twice as much, and basic composition with that ε would give a total of 5.911.

The gain matters most when the number of analyses grows with the problem, as in the next example.

Example 6.2 (Mean estimation with the Laplace mechanism)**.** Each of n people holds a vector $x_i \in \{0, 1\}^d$, and the analyst wants the mean $f(D) = \frac{1}{n}\sum_i x_i$. Recall the Laplace mechanism: if f has ℓ_1 sensitivity $\Delta_1 f$, meaning that $\|f(D) - f(D')\|_1 \le \Delta_1 f$ for all neighbours, then adding independent noise from the Laplace distribution with scale $\Delta_1 f/\varepsilon$ to each coordinate is ε-DP. A Laplace variable with scale b has mean 0 and variance $2b^2$, so the expected squared ℓ_2 error of the noise vector is $2db^2$.

All coordinates at once. Replacing one person changes each coordinate of the mean by at most $1/n$, so $\Delta_1 f = d/n$, and the noise has scale $d/(n\varepsilon)$. The root mean squared ℓ_2 error is $\sqrt{2d}\, d/(n\varepsilon) = \sqrt{2}\, d^{3/2}/(n\varepsilon)$.

One coordinate at a time. Each coordinate is a mechanism with sensitivity $1/n$, released with Laplace noise of scale $1/(n\varepsilon_1)$, which makes it ε_1-DP. With ε_1 from Corollary 6.1 for $k = d$ and $\delta = 0$, the d releases together are (ε, δ')-DP for $\varepsilon \le 1$. The root mean squared error is

$$\frac{\sqrt{2d}}{n\varepsilon_1} = \frac{\sqrt{2d} \cdot 2\sqrt{2d \ln(1/\delta')}}{n\varepsilon} = \frac{4d\sqrt{\ln(1/\delta')}}{n\varepsilon}. \tag{6.5}$$

The error now grows as d instead of $d^{3/2}$, which is the order the Gaussian mechanism achieves. Take $n = 100{,}000$, $\varepsilon = 1$ and $\delta' = 10^{-6}$. At $d = 1{,}000$, the two errors are 0.4472 and 0.1487. At $d = 10{,}000$, they are 14.14 and 1.487, nearly ten times apart. The price is the move from pure to approximate DP.

6.2. Reduction to four outcomes

A pair of output distributions $M_j(D)$ and $M_j(D')$ can be complicated. The first step of the proof replaces every such pair by the simplest (ε, δ)-indistinguishable pair. It has two ordinary outcomes, 0 and 1, whose probabilities differ by the factor e^{ε}, and two catastrophic outcomes, each with probability δ, which reveal which distribution the sample came from.

Definition 6.2 (The four-outcome pair)**.** For $\varepsilon \geq 0$ and $\delta \in [0, 1)$, let U and V take values in $\{0, 1, \text{I am } U, \text{I am } V\}$ with

$$\begin{aligned} &\Pr[U = 0] = \frac{e^{\varepsilon}(1-\delta)}{1+e^{\varepsilon}}, \quad \Pr[U = 1] = \frac{1-\delta}{1+e^{\varepsilon}}, \qquad \Pr[U = \text{I am } U] = \delta, \quad \Pr[U = \text{I am } V] = 0, \\ &\Pr[V = 0] = \frac{1-\delta}{1+e^{\varepsilon}}, \qquad \Pr[V = 1] = \frac{e^{\varepsilon}(1-\delta)}{1+e^{\varepsilon}}, \quad \Pr[V = \text{I am } U] = 0, \quad \Pr[V = \text{I am } V] = \delta. \end{aligned} \tag{6.6}$$

For $\delta = 0$ this is randomized response with truth probability $e^{\varepsilon}/(1+e^{\varepsilon})$, from the first chapter of this part. Here U is the report of a person whose bit is 0 and V that of a person whose bit is 1. The first check is that the pair meets the condition it stands for.

Lemma 6.1. *The pair U, V is (ε, δ)-indistinguishable.*

Proof.

1. On the outcomes 0 and 1, the ratio $\Pr[U = t]/\Pr[V = t]$ is e^{ε} or $e^{-\varepsilon}$, so $\Pr[U = t] \leq e^{\varepsilon} \Pr[V = t]$.
2. Take an event S. Sum step 1 over the outcomes of S in $\{0, 1\}$. The outcome "I am U" adds at most δ to $\Pr[U \in S]$, and "I am V" adds nothing. So $\Pr[U \in S] \leq e^{\varepsilon} \Pr[V \in S] + \delta$.
3. Swapping the labels $0 \leftrightarrow 1$ and "I am U" $\leftrightarrow$ "I am V" turns the distribution of U into that of V and back, so the same argument gives the other inequality.

□

Indistinguishability survives any further processing, as privacy does. The proof is that of post-processing in the approximate DP chapter, written for a pair of random variables.

Lemma 6.2 (Post-processing)**.** *Let A, B be (ε, δ)-indistinguishable, and let F be a randomized map whose coins are independent of A and B. Then $F(A)$ and $F(B)$ are (ε, δ)-indistinguishable.*

Proof.

1. *Functions with values in* $[0, 1]$. Let g take values in $[0, 1]$. The proof of Lemma 5.5 uses only the bound of the definition on events, so it applies to A and B unchanged and gives $\mathbb{E}[g(A)] \leq e^{\varepsilon} \mathbb{E}[g(B)] + \delta$.
2. *Post-processing.* Fix an event S and let $g(a) = \Pr[F(a) \in S]$, a probability over the coins of F. Because those coins are independent of A, $\Pr[F(A) \in S] = \mathbb{E}[g(A)]$, and likewise for B.

3. Step 1 applied to g gives $\Pr[F(A) \in S] \le e^{\varepsilon} \Pr[F(B) \in S] + \delta$. The other direction is the same with A and B swapped.

□

The four-outcome pair is the hardest (ε, δ)-indistinguishable pair to tell apart, in a strong sense. One randomized map produces any other such pair A, B from U, V.

Theorem 6.2 (Simulation by the four-outcome pair)**.** *Let $\varepsilon \ge 0$ and $\delta \in [0, 1)$, and let A, B be (ε, δ)-indistinguishable random variables on a countable space. Then there is a randomized map F such that $F(U)$ has the distribution of A and $F(V)$ has the distribution of B.*

The map sends each of the four outcomes to a distribution on the space. Some outcomes y are more than e^{ε} times as likely under A as under B. The definition bounds the total excess of A on these outcomes by δ, so "I am U" can carry it, and "I am V" carries the reverse excess. What is left is a pair whose probabilities are within a factor e^{ε} at every outcome. The outcome 0 puts more weight where A is the more likely of the two, and 1 where B is. Kairouz, Oh and Viswanath (2015) first proved the theorem, with hypothesis tests and Blackwell's theorem on the comparison of statistical experiments. The proof below builds the map directly.

Proof. Write $p(y) = \Pr[A = y]$ and $q(y) = \Pr[B = y]$.

1. *The excess.* Let $a(y) = \max(p(y) - e^{\varepsilon} q(y), 0)$ and $b(y) = \max(q(y) - e^{\varepsilon} p(y), 0)$, with totals $\alpha = \sum_y a(y)$ and $\beta = \sum_y b(y)$. Take S, the set of outcomes with $p(y) > e^{\varepsilon} q(y)$. Then $\alpha = \Pr[A \in S] - e^{\varepsilon} \Pr[B \in S] \le \delta$ by the definition. In the same way, $\beta \le \delta$.

2. *The rest.* Let $p_1(y) = p(y) - a(y) = \min(p(y), e^{\varepsilon} q(y))$ and $q_1(y) = q(y) - b(y) = \min(q(y), e^{\varepsilon} p(y))$. Then $p_1 \le e^{\varepsilon} q$ and $p_1 \le p \le e^{2\varepsilon} p$, so $p_1 \le e^{\varepsilon} q_1$. In the same way, $q_1 \le e^{\varepsilon} p_1$. The totals of p_1 and q_1 are $1 - \alpha$ and $1 - \beta$.

3. *Equal totals.* Suppose $\alpha \le \beta$; otherwise swap the roles of p and q. The totals of step 2 differ by $\sum_y (p_1(y) - q_1(y)) = \beta - \alpha$. So the outcomes with $p_1(y) > q_1(y)$ carry an excess $c = \sum_y \max(p_1(y) - q_1(y), 0) \ge \beta - \alpha$. If $\beta > \alpha$, remove the fraction $\theta = (\beta - \alpha)/c \le 1$ of that excess:
$$p_2(y) = p_1(y) - \theta \max(p_1(y) - q_1(y), 0), \tag{6.7}$$
and if $\beta = \alpha$, let $p_2 = p_1$. Let $q_2 = q_1$. Where p_2 differs from p_1, it lies between q_1 and p_1, so the bounds of step 2 hold for p_2 and q_2. Both now have the total $1 - \beta$.

4. *Scale.* Let $p_0 = t\, p_2$ and $q_0 = t\, q_2$ with $t = (1 - \delta)/(1 - \beta)$, which is in $(0, 1]$ because $\beta \le \delta < 1$. Then p_0 and q_0 have the total $1 - \delta$, satisfy $p_0 \le e^{\varepsilon} q_0$ and $q_0 \le e^{\varepsilon} p_0$, and $p_0 \le p$, $q_0 \le q$.

5. *The two catastrophic outcomes.* If $\delta > 0$, let F map "I am U" to a sample from $R_U = (p - p_0)/\delta$ and "I am V" to a sample from $R_V = (q - q_0)/\delta$. By step 4 both are non-negative, and each sums to $(1 - (1 - \delta))/\delta = 1$. If $\delta = 0$, these two outcomes never occur.

6. *The two ordinary outcomes.* If $\varepsilon = 0$, step 4 gives $p_0 = q_0$; let F map 0 and 1 to a sample from $p_0/(1-\delta)$. If $\varepsilon > 0$, define

$$P_0(y) = \frac{e^\varepsilon p_0(y) - q_0(y)}{(e^\varepsilon - 1)(1-\delta)}, \qquad P_1(y) = \frac{e^\varepsilon q_0(y) - p_0(y)}{(e^\varepsilon - 1)(1-\delta)}. \tag{6.8}$$

   Both are non-negative by step 4, and each sums to $(e^\varepsilon - 1)(1-\delta)/((e^\varepsilon - 1)(1-\delta)) = 1$. Let F map 0 to a sample from P_0 and 1 to a sample from P_1.

7. *Check.* U is 0 with probability $e^\varepsilon(1-\delta)/(1+e^\varepsilon)$, 1 with probability $(1-\delta)/(1+e^\varepsilon)$ and "I am U" with probability δ. So, for $\varepsilon > 0$, $F(U)$ equals y with probability

$$\frac{e^\varepsilon(e^\varepsilon p_0(y) - q_0(y)) + e^\varepsilon q_0(y) - p_0(y)}{(1+e^\varepsilon)(e^\varepsilon - 1)} + \delta R_U(y) = p_0(y) + p(y) - p_0(y) = p(y). \tag{6.9}$$

   For $\varepsilon = 0$, the probability is $(1-\delta)\,p_0(y)/(1-\delta) + p(y) - p_0(y) = p(y)$. In the same way, $F(V)$ equals y with probability $q(y)$.

For $\delta = 0$, steps 1 to 4 change nothing: $\alpha = \beta = 0$, $p_0 = p$ and $q_0 = q$. □

Applying Theorem 6.2 once per step of an adaptive composition reduces the whole transcript to a map of k independent copies of the four-outcome pair.

Theorem 6.3. *Let M be the adaptive composition of $M_1, \dots, M_k$ as in Theorem 6.1, and fix neighbours $D \sim D'$. Let $U_1, \dots, U_k$ be independent copies of U and $V_1, \dots, V_k$ independent copies of V. Then there is a randomized map F^* such that $F^*(U_1, \dots, U_k)$ has the distribution of $M(D)$ and $F^*(V_1, \dots, V_k)$ has the distribution of $M(D')$.*

Proof.

1. Fix a step j and an earlier transcript $y_{<j}$. The mechanism $M_j(\,\cdot\,; y_{<j})$ is (ε, δ)-DP, so its outputs $M_j(D; y_{<j})$ and $M_j(D'; y_{<j})$ are (ε, δ)-indistinguishable. Theorem 6.2 gives a randomized map $F_j^{y_{<j}}$ that turns U into $M_j(D; y_{<j})$ and V into $M_j(D'; y_{<j})$.
2. Define F^* on an input $(z_1, \dots, z_k)$ step by step. Set $y_1 = F_1(z_1)$, then $y_2 = F_2^{y_1}(z_2)$, and in general $y_j = F_j^{y_{<j}}(z_j)$, with fresh coins at each step. Output $(y_1, \dots, y_k)$.
3. By induction on j, the prefix $(y_1, \dots, y_j)$ computed from $(U_1, \dots, U_k)$ has the distribution of the first j outputs of $M(D)$. For $j = 1$ this is step 1. Given the prefix $y_{<j}$, the input U_j is still a fresh copy of U, because it is independent of $U_1, \dots, U_{j-1}$ and of the coins used so far. So $y_j = F_j^{y_{<j}}(U_j)$ has the distribution of $M_j(D; y_{<j})$, which is how M draws its j-th output.
4. The same induction with $V_1, \dots, V_k$ gives the distribution of $M(D')$.

□

The map F^* depends on D and D', but that is harmless, because privacy is a statement about one pair of neighbours at a time and Lemma 6.2 applies to any map. So it remains to show that $(U_1, \dots, U_k)$ and $(V_1, \dots, V_k)$ are $(\tilde{\varepsilon}, \tilde{\delta})$-indistinguishable.

6.3. Composing the four-outcome pair

Write $\mathbf{U} = (U_1, \dots, U_k)$ and $\mathbf{V} = (V_1, \dots, V_k)$. As in the approximate DP chapter, the privacy loss of an outcome compares how likely it is under $\mathbf{U}$ and under $\mathbf{V}$. An outcome that is possible under $\mathbf{U}$ but impossible under $\mathbf{V}$ has infinite loss, and for an outcome z that is possible under $\mathbf{V}$, the **privacy loss** is

$$L(z) = \ln \frac{\Pr[\mathbf{U} = z]}{\Pr[\mathbf{V} = z]}. \tag{6.10}$$

The plan is to show that, for z drawn from $\mathbf{U}$, the loss rarely exceeds $\tilde{\varepsilon}$. Two kinds of outcome are bad. An outcome that contains "I am U" is impossible under $\mathbf{V}$, so its loss is infinite, which happens with probability about $k\delta$. On the other outcomes, the loss is a sum of k independent terms, and Hoeffding's inequality bounds the chance that it strays far above its mean by δ'. The first step computes the loss on the outcomes with no catastrophe, in which every coordinate is 0 or 1.

Lemma 6.3. *For every $z \in \{0,1\}^k$, the privacy loss is $L(z) = \varepsilon \sum_{j=1}^{k} (1 - 2z_j)$.*

Proof.

1. The coordinates are independent, so both probabilities factor and the log of the ratio is a sum: $L(z) = \sum_{j=1}^{k} \ln\left(\Pr[U = z_j] / \Pr[V = z_j]\right)$.
2. By Definition 6.2, the ratio is e^{ε} when $z_j = 0$ and $e^{-\varepsilon}$ when $z_j = 1$. In both cases its logarithm is $\varepsilon(1 - 2z_j)$.

□

Given no catastrophic outcome, the coordinates of $\mathbf{U}$ are independent bits, so the loss is a sum of independent terms, each ε or $-\varepsilon$. Hoeffding's inequality bounds its tail.

Take k independent random variables $W_1, \dots, W_k$, each with values in an interval $[l, u]$ of length $u - l$. **Hoeffding's inequality** bounds the chance that their sum exceeds its mean by a margin $\tau \geq 0$ or more,

$$\Pr\left[\sum_{j=1}^{k} W_j \geq \sum_{j=1}^{k} \mathbb{E}[W_j] + \tau\right] \leq \exp\left(-\frac{2\tau^2}{k(u-l)^2}\right). \tag{6.11}$$

With the loss and its tail bound, the main estimate follows: k copies of the four-outcome pair are as hard to tell apart as Theorem 6.1 claims.

Proposition 6.1. *Let $\varepsilon \geq 0$, $\delta \in [0, 1)$ and $\delta' \in (0, 1)$. Then $\mathbf{U}$ and $\mathbf{V}$ are $(\tilde{\varepsilon}, \tilde{\delta})$-indistinguishable, with $\tilde{\varepsilon}$ and $\tilde{\delta}$ as in Equation 6.3.*

Proof.

1. *The events.* Let E_1 be the event that some coordinate of $\mathbf{U}$ is "I am U". Let G be the set of good outcomes, the $z \in \{0,1\}^k$ with $L(z) \leq \tilde{\varepsilon}$. Every outcome z with $\Pr[\mathbf{U} = z] > 0$ that is not in G either lies in E_1 or lies in $\{0,1\}^k$ with $L(z) > \tilde{\varepsilon}$. Outcomes containing "I am V" have probability 0 under $\mathbf{U}$.

2. *The first bad event.* Each coordinate avoids "I am U" with probability $1-\delta$, independently, so $\Pr[E_1] = 1-(1-\delta)^k$. Bernoulli's inequality $(1-\delta)^k \geq 1-k\delta$ gives $\Pr[E_1] \leq k\delta$.

3. *The coordinates given no catastrophe.* Given that U_j is not "I am U", it is 0 with probability $\frac{e^\varepsilon(1-\delta)/(1+e^\varepsilon)}{1-\delta} = \frac{e^\varepsilon}{1+e^\varepsilon}$ and 1 otherwise. Conditioned on the complement $\bar{E}_1$, the coordinates stay independent, because $\bar{E}_1$ is an intersection of independent events, one per coordinate.

4. *The mean of the loss.* By Lemma 6.3, given $\bar{E}_1$ the loss is $\sum_j W_j$ with $W_j = \varepsilon(1-2U_j)$. Each W_j is ε with probability $e^\varepsilon/(1+e^\varepsilon)$ and $-\varepsilon$ otherwise, so

$$\mathbb{E}\Big[\sum_{j=1}^{k} W_j \,\Big|\, \bar{E}_1\Big] = k\varepsilon\,\frac{e^\varepsilon - 1}{e^\varepsilon + 1}, \tag{6.12}$$

   which is the second term of $\tilde{\varepsilon}$.

5. *The second bad event.* If $\varepsilon = 0$, the loss is $0 \leq \tilde{\varepsilon}$ and there is nothing to bound. Otherwise apply Equation 6.11, conditioned on $\bar{E}_1$, with $[l,u] = [-\varepsilon,\varepsilon]$ and $\tau = \varepsilon\sqrt{2k\ln(1/\delta')}$, the first term of $\tilde{\varepsilon}$. The exponent is $2\tau^2/(4k\varepsilon^2) = \ln(1/\delta')$, so

$$\Pr\left[L(\mathbf{U}) > \tilde{\varepsilon} \,\middle|\, \bar{E}_1\right] \leq \delta'. \tag{6.13}$$

   Multiplying by $\Pr[\bar{E}_1] \leq 1$ bounds the probability that $\mathbf{U} \in \{0,1\}^k$ and $L(\mathbf{U}) > \tilde{\varepsilon}$ by δ'.

6. *Assembly.* Take any event S. Split it into $S \cap G$ and the rest. On G, each outcome has $\Pr[\mathbf{U} = z] \leq e^{\tilde{\varepsilon}} \Pr[\mathbf{V} = z]$ by the definition of L. By step 1, the rest has probability at most that of the two bad events. Steps 2 and 5 then give

$$\Pr[\mathbf{U} \in S] \leq e^{\tilde{\varepsilon}} \Pr[\mathbf{V} \in S \cap G] + k\delta + \delta' \leq e^{\tilde{\varepsilon}} \Pr[\mathbf{V} \in S] + \tilde{\delta}. \tag{6.14}$$

7. *The other direction.* Swapping the labels $0 \leftrightarrow 1$ and "I am U" $\leftrightarrow$ "I am V" in every coordinate maps $\mathbf{U}$ to $\mathbf{V}$ and back. The same argument then gives $\Pr[\mathbf{V} \in S] \leq e^{\tilde{\varepsilon}} \Pr[\mathbf{U} \in S] + \tilde{\delta}$.

□

So the term $k\delta$ of $\tilde{\delta}$ pays for a catastrophic outcome in some step, and δ' pays for an unusually long run of outcomes that favour $\mathbf{U}$. The proof of the theorem puts the pieces together.

Proof. Proof of Theorem 6.1. Fix neighbours $D \sim D'$.

1. By Theorem 6.3, $M(D)$ and $M(D')$ have the distributions of $F^*(\mathbf{U})$ and $F^*(\mathbf{V})$ for one randomized map F^*.
2. By Proposition 6.1, $\mathbf{U}$ and $\mathbf{V}$ are $(\tilde{\varepsilon},\tilde{\delta})$-indistinguishable.
3. By Lemma 6.2, so are $F^*(\mathbf{U})$ and $F^*(\mathbf{V})$. The neighbours were arbitrary, so M is $(\tilde{\varepsilon},\tilde{\delta})$-DP.

□

6.4. Summary

Table 6.1.: The chapter's results.

Result	Statement
Theorem 6.1	k adaptive (ε, δ)-DP steps are $(\varepsilon\sqrt{2k\ln(1/\delta')} + k\varepsilon\tanh(\varepsilon/2),\ k\delta + \delta')$-DP
Corollary 6.1	per-step $\varepsilon = \varepsilon_{\text{total}}/(2\sqrt{2k\ln(1/\delta')})$ gives a total of $\varepsilon_{\text{total}} \leq 1$
Example 6.2	Laplace noise per coordinate estimates a mean in $\{0,1\}^d$ with error $4d\sqrt{\ln(1/\delta')}/(n\varepsilon)$
Lemma 6.1	the four-outcome pair is (ε, δ)-indistinguishable
Lemma 6.2	indistinguishability survives post-processing
Theorem 6.2	every (ε, δ)-indistinguishable pair is a randomized map of the four-outcome pair
Theorem 6.3	an adaptive composition is a map of k independent copies of the four-outcome pair
Lemma 6.3	on $\{0,1\}^k$, the privacy loss is $\varepsilon\sum_j(1-2z_j)$
Proposition 6.1	k copies of the four-outcome pair are $(\tilde{\varepsilon}, \tilde{\delta})$-indistinguishable

6.5. References

- Cynthia Dwork, Guy N. Rothblum and Salil Vadhan, "Boosting and Differential Privacy", 2010. First proves the advanced composition theorem.
- Peter Kairouz, Sewoong Oh and Pramod Viswanath, "The Composition Theorem for Differential Privacy", 2015. Proves the simulation by the four-outcome pair and finds the exact optimal composition.
- Wassily Hoeffding, "Probability Inequalities for Sums of Bounded Random Variables", 1963. Gives the tail bound used for the privacy loss.
- Cynthia Dwork and Aaron Roth, *The Algorithmic Foundations of Differential Privacy*, 2014. Proves advanced composition with the privacy loss in Section 3.5, with $k\varepsilon(e^{\varepsilon} - 1)$ as the second term.
- Salil Vadhan, "The Complexity of Differential Privacy", 2017. Presents basic and advanced composition in Section 2.
- Jack Murtagh and Salil Vadhan, "The Complexity of Computing the Optimal Composition of Differential Privacy", 2016. Shows that the exact optimal composition is hard to compute for mixed parameters.

7. The Exponential Mechanism

The Laplace and Gaussian mechanisms release numbers. They work because a small change in a number does little harm. Many tasks instead ask for the best object from a set, such as a price or a classifier, and there a small change in the output can ruin its quality. The exponential mechanism samples each object with a probability that grows exponentially with a score. So good objects are likely, while the choice reveals little about any one person, as differential privacy (DP) requires. This chapter proves that the mechanism is private and nearly optimal, and applies it to three problems.

7.1. Selecting an object

A seller has a digital good, such as an e-book, that costs nothing to copy, and n buyers who each want one copy. Buyer i values the good at v_i and buys it if the price p is at most v_i. The seller's revenue at price p is the price times the number of buyers,

$$p \cdot \big|\{i : v_i \geq p\}\big|. \tag{7.1}$$

The seller could compute the price with the highest revenue from the valuations, but that price can reveal a single buyer. If one buyer values the good far above the others, the best price may be that buyer's valuation.

Noise added to the best price does not help, because revenue is very sensitive to the price. Take three buyers who value the good at \$1 and one who values it at \$2.50. At a price of \$1 all four buy and the revenue is \$4, but at \$1.01 only the last buyer buys and the revenue is \$1.01. So a cent of noise can cost three quarters of the revenue. It is better to treat the prices as objects with a quality, the revenue, than as numbers to perturb.

In general, the input to the selection problem has three parts:

- a dataset $D \in \mathcal{X}^n$,
- a set $\mathcal{R}$ of objects,
- a **score function** $u \colon \mathcal{X}^n \times \mathcal{R} \to \mathbb{R}$, where $u(D, r)$ says how good the object r is for the dataset D.

In the example, D is the list of valuations, $\mathcal{R}$ is the set of prices, and $u(D, p)$ is the revenue at price p. Only the dataset is private, while the objects and the score function are public. So the sensitivity measures how much the score of a fixed object can move when one record changes.

Definition 7.1 (Sensitivity of a score function)**.** The **sensitivity** of a score function u is

$$\Delta u = \sup_{r \in \mathcal{R}} \sup_{D \sim D'} \left| u(D, r) - u(D', r) \right|, \tag{7.2}$$

where the inner supremum is over neighbouring datasets, which differ in one record (the replace-one relation).

The score is a single real number, so it needs no choice of norm. The mechanism gives each object r a weight $\exp(\varepsilon u(D, r)/(2\Delta))$, where ε is the privacy parameter and Δ is any upper bound on the sensitivity Δu. Division by the sum of all weights, the normalizer, turns the weights into probabilities. For an infinite set of objects, such as an interval of prices, the weights become a density. The sum then becomes an integral against a fixed base measure μ on $\mathcal{R}$, such as length on the real line.

Definition 7.2 (Exponential mechanism)**.** Let $\varepsilon > 0$, and let $\Delta > 0$ satisfy $\Delta \geq \Delta u$. Let μ be a measure on $\mathcal{R}$, taken to be the counting measure, which gives each object weight 1, when $\mathcal{R}$ is finite. Assume that the normalizer

$$Z(D) = \int_{\mathcal{R}} \exp\left(\frac{\varepsilon \, u(D, r')}{2\Delta} \right) \mu(\mathrm{d}r') \tag{7.3}$$

is positive and finite for every D. The **exponential mechanism** M_E outputs an object $r \in \mathcal{R}$ drawn with density, with respect to μ,

$$p_D(r) = \frac{1}{Z(D)} \exp\left(\frac{\varepsilon \, u(D, r)}{2\Delta} \right). \tag{7.4}$$

For finite $\mathcal{R}$, Equation 7.4 says that $\Pr[M_E(D) = r]$ is proportional to $\exp(\varepsilon u(D, r)/(2\Delta))$, so a higher score makes an object more likely. A smaller ε or a larger Δ flattens the distribution towards uniform, which is more private and less accurate.

The definition says nothing about how to sample. For a finite set, you can compute every weight, add them up and sample, in time proportional to $|\mathcal{R}|$, which is slow when $\mathcal{R}$ is exponentially large. For an infinite set, the usual remedy is to discretize, as Section 7.4.2 does. So the exponential mechanism always gives an algorithm, but not always an efficient one.

7.2. Privacy

The score of r changes by at most Δ between neighbours, so the weight of r changes by at most a factor $e^{\varepsilon/2}$. The normalizer is a sum of such weights, so it also changes by at most $e^{\varepsilon/2}$, and the two factors together give e^{ε}.

Theorem 7.1. *The exponential mechanism is ε-DP.*

Proof.

1. Fix neighbours $D \sim D'$ and an object r. By Equation 7.4, the ratio of the densities is

$$\frac{p_D(r)}{p_{D'}(r)} = \exp\left(\frac{\varepsilon\,(u(D,r) - u(D',r))}{2\Delta}\right) \cdot \frac{Z(D')}{Z(D)}. \tag{7.5}$$

2. By Definition 7.1 and $\Delta \geq \Delta u$, the difference $u(D,r) - u(D',r)$ is at most Δ, so the first factor is at most $e^{\varepsilon/2}$.

3. For every object r', the same bound gives $u(D',r') \leq u(D,r')+\Delta$, so $\exp(\varepsilon u(D',r')/(2\Delta)) \leq e^{\varepsilon/2}\exp(\varepsilon u(D,r')/(2\Delta))$. Integrating over r' gives $Z(D') \leq e^{\varepsilon/2}Z(D)$, so the second factor is at most $e^{\varepsilon/2}$.

4. So $p_D(r) \leq e^{\varepsilon} p_{D'}(r)$ for every r. For an event $S \subseteq \mathcal{R}$, integrating over S gives $\Pr[M_E(D) \in S] \leq e^{\varepsilon}\Pr[M_E(D') \in S]$.

□

Step 3 spends the factor 2 in 2Δ, because it pays for the change in the normalizer. When the normalizer does not depend on the dataset, that half of the budget is not needed.

Corollary 7.1. *Suppose that $Z(D)$, computed with Δ in place of 2Δ, is the same for every dataset D. Then the mechanism with density proportional to $\exp(\varepsilon u(D,r)/\Delta)$ is ε-DP.*

Proof. In step 1 of the proof of Theorem 7.1, with Δ in place of 2Δ, the factor $Z(D')/Z(D)$ equals 1, and the first factor is at most $\exp(\varepsilon\Delta/\Delta) = e^{\varepsilon}$. Step 4 then applies unchanged. □

7.3. Utility

The mechanism should output an object whose score is close to the best one. Write the best score and the set of objects that attain it as

$$\mathrm{OPT}(D) = \max_{r\in\mathcal{R}} u(D,r), \qquad \mathcal{R}^*(D) = \{r \in \mathcal{R} : u(D,r) = \mathrm{OPT}(D)\}. \tag{7.6}$$

The theorem says that the output's score is rarely more than about $(2\Delta/\varepsilon)\ln|\mathcal{R}|$ below $\mathrm{OPT}(D)$. So the loss grows linearly with the sensitivity and with $1/\varepsilon$, but only logarithmically with the number of objects.

Theorem 7.2. *Let $\mathcal{R}$ be finite. For every dataset D and every $t \geq 0$,*

$$\Pr\left[u(D, M_E(D)) \leq \mathrm{OPT}(D) - \frac{2\Delta}{\varepsilon}\left(\ln\frac{|\mathcal{R}|}{|\mathcal{R}^*(D)|} + t\right)\right] \leq e^{-t}. \tag{7.7}$$

Proof.

1. Fix a threshold τ. The output has score at most τ exactly when it is one of the objects with $u(D, r) \leq \tau$, so

$$\Pr[u(D, M_E(D)) \leq \tau] = \frac{\sum_{r:u(D,r)\leq\tau} \exp(\varepsilon u(D, r)/(2\Delta))}{\sum_{r' \in \mathcal{R}} \exp(\varepsilon u(D, r')/(2\Delta))}. \tag{7.8}$$

2. *Numerator.* There are at most $|\mathcal{R}|$ terms, and each is at most $\exp(\varepsilon\tau/(2\Delta))$ because its score is at most τ.

3. *Denominator.* Every term is positive, so keeping only the $|\mathcal{R}^*(D)|$ terms with score $\mathrm{OPT}(D)$ gives a lower bound of $|\mathcal{R}^*(D)| \exp(\varepsilon \,\mathrm{OPT}(D)/(2\Delta))$.

4. Together, steps 2 and 3 give

$$\Pr[u(D, M_E(D)) \leq \tau] \leq \frac{|\mathcal{R}|}{|\mathcal{R}^*(D)|} \exp\left(\frac{\varepsilon\,(\tau - \mathrm{OPT}(D))}{2\Delta}\right). \tag{7.9}$$

5. Set $\tau = \mathrm{OPT}(D) - (2\Delta/\varepsilon)(\ln(|\mathcal{R}|/|\mathcal{R}^*(D)|) + t)$. The exponent becomes $-\ln(|\mathcal{R}|/|\mathcal{R}^*(D)|) - t$, and the right-hand side becomes e^{-t}.

□

Since at least one object attains the best score, dropping $|\mathcal{R}^*(D)|$ gives the form that is used most often.

Corollary 7.2. *Let $\mathcal{R}$ be finite. For every dataset D and every $t \geq 0$,*

$$\Pr\left[u(D, M_E(D)) \leq \mathrm{OPT}(D) - \frac{2\Delta}{\varepsilon}(\ln|\mathcal{R}| + t)\right] \leq e^{-t}. \tag{7.10}$$

Proof.

1. At least one object attains the best score, so $|\mathcal{R}^*(D)| \geq 1$ and $\ln(|\mathcal{R}|/|\mathcal{R}^*(D)|) \leq \ln|\mathcal{R}|$. The threshold in Equation 7.10 is therefore at most the threshold of Theorem 7.2 for the same t, so the event here is contained in the event there, and its probability is at most e^{-t}.

□

7.4. Applications

Each application below follows the same recipe. It chooses the set of objects and the score function, bounds the sensitivity, and reads off privacy and accuracy from Theorem 7.1 and Corollary 7.2.

7.4.1. The Laplace mechanism

Let $f\colon \mathcal{X}^n \to \mathbb{R}$ have sensitivity Δf, which means that $|f(D) - f(D')| \le \Delta f$ for all neighbours. Take the real line as the set of objects, with Lebesgue measure, and score each candidate answer by its distance from the true one,

$$u(D, r) = -|f(D) - r|. \tag{7.11}$$

By the triangle inequality, $\big||f(D) - r| - |f(D') - r|\big| \le |f(D) - f(D')| \le \Delta f$, so $\Delta u \le \Delta f$. With $\Delta = \Delta f$, the density Equation 7.4 is proportional to $\exp(-\varepsilon|f(D) - r|/(2\Delta f))$, which is the density of $f(D)$ plus Laplace noise of scale $2\Delta f/\varepsilon$. So the exponential mechanism contains a Laplace mechanism, but with twice the noise scale $\Delta f/\varepsilon$ of the Laplace mechanism. Corollary 7.1 removes the extra factor.

Proposition 7.1. *Let $f\colon \mathcal{X}^n \to \mathbb{R}$ have sensitivity $\Delta f > 0$. The mechanism that outputs $f(D)$ plus noise from the Laplace distribution with scale $\Delta f/\varepsilon$ is ε-DP.*

Proof.

1. The output has density proportional to $\exp(-\varepsilon|f(D) - r|/(\Delta f)) = \exp(\varepsilon u(D, r)/(\Delta f))$, with the score above, whose sensitivity is at most Δf.
2. The normalizer is $\int_{\mathbb{R}} \exp(-\varepsilon|f(D) - r|/(\Delta f))\,\mathrm{d}r = 2\Delta f/\varepsilon$, after the substitution $u = r - f(D)$. It does not depend on D.
3. Corollary 7.1, with $\Delta = \Delta f$, applies and gives ε-DP.

□

7.4.2. Pricing a digital good

Return to the seller of Section 7.1, now with a dataset of valuations $D = (v_1, \dots, v_n) \in [0, 1]^n$ and prices in $[0, 1]$. Write γ for the best revenue over all prices,

$$\gamma = \max_{p \in [0,1]} p \cdot |\{i : v_i \ge p\}|. \tag{7.12}$$

The maximum exists, because the revenue increases linearly between consecutive valuations and drops just after each one, so it is attained at one of the v_i or at 0.

The set of all prices is infinite, and the revenue has no structure that makes sampling easy, so the mechanism runs on a grid. Fix an integer $m \ge 1$ and the step $\alpha = 1/m$, and use the m prices

$$\mathcal{R} = \{\alpha, 2\alpha, \dots, 1\}. \tag{7.13}$$

Write $\mathrm{OPT}(D)$ for the best revenue over the grid. Rounding the best price down to the grid loses a little revenue, which the next lemma bounds.

Lemma 7.1. *The best revenue on the grid satisfies $\mathrm{OPT}(D) \ge \gamma - \alpha n$.*

Proof.

1. Let p^* attain γ. If $p^* < \alpha$, then $\gamma \le p^* n < \alpha n$, so $\gamma - \alpha n < 0$. Revenue is never negative, so $\mathrm{OPT}(D) \ge 0 > \gamma - \alpha n$.
2. Otherwise, round p^* down to the grid, $p' = \alpha \lfloor p^*/\alpha \rfloor \in \mathcal{R}$, so that $p^* - \alpha < p' \le p^*$.
3. Every buyer with $v_i \ge p^*$ also has $v_i \ge p'$. So the revenue at p' is at least $p' \cdot |\{i : v_i \ge p^*\}| \ge (p^* - \alpha)|\{i : v_i \ge p^*\}| \ge \gamma - \alpha n$.

□

So the private price loses revenue twice, once to rounding and once to the random selection. The theorem adds the two losses, with a grid of about εn prices.

Theorem 7.3. *Let $\varepsilon > 0$, $m = \lceil \varepsilon n \rceil$ and $\alpha = 1/m$. The exponential mechanism with the grid $\mathcal{R}$, the revenue $u(D, p) = p \cdot |\{i : v_i \ge p\}|$ as score and $\Delta = 1$ is ε-DP. For every $D \in [0,1]^n$ and $t \ge 0$, with probability at least $1 - e^{-t}$, the revenue at its output price is at least*

$$\gamma - \frac{1 + 2\ln\lceil \varepsilon n \rceil + 2t}{\varepsilon}. \tag{7.14}$$

Proof.

1. *Sensitivity.* Changing one valuation changes the number of buyers at any price p by at most 1, and $p \le 1$. So $\Delta u \le 1 = \Delta$, and Theorem 7.1 gives ε-DP.
2. *Selection.* By Corollary 7.2 with $|\mathcal{R}| = m$, with probability at least $1 - e^{-t}$ the output has revenue at least $\mathrm{OPT}(D) - (2/\varepsilon)(\ln m + t)$.
3. *Discretization.* By Lemma 7.1, $\mathrm{OPT}(D) \ge \gamma - n/m$, and $n/m \le n/(\varepsilon n) = 1/\varepsilon$ because $m \ge \varepsilon n$.
4. Adding the two losses gives Equation 7.14.

□

The loss is $O(\log(\varepsilon n)/\varepsilon)$. When a constant fraction of buyers values the good above a constant price, γ grows linearly in n, so the loss is a vanishing fraction of the best revenue. The grid size balances the two losses: a finer grid loses less to rounding, αn, but more to selection, $(2/\varepsilon)\ln(1/\alpha)$. The next example compares the guarantee with the revenue the mechanism actually earns.

Example 7.1. Draw $n = 1{,}000$ valuations uniformly from $[0, 1]$ and take $\varepsilon = 0.5$, so that the grid has $m = 500$ prices. The best revenue is $\gamma = 267.40$, and the best grid price earns 266.88. With $t = \ln 20$, Equation 7.14 guarantees a revenue of at least $267.40 - 38.84 = 228.56$ with probability 0.95. The exact output distribution does much better. Its mean revenue is 263.86, and its 5th percentile is 256.52, a loss of 10.88, so the guarantee is about 3.6 times too pessimistic. Figure 7.1 shows the output distribution on a smaller example.

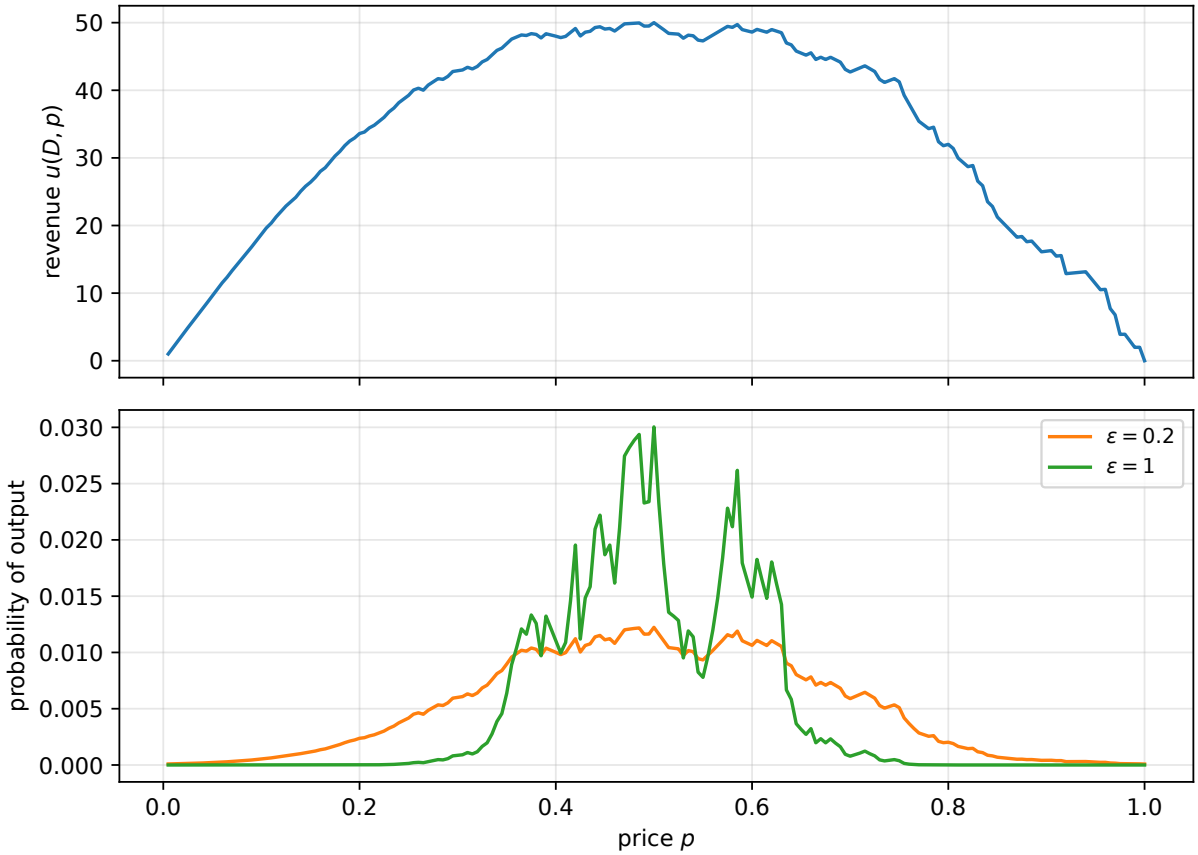


Figure 7.1.: The exponential mechanism for pricing, with 200 buyers whose valuations are uniform on $[0, 1]$ and a grid of 200 prices. Top, the revenue. Bottom, the probability of each output price for $\varepsilon = 0.2$ and $\varepsilon = 1$. A larger ε puts more weight on prices near the best revenue.

7.4.3. Private PAC learning

In the probably approximately correct (PAC) model of learning, a learner sees labelled records and must output a classifier that is accurate on new points from the same source. Here the records are points of the Boolean cube $\{0,1\}^d$, labelled 0 or 1. The definition measures accuracy by an error α and a failure probability β, so in this section α is no longer the grid step of the pricing example.

Definition 7.3 (PAC learning)**.** A **concept class** is a finite set $\mathcal{C}$ of functions $c\colon \{0,1\}^d \to \{0,1\}$. An unknown **target** $c^* \in \mathcal{C}$ and an unknown distribution $\mathcal{D}$ on $\{0,1\}^d$ generate the dataset. Each record is a point x_i, drawn independently from $\mathcal{D}$, together with its label $y_i = c^*(x_i)$. So the dataset is $D = (z_1, \dots, z_n)$ with labelled records $z_i = (x_i, y_i)$. The **error** of a function h is

$$\operatorname{err}(h) = \Pr_{x \sim \mathcal{D}}[h(x) \neq c^*(x)]. \tag{7.15}$$

A learner is (α, β)**-accurate** if, for every target and every $\mathcal{D}$, it outputs a function $\hat{h}$ with $\operatorname{err}(\hat{h}) \leq \alpha$ with probability at least $1 - \beta$.

For example, $\mathcal{C}$ might be a finite set of half-spaces, each of which labels 1 the points on one side of a hyperplane.

Without privacy, the classic learner outputs any function in $\mathcal{C}$ that labels every record correctly, and one exists because the target does. It works because, for every function at once, the fraction of mistakes on the sample is close to the error. Write that fraction as

$$\widehat{\operatorname{err}}(h) = \frac{1}{n}\big|\{i : h(x_i) \neq y_i\}\big|. \tag{7.16}$$

The tool is Hoeffding's inequality: the average of n independent random variables with values in $[0,1]$ falls below its mean by a margin $s \geq 0$ or more with probability at most $\exp(-2ns^2)$. Applied to the mistakes of one function, with a union bound over $\mathcal{C}$, it shows that the sample error is close to the true error for every function at once.

Lemma 7.2. *Let $\alpha, \beta \in (0,1)$ and $n \geq 2\ln(|\mathcal{C}|/\beta)/\alpha^2$. With probability at least $1 - \beta$ over the sample, every $h \in \mathcal{C}$ satisfies $\operatorname{err}(h) \leq \widehat{\operatorname{err}}(h) + \alpha/2$. In particular, every function in $\mathcal{C}$ that labels the whole sample correctly has error at most $\alpha/2$.*

Proof.

1. Fix $h \in \mathcal{C}$. The indicators $W_i = \mathbf{1}[h(x_i) \neq c^*(x_i)]$ are independent, lie in $[0,1]$, and have mean $\operatorname{err}(h)$, because the x_i are independent draws from $\mathcal{D}$. Their average is $\widehat{\operatorname{err}}(h)$.
2. Hoeffding's inequality with $s = \alpha/2$ gives $\Pr[\widehat{\operatorname{err}}(h) \leq \operatorname{err}(h) - \alpha/2] \leq \exp(-n\alpha^2/2) \leq \beta/|\mathcal{C}|$, by the bound on n.
3. A union bound over the $|\mathcal{C}|$ functions shows that some h fails with probability at most β.
4. A function that labels the sample correctly has $\widehat{\operatorname{err}}(h) = 0$, so its error is at most $\alpha/2$.

□

So about $\ln|\mathcal{C}|/\alpha^2$ records suffice.

For privacy, neighbouring datasets differ in one labelled record: any pair $z_i' = (x_i', y_i')$ can replace $z_i = (x_i, y_i)$. The new point need not come from $\mathcal{D}$, and its label need not equal $c^*(x_i')$. Accuracy is proved for data drawn from the model, but privacy must hold for every dataset, so that a wrong model never costs anyone their privacy.

The private learner keeps the aim of few mistakes on the sample, but chooses the function with the exponential mechanism. Its score is minus the sample error, so a function with fewer mistakes is more likely.

Theorem 7.4 (Private PAC learning)**.** *Let $\varepsilon > 0$ and $\alpha, \beta \in (0,1)$. Run the exponential mechanism with $\mathcal{R} = \mathcal{C}$, the score $u(D,h) = -\widehat{\mathrm{err}}(h)$ and $\Delta = 1/n$. It is ε-DP for every dataset. If*

$$n \geq \frac{2\ln(2|\mathcal{C}|/\beta)}{\alpha^2} \qquad \text{and} \qquad n \geq \frac{4\ln(2|\mathcal{C}|/\beta)}{\alpha\varepsilon}, \tag{7.17}$$

then it is an (α, β)-accurate learner.

Proof.

1. *Privacy.* Replacing one labelled record changes the number of mistakes of each h by at most 1, so $\Delta u \leq 1/n$. Theorem 7.1 gives ε-DP for all datasets, whatever their labels.
2. *Selection.* The target makes no mistakes, so $\mathrm{OPT} = 0$. By Corollary 7.2 with $t = \ln(2/\beta)$, with probability at least $1 - \beta/2$ over the mechanism's coins, the output $\hat{h}$ has

$$\widehat{\mathrm{err}}(\hat{h}) \leq \frac{2}{\varepsilon n}\big(\ln|\mathcal{C}| + \ln(2/\beta)\big) = \frac{2\ln(2|\mathcal{C}|/\beta)}{\varepsilon n} \leq \frac{\alpha}{2}, \tag{7.18}$$

   where the last step uses the second bound on n.
3. *Generalization.* The first bound on n is the condition of Lemma 7.2 with $\beta/2$ in place of β. So with probability at least $1 - \beta/2$ over the sample, every $h \in \mathcal{C}$, including $\hat{h}$, has $\mathrm{err}(h) \leq \widehat{\mathrm{err}}(h) + \alpha/2$.
4. By a union bound, both events hold with probability at least $1 - \beta$, and then $\mathrm{err}(\hat{h}) \leq \alpha/2 + \alpha/2 = \alpha$.

□

The first bound on n is what learning needs without privacy, and the second is the cost of privacy. Both are logarithmic in $|\mathcal{C}|$, and for ε of constant size and small α, the cost of privacy is a lower-order term. For instance, with $|\mathcal{C}| = 2^{20}$, $\alpha = \beta = 0.05$ and $\varepsilon = 1$, we have $\ln(2|\mathcal{C}|/\beta) = 17.552$, and the two bounds are 14,041 and 1,404. The price is computation, because the mechanism scores every function in $\mathcal{C}$.

7.5. Summary

Table 7.1.: The chapter's results.

Result	Statement
Definition 7.2	output r with probability proportional to $\exp(\varepsilon u(D, r)/(2\Delta))$
Theorem 7.1	the exponential mechanism is ε-DP
Corollary 7.1	with a normalizer independent of D, $\exp(\varepsilon u/\Delta)$ is ε-DP
Theorem 7.2	the score falls below $\mathrm{OPT} - \frac{2\Delta}{\varepsilon}(\ln \frac{\lvert\mathcal{R}\rvert}{\lvert\mathcal{R}^*\rvert} + t)$ with probability at most e^{-t}
Corollary 7.2	the same with $\ln \lvert\mathcal{R}\rvert$
Proposition 7.1	the Laplace mechanism is the exponential mechanism with score $-\lvert f(D) - r\rvert$
Theorem 7.3	a private price earns $\gamma - O(\log(\varepsilon n)/\varepsilon)$
Lemma 7.2	$2\ln(\lvert\mathcal{C}\rvert/\beta)/\alpha^2$ records make every sample error close to the true error
Theorem 7.4	private PAC learning with $O(\ln(\lvert\mathcal{C}\rvert/\beta)(1/\alpha^2 + 1/(\alpha\varepsilon)))$ records

7.6. References

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Part III.

Query Release and Its Limits

The second part built mechanisms that answer one question, or a few, under differential privacy (DP). This part asks what happens when an analyst wants many answers from the same dataset. Its first two chapters answer many queries for far less than one budget share per query. The third chapter lets the noise follow the dataset at hand instead of the worst case. The last proves that, under pure DP, no mechanism can do much better than the simple ones on marginals and histograms.

Table 7.2.: The chapters of this part.

Chapter	Question it answers
Private multiplicative weights	How many records are needed to answer a large set of linear queries at once?
Sparse vector	How can a stream of queries be answered when the cost should depend only on the few large ones?
Beyond global sensitivity	When can the noise follow the sensitivity of the actual dataset instead of the worst case?
Packing lower bounds	How much error must any pure DP mechanism incur on marginals and histograms?

8. Private Multiplicative Weights

An analyst wants the answers to many statistical queries about a dataset, such as the fraction of records in each cell of a table. Independent noise on each answer makes the error grow polynomially with the number of queries. Under approximate differential privacy (DP), this chapter does much better. The number of records it needs grows only with the logarithm of the number of queries and of the size of the universe. The algorithm, private multiplicative weights, comes from online learning. It learns a distribution over the universe that agrees with the data on every query, and it reads the data only through a few private measurements.

8.1. Linear queries

Records take values in a set $\mathcal{X}$, the universe, and a dataset is a list of n records. A linear query asks for the average of a bounded function over the records.

Definition 8.1 (Linear query)**.** A **linear query** is a function $q\colon \mathcal{X} \to [0,1]$ on the universe. On a dataset $D \in \mathcal{X}^n$ with records $x_1, \dots, x_n$ it takes the value

$$q(D) = \frac{1}{n} \sum_{i=1}^{n} q(x_i). \tag{8.1}$$

A counting query is the special case where q takes only the values 0 and 1, and two families of counting queries come up often. In a **histogram**, there is one query $q_s(x) = \mathbf{1}[x = s]$ for each element s of the universe, so the answers are the fractions of records equal to each s. In **one-way marginals**, the universe is $\{0,1\}^d$, and the query $q_j(x) = x_j$ for each coordinate j gives the fraction of records with feature j set to 1. This chapter allows any finite set of linear queries, however irregular.

The analyst wants every query in the set answered at once, so accuracy is the worst error over the set. A mechanism succeeds if, except with a small failure probability β, no answer is off by more than a target error α.

Definition 8.2 (Query release)**.** Let $\mathcal{Q}$ be a finite set of linear queries, $\alpha \in (0,1]$ a target error and $\beta \in (0,1)$ a failure probability. A mechanism $M\colon \mathcal{X}^n \to [0,1]^{\mathcal{Q}}$ outputs one answer $M(D)_q$ for each query $q \in \mathcal{Q}$. It answers $\mathcal{Q}$ **to accuracy** α with probability $1-\beta$ if, for every dataset D,

$$\Pr\left[\max_{q \in \mathcal{Q}} |M(D)_q - q(D)| \le \alpha\right] \ge 1 - \beta. \tag{8.2}$$

The question is how large n must be for a private mechanism to do this. The sample sizes below are sufficient conditions, and smaller is better.

The first answer uses only earlier chapters. Changing one record changes a linear query by at most $1/n$, so its **sensitivity**, the largest change over neighbouring datasets, is at most $1/n$. The Laplace mechanism adds noise from the Laplace distribution $\mathrm{Lap}(b)$ of scale b, whose density at z is $e^{-|z|/b}/(2b)$. It is ε-DP when b is the ℓ_1 sensitivity divided by ε, and its tails are $\Pr[|\mathrm{Lap}(b)| \geq tb] = e^{-t}$ for $t \geq 0$. Independent Laplace noise on every query in $\mathcal{Q}$, with the budget split evenly among them, gives the baseline that the rest of the chapter improves on.

Proposition 8.1 (Laplace baseline)**.** *Let $\varepsilon > 0$ and $\beta \in (0,1)$. Release $M(D)_q = q(D) + Z_q$ for every $q \in \mathcal{Q}$, with independent $Z_q \sim \mathrm{Lap}(|\mathcal{Q}|/(\varepsilon n))$. This mechanism is ε-DP, and it answers $\mathcal{Q}$ to accuracy α with probability $1-\beta$ whenever*

$$n \geq \frac{|\mathcal{Q}| \ln(|\mathcal{Q}|/\beta)}{\alpha\varepsilon}. \tag{8.3}$$

Proof.

1. The vector $(q(D))_{q\in\mathcal{Q}}$ changes by at most $1/n$ in each of its $|\mathcal{Q}|$ coordinates when one record changes, so its ℓ_1 sensitivity is at most $|\mathcal{Q}|/n$. The Laplace mechanism with scale $|\mathcal{Q}|/(\varepsilon n)$ is therefore ε-DP.
2. Write $b = |\mathcal{Q}|/(\varepsilon n)$ and $t = \ln(|\mathcal{Q}|/\beta)$. By the tail formula and a union bound over the $|\mathcal{Q}|$ coordinates, $\Pr[\max_q |Z_q| \geq tb] \leq |\mathcal{Q}| e^{-t} = \beta$.
3. The condition on n says exactly that $tb \leq \alpha$. So with probability at least $1-\beta$, every error is below α.

□

Approximate DP does better, through the advanced composition theorem. That theorem implies that k mechanisms run in sequence, each ε_0-DP and each possibly chosen after seeing the earlier outputs, are together (ε', δ)-DP for every $\delta \in (0,1)$, with

$$\varepsilon' = \varepsilon_0 \sqrt{2k \ln(1/\delta)} + k\varepsilon_0 \left(e^{\varepsilon_0} - 1\right). \tag{8.4}$$

To use the theorem, one starts from a target budget ε and asks how much each of the k mechanisms may spend. The next lemma gives a per-mechanism budget ε_0 that keeps ε' at most ε. It is the budget-splitting corollary of that chapter, proved again here for the form Equation 8.4.

Lemma 8.1 (Splitting a budget)**.** *Let $\varepsilon \in (0,1]$, $\delta \in (0, 1/2]$ and $k \geq 1$. The adaptive composition of k mechanisms, each ε_0-DP with*

$$\varepsilon_0 = \frac{\varepsilon}{2\sqrt{2k \ln(1/\delta)}}, \tag{8.5}$$

is (ε, δ)-DP.

Proof.

1. The first term of Equation 8.4 is $\varepsilon_0\sqrt{2k\ln(1/\delta)} = \varepsilon/2$.
2. Since $\ln(1/\delta) \geq \ln 2$, $\varepsilon \leq 1$ and $k \geq 1$, we have $\varepsilon_0 \leq 1$. The function e^x is convex, so $e^x - 1$ lies below the chord $(e-1)x \leq 2x$ on $[0,1]$. Hence the second term is at most $2k\varepsilon_0^2 = \varepsilon^2/(4\ln(1/\delta))$.
3. That is at most $\varepsilon/2$ because $\varepsilon \leq 1 \leq 2\ln 2 \leq 2\ln(1/\delta)$. Adding the two terms gives $\varepsilon' \leq \varepsilon$.

□

With one Laplace mechanism per query, so $k = |\mathcal{Q}|$, the lemma lets each query spend a budget of order $\varepsilon/\sqrt{|\mathcal{Q}|}$ instead of $\varepsilon/|\mathcal{Q}|$. So the factor $|\mathcal{Q}|$ in the baseline becomes $\sqrt{|\mathcal{Q}|}$.

Proposition 8.2 (Laplace with advanced composition)**.** *Let $\varepsilon \in (0,1]$, $\delta \in (0, 1/2]$ and $\beta \in (0,1)$. Release $M(D)_q = q(D) + Z_q$ with independent $Z_q \sim \mathrm{Lap}(1/(\varepsilon_0 n))$, where $\varepsilon_0 = \varepsilon/(2\sqrt{2|\mathcal{Q}|\ln(1/\delta)})$. This mechanism is (ε,δ)-DP, and it answers $\mathcal{Q}$ to accuracy α with probability $1-\beta$ whenever*

$$n \geq \frac{2\sqrt{2|\mathcal{Q}|\ln(1/\delta)}\,\ln(|\mathcal{Q}|/\beta)}{\alpha\varepsilon}. \tag{8.6}$$

Proof.

1. Each coordinate is the Laplace mechanism for a query of sensitivity $1/n$, so it is ε_0-DP. Lemma 8.1 with $k = |\mathcal{Q}|$ makes the whole release (ε,δ)-DP.
2. As in Proposition 8.1, $\Pr[\max_q |Z_q| \geq \ln(|\mathcal{Q}|/\beta)/(\varepsilon_0 n)] \leq \beta$, and the condition on n says that $\ln(|\mathcal{Q}|/\beta)/(\varepsilon_0 n) \leq \alpha$.

□

The Gaussian mechanism with advanced composition does a little better, $\sqrt{|\mathcal{Q}|\ln|\mathcal{Q}|\ln(1/\delta)}/(\alpha\varepsilon)$, because its lighter tails make the union bound cost $\sqrt{\ln|\mathcal{Q}|}$ instead of $\ln|\mathcal{Q}|$. These bounds are polynomial in $|\mathcal{Q}|$, and in the worst case over query sets nothing better is possible. A smaller universe helps, though, as Table 8.1 shows, up to constants.

Table 8.1.: Sufficient sample sizes for answering every query in $\mathcal{Q}$ to accuracy α.

Method	Privacy	Records needed, up to constants
Laplace, basic composition	ε-DP	$\lvert\mathcal{Q}\rvert\ln\lvert\mathcal{Q}\rvert/(\alpha\varepsilon)$, improvable to $\lvert\mathcal{Q}\rvert/(\alpha\varepsilon)$
Gaussian, advanced composition	(ε,δ)-DP	$\sqrt{\lvert\mathcal{Q}\rvert\ln\lvert\mathcal{Q}\rvert\ln(1/\delta)}/(\alpha\varepsilon)$
Noisy histogram	ε-DP	$\sqrt{\lvert\mathcal{X}\rvert\ln\lvert\mathcal{Q}\rvert}/(\alpha\varepsilon)$
Exponential mechanism over small datasets	ε-DP	$\ln\lvert\mathcal{Q}\rvert\ln\lvert\mathcal{X}\rvert/(\alpha^3\varepsilon)$, up to logarithmic factors
Private multiplicative weights	(ε,δ)-DP	$\ln\lvert\mathcal{Q}\rvert\sqrt{\ln\lvert\mathcal{X}\rvert\ln(1/\delta)}/(\alpha^2\varepsilon)$, up to logarithmic factors

The noisy histogram releases one noisy count per element of the universe and answers every query from it. So it pays polynomially in $|\mathcal{X}|$, which is 2^d for d binary features. The fourth row is logarithmic in both $|\mathcal{Q}|$ and $|\mathcal{X}|$, but its running time is $|\mathcal{X}|^{O(\ln|\mathcal{Q}|/\alpha^2)}$. Private multiplicative weights keeps both logarithms and runs in time polynomial in $|\mathcal{Q}|$, $|\mathcal{X}|$ and n. This one outputs synthetic data, and for such algorithms that is close to the best possible under standard cryptographic assumptions.

8.2. Learning from experts

Multiplicative weights is an algorithm for a game of repeated predictions, which has nothing to do with privacy. This section develops it from the simplest case.

A group of N experts makes predictions over T days. Each day, every expert predicts whether a stock goes up or down, and the algorithm sees these predictions before it makes its own. Then the true outcome is revealed, and a wrong prediction counts as a **mistake**. Suppose first that one expert, whose identity is unknown, is always right. Then the algorithm can stop listening to any expert as soon as it errs.

Proposition 8.3 (Halving)**.** *Predict with the majority of the experts that have never erred, breaking ties towards "down", and discard every expert that errs. If some expert never errs, this algorithm makes at most* $\log_2 N$ *mistakes.*

Proof.

1. Let S^t be the set of experts still kept on day t, so $|S^1| = N$.
2. On a day with a mistake, the experts who predicted like the algorithm are wrong. They are at least half of S^t, because the algorithm followed the majority. All of them are discarded, so $|S^{t+1}| \le |S^t|/2$.
3. The perfect expert is never discarded, so $|S^t| \ge 1$ on every day. After M mistakes, $1 \le N/2^M$, so $M \le \log_2 N$.

□

The algorithm is measured against the best expert in hindsight. Its **regret** is its total loss minus the total loss of the best single expert. Here the best expert makes no mistakes, so the regret is at most $\log_2 N$.

Now suppose instead that the best expert makes OPT mistakes. Then halving fails, because the best expert may err on the first day and be discarded, after which the rest may all be wrong forever. The repair is to lower trust instead of removing it.

Proposition 8.4 (Weighted majority)**.** *Give each expert weight* 1*. Each day, predict with the side whose experts have the larger total weight, breaking ties towards "down", and then halve the weight of every expert that erred. If the best expert makes* OPT *mistakes, the algorithm makes* M *mistakes with*

$$M \le \frac{\mathrm{OPT} + \log_2 N}{\log_2(4/3)} \le 2.41\left(\mathrm{OPT} + \log_2 N\right). \tag{8.7}$$

Proof.

1. Let W^t be the total weight at the start of day t, so $W^1 = N$. Weights never increase.
2. *Upper bound.* On a day with a mistake, the side the algorithm followed carries at least half of W^t, and all of its weight is halved. So $W^{t+1} \le W^t - W^t/4 = \frac{3}{4}W^t$. Over T days, $W^{T+1} \le N(3/4)^M$.
3. *Lower bound.* The best expert's weight is halved exactly OPT times, so $W^{T+1} \ge (1/2)^{\mathrm{OPT}}$.
4. Combining the bounds gives $(4/3)^M \le N\,2^{\mathrm{OPT}}$. Taking $\log_2$ of both sides gives the first inequality, and $1/\log_2(4/3) = 2.409\ldots$ gives the second.

□

The guarantee is competitive with the best expert, but its regret, $1.41\,\mathrm{OPT} + 2.41\log_2 N$, grows with OPT. A randomized algorithm with gentler updates removes the factor in front of OPT.

The general game has losses instead of mistakes. In each of T rounds, every expert i suffers a loss $\ell_i^t \in [-1, 1]$, chosen by an adversary, and a negative loss is a gain. A randomized algorithm picks one expert and suffers the loss of that expert. It is cleaner to let the algorithm pick a distribution p^t over the experts and suffer the average loss $\ell^t \cdot p^t = \sum_i p_i^t \ell_i^t$. Multiplicative weights keeps one weight per expert and plays the distribution proportional to the weights. After each round it multiplies each weight by a factor that falls as the loss of that expert rises.

Definition 8.3 (Multiplicative weights)**.** Fix a learning rate $\eta \in (0, 1/2]$. Start with weights $w_i^1 = 1$ for $i = 1, \ldots, N$. In round $t = 1, 2, \ldots$:

1. Play the distribution $p_i^t = w_i^t / W^t$, where $W^t = \sum_{i=1}^N w_i^t$.
2. Observe the loss vector $\ell^t \in [-1, 1]^N$, which may depend on $p^1, \ldots, p^t$, and suffer $\ell^t \cdot p^t$.
3. Update $w_i^{t+1} = w_i^t\,(1 - \eta \ell_i^t)$ for every i.

The factor $1 - \eta\ell_i^t$ lies in $[1/2, 3/2]$, so the weights stay positive, and an expert with a positive loss loses weight while one with a negative loss gains it. The analysis takes the logarithm of the total weight, which turns the products of factors $1 - \eta\ell_i^t$ into sums of $\ln(1 - \eta\ell_i^t)$. So it needs two bounds on that logarithm, one from above and one from below.

Lemma 8.2. *For every $x < 1$, $\ln(1 - x) \le -x$. For every $x \in [-1/2, 1/2]$, $\ln(1 - x) \ge -x - x^2$.*

Proof.

1. The function $\ln(1 - x)$ is concave, so it lies below its tangent line at 0, which is $-x$.
2. Let $h(x) = \ln(1 - x) + x + x^2$. Then $h(0) = 0$ and $h'(x) = -\frac{1}{1-x} + 1 + 2x = \frac{x(1-2x)}{1-x}$.
3. On $[0, 1/2]$, $h' \ge 0$, so $h(x) \ge h(0) = 0$. On $[-1/2, 0]$, $h' \le 0$, so h decreases to $h(0) = 0$ and is non-negative there too.

□

The main result compares the total loss of multiplicative weights with the total loss of any fixed distribution p chosen in hindsight. The difference is the regret, and the theorem bounds it by a term that grows with η and a term that shrinks with it.

Theorem 8.1 (Regret of multiplicative weights)**.** *Let $p^1, p^2, \ldots$ be the distributions that multiplicative weights plays. For every learning rate $\eta \in (0, 1/2]$, every number of rounds $T \geq 1$, every sequence of loss vectors $\ell^1, \ldots, \ell^T$ in $[-1, 1]^N$ and every fixed distribution p over the experts,*

$$\sum_{t=1}^{T} \ell^t \cdot p^t \leq \sum_{t=1}^{T} \ell^t \cdot p + \eta T + \frac{\ln N}{\eta}. \tag{8.8}$$

If $N \geq 2$, $T \geq 4 \ln N$ and $\eta = \sqrt{\ln N / T}$, the last two terms add up to $2\sqrt{T \ln N}$.

Proof.

1. *Upper bound on the total weight.* By the update rule and $p_i^t = w_i^t / W^t$,

$$W^{t+1} = \sum_{i=1}^{N} w_i^t (1 - \eta \ell_i^t) = W^t \left(1 - \eta \, \ell^t \cdot p^t\right). \tag{8.9}$$

Unrolling from $W^1 = N$ and applying the first bound of Lemma 8.2 to each factor,

$$\ln W^{T+1} = \ln N + \sum_{t=1}^{T} \ln \left(1 - \eta \, \ell^t \cdot p^t\right) \leq \ln N - \eta \sum_{t=1}^{T} \ell^t \cdot p^t. \tag{8.10}$$

2. *Lower bound through one expert.* The total weight is at least the weight of any expert i. Each $\eta \ell_i^t$ lies in $[-1/2, 1/2]$, so the second bound of Lemma 8.2 applies, and $(\ell_i^t)^2 \leq 1$:

$$\ln W^{T+1} \geq \ln w_i^{T+1} = \sum_{t=1}^{T} \ln(1 - \eta \ell_i^t) \geq -\eta \sum_{t=1}^{T} \ell_i^t - \eta^2 T. \tag{8.11}$$

3. *Lower bound through a distribution.* Multiply the inequality of step 2 by p_i and sum over i. The weights p_i sum to 1, so $\ln W^{T+1} \geq -\eta \sum_t \ell^t \cdot p - \eta^2 T$.
4. Combining steps 1 and 3 gives $-\eta \sum_t \ell^t \cdot p - \eta^2 T \leq \ln N - \eta \sum_t \ell^t \cdot p^t$. Dividing by η and rearranging gives Equation 8.8.
5. With $\eta = \sqrt{\ln N / T}$, both ηT and $\ln N / \eta$ equal $\sqrt{T \ln N}$. The condition $N \geq 2$ makes $\eta > 0$, and $T \geq 4 \ln N$ makes $\eta \leq 1/2$.

□

The regret is of order $\sqrt{T}$, while the losses themselves can add up to order T. So on average over rounds the algorithm does as well as the best fixed distribution, up to a term that vanishes as $\sqrt{\ln N / T}$. With p a point mass, the distribution that puts all its weight on one expert, the theorem covers the randomized algorithm that picks one expert.

Corollary 8.1. *Let $N \geq 2$. Fix losses $\ell^1, \dots, \ell^T \in [-1,1]^N$ in advance, with $T \geq 4 \ln N$, and let $L_i = \sum_t \ell_i^t$. Pick expert i_t with probability $p_{i_t}^t$, where p^t comes from Definition 8.3 with $\eta = \sqrt{\ln N / T}$. Then*

$$\mathbb{E}\Big[\sum_{t=1}^{T} \ell_{i_t}^t\Big] \leq \min_i L_i + 2\sqrt{T \ln N}. \tag{8.12}$$

Proof. The losses are fixed, so each p^t is fixed too, and $\mathbb{E}[\ell_{i_t}^t] = \ell^t \cdot p^t$. Sum over t and apply Theorem 8.1 with p the point mass on the best expert. □

8.3. Multiplicative weights for queries

To use the game for query release, the experts become the elements of the universe, and the distribution p^t becomes a candidate dataset. To compare a candidate with the true dataset, we also write the true dataset as a distribution over the universe.

Definition 8.4 (Histogram of a dataset)**.** The **histogram** of $D \in \mathcal{X}^n$ is the distribution p_D on $\mathcal{X}$ with $p_D(x) = |\{i : x_i = x\}|/n$, the fraction of records equal to x. For a linear query q and a distribution p on $\mathcal{X}$, write $\langle q, p\rangle = \sum_{x \in \mathcal{X}} q(x)p(x)$.

Grouping the records by value gives $q(D) = \langle q, p_D\rangle$. So answering the queries means finding a distribution $\hat{p}$ with $\langle q, \hat{p}\rangle$ close to $\langle q, p_D\rangle$ for every $q \in \mathcal{Q}$.

Picture a game in rounds. An adversary picks a query $q^t \in \mathcal{Q}$ after the algorithm commits to a distribution p^t, and the algorithm pays its error $|\langle q^t, p^t\rangle - \langle q^t, p_D\rangle|$. This looks pointless, since the algorithm could play $p^t = p_D$ from the start. But that move reads the whole dataset, while multiplicative weights reads it only through one query per round and the sign of its error, which are cheap to privatize. And a bound on the total error turns into a bound on the number of rounds with a large error, as shown below.

The loss vector comes from the error itself. Write the error as a function of the candidate, $f(p') = |\langle q^t, p'\rangle - \langle q^t, p_D\rangle|$. This function is convex, and $f(p_D) = 0$. Where the error is not zero, its gradient at p^t is $s_t q^t$, where s_t is the sign of $\langle q^t, p^t - p_D\rangle$, taken as $+1$ at 0. So the loss vector to feed the experts is $\ell^t = s_t q^t$, and the next lemma shows that it measures the error exactly.

Lemma 8.3. *Let q be a linear query, p' and p distributions on $\mathcal{X}$, and s the sign of $\langle q, p' - p\rangle$, with $s = +1$ if it is 0. The vector $\ell = sq$ lies in $[-1,1]^{\mathcal{X}}$, and*

$$\ell \cdot (p' - p) = |\langle q, p'\rangle - \langle q, p\rangle|. \tag{8.13}$$

Proof. The entries of q lie in $[0,1]$, so those of sq lie in $[-1,1]$. By linearity, $\ell \cdot (p' - p) = s\langle q, p' - p\rangle$, which is the absolute value of $\langle q, p' - p\rangle$ by the choice of s. □

With these losses, the regret of multiplicative weights against the fixed distribution p_D is exactly the total error of the candidates. So the regret bound also bounds that total.

Theorem 8.2. *Run multiplicative weights over the experts $\mathcal{X}$ with learning rate $\eta \in (0, 1/2]$, and let p^t be the distribution it plays in round t. Feed it the losses $\ell^t = s_t q^t$, where $q^1, q^2, \ldots$ are any queries in $\mathcal{Q}$ and s_t is the sign of $\langle q^t, p^t - p_D \rangle$. After any number U of rounds,*

$$\sum_{t=1}^{U} |\langle q^t, p^t \rangle - q^t(D)| \le \eta U + \frac{\ln |\mathcal{X}|}{\eta}. \tag{8.14}$$

With $|\mathcal{X}| \ge 2$, $U \ge 4 \ln |\mathcal{X}|$ and $\eta = \sqrt{\ln |\mathcal{X}|/U}$, the right-hand side is $2\sqrt{U \ln |\mathcal{X}|}$.

Proof. By Lemma 8.3, the left-hand side equals $\sum_t \ell^t \cdot p^t - \sum_t \ell^t \cdot p_D$. Apply Theorem 8.1 with $N = |\mathcal{X}|$ and the fixed distribution $p = p_D$. □

This bounds a total, while the goal is one distribution that is accurate on every query. Call round t a **mistake** if the error there exceeds α. If the adversary always picks a query on which the current distribution makes a mistake, the total grows by more than α per round, while the regret bound grows only by η per round. So the mistakes cannot go on for long. To use this, the algorithm plays the adversary against itself and always chooses the hardest query it can find.

Definition 8.5 (Multiplicative weights for queries)**.** Given D, $\mathcal{Q}$ and $\alpha \in (0, 1]$, set $\eta = \alpha/2$ and let p^1 be uniform on $\mathcal{X}$. In round $t = 1, 2, \ldots$:

1. If $|\langle q, p^t \rangle - q(D)| \le \alpha$ for every $q \in \mathcal{Q}$, output p^t and stop.
2. Otherwise pick $q^t \in \mathcal{Q}$ with $|\langle q^t, p^t \rangle - q^t(D)| > \alpha$, and let s_t be the sign of $\langle q^t, p^t \rangle - q^t(D)$.
3. Update $p^{t+1}(x) \propto p^t(x)\,(1 - \eta s_t q^t(x))$.

In step 3, the symbol $\propto$ means that the right-hand side is divided by its sum over x, so that p^{t+1} is again a distribution. Every update adds more than α to the total error, while the bound on that total grows by only $\eta = \alpha/2$ per round. So the updates must stop soon.

Corollary 8.2 (Mistake bound)**.** *Let $|\mathcal{X}| \ge 2$. The algorithm of Definition 8.5 stops after fewer than $4 \ln |\mathcal{X}|/\alpha^2$ updates, and its output $\hat{p}$ satisfies $|\langle q, \hat{p} \rangle - q(D)| \le \alpha$ for every $q \in \mathcal{Q}$.*

Proof.

1. The algorithm is multiplicative weights with $\eta = \alpha/2 \le 1/2$ and the losses of Theorem 8.2, since normalizing the weights gives the update in step 3.
2. Suppose it makes U updates. Each of them has error above α, so by Theorem 8.2

$$\alpha U < \frac{\alpha}{2} U + \frac{2 \ln |\mathcal{X}|}{\alpha}, \qquad \text{that is,} \qquad U < \frac{4 \ln |\mathcal{X}|}{\alpha^2}. \tag{8.15}$$

3. So the algorithm stops, which happens only at step 1, where every query is within α.

□

The bound is a worst case, and Figure 8.1 runs the algorithm on a small example where the number of updates stays far below it.

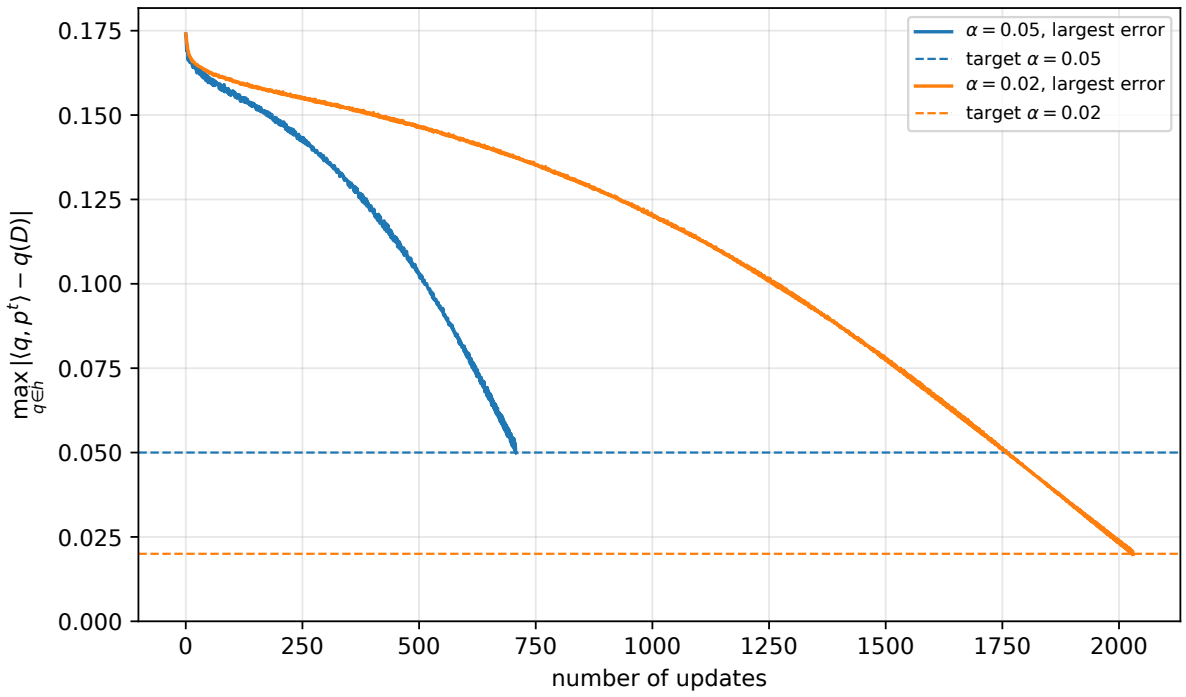


Figure 8.1.: Multiplicative weights for queries on the universe $\{0,1\}^8$, with the 128 one-way and two-way marginal queries $\mathbf{1}[x_j = a]$ and $\mathbf{1}[x_j = a, x_k = b]$. The dataset has 2,000 records whose eight features copy a hidden fair bit with probability 0.8. Each round picks the query with the largest error. The algorithm stops after about 700 updates for $\alpha = 0.05$ and about 2,000 for $\alpha = 0.02$, against the bounds $4 \ln 256/\alpha^2$ of 8,872 and 55,452.

8.4. Private multiplicative weights

The algorithm of Definition 8.5 reads the data in two places: it picks a query with a large error, and it reads the sign of that error. Both have private versions. The exponential mechanism picks a query with nearly the largest error, and the Laplace mechanism can measure the error itself, which has sensitivity $1/n$.

The exponential mechanism takes a finite set $\mathcal{R}$ of candidates and a score $u(D, r)$ whose sensitivity in D is at most Δ for each r. It outputs r with probability proportional to $\exp(\varepsilon u(D, r)/(2\Delta))$. It is ε-DP, and for every $t \geq 0$,

$$\Pr\left[u(D, \hat{r}) \leq \max_{r \in \mathcal{R}} u(D, r) - \frac{2\Delta}{\varepsilon}(\ln|\mathcal{R}| + t)\right] \leq e^{-t}, \tag{8.16}$$

where $\hat{r}$ is its output.

The private algorithm replaces the two steps of Definition 8.5 that read the data by these two mechanisms. It also fixes the number of rounds T in advance, so that the privacy budget can be split among them.

Definition 8.6 (Private multiplicative weights)**.** The inputs are a dataset $D \in \mathcal{X}^n$, a finite set $\mathcal{Q}$ of linear queries, an accuracy $\alpha \in (0, 1]$ and privacy parameters $\varepsilon \in (0, 1]$ and $\delta \in (0, 1/2]$. Set

$$T = \left\lceil \frac{64 \ln|\mathcal{X}|}{\alpha^2} \right\rceil, \qquad \eta = \frac{\alpha}{8}, \qquad \varepsilon_0 = \frac{\varepsilon}{4\sqrt{T \ln(1/\delta)}}, \tag{8.17}$$

and let p^1 be uniform on $\mathcal{X}$. In round $t = 1, \ldots, T$:

1. Choose $q^t \in \mathcal{Q}$ with the exponential mechanism, with privacy parameter ε_0 and score $u_t(D, q) = |\langle q, p^t\rangle - q(D)|$, whose sensitivity is $1/n$.
2. Measure $y_t = \langle q^t, p^t\rangle - q^t(D) + Z_t$, with $Z_t \sim \mathrm{Lap}(1/(\varepsilon_0 n))$.
3. If $|y_t| \leq \alpha/2$, output p^t and stop.
4. Otherwise let s_t be the sign of y_t and update $p^{t+1}(x) \propto p^t(x)\,(1 - \eta s_t q^t(x))$.

If no round stops, output p^{T+1}.

The score u_t has sensitivity $1/n$ because $\langle q, p^t\rangle$ does not depend on D, and $q(D)$ changes by at most $1/n$. The threshold $\alpha/2$ in step 3 sits between the errors that must cause an update and those small enough to stop on. It leaves room on both sides for the noise. Every access to the data goes through one of the two mechanisms, so privacy follows from composing them over the T rounds.

Theorem 8.3. *Private multiplicative weights is (ε, δ)-DP.*

Proof.

1. In round t, the distribution p^t is a function of the earlier outputs $q^1, y_1, \dots, q^{t-1}, y_{t-1}$ alone. So each round runs two mechanisms chosen from earlier outputs: the exponential mechanism, which is ε_0-DP, and the Laplace mechanism on a query of sensitivity $1/n$ with scale $1/(\varepsilon_0 n)$, which is also ε_0-DP.
2. A round after the algorithm stops can be treated as a mechanism with a fixed output, which is 0-DP. So the transcript of all T rounds is an adaptive composition of $k = 2T$ mechanisms, each ε_0-DP.
3. The value $\varepsilon_0 = \varepsilon/(4\sqrt{T\ln(1/\delta)})$ equals $\varepsilon/(2\sqrt{2k\ln(1/\delta)})$ for $k = 2T$. Lemma 8.1 makes the transcript (ε, δ)-DP.
4. The output is a function of the transcript, so it is (ε, δ)-DP by post-processing.

□

Basic composition would also work, and it would even give pure DP, but only with $\varepsilon_0 = \varepsilon/(2T)$. Then the condition on n in the next theorem would grow like T instead of $\sqrt{T}$, which means $1/\alpha^3$ instead of $1/\alpha^2$.

Accuracy needs enough records for the errors of both mechanisms to stay below $\alpha/4$ in every round. Then each update still makes progress, and the algorithm stops on a distribution that is accurate on every query.

Theorem 8.4 (Accuracy of private multiplicative weights)**.** *Let $|\mathcal{X}| \geq 2$ and $\beta \in (0,1)$, and let the other parameters be as in Definition 8.6. If*

$$n \geq \frac{32\sqrt{T\ln(1/\delta)}\,(\ln|\mathcal{Q}| + \ln(2T/\beta))}{\alpha\varepsilon}, \tag{8.18}$$

then with probability at least $1-\beta$ the output $\hat{p}$ satisfies $|\langle q, \hat{p}\rangle - q(D)| \leq \alpha$ for every $q \in \mathcal{Q}$.

Proof.

1. *The good event.* Let G be the event that in every round t that runs, the chosen query has $u_t(D, q^t) \geq \max_q u_t(D, q) - \alpha/4$ and the noise has $|Z_t| < \alpha/4$. Condition Equation 8.18 is the same as $\frac{2}{n\varepsilon_0}(\ln|\mathcal{Q}| + \ln(2T/\beta)) \leq \frac{\alpha}{4}$, by the value of ε_0. So Equation 8.16 with $t = \ln(2T/\beta)$ bounds the first failure in each round by $\beta/(2T)$. The same condition gives $\ln(2T/\beta)/(\varepsilon_0 n) \leq \alpha/4$, so the Laplace tail bounds the second failure by $\beta/(2T)$. A union bound over the two failures and the T rounds gives $\Pr[G] \geq 1-\beta$. The rest of the proof assumes G.
2. *Every update makes progress.* In a round that updates, $|y_t| > \alpha/2 > |Z_t|$. So the true error $\langle q^t, p^t\rangle - q^t(D) = y_t - Z_t$ has the sign s_t of y_t, and its absolute value exceeds $\alpha/2 - \alpha/4 = \alpha/4$.
3. *Few updates.* By step 2, the update is the one of Theorem 8.2, with $\eta = \alpha/8 \leq 1/2$. If U rounds update, each has error above $\alpha/4$, and the theorem gives

$$\frac{\alpha}{4}U < \frac{\alpha}{8}U + \frac{8\ln|\mathcal{X}|}{\alpha}, \qquad \text{that is,} \qquad U < \frac{64\ln|\mathcal{X}|}{\alpha^2} \leq T. \tag{8.19}$$

So the T rounds cannot all update, and the algorithm stops at step 3 in some round t. 4. *The output is accurate.* In that round, $|y_t| \leq \alpha/2$ and $|Z_t| < \alpha/4$, so $u_t(D, q^t) < 3\alpha/4$. On G, every query then has $u_t(D, q) \leq u_t(D, q^t) + \alpha/4 < \alpha$, and the output is $\hat{p} = p^t$. □

Since $\ln|\mathcal{X}|/\alpha^2 \geq \ln 2 > 1/2$, rounding up adds less than $2\ln|\mathcal{X}|/\alpha^2$ and $T \leq 66\ln|\mathcal{X}|/\alpha^2$. So condition Equation 8.18 asks for

$$n = O\left(\frac{\sqrt{\ln|\mathcal{X}|\,\ln(1/\delta)}\,(\ln|\mathcal{Q}| + \ln\frac{\ln|\mathcal{X}|}{\alpha\beta})}{\alpha^2\varepsilon}\right) \tag{8.20}$$

records, which is the last row of Table 8.1. It is logarithmic in both $|\mathcal{Q}|$ and $|\mathcal{X}|$, and the price is a worse dependence on α, $1/\alpha^2$ instead of $1/\alpha$. Each round evaluates every query on p^t and on D, so the running time is $O(T|\mathcal{Q}|(|\mathcal{X}| + n))$, linear in $|\mathcal{X}|$.

The output is a distribution $\hat{p}$ over the universe, a **synthetic dataset** that agrees with the original on every query in $\mathcal{Q}$ and that an analyst can use like the original. The logarithms pay off only when the number of queries is large, as a comparison of the three sufficient sample sizes shows.

Example 8.1. Take records with $d = 20$ binary features, so $|\mathcal{X}| = 2^{20}$, and $\alpha = 0.05$, $\varepsilon = 1$, $\delta = 10^{-6}$, $\beta = 0.05$. Private multiplicative weights runs for at most $T = 354{,}892$ rounds. For $|\mathcal{Q}| = 10^6$ queries, the sufficient sample sizes of Proposition 8.1, Proposition 8.2 and Theorem 8.4 are about 3.4×10^8, 3.5×10^6 and 4.3×10^7 records. For $|\mathcal{Q}| = 10^9$ they become 4.7×10^{11}, 1.6×10^8 and 5.3×10^7. Private multiplicative weights wins only for very many queries, because the constants in Equation 8.18 are large and its accuracy costs $1/\alpha^2$.

8.5. Summary

Table 8.2.: The chapter's results.

Result	Statement
Proposition 8.1	Laplace noise on every query: $n \geq \lvert\mathcal{Q}\rvert\ln(\lvert\mathcal{Q}\rvert/\beta)/(\alpha\varepsilon)$ under ε-DP
Lemma 8.1	k mechanisms at $\varepsilon/(2\sqrt{2k\ln(1/\delta)})$ each compose to (ε,δ)-DP
Proposition 8.2	Laplace with advanced composition: $n \geq 2\sqrt{2\lvert\mathcal{Q}\rvert\ln(1/\delta)}\ln(\lvert\mathcal{Q}\rvert/\beta)/(\alpha\varepsilon)$
Proposition 8.3	with a perfect expert, halving makes at most $\log_2 N$ mistakes
Proposition 8.4	weighted majority makes at most $2.41(\mathrm{OPT} + \log_2 N)$ mistakes
Theorem 8.1	multiplicative weights has regret $\eta T + \ln N/\eta$, or $2\sqrt{T\ln N}$
Theorem 8.2	with losses $s_t q^t$, the total query error is at most $2\sqrt{U\ln\lvert\mathcal{X}\rvert}$
Corollary 8.2	fewer than $4\ln\lvert\mathcal{X}\rvert/\alpha^2$ updates, then every query is within α
Theorem 8.3	private multiplicative weights is (ε,δ)-DP

Result	Statement
Theorem 8.4	accuracy α from $n = O(\sqrt{\log\|\mathcal{X}\| \log(1/\delta)} \log\|\mathcal{Q}\|/(\alpha^2\varepsilon))$ records, up to lower-order logarithmic terms

8.6. References

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9. Sparse Vector

Queries often arrive one at a time, and each must be answered before the next is seen. Under differential privacy (DP), answering each of them with its own noise makes the error grow polynomially with the number of queries. But often the analyst only needs to know which queries exceed a threshold, and few of them do. This chapter shows, with the sparse vector technique, that the cost then depends on the number of large queries and only logarithmically on the total. Its privacy proof rests on noise on the threshold as well as on the queries. At the end, the technique turns private multiplicative weights into an algorithm that answers queries as they arrive.

9.1. Answering queries online

The setting is a stream of queries $f_1, f_2, \ldots, f_k \colon \mathcal{X}^n \to \mathbb{R}$, each with sensitivity at most 1. Two datasets D and D' are neighbours, written $D \sim D'$, when they differ by the replacement of one record. Sensitivity at most 1 means that $|f_i(D) - f_i(D')| \leq 1$ for every such pair. Counting queries, which count the records with some property, are the typical example. Query f_i arrives at step i and must be answered before f_{i+1} is revealed.

The Laplace mechanism answers each query with noise from the Laplace distribution $\mathrm{Lap}(b)$ of scale b, whose density at z is $e^{-|z|/b}/(2b)$. For a sensitivity-1 query it is ε-DP when $b = 1/\varepsilon$, and its tails are $\Pr[|\mathrm{Lap}(b)| \geq tb] = e^{-t}$. Under pure DP with a total budget ε, basic composition forces the scale k/ε on each of k answers, so the error is of order k/ε. Under approximate DP, the advanced composition theorem brings this down to order $\sqrt{k \ln(1/\delta)}/\varepsilon$. Both are polynomial in k, and in the worst case no algorithm that answers every query can do much better.

A weaker question has a cheaper answer: which queries are large? If all k queries were known in advance, the exponential mechanism with score $f_i(D)$ would pick a query whose value is within $2(\ln k + t)/\varepsilon$ of the largest, except with probability e^{-t}. Repeating it c times with budget ε/c each finds c large queries with error of order $c \ln k/\varepsilon$, or $\sqrt{c \ln(1/\delta)} \ln k/\varepsilon$ with advanced composition. That is logarithmic in k, but it needs the whole list. Online, the first query cannot be compared with queries not yet seen, so the question becomes absolute.

> Given a public threshold τ, report the first c queries in the stream with $f_i(D) \geq \tau$, and stop.

The idea is to run the Laplace mechanism on each query and report only whether the noisy value exceeds a noisy threshold $\hat{\tau}$, not τ itself. The details matter, because several published variants are not private.

9.2. AboveThreshold

The case $c = 1$ finds the first large query and halts. AboveThreshold draws one noisy threshold at the start, compares each noisy query with it, and answers with one of two symbols. The symbol $\top$ means "above the threshold" and $\perp$ means "below".

Definition 9.1 (AboveThreshold)**.** The inputs are a dataset D, a stream of sensitivity-1 queries $f_1, f_2, ...$, a threshold $\tau \in \mathbb{R}$ and $\varepsilon > 0$.

1. Draw $\hat{\tau} = \tau + \mathrm{Lap}(2/\varepsilon)$.
2. For each query f_i in turn, draw $\nu_i \sim \mathrm{Lap}(4/\varepsilon)$. If $f_i(D) + \nu_i \geq \hat{\tau}$, output $\top$ and halt. Otherwise output $\perp$.

All noise draws are independent.

Nothing in the algorithm depends on the number of queries, so the stream may even be infinite, and the privacy guarantee does not change. The accuracy guarantee does, because more noise draws mean a larger largest one.

The queries can be fixed in advance, or an analyst who sees the earlier answers can choose them. Before the algorithm halts every answer is $\perp$, so in the second case each query is still a fixed function of the analyst's own coins, which do not depend on the dataset. The proof below applies to the stream fixed by each value of those coins, and averaging the inequality over the coins keeps it.

Theorem 9.1. *AboveThreshold is ε-DP.*

Proof.

1. Fix neighbours $D \sim D'$ and a stream, and write Y and Y' for the outputs on D and D'. The outputs are $y = \perp^{t-1}\top$, which halts at query t, and the output that never halts: $\perp^k$ for a stream of finite length k, or $\perp^\infty$ for an infinite one. These form a countable set, so by the pointwise lemma for discrete outputs it suffices to show $\Pr[Y = y] \leq e^\varepsilon \Pr[Y' = y]$ for each of them.
2. *Fix the noise of the first $t-1$ queries.* Condition on $\nu_i = v_i$ for $i < t$, the same values on both datasets, and let

$$g(D) = \max_{i<t} \left(f_i(D) + v_i\right), \tag{9.1}$$

with $g = -\infty$ if $t = 1$. With the v_i fixed, g is a number, not a random variable. Each f_i has sensitivity 1, so $|g(D) - g(D')| \leq 1$. 3. The output is y exactly when every earlier query falls below $\hat{\tau}$ and query t does not, that is, when $g(D) < \hat{\tau} \leq f_t(D) + \nu_t$. Write ϕ_2 and ϕ_4 for the densities of $\hat{\tau}$ and ν_t, whose subscripts recall the scales $2/\varepsilon$ and $4/\varepsilon$. Then

$$\Pr[Y = y] = \iint \phi_2(u)\, \phi_4(v)\, \mathbf{1}[g(D) < u \leq f_t(D) + v]\, \mathrm{d}u\, \mathrm{d}v. \tag{9.2}$$

4. *Change variables to move from D to D'.* Substitute

$$u = u' + g(D) - g(D'), \qquad v = v' + g(D) - g(D') + f_t(D') - f_t(D). \tag{9.3}$$

This is a shift, so $\mathrm{d}u\,\mathrm{d}v = \mathrm{d}u'\,\mathrm{d}v'$. The condition $g(D) < u$ becomes $g(D') < u'$, and $u \le f_t(D)+v$ becomes $u' \le f_t(D') + v'$, so the indicator becomes $\mathbf{1}[g(D') < u' \le f_t(D') + v']$. 5. *Pay for the shifts.* By step 2, $|u - u'| \le 1$. By step 2 and the sensitivity of f_t, $|v - v'| \le 2$. A Laplace density of scale b changes by at most a factor $e^{s/b}$ when its argument moves by s, so $\phi_2(u) \le e^{\varepsilon/2}\phi_2(u')$ and $\phi_4(v) \le e^{2\varepsilon/4}\phi_4(v') = e^{\varepsilon/2}\phi_4(v')$. Hence

$$\Pr[Y = y] \le e^{\varepsilon} \iint \phi_2(u')\,\phi_4(v')\,\mathbf{1}\big[g(D') < u' \le f_t(D') + v'\big]\,\mathrm{d}u'\,\mathrm{d}v' = e^{\varepsilon}\Pr[Y' = y], \tag{9.4}$$

where both probabilities are conditional on the fixed $v_1, \dots, v_{t-1}$. 6. The v_i have the same distribution on both datasets, so averaging step 5 over them gives the unconditional inequality. 7. *The output that never halts.* It occurs exactly when $\hat{\tau} > \sup_i (f_i(D) + \nu_i)$, the supremum over all queries of the stream. Fixing all ν_i makes this supremum a number $g(D)$, possibly $+\infty$, with $|g(D) - g(D')| \le 1$ when it is finite, and the output has probability 0 on both datasets when it is not. Shifting u by at most 1 as in steps 4 and 5 gives the factor $e^{\varepsilon/2} \le e^{\varepsilon}$. □

The two noises do different jobs. The noise on $\hat{\tau}$ pays, in one draw, for all the $\perp$ answers: the proof fixes their noise and absorbs the change in their maximum by shifting the threshold. The noise ν_t pays for the single $\top$. So the privacy cost does not grow with the number of $\perp$ answers.

The output carries no numbers, so accuracy means making no mistakes, with some slack around the threshold.

Definition 9.2 (Accuracy for threshold answers)**.** Let an algorithm answer a stream of queries $f_1, f_2, \dots$ with $\top$ or $\perp$, compared with a threshold τ, and write a_i for its answer to f_i. Let $\alpha \ge 0$ be a slack and $\beta \in (0,1)$ a failure probability. The algorithm is (α, β)**-accurate** on D if, with probability at least $1 - \beta$, every answered query satisfies

$$a_i = \top \implies f_i(D) \ge \tau - \alpha \qquad \text{and} \qquad a_i = \perp \implies f_i(D) \le \tau + \alpha. \tag{9.5}$$

In words, no query more than α below the threshold gets a $\top$, and no query more than α above it gets a $\perp$. AboveThreshold meets this definition with a slack α that grows only with the logarithm of the number of queries. The theorem also asks that every query before the last be at least α below the threshold, so that the algorithm does not halt early.

Theorem 9.2 (Accuracy of AboveThreshold)**.** *Let $\beta \in (0,1)$, and let $f_1, \dots, f_k$ be a stream with $f_i(D) \le \tau - \alpha$ for every $i < k$, where*

$$\alpha = \frac{8(\ln k + \ln(2/\beta))}{\varepsilon}. \tag{9.6}$$

Then AboveThreshold is (α, β)-accurate on D, and with probability at least $1 - \beta$ it does not halt before query k.

Proof.

1. *The good event.* Let G be the event that $|\hat{\tau} - \tau| < \alpha/2$ and $|\nu_i| < \alpha/2$ for every $i \le k$.
2. *Accuracy on G.* If $a_i = \top$, then $f_i(D) \ge \hat{\tau} - \nu_i > (\tau - \alpha/2) - \alpha/2 = \tau - \alpha$. If $a_i = \bot$, then $f_i(D) < \hat{\tau} - \nu_i < \tau + \alpha$.
3. *No early halt on G.* By step 2, a $\top$ at query i needs $f_i(D) > \tau - \alpha$, which the hypothesis rules out for $i < k$.
4. *G is likely.* The threshold noise has scale $2/\varepsilon$, so the tail formula gives $\Pr[|\hat{\tau} - \tau| \ge \alpha/2] = e^{-\varepsilon\alpha/4}$. Each ν_i has scale $4/\varepsilon$, so a union bound gives $\Pr[\max_{i \le k} |\nu_i| \ge \alpha/2] \le k e^{-\varepsilon\alpha/8}$.
5. With Equation 9.6, $k e^{-\varepsilon\alpha/8} = k \cdot \frac{1}{k} \cdot \frac{\beta}{2} = \beta/2$. Also $e^{-\varepsilon\alpha/4} = (\beta/(2k))^2 \le \beta/2$. So $\Pr[G] \ge 1 - \beta$.

□

The error grows only logarithmically in k, like that of the exponential mechanism, but the queries can arrive online.

Example 9.1. Take $k = 10^6$ counting queries, $\varepsilon = 1$ and $\beta = 0.05$. Then Equation 9.6 gives $\alpha = 8(\ln 10^6 + \ln 40) = 8(13.82 + 3.69) = 140.0$. On a dataset of $n = 10^5$ records, a query is placed correctly unless its count lies within 140 of the threshold. Answering every query with the Laplace mechanism and basic composition would instead add noise of scale 10^6, ten times the size of the dataset.

9.3. Sparse vector

To find c large queries, restart AboveThreshold after each $\top$, with a fresh threshold, until c of them have been found.

Definition 9.3 (Sparse vector)**.** The inputs are those of AboveThreshold, a count $c \ge 1$ of $\top$ answers to find, and a parameter δ, with either $\delta = 0$ or $\delta \in (0, 1/2]$. Set the Laplace noise scale b to

$$b = \frac{2c}{\varepsilon} \text{ if } \delta = 0, \qquad b = \frac{\sqrt{32c \ln(1/\delta)}}{\varepsilon} \text{ if } \delta > 0. \tag{9.7}$$

1. Draw $\hat{\tau} = \tau + \mathrm{Lap}(b)$ and set count $= 0$.
2. For each query f_i in turn, draw $\nu_i \sim \mathrm{Lap}(2b)$. If $f_i(D) + \nu_i < \hat{\tau}$, output $\bot$. Otherwise output $\top$ and add 1 to count. If count $= c$, halt. If not, draw a fresh $\hat{\tau} = \tau + \mathrm{Lap}(b)$ and continue.

Between two $\top$ answers the algorithm runs AboveThreshold with parameter $\varepsilon' = 2/b$, whose noise scales are $2/\varepsilon' = b$ and $4/\varepsilon' = 2b$. The fresh threshold matters, because reusing $\hat{\tau}$ after a $\top$ would break this equivalence, and with it the proof. With the equivalence, sparse vector is a composition of c runs of AboveThreshold, and its privacy follows from the composition theorems.

Theorem 9.3. *Sparse vector with $\delta = 0$ is ε-DP for every $\varepsilon > 0$. With $\varepsilon \in (0, 1]$ and $\delta \in (0, 1/2]$ it is (ε, δ)-DP.*

Proof.

1. The output is the concatenation of at most c runs of AboveThreshold, each with parameter $\varepsilon' = 2/b$ and fresh noise. Each run takes the rest of the stream, which may depend on the earlier outputs, so the whole is an adaptive composition of c mechanisms. A run that never starts, because the algorithm halted, has a fixed output. Each run is ε'-DP by Theorem 9.1.
2. If $\delta = 0$, then $\varepsilon' = \varepsilon/c$, and basic composition gives $c\varepsilon' = \varepsilon$.
3. If $\delta > 0$, then $\varepsilon' = 2\varepsilon/\sqrt{32c\ln(1/\delta)} = \varepsilon/(2\sqrt{2c\ln(1/\delta)})$. The budget-splitting lemma says that m adaptively composed mechanisms, each $\varepsilon/(2\sqrt{2m\ln(1/\delta)})$-DP, are together (ε, δ)-DP when $\varepsilon \le 1$ and $\delta \le 1/2$. Apply it with $m = c$.

□

Accuracy is proved as for AboveThreshold. The only change is that the union bound now runs over every threshold drawn, one per run, as well as over the noise on each query.

Theorem 9.4 (Accuracy of sparse vector)**.** *Let $\beta \in (0, 1)$, let $f_1, \dots, f_k$ be a stream, and let*

$$\alpha = 4b\ln(2k/\beta). \tag{9.8}$$

With probability at least $1 - \beta$, every answered query satisfies the two implications of Definition 9.2. If moreover fewer than c of the queries have $f_i(D) > \tau - \alpha$, the algorithm answers all k queries. In terms of the parameters, the accuracy is

$$\alpha = \frac{8c(\ln k + \ln(2/\beta))}{\varepsilon} \quad \text{if } \delta = 0, \qquad \alpha = \frac{\sqrt{512\,c\ln(1/\delta)}\,(\ln k + \ln(2/\beta))}{\varepsilon} \quad \text{if } \delta > 0. \tag{9.9}$$

Proof.

1. Let G be the event that every threshold drawn and every ν_i with $i \le k$ lies within $\alpha/2$ of its centre. On G, steps 2 and 3 of the proof of Theorem 9.2 apply to each query, with whichever threshold is current. So the implications hold, and a $\top$ needs $f_i(D) > \tau - \alpha$.
2. If fewer than c queries have $f_i(D) > \tau - \alpha$, then on G there are fewer than c answers $\top$, and the algorithm never halts.
3. The query noise has scale $2b$, and $\alpha/(4b) = \ln(2k/\beta)$. A union bound over the k draws bounds its failure by $ke^{-\alpha/(4b)} = \beta/2$.
4. At most $k+1$ thresholds are drawn, each with scale b. Their failure is at most $(k+1)e^{-\alpha/(2b)} = (k+1)\beta^2/(4k^2) \le \beta/2$, since $k + 1 \le 2k$ and $\beta \le 1$.
5. Substituting the two values of b into $4b\ln(2k/\beta)$ gives Equation 9.9, because $4\sqrt{32} = \sqrt{512}$.

□

With $c = 1$ and $\delta = 0$ this is Theorem 9.2 again. The error is polynomial in the number c of large queries and logarithmic in the total number k. So sparse vector can scan a very long stream, as long as few of its queries are large.

9.4. Numeric sparse vector

Sparse vector already pays in proportion to the number c of large queries. A noisy value for each of those c queries takes c more Laplace mechanisms, which cost the same order of budget. So sparse vector can also release the values of the large queries for a constant factor.

Definition 9.4 (Numeric sparse vector)**.** Run sparse vector with $\delta = 0$ and budget $\varepsilon/2$, so $b = 4c/\varepsilon$. Whenever it would output $\top$ for query f_i, output instead $a_i = f_i(D) + \mathrm{Lap}(2c/\varepsilon)$, with fresh noise.

Each answer is now either $\perp$ or a number. The guarantee combines the two implications of Definition 9.2 with an error bound on every released number.

Theorem 9.5. *Numeric sparse vector is ε-DP. Let $f_1, \dots, f_k$ be a stream, $\beta \in (0,1)$ and $\alpha = 16c\ln(4k/\beta)/\varepsilon$. With probability at least $1-\beta$, every answered query satisfies the following, where $a_i \in \mathbb{R}$ means that the answer to f_i is a released number:*

$$a_i = \perp \implies f_i(D) \le \tau + \alpha, \qquad a_i \in \mathbb{R} \implies f_i(D) \ge \tau - \alpha \ \text{ and } \ |a_i - f_i(D)| \le \alpha, \tag{9.10}$$

and if fewer than c queries have $f_i(D) > \tau - \alpha$, all k are answered.

Proof.

1. *Privacy.* The algorithm is an adaptive composition of at most c runs of AboveThreshold, each with parameter $2/b = \varepsilon/(2c)$, and at most c Laplace mechanisms. Each Laplace mechanism measures one sensitivity-1 query, chosen from the earlier outputs, with scale $2c/\varepsilon$, so it is $\varepsilon/(2c)$-DP. Basic composition over the $2c$ mechanisms gives ε.
2. *Accuracy of the labels.* Theorem 9.4 with $b = 4c/\varepsilon$ and failure probability $\beta/2$ gives accuracy $4b\ln(4k/\beta) = \alpha$ for the labels, and no early halt, with probability at least $1 - \beta/2$.
3. *Accuracy of the values.* At most k values are released, one per query. By the tail formula and a union bound, some noise exceeds α with probability at most $k\,e^{-\alpha\varepsilon/(2c)} = k\,(\beta/(4k))^8 \le \beta/2$.

□

The fresh noise in the released value matters. Releasing $f_i(D) + \nu_i$, the value already compared with the threshold, is not private, and published variants that did so were wrong.

9.5. Online private multiplicative weights

The previous chapter built private multiplicative weights for a fixed set of linear queries $q\colon \mathcal{X} \to [0,1]$, with $q(D) = \frac{1}{n}\sum_i q(x_i)$. This section uses four of its facts. For a distribution p on $\mathcal{X}$, write $\langle q, p\rangle = \sum_x q(x)p(x)$ for the average of q under p. The histogram p_D of the dataset, the fraction of records equal to each x, satisfies $q(D) = \langle q, p_D\rangle$. Multiplicative weights over the experts $\mathcal{X}$ with learning rate η and losses $s_t q^t$, where s_t is the sign of the error $\langle q^t, p^t\rangle - q^t(D)$, has total error at most $\eta U + \ln|\mathcal{X}|/\eta$ after U updates. And the offline algorithm used the exponential mechanism to find a query with a large error.

Online, the exponential mechanism cannot be used, because the queries arrive one at a time, so sparse vector replaces it. The algorithm keeps a distribution $\hat{p}$ and answers each query from it while it is accurate. When sparse vector reports a large error, the algorithm answers with a noisy true value and updates $\hat{p}$.

These queries have sensitivity $1/n$, not 1. Sparse vector on f_i/Δ with threshold τ/Δ handles sensitivity Δ. It is the same algorithm with every noise scale multiplied by Δ, so it has the same privacy and Δ times the accuracy α. The algorithm runs this rescaled sparse vector on the errors of $\hat{p}$, with threshold $\alpha/2$. Its count c of large errors is set to the bound on the number of updates from the previous chapter, and ε_0 is the budget of each of the $2c$ mechanisms it may run.

Definition 9.5 (Online private multiplicative weights)**.** The inputs are a dataset $D \in \mathcal{X}^n$, a stream of linear queries $f_1, f_2, \ldots$, an accuracy $\alpha \in (0,1]$, and $\varepsilon \in (0,1]$ and $\delta \in (0, 1/2]$. Set

$$c = \left\lceil \frac{64 \ln|\mathcal{X}|}{\alpha^2} \right\rceil, \qquad \eta = \frac{\alpha}{8}, \qquad \varepsilon_0 = \frac{\varepsilon}{4\sqrt{c \ln(1/\delta)}}, \tag{9.11}$$

let $\hat{p}$ be uniform on $\mathcal{X}$, set count $= 0$, and draw $\hat{\tau} = \alpha/2 + \mathrm{Lap}(2/(\varepsilon_0 n))$. For each query f_i in turn:

1. Let $e_i = |\langle f_i, \hat{p}\rangle - f_i(D)|$ and draw $\nu_i \sim \mathrm{Lap}(4/(\varepsilon_0 n))$.
2. If $e_i + \nu_i < \hat{\tau}$, answer $\langle f_i, \hat{p}\rangle$.
3. Otherwise answer $a_i = f_i(D) + Z_i$ with $Z_i \sim \mathrm{Lap}(1/(\varepsilon_0 n))$. Let s_i be the sign of $\langle f_i, \hat{p}\rangle - a_i$, update $\hat{p}(x) \propto \hat{p}(x)(1 - \eta s_i f_i(x))$, and add 1 to count. If count $= c$, halt. Otherwise draw a fresh $\hat{\tau} = \alpha/2 + \mathrm{Lap}(2/(\varepsilon_0 n))$.

Steps 1 to 3 follow numeric sparse vector, run on the errors e_i with threshold $\alpha/2$, except that the fresh measurement in step 3 is of $f_i(D)$ rather than of e_i. The answer in step 2 uses only $\hat{p}$, which depends on the earlier outputs alone, so it costs no privacy.

Theorem 9.6. *Online private multiplicative weights is (ε, δ)-DP.*

Proof.

1. Between two updates $\hat{p}$ is fixed, so the errors e_i form a stream of queries determined by the earlier outputs and the analyst. Each e_i has sensitivity $1/n$, since $\langle f_i, \hat{p}\rangle$ does not depend on D and $f_i(D)$ changes by at most $1/n$.

2. Each stretch between updates is therefore AboveThreshold with parameter ε_0, rescaled to sensitivity $1/n$, and so ε_0-DP by Theorem 9.1. Each measurement in step 3 is the Laplace mechanism on a query of sensitivity $1/n$ with scale $1/(\varepsilon_0 n)$, so it is also ε_0-DP. The answers of step 2 and the updates are post-processing.
3. There are at most c stretches and c measurements, so the algorithm is an adaptive composition of $2c$ mechanisms, each ε_0-DP. Since $\varepsilon_0 = \varepsilon/(2\sqrt{2 \cdot 2c \ln(1/\delta)})$, the budget-splitting lemma with $m = 2c$ mechanisms makes it (ε, δ)-DP.

□

Accuracy follows the proof for the offline algorithm. When all the noise is small, every answer from $\hat{p}$ is accurate, every measured answer is accurate, and every update makes progress, so the count of updates never reaches c.

Theorem 9.7 (Accuracy of online private multiplicative weights)**.** *Let $|\mathcal{X}| \geq 2$, $\beta \in (0, 1)$ and a stream of k queries, and let the other parameters be as in Definition 9.5. If*

$$n \geq \frac{128\sqrt{c \ln(1/\delta)}\, \ln\left((2c + k)/\beta\right)}{\alpha\varepsilon}, \tag{9.12}$$

then with probability at least $1 - \beta$ the algorithm answers all k queries, each within α of $f_i(D)$.

Proof.

1. *The good event.* Let G be the event that every threshold noise, every ν_i and every Z_i is less than $\alpha/8$ in absolute value. These are at most $2c + k$ draws, because there are at most c thresholds and c measurements. The widest scale is $4/(\varepsilon_0 n)$, so each draw fails with probability at most $e^{-\alpha\varepsilon_0 n/32}$. Condition Equation 9.12 is the same as $\alpha\varepsilon_0 n/32 \geq \ln((2c+k)/\beta)$, by the value of ε_0. A union bound gives $\Pr[G] \geq 1 - \beta$, and the rest assumes G.
2. *Answers from $\hat{p}$.* In step 2, $e_i < \hat{\tau} - \nu_i < \alpha/2 + \alpha/8 + \alpha/8 = 3\alpha/4$, so the answer is within α.
3. *Measured answers.* In step 3, $|a_i - f_i(D)| = |Z_i| < \alpha/8$. Also $e_i \geq \hat{\tau} - \nu_i > \alpha/2 - \alpha/4 = \alpha/4$. The difference $\langle f_i, \hat{p}\rangle - a_i$ is the true error minus Z_i, and $|Z_i| < \alpha/8 < e_i$, so s_i is the sign of the true error.
4. *Few updates.* By step 3, the updates are those of multiplicative weights for queries with $\eta = \alpha/8$, and each has error above $\alpha/4$. If there are U of them, the regret bound gives

$$\frac{\alpha}{4}U < \frac{\alpha}{8}U + \frac{8 \ln|\mathcal{X}|}{\alpha}, \qquad \text{that is,} \qquad U < \frac{64 \ln|\mathcal{X}|}{\alpha^2} \leq c. \tag{9.13}$$

So the count never reaches c, the algorithm never halts, and all k queries are answered. □

As in the previous chapter, $c \leq 66 \ln|\mathcal{X}|/\alpha^2$. So condition Equation 9.12 asks for

$$n = O\left(\frac{\sqrt{\ln|\mathcal{X}|\, \ln(1/\delta)}\, \ln\frac{k + \ln|\mathcal{X}|/\alpha^2}{\beta}}{\alpha^2\varepsilon}\right) \tag{9.14}$$

records. This matches the offline algorithm, with the length k of the stream in place of the number $|\mathcal{Q}|$ of queries.

9.6. Summary

Table 9.1.: The chapter's results.

Result	Statement
Theorem 9.1	AboveThreshold, with noise on the threshold and on each query, is ε-DP
Theorem 9.2	it finds the first large query with error $8(\ln k + \ln(2/\beta))/\varepsilon$
Theorem 9.3	sparse vector, c restarts with fresh thresholds, is ε-DP or (ε, δ)-DP
Theorem 9.4	error $8c(\ln k + \ln(2/\beta))/\varepsilon$, or $\sqrt{512c\ln(1/\delta)}(\ln k + \ln(2/\beta))/\varepsilon$
Theorem 9.5	releasing the large values with fresh noise costs a constant factor
Theorem 9.6	online private multiplicative weights is (ε, δ)-DP
Theorem 9.7	it answers k online queries to accuracy α from $n = O(\sqrt{\log\|\mathcal{X}\|\log(1/\delta)}\log k/(\alpha^2\varepsilon))$ records, up to lower-order logarithmic terms

9.7. References

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10. Beyond Global Sensitivity

The Laplace mechanism adds noise scaled to the global sensitivity of a function, the largest change one record can cause on any pair of neighbouring datasets. For many functions that worst case is far from the dataset at hand, or infinite. This chapter asks whether the noise can be scaled to the dataset actually held instead, while keeping differential privacy (DP).

Done naively, it cannot, because the amount of noise then depends on the data and leaks it. Four techniques make the idea work. Propose-test-release checks privately that the dataset is far from any dataset where the function is sensitive. A stability argument built on it finds the most frequent element and every count of a histogram, whatever the size of the domain. The last two techniques bound the sensitivity privately, or smooth it, before they use it.

10.1. Local sensitivity

Throughout the chapter, $f\colon \mathcal{X}^n \to \mathbb{R}$ is a real-valued function of the dataset. Its **global sensitivity** is the largest change that replacing one record can cause, over neighbouring datasets $D \sim D'$ (the replace-one relation),

$$\Delta f = \max_{D \sim D'} |f(D) - f(D')|, \tag{10.1}$$

and the Laplace mechanism adds to $f(D)$ a noise term drawn from $\mathrm{Lap}(\Delta f/\varepsilon)$, which makes the release ε-DP. The maximum ranges over all pairs of neighbours, including datasets that look nothing like the one held. Fixing the first dataset gives a local version.

Definition 10.1 (Local sensitivity)**.** The **local sensitivity** of f at a dataset $D \in \mathcal{X}^n$ is

$$\mathrm{LS}_f(D) = \max_{D' \sim D} |f(D) - f(D')|. \tag{10.2}$$

The global sensitivity is the largest local sensitivity, $\Delta f = \max_D \mathrm{LS}_f(D)$, and the local sensitivity can be much smaller. The natural mechanism releases $f(D) + \mathrm{Lap}(\mathrm{LS}_f(D)/\varepsilon)$. It fails, because the scale of the noise is itself a function of the data. The distance between the two closest records shows how the local sensitivity can jump from one dataset to a neighbour.

Example 10.1. Let $\mathcal{X} = \mathbb{R}$, $n = 3$, and let $f(D)$ be the distance between the two closest records. On the real line f has infinite global sensitivity. For example, $(0, c, 2c)$ and its neighbour $(0, 0, 2c)$ have f equal to c and 0, for every $c > 0$, so the Laplace mechanism does not apply. Take $D = (0, 0, 0)$. Replacing any one record leaves two records at 0, so f stays 0 and $\mathrm{LS}_f(D) = 0$. Now take the neighbour $D' = (0, 0, 5)$. It also has $f(D') = 0$, but replacing one of its zeros by 10 gives $(0, 10, 5)$ with closest distance 5, so $\mathrm{LS}_f(D') \geq 5$.

On these two neighbours, the naive mechanism adds no noise to one and some noise to the other, and that difference alone gives the dataset away.

Proposition 10.1. *The mechanism $M(D) = f(D) + \mathrm{Lap}(\mathrm{LS}_f(D)/\varepsilon)$, where* $\mathrm{Lap}(0)$ *means no noise, is not ε-DP for any finite ε when f is the closest-pair distance of Example 10.1.*

Proof.

1. On $D = (0, 0, 0)$ the local sensitivity is 0, so $M(D) = 0$ with probability 1.
2. On the neighbour $D' = (0, 0, 5)$ the local sensitivity is positive, so $M(D')$ has a continuous distribution and $\Pr[M(D') = 0] = 0$.
3. So the event $\{0\}$ has probability 1 on D and 0 on D', and no factor e^{ε} with finite ε can bound the first by the second.

□

The failure is that D has low local sensitivity while a neighbour has high local sensitivity, and the noise level tells them apart. So the local sensitivity is itself a statistic of the data, which leaks even through the scale of the noise. Each technique below accounts for the sensitivity of nearby datasets, not just of D.

10.2. Propose-test-release

The first technique proposes a bound b on the local sensitivity and tests privately that the bound holds at D and at every dataset near D. Only if the test passes does it release $f(D)$ with noise scaled to b. The test measures how many records must change before the bound fails. The **Hamming distance** $d(D, D')$ between two datasets is the number of records in which they differ.

Definition 10.2 (Distance to instability)**.** For a proposed bound $b \geq 0$ on the local sensitivity LS_f, the **distance to instability** of D is

$$\gamma_b(D) = \min\{\, d(D, D') : D' \in \mathcal{X}^n,\ \mathrm{LS}_f(D') > b \,\}, \tag{10.3}$$

the fewest records that must change to reach a dataset whose local sensitivity exceeds b, where d is the Hamming distance. If no dataset has local sensitivity above b, set $\gamma_b(D) = \infty$ for every D.

So $\gamma_b(D) = 0$ exactly when $\mathrm{LS}_f(D) > b$. The distance is a count of records, and that gives it sensitivity 1 whatever f is.

Lemma 10.1. *If γ_b is finite, then $|\gamma_b(D) - \gamma_b(D')| \leq 1$ for all neighbours $D \sim D'$.*

Proof.

1. If some dataset W has $\mathrm{LS}_f(W) > b$, every dataset is within distance n of W, so γ_b is finite everywhere.

2. Let W attain the minimum for D', so $d(D', W) = \gamma_b(D')$ and $\mathrm{LS}_f(W) > b$. The Hamming distance satisfies the triangle inequality, so $d(D, W) \le d(D, D') + d(D', W) = 1 + \gamma_b(D')$. Hence $\gamma_b(D) \le \gamma_b(D') + 1$.
3. Swapping the roles of D and D' gives the other inequality.

□

Because the distance has sensitivity 1, the Laplace mechanism can release it privately, and that noisy distance is the test. The threshold of the test is set so that a dataset whose local sensitivity exceeds b passes only with probability $\delta/2$.

Definition 10.3 (Propose-test-release)**.** The mechanism takes a proposed bound $b > 0$ and privacy parameters $\varepsilon > 0$ and $\delta \in (0, 1)$. On input D, it does three steps.

1. compute $\hat{\gamma} = \gamma_b(D) + Z_1$ with $Z_1 \sim \mathrm{Lap}(1/\varepsilon)$;
2. if $\hat{\gamma} \le \ln(1/\delta)/\varepsilon$, output $\perp$, meaning "no answer";
3. otherwise output $f(D) + Z_2$ with $Z_2 \sim \mathrm{Lap}(b/\varepsilon)$, independent of Z_1.

The test spends one budget of ε, and the release spends another. The test can wrongly pass on a sensitive dataset, and δ pays for that. The release step is the naive mechanism of Proposition 10.1 with the scale b in place of $\mathrm{LS}_f(D)$. It is safe now because b is fixed in advance. The decision to use it depends on the data only through the private estimate $\hat{\gamma}$.

Theorem 10.1. *Propose-test-release is* $(2\varepsilon, \delta)$*-DP.*

Proof. Fix neighbours $D \sim D'$ and an event $S \subseteq \mathbb{R} \cup \{\perp\}$. Write $\tau = \ln(1/\delta)/\varepsilon$ for the threshold. If γ_b is infinite at every dataset, the test always passes and $b \ge \Delta f$, so the output is the Laplace mechanism and is ε-DP. Assume from now on that γ_b is finite.

1. *The test is ε-DP.* By Lemma 10.1, γ_b has sensitivity 1, so $\hat{\gamma}$ is the Laplace mechanism and is ε-DP. The events $\{\hat{\gamma} \le \tau\}$ and $\{\hat{\gamma} > \tau\}$ are post-processing of $\hat{\gamma}$, so each has probability on D at most e^{ε} times its probability on D'.
2. *Case* $\mathrm{LS}_f(D) > b$. Then $\gamma_b(D) = 0$, and the test passes only if $Z_1 > \tau$, which has probability $\frac{1}{2}e^{-\varepsilon\tau} = \delta/2$. Splitting on whether the output is $\perp$ and using step 1,

$$\Pr[M(D) \in S] \le \Pr[M(D) = \perp]\,\mathbf{1}[\perp \in S] + \Pr[M(D) \ne \perp] \le e^{\varepsilon}\Pr[M(D') \in S] + \tfrac{\delta}{2}. \quad (10.4)$$

3. *Case* $\mathrm{LS}_f(D) \le b$. Then $|f(D) - f(D')| \le \mathrm{LS}_f(D) \le b$, so the release step is the Laplace mechanism for a pair at distance at most b, with scale b/ε. Its density at any y changes by a factor at most e^{ε} between D and D'. Write $S_{\mathbb{R}} = S \cap \mathbb{R}$. The two steps use independent noise, so

$$\Pr[M(D) \in S] = \Pr[\hat{\gamma} \le \tau]\,\mathbf{1}[\perp \in S] + \Pr[\hat{\gamma} > \tau]\,\Pr[f(D) + Z_2 \in S_{\mathbb{R}}]. \quad (10.5)$$

Each probability on the right is at most e^{ε} times the same probability on D', by step 1 and by the Laplace mechanism. So $\Pr[M(D) \in S] \le e^{2\varepsilon}\Pr[M(D') \in S]$.

4. In both cases $\Pr[M(D) \in S] \leq e^{2\varepsilon} \Pr[M(D') \in S] + \delta$. The pair was arbitrary, so this holds for every ordered pair of neighbours.

□

The test passes with high probability on datasets far from instability, because if $\gamma_b(D) \geq 2\ln(1/\delta)/\varepsilon$, then the output is $\perp$ only if $Z_1 \leq -\ln(1/\delta)/\varepsilon$, which has probability $\delta/2$. The price is computation, because γ_b is a minimum over all $|\mathcal{X}|^n$ datasets. So the distance to instability has to be worked out by hand for each function, and the median is the standard case where this is efficient.

10.3. Stability: the mode and the histogram

Propose-test-release is most striking with the bound $b = 0$. Then the test checks that the answer does not change at all near D, and the release step adds no noise. The mode, the most frequent element, is the standard example. If all n records equal the same value v, the mode stays v until about $n/2$ records change, so the mode is far from instability.

Let $\mathcal{X}$ be any set with a fixed total order, used only to break ties. Write $c_x(D)$ for the number of records of D equal to x, and $c_{(1)}(D) \geq c_{(2)}(D)$ for the largest and second-largest of these counts, with $c_{(2)}(D) = 0$ if all records are equal. The mode $\mathrm{mode}(D)$ is the element with the largest count, the least one in the order among ties. To change the mode, records must move from the most frequent element to the runner-up until it overtakes. So half the gap between the two largest counts is, up to 1, the distance to instability, and it is computed in one pass over the data whatever the domain.

Lemma 10.2. *Let $g(D) = (c_{(1)}(D) - c_{(2)}(D))/2$. For all neighbours $D \sim D'$:*

1. $|g(D) - g(D')| \leq 1$;
2. *if $g(D) > 1$, then* $\mathrm{mode}(D') = \mathrm{mode}(D)$.

Proof.

1. Replacing one record lowers one count by 1 and raises another by 1, so every count changes by at most 1.
2. The j-th largest count is the maximum, over sets S of j elements, of $\min_{x \in S} c_x$. If every count rises by at most 1, each such minimum rises by at most 1, and so does the maximum. The same holds for decreases, so $c_{(1)}$ and $c_{(2)}$ each change by at most 1, and g changes by at most $(1 + 1)/2 = 1$. This is claim 1.
3. For claim 2, let $a = \mathrm{mode}(D)$. In D', the count of a is at least $c_{(1)}(D) - 1$, and every other count is at most $c_{(2)}(D) + 1$. The condition $g(D) > 1$ means $c_{(1)}(D) - 1 > c_{(2)}(D) + 1$, so a has the strictly largest count in D'.

□

The stable mode is propose-test-release with $b = 0$, with the gap g in place of the distance to instability. Its threshold has an extra 1 because claim 2 protects the mode only when $g(D) > 1$.

Definition 10.4 (Stable mode)**.** Given $\varepsilon > 0$ and $\delta \in (0, 1)$, on input D, compute $\hat{g} = g(D) + Z$ with $Z \sim \mathrm{Lap}(1/\varepsilon)$, where $g(D)$ is half the gap between the two largest counts. If $\hat{g} \le 1 + \ln(1/\delta)/\varepsilon$, output $\perp$, and otherwise output $\mathrm{mode}(D)$ exactly.

The guarantee has two parts. The test costs δ in privacy, and it fails, by outputting $\perp$, only on datasets whose two largest counts are close.

Theorem 10.2. *The stable mode is (ε, δ)-DP. On every dataset whose two largest counts differ by at least $2 + 4\ln(1/\delta)/\varepsilon$, it outputs the mode with probability at least $1 - \delta/2$.*

Proof. Fix neighbours $D \sim D'$ and an event S, and write $\tau = 1 + \ln(1/\delta)/\varepsilon$.

1. *The gap is ε-DP.* By Lemma 10.2, claim 1, $\hat{g}$ is the Laplace mechanism for a function of sensitivity 1, so it is ε-DP.
2. *Case* $\mathrm{mode}(D') = \mathrm{mode}(D)$. The output on both datasets is the same function of $\hat{g}$, namely $\perp$ below the threshold and the common mode above it. Post-processing gives $\Pr[M(D) \in S] \le e^{\varepsilon} \Pr[M(D') \in S]$.
3. *Case* $\mathrm{mode}(D') \neq \mathrm{mode}(D)$. By Lemma 10.2, claim 2, $g(D) \le 1$. The mode is released only if $Z > \tau - g(D) \ge \ln(1/\delta)/\varepsilon$, which has probability at most $\delta/2$. As in the proof of Theorem 10.1, $\Pr[M(D) \in S] \le e^{\varepsilon} \Pr[M(D') \in S] + \delta/2$.
4. *Accuracy.* If $c_{(1)} - c_{(2)} \ge 2 + 4\ln(1/\delta)/\varepsilon$, then $g(D) \ge \tau + \ln(1/\delta)/\varepsilon$. The output is $\perp$ only if $Z \le -\ln(1/\delta)/\varepsilon$, which has probability $\delta/2$.

□

With typical parameters, the gap the theorem needs is a few dozen records.

Example 10.2. With $\varepsilon = 1$ and $\delta = 10^{-6}$, the threshold is $1 + \ln(10^6) = 14.82$. A gap of $2 + 4 \times 13.82 = 57.3$, so 58 records, between the most frequent element and the runner-up is enough. The domain plays no part. It may be the set of all strings, which is infinite.

The stable mode costs ε, not 2ε, because the release step adds no noise and spends no budget. Its accuracy depends on the gap, not on the domain, and under pure DP no such bound is possible. The packing chapter shows that an ε-DP algorithm that finds the mode of a dataset over k elements needs at least $\ln(k/2)/\varepsilon$ records, even when all records are equal. So approximate DP replaces the cost $\ln k$ of a large domain by $\ln(1/\delta)$.

The same idea estimates every count at once. The algorithm adds noise only to the counts of elements that appear in the data. It reports only those whose noisy count clears a threshold. An element that appears once in D and not at all in a neighbour is the only danger, and the threshold hides it except with probability $\delta/2$.

Definition 10.5 (Stability-based histogram)**.** Given $\varepsilon > 0$ and $\delta \in (0, 1)$, let $\tau = 1 + 2\ln(1/\delta)/\varepsilon$. On input D, for each element x whose count $c_x(D)$ of records is positive, draw an independent $Z_x \sim \mathrm{Lap}(2/\varepsilon)$ and set $\hat{c}_x = c_x(D) + Z_x$. Output the list of pairs $(x, \hat{c}_x)$ with $\hat{c}_x > \tau$. An element not on the list has estimate $\hat{c}_x = 0$.

The noise has scale $2/\varepsilon$ because replacing one record changes two counts. The threshold costs δ in privacy. In accuracy, it adds at most τ to the noise on each count, because a count is reported as 0 only when its noisy value is at most τ.

Theorem 10.3. *The stability-based histogram is (ε, δ)-DP. For every $\beta \in (0,1)$, with probability at least $1-\beta$, every element $x \in \mathcal{X}$ has*

$$|\hat{c}_x - c_x(D)| \le \tau + \frac{2}{\varepsilon} \ln \frac{n}{\beta}. \tag{10.6}$$

Proof. Privacy. Let D' be D with one record equal to a replaced by $a' \neq a$. Let C be the set of elements that appear in both datasets. Any element that appears only in D is a, with count 1, and any element that appears only in D' is a', with count 1. Write $O(D) = O_C(D) \cup O_A(D)$ for the output list on D, where O_C is the part of the list from C and O_A the part from elements only in D, which can only be a. Define $O_C(D')$ and $O_{A'}(D')$ in the same way, where $O_{A'}$ can only contain a'. Fix an event S and let $p_C = \Pr[O_C(D') \in S]$.

1. *The extra element of D.* An element with count 1 is listed only if its noise exceeds $\tau - 1 = 2\ln(1/\delta)/\varepsilon$, which has probability $\frac{1}{2}e^{-\ln(1/\delta)} = \delta/2$. When $O_A(D)$ is empty, $O(D) = O_C(D)$, so $\Pr[O(D) \in S] \le \Pr[O_C(D) \in S] + \delta/2$.
2. *The common elements.* On C, the counts of D and D' differ by at most 1 at a and at a', so in ℓ_1 by at most 2. Laplace noise of scale $2/\varepsilon$ makes the noisy counts on C ε-DP, and the list is post-processing of them. So $\Pr[O_C(D) \in S] \le \min(1, e^{\varepsilon} p_C)$.
3. *The extra element of D'.* The parts $O_C(D')$ and $O_{A'}(D')$ are independent, and $O_{A'}(D')$ is empty with probability at least $1 - \delta/2$ by the computation of step 1. When it is empty, $O(D') = O_C(D')$. So $\Pr[O(D') \in S] \ge (1 - \delta/2)\, p_C$.
4. *Combining.* If $e^{\varepsilon} p_C \le 1$, then $\min(1, e^{\varepsilon} p_C) - e^{\varepsilon}(1-\delta/2)p_C = e^{\varepsilon} p_C\, \delta/2 \le \delta/2$. Otherwise $1 - e^{\varepsilon}(1-\delta/2)p_C < 1 - (1-\delta/2) = \delta/2$. In both cases, steps 1 to 3 give

$$\Pr[O(D) \in S] \le e^{\varepsilon}(1-\delta/2)\, p_C + \tfrac{\delta}{2} + \tfrac{\delta}{2} \le e^{\varepsilon} \Pr[O(D') \in S] + \delta. \tag{10.7}$$

Accuracy.

5. At most n elements appear in D. A Laplace variable of scale $2/\varepsilon$ exceeds s in absolute value with probability $e^{-\varepsilon s/2}$, so by a union bound all of them satisfy $|Z_x| \le (2/\varepsilon)\ln(n/\beta)$ with probability at least $1-\beta$.
6. On that event, take any element x. If it is listed, its error is $|Z_x|$. If it appears in D but is not listed, then $c_x + Z_x \le \tau$, so its error is $c_x \le \tau + |Z_x|$. If it does not appear, both its count and its estimate are 0. In every case the error is at most Equation 10.6.

□

In the usual regime $\delta \le 1/n$, the bound Equation 10.6 is at most $1 + (2/\varepsilon)(2\ln(1/\delta) + \ln(1/\beta))$. So every count is accurate to within a number of order $\ln(1/\delta)/\varepsilon$, whatever the size of the domain. The Laplace histogram under pure DP pays $\ln|\mathcal{X}|$ instead, and by the packing chapter pure DP cannot avoid that cost.

10.4. Privately bounding the local sensitivity

Propose-test-release guesses a bound and checks it. A second technique estimates the local sensitivity itself, privately, and uses the estimate as the noise scale. It applies when the local sensitivity does not change fast between neighbours, that is, when LS_f has bounded global sensitivity.

Definition 10.6 (Noisy local sensitivity)**.** Suppose that the local sensitivity changes by at most $\lambda > 0$ between neighbours, that is, $|\mathrm{LS}_f(D) - \mathrm{LS}_f(D')| \le \lambda$ for all $D \sim D'$. Given $\varepsilon > 0$ and $\delta \in (0, 1)$, on input D:

1. compute $\hat{s} = \mathrm{LS}_f(D) + Z_1 + \lambda \ln(1/\delta)/\varepsilon$ with $Z_1 \sim \mathrm{Lap}(\lambda/\varepsilon)$;
2. if $\hat{s} \le 0$, output $\perp$, and otherwise output $f(D) + Z_2$ with $Z_2 \sim \mathrm{Lap}(\hat{s}/\varepsilon)$.

The shift $\lambda \ln(1/\delta)/\varepsilon$ makes the estimate $\hat{s}$ an overestimate of $\mathrm{LS}_f(D)$ except with probability $\delta/2$. On that good event, the noise is large enough for privacy. Paying both for the noisy estimate and for the release gives the factor 2ε in the next theorem.

Theorem 10.4. *Let $\lambda > 0$ satisfy $|\mathrm{LS}_f(D) - \mathrm{LS}_f(D')| \le \lambda$ for all neighbours $D \sim D'$, and let $\varepsilon > 0$ and $\delta \in (0, 1)$. Then noisy local sensitivity with these parameters is $(2\varepsilon, \delta)$-DP.*

Proof. Fix neighbours $D \sim D'$. It suffices to prove the guarantee for the pair $(\hat{s}, \text{output})$, since the output alone is post-processing of it.

1. *The good event.* Let $G = \{\hat{s} \ge \mathrm{LS}_f(D)\}$, computed with the data D. It fails only if $Z_1 < -\lambda \ln(1/\delta)/\varepsilon$, which has probability $\delta/2$.
2. *Density of the scale.* The function LS_f has sensitivity λ, so the density of $\hat{s}$ on D is at most e^{ε} times its density on D', at every value s.
3. *Density of the release.* Fix a value s with $s \ge \mathrm{LS}_f(D)$ and $s > 0$. Then $|f(D) - f(D')| \le \mathrm{LS}_f(D) \le s$, so the Laplace density of scale s/ε at any y changes by a factor at most $e^{\varepsilon |f(D) - f(D')|/s} \le e^{\varepsilon}$ between D and D'.
4. *Combining.* On values of $\hat{s}$ in G, the joint density of $(\hat{s}, \text{output})$ on D is at most $e^{2\varepsilon}$ times that on D', by steps 2 and 3. For an event S, split on G:

$$\Pr[(\hat{s}, M(D)) \in S] \le e^{2\varepsilon} \Pr[(\hat{s}, M(D')) \in S] + \Pr[\text{not } G] \le e^{2\varepsilon} \Pr[(\hat{s}, M(D')) \in S] + \tfrac{\delta}{2}. \quad (10.8)$$

□

For each function, the local sensitivity at D and a bound λ on how fast it changes must be worked out by hand. The first takes up to $n(|\mathcal{X}| - 1)$ evaluations of f, one per neighbour. For some functions it is much faster, such as the mode, whose local sensitivity depends only on the two largest counts. The technique is simpler to run than propose-test-release, but it applies to fewer functions, because it needs the local sensitivity to change slowly everywhere.

10.5. Smooth sensitivity

The last technique also lets the local sensitivity grow away from D, but discounts each dataset by its distance. Take the closest-pair function of Example 10.1 on the bounded domain $\mathcal{X} = [0, 10]$, so that its global sensitivity is finite. The dataset $D = (0, 0, 0)$ still has local sensitivity 0, while its neighbour $(0, 0, 5)$ has local sensitivity at least 5. A safe noise scale at D must account for that neighbour, but less than for D itself.

Definition 10.7 (Smooth sensitivity)**.** For a smoothing parameter $s > 0$, the **smooth sensitivity** of f at D discounts the local sensitivity of every dataset D' by a factor e^{-s} for each record in which D' differs from D:

$$S_{f,s}(D) = \sup_{D' \in \mathcal{X}^n} \mathrm{LS}_f(D')\, e^{-s\, d(D,D')}, \tag{10.9}$$

where d is the Hamming distance.

The supremum ranges over all datasets, not only neighbours. Every dataset is within distance n of every other, so $e^{-sn}\Delta f \le S_{f,s}(D) \le \Delta f$. The smooth sensitivity is therefore finite exactly when the global sensitivity is, and it can be smaller by a factor up to e^{sn}. Two properties follow from the definition, and they are all that privacy needs.

Lemma 10.3. *For every D, $S_{f,s}(D) \ge \mathrm{LS}_f(D)$. For all neighbours $D \sim D'$, $S_{f,s}(D) \le e^s S_{f,s}(D')$.*

Proof.

1. The term $D' = D$ in the supremum is $\mathrm{LS}_f(D)$, so the supremum is at least that.
2. For any dataset W, the triangle inequality gives $d(D, W) \ge d(D', W) - 1$, so $\mathrm{LS}_f(W)e^{-s\, d(D,W)} \le e^s\, \mathrm{LS}_f(W)e^{-s\, d(D',W)} \le e^s S_{f,s}(D')$. Taking the supremum over W gives the second claim.

□

Laplace noise scaled to the smooth sensitivity is not enough for pure DP. The scale moves by a factor up to e^s between neighbours, and far in the tail a Laplace density changes by much more than that when its scale changes. Noise with polynomial tails changes little under rescaling, and it works. The standard Cauchy distribution is such a noise, with density at a point z equal to

$$h(z) = \frac{1}{\pi(1 + z^2)}. \tag{10.10}$$

Theorem 10.5 (Smooth sensitivity with Cauchy noise)**.** *Let f be non-constant with finite global sensitivity Δf, let $\varepsilon > 0$, and write $S^* = S_{f,\varepsilon/2}$. The mechanism*

$$M(D) = f(D) + \frac{2\, S^*(D)}{\varepsilon}\, Z, \qquad Z \sim \textit{standard Cauchy}, \tag{10.11}$$

is ε-DP.

Proof.

1. *The scale is positive and finite.* Some dataset W_0 has $\mathrm{LS}_f(W_0) > 0$ because f is not constant, and every D is within distance n of it, so $S^*(D) \geq \mathrm{LS}_f(W_0)e^{-\varepsilon n/2} > 0$. Every term of the supremum is at most Δf, so $S^*(D) \leq \Delta f < \infty$.
2. *Two facts about h.* The derivative of $\ln(1+z^2)$ is $2z/(1+z^2)$, which lies in $[-1,1]$, so $h(z)/h(z+u) \leq e^{|u|}$ for all z and u. For every real κ and v, $e^{-\kappa}h(v)/h(e^{\kappa}v) = e^{-\kappa}(1+e^{2\kappa}v^2)/(1+v^2) \leq e^{-\kappa}\max(1, e^{2\kappa}) = e^{|\kappa|}$.
3. *The ratio of densities.* Fix neighbours $D \sim D'$ and write $a = \varepsilon/2$. The density of $M(D)$ at y is $(a/S^*(D))\, h(a(y-f(D))/S^*(D))$. Set $u = a(y-f(D))/S^*(D)$, $w = a(f(D)-f(D'))/S^*(D)$ and $e^{\kappa} = S^*(D)/S^*(D')$. Then $a(y-f(D'))/S^*(D') = e^{\kappa}(u+w)$, and the ratio of the densities on D and D' is

$$e^{-\kappa}\,\frac{h(u)}{h(e^{\kappa}(u+w))} = \frac{h(u)}{h(u+w)} \cdot \frac{e^{-\kappa}\,h(u+w)}{h(e^{\kappa}(u+w))} \leq e^{|w|}\,e^{|\kappa|}, \tag{10.12}$$

by the two facts of step 2.

4. *Bounding the exponents.* By Lemma 10.3, $|f(D)-f(D')| \leq \mathrm{LS}_f(D) \leq S^*(D)$, so $|w| \leq a = \varepsilon/2$. By the same lemma applied in both directions, $|\kappa| \leq \varepsilon/2$. The densities therefore differ by a factor at most e^{ε}, and integrating over any event gives ε-DP.

□

The Cauchy distribution has such heavy tails that it has no mean. Its tail is $\Pr[|Z| > z] = \frac{2}{\pi}\arctan(1/z) \leq 2/(\pi z)$, so the error exceeds $4S^*(D)/(\pi\varepsilon\beta)$ with probability at most β. That bound grows as $1/\beta$, while the exponential tails of Laplace or Gaussian noise give $\ln(1/\beta)$. Those two noises, scaled to the smooth sensitivity with a smaller smoothing parameter, give approximate DP with lighter tails. Computing $S_{f,s}$ means a maximum over all datasets, which is hard in general, but for the median it takes polynomial time.

10.6. Summary

Table 10.1.: The chapter's results.

Result	Statement
Proposition 10.1	Laplace noise scaled to the local sensitivity is not private
Lemma 10.1	the distance to instability has sensitivity 1
Theorem 10.1	propose-test-release is $(2\varepsilon, \delta)$-DP
Theorem 10.2	the stable mode is (ε, δ)-DP and exact when the top two counts differ by $2 + 4\ln(1/\delta)/\varepsilon$
Theorem 10.3	the stability-based histogram is (ε, δ)-DP with every count within $\tau + (2/\varepsilon)\ln(n/\beta)$

Result	Statement
Theorem 10.4	a noisy overestimate of a Lipschitz local sensitivity gives $(2\varepsilon, \delta)$-DP
Lemma 10.3	smooth sensitivity bounds the local sensitivity and changes by at most e^s between neighbours
Theorem 10.5	Cauchy noise of scale $2S_{f,\varepsilon/2}(D)/\varepsilon$ gives ε-DP when $0 < \Delta f < \infty$

10.7. References

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11. Packing Lower Bounds

Every chapter so far has given an algorithm and proved how much data it needs. This chapter asks whether a cleverer algorithm could need less. For pure differential privacy (DP), a short argument called packing answers it. An accurate algorithm must put its output in a different region for each of many datasets. But a private one barely moves its output distribution between them. Both can hold only when the dataset is large enough.

The resulting lower bounds hold for every ε-DP algorithm, however it is built. On one-way marginals and histograms, they show that the simple Laplace-noise algorithms are optimal up to a constant or a logarithmic factor. They also show that pure DP costs strictly more than approximate DP there.

11.1. Two problems and their upper bounds

Both problems ask for many counting queries at once, answered as fractions of the dataset. The first has records that are d yes-or-no answers, and asks for the fraction of yes answers to each question.

Definition 11.1 (One-way marginals)**.** Let $\mathcal{X} = \{0,1\}^d$. For a record x_i, write x_{ij} for its coordinate j. The **one-way marginals** of a dataset $D \in \mathcal{X}^n$ are the d averages

$$f_j(D) = \frac{1}{n}\sum_{i=1}^{n} x_{ij}, \qquad j = 1, \dots, d. \tag{11.1}$$

The second problem has records that are symbols from a finite set, and asks for the fraction of records equal to each symbol.

Definition 11.2 (Histogram)**.** Let $\mathcal{X} = [k] = \{1, \dots, k\}$. The **normalized histogram** of a dataset $D \in [k]^n$ is the vector $h(D) \in [0,1]^k$ with coordinates

$$h_j(D) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}[x_i = j], \qquad j = 1, \dots, k, \tag{11.2}$$

where $\mathbf{1}[x_i = j]$ is 1 if record i equals j and 0 otherwise.

For both problems, an answer y is judged by its worst coordinate, the ℓ_∞ error $\max_j |y_j - f_j(D)|$. So an answer with error at most α gets every query right to within α at once.

The upper bounds come from the Laplace mechanism. Suppose the vector of answers f has ℓ_1 sensitivity at most $\Delta_1 f$, which means that $\sum_j |f_j(D) - f_j(D')| \le \Delta_1 f$ for all neighbours $D \sim D'$, datasets that differ in one record (the replace-one relation). The Laplace mechanism releases $f(D) + (Z_1, \dots, Z_d)$, where the noise terms Z_j are independent Laplace variables with scale $\Delta_1 f/\varepsilon$, and it is ε-DP. The analysis needs only the tail of a Laplace variable Z with scale b, for every $s \ge 0$,

$$\Pr[|Z| \ge s] = e^{-s/b}. \tag{11.3}$$

Computing the ℓ_1 sensitivity of each problem and applying this tail to every coordinate gives the two upper bounds that the rest of the chapter tries to beat.

Proposition 11.1. *Let $\varepsilon > 0$, $\alpha > 0$ and $\beta \in (0,1)$. With probability at least $1 - \beta$:*

1. *the Laplace mechanism for the d one-way marginals, with scale $d/(n\varepsilon)$, has error at most $d \ln(d/\beta)/(n\varepsilon)$;*
2. *the Laplace mechanism for the histogram over $[k]$, with scale $2/(n\varepsilon)$, has error at most $2\ln(k/\beta)/(n\varepsilon)$.*

Both are ε-DP. So error at most α with probability at least $1/2$ needs only

$$n \ge \frac{d \ln(2d)}{\alpha\varepsilon} \quad \textit{for marginals}, \qquad n \ge \frac{2\ln(2k)}{\alpha\varepsilon} \quad \textit{for histograms}. \tag{11.4}$$

Proof.

1. *Sensitivity of the marginals.* Replacing one record changes each average f_j by at most $1/n$, so the ℓ_1 sensitivity of $(f_1, \dots, f_d)$ is at most d/n. The Laplace mechanism with scale $d/(n\varepsilon)$ is therefore ε-DP.
2. *Sensitivity of the histogram.* Replacing a record equal to a by a record equal to $a' \ne a$ lowers h_a by $1/n$ and raises $h_{a'}$ by $1/n$, so the ℓ_1 sensitivity of h is $2/n$. The scale $2/(n\varepsilon)$ gives ε-DP.
3. *Union bound.* For m Laplace variables with scale b, the chance that some $|Z_j|$ is at least s is at most $me^{-s/b}$. Setting this equal to β gives $s = b\ln(m/\beta)$. With $m = d$ and $b = d/(n\varepsilon)$ this is claim 1, and with $m = k$ and $b = 2/(n\varepsilon)$ it is claim 2.
4. *Sample size.* Each error bound is at most α exactly when n satisfies Equation 11.4 with $\beta = 1/2$.

□

Approximate DP does better on both problems. For the marginals, the Gaussian mechanism of the approximate DP chapter uses the ℓ_2 sensitivity $\sqrt{d}/n$ instead of the ℓ_1 sensitivity d/n, and needs n only of order $\sqrt{d\ln(1/\delta)\ln d}/(\alpha\varepsilon)$, a factor of about $\sqrt{d}$ less. For histograms, the stability-based histogram of the previous chapter needs n of order $\ln(1/\delta)/(\alpha\varepsilon)$ when $\delta \le 1/n$, with no dependence on k at all. The rest of the chapter shows that under pure DP neither gain is possible.

11.2. The packing argument

The name comes from packing balls into a box, where only a limited number fit because the balls cannot overlap. Here the balls are regions of the output space that do not overlap, one for each of many datasets. Accuracy forces each dataset to send its output into its own region, and privacy forces every dataset to give every region some probability. The probabilities of disjoint regions add up to at most 1, so only a limited number of datasets can be packed.

The tool that connects datasets far apart is group privacy, from the Laplace mechanism chapter. The **Hamming distance** between two datasets in $\mathcal{X}^n$ is the number of records in which they differ. Group privacy says that an ε-DP mechanism M is $t\varepsilon$-DP for datasets at Hamming distance at most t. So for such datasets D and D' and every event S,

$$\Pr[M(D) \in S] \le e^{t\varepsilon} \Pr[M(D') \in S]. \tag{11.5}$$

11.2.1. A first example: the mode

The simplest instance of the argument is the mode of a dataset over $[k]$, the most frequent symbol. Take the k datasets $D_1, \dots, D_k$, where D_j consists of n copies of the symbol j. The mode of D_j is j, and any two of these datasets are at Hamming distance n. An accurate mechanism must answer j on D_j at least half the time, for every j. Group privacy then forces each answer j to keep some probability on every other dataset as well, and k such probabilities cannot all fit into a total of 1 unless n is large.

Proposition 11.2 (Lower bound for the mode)**.** *Let $k \ge 2$ and $\varepsilon > 0$, and for each $j \in [k]$ let D_j be the dataset of n copies of the symbol j. Let $M \colon [k]^n \to [k]$ be ε-DP, and suppose that $\Pr[M(D_j) = j] \ge 1/2$ for every $j \in [k]$. Then*

$$n \ge \frac{\ln(k/2)}{\varepsilon}. \tag{11.6}$$

Proof.

1. *Pigeonhole.* Fix any j. The k probabilities $\Pr[M(D_j) = \ell]$, for $\ell \in [k]$, sum to 1. So at least one symbol ℓ has $\Pr[M(D_j) = \ell] \le 1/k$, since otherwise the sum would exceed 1.
2. *Accuracy.* The mechanism is correct on D_ℓ with probability at least $1/2$, so $\Pr[M(D_\ell) = \ell] \ge 1/2$.
3. *Group privacy.* The datasets D_ℓ and D_j are at Hamming distance at most n. By Equation 11.5 with the event $\{\ell\}$,

$$\frac{1}{2} \le \Pr[M(D_\ell) = \ell] \le e^{n\varepsilon} \Pr[M(D_j) = \ell] \le \frac{e^{n\varepsilon}}{k}. \tag{11.7}$$

4. *Rearranging.* The chain gives $e^{n\varepsilon} \ge k/2$, and taking logarithms gives $n\varepsilon \ge \ln(k/2)$.

□

So a private mode needs a number of records of order $\ln k/\varepsilon$. With $k = 1{,}000$ symbols and $\varepsilon = 1$, the bound is $\ln 500 = 6.21$, so no 1-DP algorithm finds the mode reliably from 6 records.

11.2.2. The packing theorem

The proof used three things: many datasets, disjoint sets of correct answers, and group privacy across a bounded distance. The theorem below keeps exactly these. It also lets the datasets be close to one common centre D, at distance t that may be much smaller than n, which the histogram bound will need.

Theorem 11.1 (Packing)**.** *Let $D_1, \dots, D_m \in \mathcal{X}^n$ be datasets, each at Hamming distance at most t from a fixed dataset $D \in \mathcal{X}^n$. Let $S_1, \dots, S_m$ be pairwise disjoint subsets of the output space $\mathcal{Y}$. If $M \colon \mathcal{X}^n \to \mathcal{Y}$ is ε-DP and, for some $p > 0$,*

$$\Pr[M(D_\ell) \in S_\ell] \geq p \qquad \text{for every } \ell \in [m], \tag{11.8}$$

then

$$\frac{1}{m} \geq p\, e^{-t\varepsilon}. \tag{11.9}$$

Proof.

1. *Accuracy and group privacy.* Fix ℓ. The datasets D_ℓ and D are at Hamming distance at most t, so Equation 11.5 gives $p \leq \Pr[M(D_\ell) \in S_\ell] \leq e^{t\varepsilon} \Pr[M(D) \in S_\ell]$. Hence $\Pr[M(D) \in S_\ell] \geq pe^{-t\varepsilon}$.
2. *Disjointness.* The sets S_ℓ are disjoint, so the probability of their union is the sum of their probabilities:

$$1 \geq \Pr\left[M(D) \in \bigcup_{\ell=1}^{m} S_\ell\right] = \sum_{\ell=1}^{m} \Pr[M(D) \in S_\ell] \geq m\, p\, e^{-t\varepsilon}. \tag{11.10}$$

3. *Rearranging.* Dividing by m gives Equation 11.9.

□

Taking logarithms, the theorem says that $t\varepsilon \geq \ln(mp)$. With the trivial distance $t = n$ and $p = 1/2$, this reads $n \geq \ln(m/2)/\varepsilon$, which is Proposition 11.2 again with $m = k$. A lower bound on n therefore needs a packing with many datasets m, accuracy regions that do not overlap, and a small distance t in terms of n.

11.3. One-way marginals

To make the answers on different datasets as different as possible, take datasets made of one repeated record. For each $w \in \{0,1\}^d$, let D_w consist of n copies of w. Its marginals are $f(D_w) = w$, a corner of the cube $[0,1]^d$, and there are 2^d corners. Two distinct corners differ by 1 in some coordinate, so an answer with every error below $1/2$ points to a single corner. The packing theorem then applies with 2^d datasets.

Theorem 11.2 (Lower bound for one-way marginals)**.** *Let $\varepsilon > 0$. Let $M \colon (\{0,1\}^d)^n \to \mathbb{R}^d$ be ε-DP, and suppose that on every dataset D, with probability at least $1/2$, it answers every one-way marginal with error less than $1/2$:*

$$\Pr\left[\max_j |M(D)_j - f_j(D)| < \tfrac{1}{2}\right] \geq \tfrac{1}{2}. \tag{11.11}$$

Then

$$n \geq \frac{(d-1)\ln 2}{\varepsilon}. \tag{11.12}$$

Proof.

1. *The packing.* Use the $m = 2^d$ datasets D_w. For each w, let $S_w = \{y \in \mathbb{R}^d : |y_j - w_j| < 1/2 \text{ for all } j\}$ be the open ℓ_∞ ball of radius $1/2$ around the corner w. Since $f(D_w) = w$, the hypothesis says exactly that $\Pr[M(D_w) \in S_w] \geq 1/2$.
2. *Disjointness.* Let $w \neq w'$ differ in coordinate j, so $|w_j - w'_j| = 1$. For any y, the triangle inequality gives $|y_j - w_j| + |y_j - w'_j| \geq 1$, so the two distances cannot both be less than $1/2$. Hence S_w and $S_{w'}$ are disjoint.
3. *Distance.* Any two datasets in $\mathcal{X}^n$ differ in at most n records, so every D_w is within Hamming distance $t = n$ of $D = D_0$, the all-zero dataset.
4. *Packing.* Theorem 11.1 with $m = 2^d$, $p = 1/2$ and $t = n$ gives $2^{-d} \geq \frac{1}{2}e^{-n\varepsilon}$. Taking logarithms, $n\varepsilon \geq d\ln 2 - \ln 2$.

□

So $n = \Omega(d/\varepsilon)$ records are necessary for error below $1/2$ under ε-DP, against the upper bound $d\ln(2d)/(\alpha\varepsilon)$ of Proposition 11.1. Under approximate DP, n of order $\sqrt{d}$ suffices, so pure DP costs a factor $\sqrt{d}$ more. A matching lower bound of order $\sqrt{d}$ under approximate DP also holds. Packing cannot give it, because group privacy under (ε, δ)-DP degrades too fast with t, and the proof needs fingerprinting codes instead.

The bound above does not depend on α, because the packing uses corners of the cube, which are far apart. Datasets with only a few records changed have marginals closer together, and they give the factor $1/\alpha$.

Corollary 11.1. *Let $\varepsilon > 0$ and $\alpha \in (0, 1/2)$. If an ε-DP mechanism $M \colon (\{0,1\}^d)^n \to \mathbb{R}^d$ answers every one-way marginal with error at most α, with probability at least $1/2$ on every dataset, then*

$$n \geq \frac{(d-1)\ln 2 - \varepsilon}{2\alpha\varepsilon}. \tag{11.13}$$

Proof.

1. *The packing.* Let $t = \lfloor 2\alpha n \rfloor + 1$, so that $t/n > 2\alpha$, and $t \leq n$ because $2\alpha < 1$. For each $w \in \{0,1\}^d$, let D_w consist of t copies of w and $n - t$ copies of the all-zero record. Its marginals are $f(D_w) = (t/n)\, w$.
2. *Disjointness.* Let S_w be the closed ℓ_∞ ball of radius α around $(t/n)\, w$, so that $\Pr[M(D_w) \in S_w] \geq 1/2$ by hypothesis. If $w \neq w'$ differ in coordinate j, the centres differ by $t/n > 2\alpha$ in that coordinate, and two closed intervals of half-width α whose centres are more than 2α apart do not meet. So the sets S_w are pairwise disjoint.
3. *Distance.* Each D_w differs from the all-zero dataset in at most t records.
4. *Packing.* Theorem 11.1 with $m = 2^d$ and $p = 1/2$ gives $\ln(2^{d-1}) \leq t\varepsilon \leq (2\alpha n + 1)\varepsilon$. Rearranging gives the claim.

□

For concrete parameters, the lower bound and the Laplace upper bound of Proposition 11.1 are within a modest factor of each other.

Example 11.1. With $d = 1{,}000$ questions, $\alpha = 0.1$ and $\varepsilon = 1$, Corollary 11.1 says that $n \geq 3{,}458$ is necessary, and Equation 11.4 says that $n \geq 76{,}010$ suffices. The ratio, 22.0, is the factor $\ln(2d) = 7.60$ times constants, and it grows only logarithmically with d.

11.4. Histograms

For histograms, the corner construction gives only the mode bound $\ln(k/2)/\varepsilon$, with no α. The factor $1/\alpha$ comes from datasets that are all close to one centre, as in Corollary 11.1. Take the centre D_1 to be n copies of the symbol 1, and move a small block of records to another symbol.

Theorem 11.3 (Lower bound for histograms)**.** *Let $k \geq 2$, $\varepsilon > 0$ and $\alpha \in (0, 1/2)$. If an ε-DP mechanism $M \colon [k]^n \to \mathbb{R}^k$ estimates the normalized histogram with error at most α,*

$$\Pr\Big[\max_j |M(D)_j - h_j(D)| \leq \alpha\Big] \geq \tfrac{1}{2} \qquad \text{for every } D \in [k]^n, \tag{11.14}$$

then

$$n \geq \frac{\ln(k/2) - \varepsilon}{2\alpha\varepsilon}. \tag{11.15}$$

Proof.

1. *The packing.* Let $t = \lfloor 2\alpha n \rfloor + 1$, so that $t/n > 2\alpha$, and $t \le n$ because $2\alpha < 1$. Let D_1 be n copies of the symbol 1, so its histogram puts all its mass on the symbol 1. For $\ell = 2, \dots, k$, let D_ℓ consist of t copies of ℓ and $n - t$ copies of 1. Its histogram $h(D_\ell)$ puts mass t/n on the symbol ℓ, mass $1 - t/n$ on the symbol 1, and nothing elsewhere.
2. *Accuracy regions.* Let S_ℓ be the closed ℓ_∞ ball of radius α around $h(D_\ell)$. The hypothesis applied to the dataset D_ℓ says that $\Pr[M(D_\ell) \in S_\ell] \ge 1/2$.
3. *Disjointness.* Take $\ell \ne \ell'$. At least one of them is not 1, say $\ell \ne 1$. In coordinate ℓ, the centre $h(D_\ell)$ has value t/n and the centre $h(D_{\ell'})$ has value 0, because $D_{\ell'}$ contains only the symbols 1 and ℓ'. These differ by more than 2α, so S_ℓ and $S_{\ell'}$ are disjoint.
4. *Distance.* Each D_ℓ differs from D_1 in at most t records, the ones not equal to 1.
5. *Packing.* Theorem 11.1 with $m = k$, $p = 1/2$ and centre D_1 gives $1/k \ge \frac{1}{2} e^{-t\varepsilon}$, that is, $\ln(k/2) \le t\varepsilon$. Since $t \le 2\alpha n + 1$, this gives $\ln(k/2) \le (2\alpha n + 1)\varepsilon$, which rearranges to Equation 11.15.

□

Figure 11.1 shows the construction for $k = 5$, $n = 20$ and $\alpha = 0.1$, where $t = 5$ and each moved block holds a quarter of the records. In coordinate ℓ, the band of acceptable answers for D_ℓ starts at 0.15, above the band of every other dataset, which ends at 0.1.

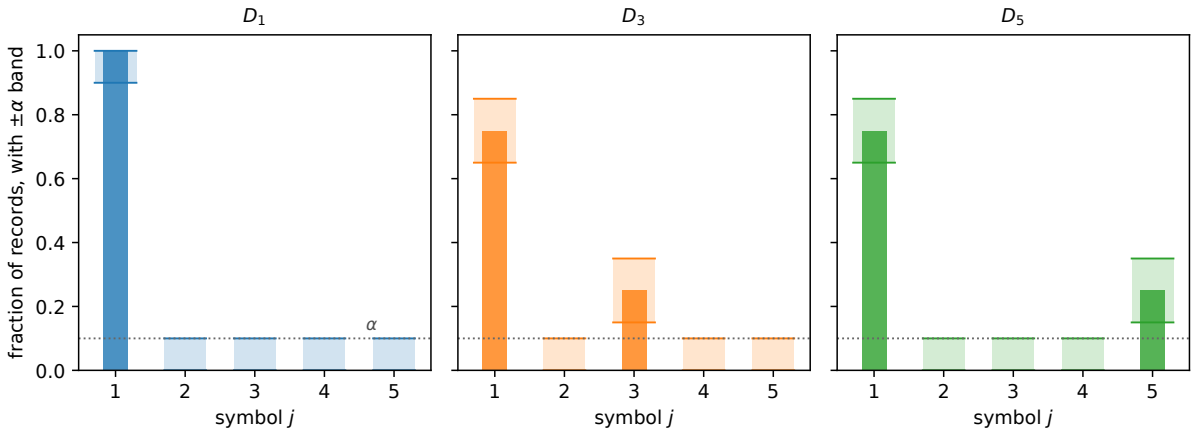


Figure 11.1.: The packing behind Theorem 11.3, for $k = 5$, $n = 20$ and $\alpha = 0.1$. Each panel shows the normalized histogram of one dataset as bars and the band of answers within α of each count as a shaded box. Three of the five datasets are shown. Any two datasets have non-overlapping bands in at least one coordinate, so no answer is acceptable for both.

The choice of t produces the factor $1/\alpha$. Moving the whole dataset, $t = n$, would also give disjoint regions, but then the packing theorem would bound only $n\varepsilon$, as in Proposition 11.2. A block of about $2\alpha n$ records is just enough to separate the answers, and group privacy pays $e^{t\varepsilon}$ for those t records only.

Example 11.2. With $k = 10^6$ symbols, $\alpha = 0.01$ and $\varepsilon = 1$, Equation 11.15 says that $n \ge 607$ is necessary, and the Laplace histogram of Proposition 11.1 succeeds once $n \ge 2{,}902$. The ratio is 4.8, and as k grows it falls towards 4, because the upper bound is $2\ln(2k)/(\alpha\varepsilon)$ and the lower

bound is about $\ln k/(2\alpha\varepsilon)$. The union bound in Proposition 11.1 is nearly tight. In simulation with $k = 10^4$ and $n = 1{,}981$, the value Equation 11.4 gives for that k, the Laplace histogram met error α in 217 of 400 runs, 54%, against the 50% that the bound guarantees.

So under pure DP the Laplace histogram is optimal up to a constant factor, and the $\ln k$ in its sample size cannot be removed. The stability-based histogram, which replaces $\ln k$ by $\ln(1/\delta)$, shows that this cost belongs to pure DP and not to the problem.

11.5. Summary

Table 11.1.: The chapter's results. All lower bounds hold for every ε-DP mechanism that meets the accuracy requirement with probability at least $1/2$.

Result	Statement
Proposition 11.1	Laplace noise answers d marginals with $n \geq d\ln(2d)/(\alpha\varepsilon)$ and a k-bin histogram with $n \geq 2\ln(2k)/(\alpha\varepsilon)$
Proposition 11.2	an ε-DP mode that is right half the time needs $n \geq \ln(k/2)/\varepsilon$
Theorem 11.1	m datasets within distance t of a centre, with disjoint accuracy regions hit with probability p, force $1/m \geq pe^{-t\varepsilon}$
Theorem 11.2	error below $1/2$ on all one-way marginals needs $n \geq (d-1)\ln 2/\varepsilon$
Corollary 11.1	error at most α on all one-way marginals needs $n \geq ((d-1)\ln 2 - \varepsilon)/(2\alpha\varepsilon)$
Theorem 11.3	error at most α on a k-bin histogram needs $n \geq (\ln(k/2) - \varepsilon)/(2\alpha\varepsilon)$

11.6. References

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- Thomas Steinke and Jonathan Ullman, "Between Pure and Approximate Differential Privacy", 2016. Tight bounds for one-way marginals across pure and approximate DP.

- Salil Vadhan, "The Complexity of Differential Privacy", 2017. Section 5 presents packing lower bounds in the generality used here.

Part IV.

Private Learning and Statistics

The earlier parts released statistics of a fixed dataset under differential privacy (DP). This part turns to the goals of machine learning and statistics, which are to fit models and to learn about the population that the data comes from. It starts with the question of what privacy must protect, because a model can predict a person's sensitive label without any use of that person's record. Then it trains convex models and neural networks privately, and finds how many samples a private estimate of a population mean needs. Last, it shows that a private mechanism lets one dataset answer adaptively chosen questions without overfitting. Table 11.2 gives the question of each chapter.

Table 11.2.: The chapters of this part and the questions they answer.

Chapter	Question it answers
What differential privacy protects	Which inferences from a trained model are privacy violations, and which are learning at work?
Private empirical risk minimization	How can a convex model be fitted privately, and at what cost in accuracy?
Private deep learning	How are neural networks trained privately, and how tightly can their privacy be accounted?
Private mean estimation	How many samples does a private estimate of a population mean need?
Adaptive data analysis	How can one dataset answer adaptively chosen questions without overfitting?

12. What Differential Privacy Protects

Machine learning on sensitive data often gives results that feel invasive, but not each of them is a privacy violation. This chapter gives a test that separates the inferences that differential privacy (DP) must prevent from those it allows. Two facts support the test. A private learner can still find a rule that predicts a person's sensitive label, but it cannot memorize a secret that is only in one person's records.

12.1. What the definition promises

Recall the definition from the randomized response chapter. Two datasets $D \sim D'$ are neighbours if they differ by the replacement of one record. A mechanism M is ε-DP if, for all such pairs and all sets S of outputs,

$$\Pr[M(D) \in S] \leq e^{\varepsilon} \Pr[M(D') \in S]. \tag{12.1}$$

So the output changes little when one record changes, and the definition says nothing more. Two properties from the Laplace mechanism chapter turn the definition into a statement about attackers. **Post-processing** says that a function of the output of an ε-DP mechanism, computed with coins independent of the data, is again ε-DP. **Group privacy** says that if D and D' differ in some number k of entries, then

$$\Pr[M(D) \in S] \leq e^{k\varepsilon} \Pr[M(D') \in S]. \tag{12.2}$$

Write $D^{(i \leftarrow z)}$ for the dataset D with entry i replaced by a record z. An attacker is any function $\mathcal{A}$ of the output, possibly randomized, that answers 1 when it believes it has learned something harmful. The attacker may know every other record and any side information, because all of that is fixed in $\mathcal{A}$. The first fact compares its chance to answer 1 on the real dataset and on the dataset with one record replaced.

Proposition 12.1 (What an attacker gains from one record)**.** *Let M be ε-DP, and let $\mathcal{A}$ be a randomized function of the output whose coins are independent of the coins of M. Then for every dataset D, every index i and every record z,*

$$\Pr\left[\mathcal{A}(M(D)) = 1\right] \leq e^{\varepsilon} \Pr\left[\mathcal{A}(M(D^{(i \leftarrow z)})) = 1\right]. \tag{12.3}$$

Proof.

1. The composed algorithm $\mathcal{A} \circ M$ is ε-DP by post-processing.

2. The datasets D and $D^{(i \leftarrow z)}$ differ in at most one entry, so they are neighbours or equal. The definition, applied to the event $\{1\}$, gives the inequality.

□

So a harm that the output causes a person is about as likely when a placeholder replaces their record. The definition does not try to stop that harm. This gives three questions to ask of an alleged violation.

1. Is the result substantially different if one person's record is changed or left out?
2. Is the harmful inference the same thing as the accuracy of the learned model?
3. If the inference still feels wrong, at what point, and by whom, was the information given away?

The second question matters most. If an inference about a person is exactly what an accurate model predicts, the only way to prevent it is to release an inaccurate model, for example one that ignores the data. That is always possible and always useless.

12.2. Learning a rule is not a violation

A learning algorithm takes a training set of pairs (x_i, y_i), where x_i is a **feature vector** and y_i a **label**. It outputs a **classifier**, a function that maps a feature vector to a predicted label. In the private setting, the training set is the sensitive dataset, and the classifier is the output that must be differentially private. The test points that measure its accuracy are not protected.

Take a simple training set of real pairs,

$$(0, 1),\ (2, 3),\ (5, 6),\ (-6, -5),\ (0.5, 1.5),\ (\pi, \pi + 1),\ \ldots \tag{12.4}$$

in which every label is one more than its feature. Any reasonable learner finds the rule $y = x + 1$. At $x = 4$, which is not in the training set, it predicts 5. At $x = \pi$, it predicts $\pi + 1$, which is the true label of a training record. That is not a violation, because the learner found a fact about the population. The prediction at $x = \pi$ is the same if the record $(\pi, \pi + 1)$ was never collected.

A private algorithm makes this exact. Each record gives an offset $y_i - x_i$, and a rule $y = x + b$ is a choice of offset. The stability-based histogram of the chapter on beyond global sensitivity releases the offsets that many records share, although offsets are real numbers with infinitely many possible bins. Its guarantee is approximate DP, (ε, δ)-DP, from the approximate DP chapter, which adds a slack δ to the bound of the definition:

$$\Pr[M(D) \in S] \le e^{\varepsilon} \Pr[M(D') \in S] + \delta \quad \text{for all } D \sim D' \text{ and all } S. \tag{12.5}$$

The algorithm adds Laplace noise to the count of each offset and releases the offsets whose noisy count is above a threshold.

Definition 12.1 (Stable histogram of offsets)**.** Let $\varepsilon > 0$ and $\delta \in (0, 1)$, and set the threshold

$$\tau = 1 + \frac{2\ln(1/\delta)}{\varepsilon}. \tag{12.6}$$

On input $D = ((x_1, y_1), \ldots, (x_n, y_n))$, a list of n records with real entries, let $c_b = |\{i : y_i - x_i = b\}|$ be the number of records whose offset is b. For each b with $c_b > 0$, draw an independent Z_b from the Laplace distribution $\mathrm{Lap}(2/\varepsilon)$ with scale $2/\varepsilon$, and output the set

$$M(D) = \{\, b : c_b > 0 \text{ and } c_b + Z_b > \tau \,\}. \tag{12.7}$$

This is the stability-based histogram on the offsets $y_i - x_i$, with the noisy counts dropped from its output. Bins with count zero are never released, and the threshold τ stops the release of a bin that one record holds, except with small probability. So the guarantee of the histogram carries over.

Theorem 12.1. *The stable histogram of offsets is (ε, δ)-DP.*

Proof.

1. Replacing one record changes at most one offset, so when $D \sim D'$ the two lists of offsets are neighbours or equal.
2. The stability-based histogram is (ε, δ)-DP on datasets of offsets, by the theorem of that chapter.
3. The output $M(D)$ is the set of elements on the histogram's list, a function of that output alone. So it is (ε, δ)-DP by post-processing, which holds for approximate DP as well.

□

The privacy cost does not depend on how many records follow the rule. Accuracy only needs enough of them to clear the threshold.

Proposition 12.2. *Suppose every record satisfies $y_i = x_i + 1$, let τ be the threshold of Definition 12.1, and let $\beta \in (0, 1/2)$ be a failure probability. If*

$$n \geq \tau + \frac{2}{\varepsilon} \ln \frac{1}{2\beta}, \tag{12.8}$$

then with probability at least $1 - \beta$ the output is exactly $\{1\}$, and the released rule predicts $y = x + 1$ at every feature value.

Proof.

1. Only the offset 1 has a positive count, $c_1 = n$, because all n records have that offset. So the output is either $\{1\}$ or empty.

2. The output is empty when the Laplace noise Z_1 on that count satisfies $n + Z_1 \leq \tau$. A $\mathrm{Lap}(2/\varepsilon)$ variable Z has $\Pr[Z \leq -s] = \frac{1}{2}e^{-\varepsilon s/2}$ for every $s \geq 0$. With $s = n - \tau$, which is not negative because $n \geq \tau$, this gives

$$\Pr[Z_1 \leq -(n - \tau)] = \tfrac{1}{2}e^{-\varepsilon(n-\tau)/2}. \tag{12.9}$$

3. The assumed bound on n makes this at most β.

□

Example 12.1. Take $\varepsilon = 1$, $\delta = 10^{-6}$ and $\beta = 0.01$. With these values the threshold is $\tau = 1 + 2\ln(10^6) = 1 + 2 \times 13.82 = 28.63$. The bound on n adds $2\ln 50 = 7.82$ to it, which gives 36.45. So 37 records that follow the rule are enough for the private algorithm to output it with probability at least 0.99.

So Proposition 12.1 protects the record $(\pi, \pi + 1)$, but the rule predicts the label of that record exactly. The prediction follows from the rule, and the rule is a fact about nature. If this still feels like a leak, the third question finds the point of the leak. It is the moment when someone revealed that the person's feature value is π, because after that anyone who knows the rule can calculate the label.

12.3. Inferences from what people reveal

The same pattern explains examples that people often present as privacy failures. Take a smoker who must decide whether to join a study of whether smoking causes cancer, and fears higher insurance premiums if the study publishes the link. This is the rule $y = x + 1$ in another form. The feature is "smokes", the label is "likely to develop cancer", and the study reaches the same conclusion without this person. The only way to prevent the harm is not to run the study. The person gave up privacy when their smoking became known.

12.4. What a violation looks like

A violation is an inference that depends on the presence of the person's own sensitive record in the dataset.

The Netflix Prize. Netflix released movie ratings with dates under pseudonymous user identifiers. Narayanan and Shmatikov linked them to public reviews on the Internet Movie Database, and so found films that the matched users had rated but never reviewed in public. Without a user's record, that reidentification is impossible, so it fails the first question. In contrast, an analyst who sees that a user reviewed the first and third films of a trilogy may guess that they watched the second. That guess uses only what the user chose to publish, so it is not a violation.

Memorized secrets. Carlini and coauthors trained language models on text with a secret, such as a nine-digit identification number, that one person had typed several times. The model gave the secret an unusually high likelihood, and a search guided by those scores could extract it. This is a violation. A model that learned only the language has no reason to prefer one nine-digit

number to another, so the preference comes from that person's records. DP prevents it, with group privacy for the repetitions, as the next proposition shows.

Proposition 12.3 (Secrets are not memorized)**.** *Let $\mathcal{S}$ be a finite set of N possible secrets. For each $s \in \mathcal{S}$, let $D(s)$ be a training set in which k entries depend on s and the other entries do not. Let M be ε-DP, and let G be any guessing rule, with coins independent of M, that maps the trained model to an element of $\mathcal{S}$. If the secret s is drawn uniformly from $\mathcal{S}$, then*

$$\Pr\left[G(M(D(s))) = s\right] \leq \frac{e^{k\varepsilon}}{N}. \tag{12.10}$$

Proof.

1. Let D_0 be the training set with the k entries that depend on the secret replaced by fixed placeholder records. For every s, the datasets $D(s)$ and D_0 differ in at most k entries.
2. By post-processing and group privacy, $\Pr[G(M(D(s))) = s] \leq e^{k\varepsilon} \Pr[G(M(D_0)) = s]$ for every s.
3. The guess on D_0 does not depend on s, and it equals at most one secret, so $\sum_{s \in \mathcal{S}} \Pr[G(M(D_0)) = s] \leq 1$.
4. The average of step 2 over the uniform secret, with step 3, gives

$$\Pr\left[G(M(D(s))) = s\right] = \frac{1}{N} \sum_{s \in \mathcal{S}} \Pr\left[G(M(D(s))) = s\right] \leq \frac{e^{k\varepsilon}}{N}. \tag{12.11}$$

□

For a nine-digit number, $N = 10^9$. With $\varepsilon = 1$ and a secret typed $k = 4$ times, the chance of the attacker is at most $e^4/10^9 = 5.5 \times 10^{-8}$, against 10^{-9} for a blind guess. The bound grows exponentially in k. So a person who contributes many records needs a guarantee per person, not per record.

12.5. Summary

Table 12.1.: The chapter's results.

Result	Statement
Proposition 12.1	an attacker succeeds about as often when a person's record is replaced by any other
Theorem 12.1	releasing offsets whose noisy count exceeds τ is (ε, δ)-DP
Proposition 12.2	with $n \geq \tau + \frac{2}{\varepsilon} \ln \frac{1}{2\beta}$ records, the private algorithm finds $y = x + 1$
Proposition 12.3	a secret in k records of one person is guessed with probability at most $e^{k\varepsilon}/N$

12.6. References

- Cynthia Dwork and Aaron Roth, *The Algorithmic Foundations of Differential Privacy*, 2014, whose Section 2.3 discusses what the definition promises and the smoking example.
- Arvind Narayanan and Vitaly Shmatikov, "Robust De-anonymization of Large Sparse Datasets", 2008, the linkage attack on the Netflix Prize data.
- Nicholas Carlini, Chang Liu, Úlfar Erlingsson, Jernej Kos and Dawn Song, "The Secret Sharer: Evaluating and Testing Unintended Memorization in Neural Networks", 2019, which extracts secrets that appear in few training records.
- Abhradeep Thakurta and Adam Smith, "Differentially Private Feature Selection via Stability Arguments, and the Robustness of the Lasso", 2013, and Salil Vadhan, "The Complexity of Differential Privacy", 2017, Section 3, on stability-based algorithms such as the histogram behind Theorem 12.1.

13. Private Empirical Risk Minimization

Most of machine learning fits a model by minimizing its average loss on a training set. This chapter asks how to do that under differential privacy (DP) when the loss is convex. Noise can go into the output of a non-private solver, into the objective that the solver minimizes, or into each step of gradient descent. The chapter proves a privacy guarantee for each of the three methods. It also proves an accuracy guarantee for the third, which the next chapter scales to neural networks.

13.1. Empirical risk minimization

Each record $z = (x, y)$ is a pair of a feature vector x and a label y, and the dataset $D = (z_1, \dots, z_n)$ holds n records. A parameter vector θ in $\mathbb{R}^d$ describes a model. A **loss function** $\ell(\theta; z)$ measures how badly the model θ fits the record z. To fit a model is to choose θ from a set of allowed values so that the total loss over the dataset is small. The definition also says how to measure an algorithm that only comes close to the best choice.

Definition 13.1 (Empirical risk minimization)**.** Let $\Theta \subseteq \mathbb{R}^d$ be a nonempty, closed, convex set of allowed parameters. The empirical risk of θ on D is the total loss

$$\mathcal{L}(\theta; D) = \sum_{i=1}^{n} \ell(\theta; z_i), \tag{13.1}$$

and $\theta^* \in \Theta$ denotes a minimizer of $\mathcal{L}(\cdot\,; D)$ over Θ. An algorithm that outputs $\tilde{\theta} \in \Theta$ has **excess empirical risk** $\mathbb{E}[\mathcal{L}(\tilde{\theta}; D)] - \mathcal{L}(\theta^*; D)$, where the expectation is over the algorithm's coins only. The dataset is fixed.

The problem is called empirical risk minimization (ERM). In practice the loss on new data matters. But where a uniform convergence argument applies, a good empirical minimizer also has a small loss on new data. So this chapter studies only the empirical problem.

In this generality the problem is hopeless, so the chapter makes three assumptions. The set of allowed parameters is bounded, which its diameter measures. The loss is convex, so that a local minimum is a global one, and Lipschitz, so that it cannot change quickly. The definitions below make these terms precise, with the subgradient, which replaces the gradient where a convex function has a kink.

Definition 13.2 (Diameter, convexity, Lipschitz continuity)**.**

- The **diameter** of Θ is $\|\Theta\|_2 = \sup_{\theta, \theta' \in \Theta} \|\theta - \theta'\|_2$.

- A function f on a convex set K is **convex** if $f(t\theta + (1-t)\theta') \le tf(\theta) + (1-t)f(\theta')$ for all $\theta, \theta' \in K$ and $t \in [0,1]$.
- A function f on K is **L-Lipschitz** if $|f(\theta) - f(\theta')| \le L\|\theta - \theta'\|_2$ for all $\theta, \theta' \in K$.
- A vector g is a **subgradient** of a convex f at θ if $f(\theta') \ge f(\theta) + \langle g, \theta' - \theta\rangle$ for all θ'. Where f is differentiable, its gradient $\nabla f(\theta)$, the vector of partial derivatives, is the only subgradient.

So the chapter assumes that Θ has finite diameter. It also assumes that, for every record z, the loss $\theta \mapsto \ell(\theta; z)$ is convex and L-Lipschitz on $\mathbb{R}^d$. A convex function that is finite on all of $\mathbb{R}^d$ has a subgradient at every point. Write $\nabla\ell(\theta; z)$ for a gradient, or a subgradient where ℓ is not differentiable. Lipschitz continuity matters to privacy because it bounds every gradient, and so the effect of one record on a gradient step.

Lemma 13.1. *If a convex function f is L-Lipschitz on $\mathbb{R}^d$, then every subgradient g of f satisfies $\|g\|_2 \le L$.*

Proof.

1. Suppose $g \neq 0$, take any point θ, and set $u = g/\|g\|_2$. The definition of a subgradient at $\theta' = \theta + u$ gives $f(\theta + u) - f(\theta) \ge \langle g, u\rangle = \|g\|_2$.
2. The Lipschitz condition gives $f(\theta + u) - f(\theta) \le L\|u\|_2 = L$. The two bounds together give $\|g\|_2 \le L$.

□

The most common losses are convex, and most of them are Lipschitz when the features are bounded.

Example 13.1 (Common losses)**.** Take $x, \theta \in \mathbb{R}^d$.

- **Linear regression**, $y \in \mathbb{R}$: the squared loss $\ell(\theta; x, y) = (\langle x, \theta\rangle - y)^2$. It is convex, and Lipschitz only on bounded sets.
- **Logistic regression**, $y \in \{-1, 1\}$: $\ell(\theta; x, y) = \ln\left(1 + e^{-y\langle x, \theta\rangle}\right)$. Its gradient is $-y\,x/(1 + e^{y\langle x,\theta\rangle})$, of norm at most $\|x\|_2$, so it is $\|x\|_2$-Lipschitz.
- **Geometric median**, no label: $\ell(\theta; x) = \|\theta - x\|_2$. By the triangle inequality it is 1-Lipschitz. In one dimension it is $|\theta - x|$, and its minimizer is the median.
- **Support vector machine (SVM)**, $y \in \{-1, 1\}$: the hinge loss $\ell(\theta; x, y) = \max(0, 1 - y\langle x, \theta\rangle)$. It is a maximum of two functions that are linear in θ, so it is convex and $\|x\|_2$-Lipschitz.

Stronger assumptions, such as strong convexity or smoothness of the loss, give faster rates, but the chapter does not need them. Until Section 13.5, the solver of this convex problem is a black box.

13.2. Why the minimizer is not private

An exact minimizer can reveal records outright, and the median is the simplest case.

Proposition 13.1. *Let $d = 1$, $\ell(\theta; x) = |\theta - x|$, records in $[-R, R]$ with $R > 0$, $\Theta = [-R, R]$, and n odd. Then the empirical risk has a unique minimizer, the median $x_{(m)}$ with $m = (n+1)/2$, where $x_{(1)} \le \ldots \le x_{(n)}$ are the sorted records. The algorithm that outputs it is not (ε, δ)-DP for any ε and any $\delta < 1$.*

Proof.

1. Away from the records, the function $\theta \mapsto \sum_i |\theta - x_i|$ has slope equal to the number of records below θ minus the number above. For $\theta < x_{(m)}$ at least m records lie above, so the slope is at most $(m-1) - m = -1$. For $\theta > x_{(m)}$ the slope is at least 1 in the same way. The function is continuous, so $x_{(m)}$ is its unique minimizer.
2. On the dataset with every record equal to $-R$ the output is $-R$, and on the dataset with every record equal to R it is R. A path of n replacements joins the two datasets, so some neighbours $D \sim D'$ along it have different outputs $v \neq v'$.
3. The algorithm is deterministic. For the event $S = \{v\}$ this gives $\Pr[M(D) \in S] = 1$ and $\Pr[M(D') \in S] = 0$, and $1 \le e^{\varepsilon} \cdot 0 + \delta$ fails for every $\delta < 1$.

□

To privatize each record before any non-private algorithm is local DP, which needs far more data for the same accuracy. So the three methods below add noise after the data is pooled.

13.3. Output perturbation

The obvious method computes the minimizer and adds noise scaled to its sensitivity. But Proposition 13.1 shows that the minimizer can have large sensitivity, and regularization is the fix.

Polynomial regression shows the problem. With records $(x_i, y_i) \in \mathbb{R}^2$ and the squared loss $\ell(\theta; x, y) = \left(\sum_{j=0}^{p} \theta_j x^j - y\right)^2$, the model is a polynomial of degree p. If the degree is as high as $p = n - 1$, then for distinct features the minimizer passes through every point. Its excess risk is zero, but it fits new data poorly and swings wildly when one point moves. These are two views of the same instability, as the left panel of Figure 13.1 shows.

Regularization adds a penalty that favours simple models. For a function N of the parameters alone, called the **regularizer**, and a weight $\lambda > 0$, the regularized objective is

$$F_D(\theta) = \mathcal{L}(\theta; D) + \lambda N(\theta). \tag{13.2}$$

At $\lambda = 0$ there is no regularization, and as $\lambda \to \infty$ the data is ignored. The usual choice is $N(\theta) = \frac{1}{2}\|\theta\|_2^2$. Its effect on sensitivity comes from strong convexity, which asks that the function curve upward by at least a fixed amount everywhere. For the regularized objective, that amount is the weight λ.

Definition 13.3 (Strong convexity)**.** For $\lambda > 0$, a function f on a convex set K is **λ-strongly convex** if, for all $\theta, \theta' \in K$ and $t \in [0,1]$,

$$f(t\theta + (1-t)\theta') \le tf(\theta) + (1-t)f(\theta') - \frac{\lambda}{2}\, t(1-t)\, \|\theta - \theta'\|_2^2. \tag{13.3}$$

The function $\frac{1}{2}\|\theta\|_2^2$ is 1-strongly convex, with equality, by the identity $t\|a\|^2 + (1-t)\|b\|^2 - \|ta + (1-t)b\|^2 = t(1-t)\|a-b\|^2$. A λ-strongly convex function plus a convex function is λ-strongly convex. The next lemma shows that such a function grows at least quadratically away from its minimizer.

Lemma 13.2. *Let f be λ-strongly convex on a convex set K, with a minimizer $\hat{\theta}$ over K. Then for every $\theta \in K$,*

$$f(\theta) - f(\hat{\theta}) \ge \frac{\lambda}{2}\|\theta - \hat{\theta}\|_2^2. \tag{13.4}$$

In particular, the minimizer is unique.

Proof.

1. For $t \in (0,1)$, the point $\hat{\theta} + t(\theta - \hat{\theta})$ lies in K, so its value is at least $f(\hat{\theta})$.
2. Strong convexity, applied with weight t on θ, bounds that value by $tf(\theta) + (1-t)f(\hat{\theta}) - \frac{\lambda}{2}t(1-t)\|\theta - \hat{\theta}\|_2^2$.
3. The two bounds together, divided by t, give $f(\theta) - f(\hat{\theta}) \ge \frac{\lambda}{2}(1-t)\|\theta - \hat{\theta}\|_2^2$. Letting $t \to 0$ gives the claim.
4. A second minimizer θ would have $f(\theta) - f(\hat{\theta}) = 0$, which forces $\theta = \hat{\theta}$.

□

This growth keeps the regularized minimizer stable. When one record changes, the two regularized objectives differ by a Lipschitz function. Each objective grows quadratically around its own minimizer, so the two minimizers cannot be far apart.

Theorem 13.1 (Sensitivity of the regularized minimizer)**.** *Let Θ be nonempty, closed, convex and bounded. Suppose that for every record z the loss $\ell(\cdot\,; z)$ is convex and L-Lipschitz on Θ, and that N is continuous and 1-strongly convex. For each dataset D, let θ_D be the minimizer of $F_D = \mathcal{L}(\cdot\,; D) + \lambda N$ over Θ. Then for all neighbours $D \sim D'$, which differ in one replaced record,*

$$\|\theta_D - \theta_{D'}\|_2 \le \frac{2L}{\lambda}. \tag{13.5}$$

Proof.

1. F_D is continuous on the closed bounded set Θ, so it has a minimizer, and it is λ-strongly convex, so the minimizer is unique by Lemma 13.2. The same holds for $F_{D'}$.

2. Write $\theta = \theta_D$ and $\theta' = \theta_{D'}$. Lemma 13.2 applied to each function gives

$$F_D(\theta') - F_D(\theta) \geq \frac{\lambda}{2}\|\theta - \theta'\|_2^2, \qquad F_{D'}(\theta) - F_{D'}(\theta') \geq \frac{\lambda}{2}\|\theta - \theta'\|_2^2. \tag{13.6}$$

3. Add the two inequalities. The left side becomes $h(\theta') - h(\theta)$, where $h = F_D - F_{D'}$, and the right side becomes $\lambda\|\theta - \theta'\|_2^2$.
4. If D and D' differ in record j, then $h(\cdot) = \ell(\cdot\,; z_j) - \ell(\cdot\,; z'_j)$ is a difference of two L-Lipschitz functions, so $h(\theta') - h(\theta) \leq 2L\|\theta - \theta'\|_2$.
5. So $\lambda\|\theta - \theta'\|_2^2 \leq 2L\|\theta - \theta'\|_2$, and dividing by $\lambda\|\theta - \theta'\|_2$ gives the claim. If $\theta = \theta'$, there is nothing to prove.

□

The bound depends on neither n nor the dimension, and the Gaussian mechanism of the approximate DP chapter does the rest. Take a function f whose ℓ_2 sensitivity, the largest change in Euclidean norm between neighbours, is $\Delta_2 f$. The mechanism adds Gaussian noise $\mathcal{N}(0, \sigma^2 I_d)$ with standard deviation σ in each coordinate. For every $\varepsilon > 0$ and $\delta \in (0, 1)$ it is (ε, δ)-DP, the relaxation that allows an additive slack δ, when

$$\sigma = \frac{\Delta_2 f}{\varepsilon}\sqrt{2\ln(1/\delta) + \varepsilon}. \tag{13.7}$$

The mechanism applied to the regularized minimizer $f(D) = \theta_D$, with $\Delta_2 f = 2L/\lambda$, gives the first private algorithm.

Theorem 13.2 (Output perturbation)**.** *Under the assumptions of Theorem 13.1, let $\varepsilon > 0$ and $\delta \in (0, 1)$. The algorithm that outputs $\tilde{\theta} = \theta_D + b$, where θ_D is the regularized minimizer, with $b \sim \mathcal{N}(0, \sigma^2 I_d)$ and*

$$\sigma = \frac{2L}{\lambda} \cdot \frac{\sqrt{2\ln(1/\delta) + \varepsilon}}{\varepsilon}, \tag{13.8}$$

is (ε, δ)-DP. So is its projection onto Θ.

Proof. The map $D \mapsto \theta_D$ has ℓ_2 sensitivity at most $2L/\lambda$ by Theorem 13.1, so the Gaussian mechanism gives the guarantee. The projection is post-processing. □

Output perturbation is simple and works with any exact solver. But its accuracy is often weak, because the regularization that privacy needs can be far from the one that fits the data best.

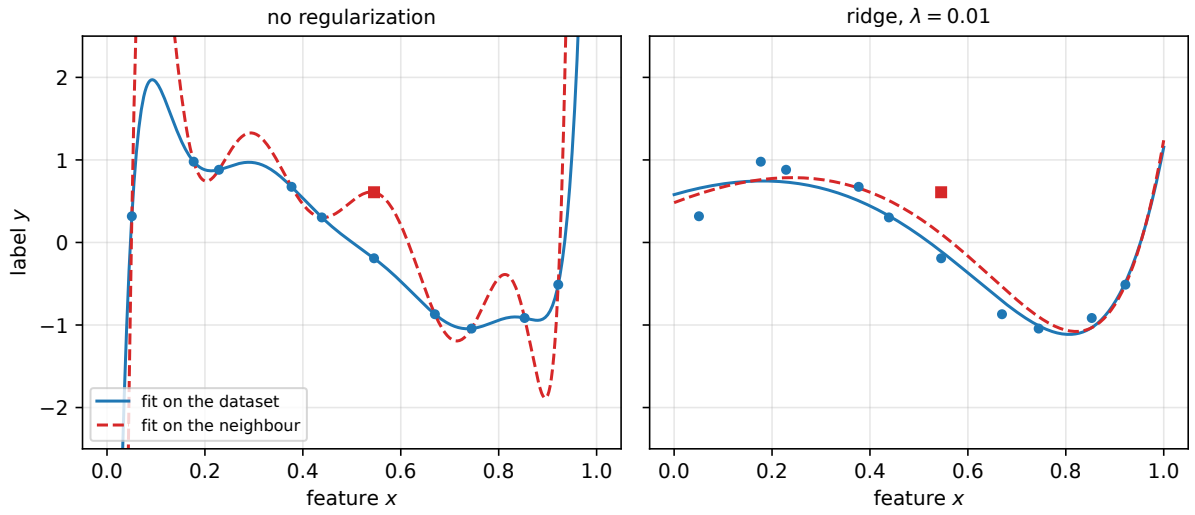


Figure 13.1.: Polynomials of degree 9 fitted by least squares to 10 points, on a dataset and on a neighbour with one label raised (the square). Without regularization, both fits pass through every point, and one moved point changes the whole curve. With the penalty $\frac{\lambda}{2}\|\theta\|_2^2$, the two fits stay close.

13.4. Objective perturbation

The second method adds noise to the objective instead of the output. It minimizes

$$J_b(\theta) = \mathcal{L}(\theta; D) + \frac{\lambda}{2}\|\theta\|_2^2 + \langle b, \theta \rangle \tag{13.9}$$

for a random vector b, and releases the exact minimizer. The linear term rewards or penalizes each direction of the parameter space by a random amount.

The proof below is for losses of **generalized linear** form, such as logistic regression and smoothed versions of the hinge loss. Such a loss is $\ell(\theta; x, y) = \phi(\langle x, \theta \rangle, y)$. For each label y, the scalar function $u \mapsto \phi(u, y)$ is convex and twice continuously differentiable, with derivatives $|\phi'| \le L$ and $0 \le \phi'' \le \beta$. The features satisfy $\|x\|_2 \le 1$. So the gradient $\nabla \ell = \phi'(\langle x, \theta \rangle, y)\, x$ has norm at most L. The Hessian, the matrix of second derivatives, is $\nabla^2 \ell = \phi''(\langle x, \theta \rangle, y)\, xx^\top$, with rank at most one and eigenvalues in $[0, \beta]$.

The proof compares the densities of the output on two neighbouring datasets. So it needs a criterion in terms of densities, from the approximate DP chapter. Write p_D for the density of the output when the dataset is D. The **privacy loss** of an output θ is $\ln\left(p_D(\theta)/p_{D'}(\theta)\right)$, which is large when θ is much more likely under D than under D'. The criterion says that M is (ε, δ)-DP if, for every ordered pair of neighbours, an output drawn from $M(D)$ has privacy loss above ε with probability at most δ.

Theorem 13.3 (Objective perturbation)**.** *Let $\Theta = \mathbb{R}^d$, and let ℓ be a generalized linear loss as above, with $|\phi'| \le L$, $0 \le \phi'' \le \beta$ and features of norm at most 1. Let $\varepsilon \in (0, 4]$ and $\delta \in (0, 1)$, and set*

$$\lambda \geq \frac{\beta}{e^{\varepsilon/2}-1}, \qquad \sigma = \frac{4L(1+\sqrt{2\ln(1/\delta)})}{\varepsilon}. \tag{13.10}$$

Then the algorithm that draws $b \sim \mathcal{N}(0, \sigma^2 I_d)$ and outputs the exact minimizer $\tilde{\theta}$ of J_b is (ε, δ)-DP.

Proof.

1. *The output as a function of the noise.* For each b, the function J_b is continuous and λ-strongly convex on $\mathbb{R}^d$, so it has a unique minimizer, where its gradient vanishes. So $\tilde{\theta} = \theta$ exactly when $b = g_D(\theta)$, where $g_D(\theta) = -\nabla\mathcal{L}(\theta; D) - \lambda\theta$. Every θ is the minimizer for the noise $g_D(\theta)$, and each noise has one minimizer, so g_D is a bijection of $\mathbb{R}^d$ and $\tilde{\theta} = g_D^{-1}(b)$.
2. *The density of the output.* The map g_D is continuously differentiable with Jacobian $-(H_D(\theta)+\lambda I)$, where $H_D(\theta) = \sum_i \nabla^2\ell(\theta; z_i)$ is positive semidefinite. By the change of variables formula, $\tilde{\theta}$ has density
$$p_D(\theta) = \varphi_\sigma(g_D(\theta)) \det(H_D(\theta) + \lambda I), \tag{13.11}$$
where φ_σ is the density of $\mathcal{N}(0, \sigma^2 I_d)$.
3. *The determinants.* Let D and D' differ in the last record, $z = (x, y)$ in D and $z' = (x', y')$ in D'. Fix θ and write $A = \sum_{i<n} \nabla^2\ell(\theta; z_i) + \lambda I$, which satisfies $A \succeq \lambda I$. Then $H_D + \lambda I = A + a\,xx^\top$ with $a = \phi''(\langle x, \theta\rangle, y) \in [0, \beta]$. The matrix determinant lemma gives
$$\det(A + a\,xx^\top) = \det(A)\,(1 + a\,x^\top A^{-1}x) \in \big[\det A,\ (1+\beta/\lambda)\det A\big], \tag{13.12}$$
because $0 \leq x^\top A^{-1}x \leq \|x\|_2^2/\lambda \leq 1/\lambda$. The same holds for D'. So the ratio of the two determinants is at most $1 + \beta/\lambda \leq e^{\varepsilon/2}$, by the choice of λ.
4. *The Gaussian factors.* The two noise vectors that lead to θ differ by $u = g_{D'}(\theta) - g_D(\theta) = \nabla\ell(\theta; z) - \nabla\ell(\theta; z')$. This vector has norm at most $2L$, and it lies in the subspace V spanned by x and x', which does not depend on θ. With $b = g_D(\theta)$ and P_V the orthogonal projection onto V,
$$\ln \frac{\varphi_\sigma(b)}{\varphi_\sigma(b+u)} = \frac{2\langle b, u\rangle + \|u\|_2^2}{2\sigma^2} \leq \frac{2L\,\|P_V b\|_2}{\sigma^2} + \frac{2L^2}{\sigma^2}, \tag{13.13}$$
by the Cauchy–Schwarz inequality applied to $\langle b, u\rangle = \langle P_V b, u\rangle$.
5. *The privacy loss.* When the output is drawn from $M(D)$, the vector $g_D(\tilde{\theta})$ is the noise b itself. Steps 2 to 4 bound the privacy loss at the output by
$$\ln \frac{p_D(\tilde{\theta})}{p_{D'}(\tilde{\theta})} \leq \frac{\varepsilon}{2} + \frac{2L\,\|P_V b\|_2}{\sigma^2} + \frac{2L^2}{\sigma^2}. \tag{13.14}$$
6. *The tail.* The subspace V has dimension $d_V \leq 2$, so $\|P_V b\|_2^2/\sigma^2$ is chi-squared with d_V degrees of freedom. Its tail is at most that of two degrees of freedom, which is $e^{-r/2}$ beyond r. Write $s = \sqrt{2\ln(1/\delta)}$ for the cut-off. Then
$$\Pr\big[\|P_V b\|_2 > \sigma s\big] \leq e^{-s^2/2} = \delta. \tag{13.15}$$
7. Outside that event, substituting $\sigma = 4L(1+s)/\varepsilon$ into step 5 gives
$$\ln \frac{p_D(\tilde{\theta})}{p_{D'}(\tilde{\theta})} \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2}\left(\frac{s}{1+s} + \frac{\varepsilon}{4(1+s)^2}\right) \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2}\cdot\frac{s+1}{1+s} = \varepsilon, \tag{13.16}$$
where the second step uses $\varepsilon \leq 4$ and $1/(1+s)^2 \leq 1/(1+s)$.

8. The privacy loss criterion turns this bound into (ε, δ)-DP.

□

The theorem has a practical weakness. It holds only for the exact minimizer, and a real solver never returns one. Iterative solvers such as gradient descent only approach the minimizer. Floating-point arithmetic cannot represent most real numbers, and even a naive floating-point Laplace mechanism leaks the data through the low-order bits of its output (Mironov). Later versions of objective perturbation need only an approximate minimizer.

13.5. Gradient perturbation

The third method, now the standard one, runs gradient descent and adds noise to each gradient. Its analysis starts from the non-private algorithm. Write $\Pi_\Theta(u) = \arg\min_{\theta \in \Theta} \|\theta - u\|_2$ for the **projection** onto Θ, the closest point of Θ, which exists and is unique because Θ is closed and convex.

The definition leaves the step direction G_t random, so that one analysis covers exact gradients, gradients of one sampled record, and the noisy gradients of the private algorithm.

Definition 13.4 (Projected stochastic gradient descent)**.** Fix a start $\theta_0 \in \Theta$, a number of steps T and a learning rate $\eta > 0$, also called the step size. For $t = 1, \dots, T$, draw a random vector G_t and set

$$\theta_t = \Pi_\Theta(\theta_{t-1} - \eta\, G_t). \tag{13.17}$$

Output the average $\bar{\theta} = \frac{1}{T} \sum_{t=0}^{T-1} \theta_t$. With $G_t = \nabla \mathcal{L}(\theta_{t-1}; D)$ this is projected **gradient descent**. With $G_t = n\, \nabla \ell(\theta_{t-1}; z_{I_t})$, for an index I_t drawn uniformly from $\{1, \dots, n\}$, it is projected stochastic gradient descent (SGD).

A step moves against the gradient, the direction of steepest increase. The projection brings a step that leaves Θ back to the nearest allowed point. The convergence proof compares each iterate with the minimizer, and for that it uses the fact that projection never moves a point farther from any point of Θ.

Lemma 13.3. *For every $u \in \mathbb{R}^d$ and every $\theta \in \Theta$, $\|\Pi_\Theta(u) - \theta\|_2 \le \|u - \theta\|_2$.*

Proof.

1. Let $v = \Pi_\Theta(u)$. For $t \in (0, 1]$ the point $v + t(\theta - v)$ is in Θ, so it is no closer to u than v: $\|u - v - t(\theta - v)\|^2 \ge \|u - v\|^2$. Expanding, dividing by t and letting $t \to 0$ gives $\langle u - v, \theta - v \rangle \le 0$.
2. Then $\|u - \theta\|^2 = \|u - v\|^2 + \|v - \theta\|^2 + 2\langle u - v, v - \theta \rangle \ge \|v - \theta\|^2$, because the last term is nonnegative by step 1.

□

The convergence theorem asks only that each G_t is right on average and not too large.

Theorem 13.4 (Convergence of projected SGD)**.** *Let F be convex on $\mathbb{R}^d$, and let θ^* minimize F over Θ. Suppose that, given $\theta_0, \dots, \theta_{t-1}$, the conditional mean $\mathbb{E}[G_t \mid \theta_0, \dots, \theta_{t-1}]$ is a subgradient of F at θ_{t-1}, and that $\mathbb{E}\|G_t\|_2^2 \le G^2$ for every t. With learning rate $\eta = \|\Theta\|_2/(G\sqrt{T})$,*

$$\mathbb{E}[F(\bar{\theta})] - F(\theta^*) \le \frac{\|\Theta\|_2 \, G}{\sqrt{T}}. \tag{13.18}$$

Proof.

1. By Lemma 13.3 with $\theta = \theta^*$,
$$\|\theta_t - \theta^*\|^2 \le \|\theta_{t-1} - \eta G_t - \theta^*\|^2 = \|\theta_{t-1} - \theta^*\|^2 - 2\eta\langle G_t, \theta_{t-1} - \theta^*\rangle + \eta^2\|G_t\|^2. \tag{13.19}$$
2. Condition on the past and let g be the conditional mean of G_t, a subgradient at θ_{t-1}. The definition of a subgradient gives $\langle g, \theta_{t-1} - \theta^*\rangle \ge F(\theta_{t-1}) - F(\theta^*)$. Taking expectations in step 1,
$$\mathbb{E}[F(\theta_{t-1})] - F(\theta^*) \le \frac{\mathbb{E}\|\theta_{t-1} - \theta^*\|^2 - \mathbb{E}\|\theta_t - \theta^*\|^2}{2\eta} + \frac{\eta G^2}{2}. \tag{13.20}$$
3. Summing over $t = 1, \dots, T$, the distances telescope. The first is at most $\|\Theta\|_2^2$ and the last is nonnegative, so
$$\sum_{t=0}^{T-1} \left(\mathbb{E}[F(\theta_t)] - F(\theta^*)\right) \le \frac{\|\Theta\|_2^2}{2\eta} + \frac{T\eta G^2}{2}. \tag{13.21}$$
4. The average $\bar{\theta}$ lies in Θ, and convexity gives $F(\bar{\theta}) \le \frac{1}{T}\sum_t F(\theta_t)$. Step 3 divided by T, with the value of η, makes both terms $\|\Theta\|_2 G/(2\sqrt{T})$.

□

For the empirical risk, the theorem gives the number of steps that each method needs. The bound G on the step comes from the bound L on every gradient.

Corollary 13.1. *Suppose that Θ has finite diameter $\|\Theta\|_2$ and that every loss $\ell(\cdot\,; z)$ is convex and L-Lipschitz on $\mathbb{R}^d$. Then gradient descent and SGD both reach excess empirical risk at most α after $T = \lceil n^2 L^2 \|\Theta\|_2^2/\alpha^2 \rceil$ steps. Gradient descent evaluates nT gradients of ℓ, and SGD evaluates T.*

Proof.

1. For gradient descent, $G_t = \sum_i \nabla\ell(\theta_{t-1}; z_i)$ is a subgradient of $\mathcal{L}$, and by Lemma 13.1 its norm is at most nL.
2. For SGD, the conditional mean of $G_t = n\nabla\ell(\theta_{t-1}; z_{I_t})$ is $\frac{1}{n}\sum_i n\nabla\ell(\theta_{t-1}; z_i)$, the same subgradient. Its second moment is $\frac{1}{n}\sum_i n^2\|\nabla\ell(\theta_{t-1}; z_i)\|^2 \le n^2L^2$.
3. In both cases $G = nL$, and Theorem 13.4 gives excess risk at most $\|\Theta\|_2 nL/\sqrt{T} \le \alpha$.

□

For a constant α, gradient descent costs about n^3 gradient evaluations and SGD about n^2. SGD is also easier to make private, because each step touches one record. Private SGD adds Gaussian noise to each sampled gradient, at the level σ that the privacy proof below needs, and its learning rate allows for the extra noise.

Definition 13.5 (Private SGD)**.** Let $\varepsilon \in (0, 1]$ and $\delta \in (0, 1)$. Run projected SGD for $T = n^2$ steps with

$$G_t = n\,\nabla \ell(\theta_{t-1}; z_{I_t}) + b_t, \qquad b_t \sim \mathcal{N}(0, \sigma^2 I_d), \qquad \sigma = \frac{16\,nL\sqrt{\ln(3.3n/\delta)\,\ln(2/\delta)}}{\varepsilon}, \tag{13.22}$$

with fresh, independent indices I_t and noise b_t, and learning rate $\eta = \|\Theta\|_2/(G\sqrt{T})$ where $G^2 = n^2L^2 + d\sigma^2$.

Accuracy is the easier half, because the noise has mean zero and only makes the second moment larger.

Theorem 13.5. *With $\ell(\cdot\,; z)$ convex and L-Lipschitz on $\mathbb{R}^d$, private SGD satisfies*

$$\mathbb{E}[\mathcal{L}(\bar{\theta}; D)] - \mathcal{L}(\theta^*; D) \le \|\Theta\|_2 \sqrt{L^2 + \frac{d\sigma^2}{n^2}} \le \|\Theta\|_2\, L \left(1 + \frac{16\sqrt{d\,\ln(3.3n/\delta)\ln(2/\delta)}}{\varepsilon}\right). \tag{13.23}$$

Proof.

1. Given the past, b_t has mean zero and is independent of I_t, so the conditional mean of G_t is the subgradient of step 2 of Corollary 13.1.
2. Expanding the square,
$$\mathbb{E}\|G_t\|^2 = n^2\,\mathbb{E}\|\nabla \ell(\theta_{t-1}; z_{I_t})\|^2 + 2n\,\mathbb{E}\langle \nabla \ell(\theta_{t-1}; z_{I_t}), b_t\rangle + \mathbb{E}\|b_t\|^2 \le n^2L^2 + 0 + d\sigma^2. \tag{13.24}$$
The first term is bounded by Lemma 13.1. The middle term vanishes because b_t is independent of the gradient and has mean zero. The last is a sum of d squared Gaussians of variance σ^2.
3. Theorem 13.4 with $T = n^2$ gives $\|\Theta\|_2 G/n$, which is the first bound. The second uses $\sqrt{a+b} \le \sqrt{a} + \sqrt{b}$ and the value of σ.

□

Up to logarithmic factors, the excess risk is $O(\|\Theta\|_2 L\sqrt{d\ln(1/\delta)}/\varepsilon)$, so the average excess risk per record falls as $1/n$. The price of privacy is the factor $\sqrt{d}$, and Bassily, Smith and Thakurta proved that some dependence of this kind is unavoidable. But the constant 16 comes from a loose chain of bounds. For example, with $n = 10{,}000$, $\varepsilon = 1$ and $\delta = 10^{-6}$, the noise is $\sigma = 299.9\,nL$. With $d = 100$, the bound per record is $\|\Theta\|_2 L(1 + 2999)/10{,}000 = 0.30\,\|\Theta\|_2 L$. The next chapter replaces this chain with a much tighter one.

The privacy proof uses three tools. The first is the Gaussian mechanism, restated above. The second is **amplification by subsampling**, which says that a mechanism run on one random record is more private than the same mechanism run on a fixed record.

Lemma 13.4 (Amplification by sampling one record)**.** *Let M_0 be a mechanism whose input is a single record and which satisfies $\Pr[M_0(z) \in S] \le e^{\varepsilon_0} \Pr[M_0(z') \in S] + \delta_0$ for all records z, z' and all events S. Let $M(D) = M_0(z_I)$ with I uniform on $\{1, \dots, n\}$. Then M is $(\varepsilon_1, \delta_0/n)$-DP with*

$$\varepsilon_1 = \ln\left(1 + \frac{e^{\varepsilon_0} - 1}{n}\right) \le \frac{e^{\varepsilon_0} - 1}{n}. \tag{13.25}$$

Proof.

1. Let D and D' differ in record j, fix an event S, and write $p_i = \Pr[M_0(z_i) \in S]$ and $p'_j = \Pr[M_0(z'_j) \in S]$. Let r be the average of p_i over $i \ne j$, and write $q = 1/n$ for the sampling rate. Then
$$P = \Pr[M(D) \in S] = (1-q)r + qp_j, \qquad P' = \Pr[M(D') \in S] = (1-q)r + qp'_j. \tag{13.26}$$
2. The hypothesis, applied to z_j against z'_j and against each z_i and then averaged, gives two bounds:
$$p_j - p'_j \le (e^{\varepsilon_0} - 1)p'_j + \delta_0, \qquad p_j - p'_j \le e^{\varepsilon_0} r - p'_j + \delta_0. \tag{13.27}$$
3. Let $c = e^{\varepsilon_0} - 1$ and take the convex combination of the two bounds with weights w and $1 - w$, where $1 - w = c(1-q)/e^{\varepsilon_0} \in [0, 1)$. The coefficient of r becomes $c(1-q)$, and that of p'_j becomes $we^{\varepsilon_0} - 1 = cq$. So
$$p_j - p'_j \le c((1-q)r + qp'_j) + \delta_0 = cP' + \delta_0. \tag{13.28}$$
4. Then $P = P' + q(p_j - p'_j) \le (1 + qc)P' + q\delta_0$, which is the claim. The inequality $\ln(1+x) \le x$ gives the simpler bound on ε_1.

□

For $\varepsilon_0 \le 1$, convexity of the exponential gives $e^{\varepsilon_0} - 1 \le (e-1)\varepsilon_0 \le 2\varepsilon_0$. So a sample of one record of n cuts the privacy parameter by a factor of about $n/2$.

The third tool is advanced composition, from the advanced composition chapter. Take k mechanisms in sequence, each $(\varepsilon_0, \delta_0)$-DP and each chosen adaptively from the outputs of the previous ones. Their sequence is $(\varepsilon, k\delta_0 + \delta')$-DP for every $\delta' \in (0, 1)$, where

$$\varepsilon = \varepsilon_0 \sqrt{2k \ln(1/\delta')} + k\varepsilon_0 \frac{e^{\varepsilon_0} - 1}{e^{\varepsilon_0} + 1} \le \varepsilon_0 \sqrt{2k \ln(1/\delta')} + \frac{k\varepsilon_0^2}{2}, \tag{13.29}$$

and the inequality uses $(e^x - 1)/(e^x + 1) = \tanh(x/2) \le x/2$ for $x \ge 0$.

The proof uses the three tools in turn. The Gaussian noise protects the sampled record in one step, the random choice of that record makes the step more private, and advanced composition adds up the steps.

Theorem 13.6 (Privacy of private SGD)**.** *For $\varepsilon \in (0, 1]$ and $\delta \in (0, 1)$, private SGD is (ε, δ)-DP, even if it releases every iterate $\theta_1, \dots, \theta_T$.*

Proof.

1. *One step, fixed index.* Fix the previous iterate and the index i. The step releases a post-processing of $n\nabla\ell(\theta_{t-1}; z_i) + b_t$. As a function of the single record z_i, the vector $n\nabla\ell$ has ℓ_2 sensitivity at most $2nL$, by Lemma 13.1. Set $\varepsilon_0 = \varepsilon/(4\sqrt{2\ln(2/\delta)})$, which is below 1, and $\delta_0 = \delta/(2n)$. The Gaussian mechanism is $(\varepsilon_0, \delta_0)$-DP with respect to z_i at the noise level

$$\sigma_0 = \frac{2nL}{\varepsilon_0}\sqrt{2\ln(1/\delta_0) + \varepsilon_0} \le \frac{2nL}{\varepsilon_0}\sqrt{2\ln\frac{2\sqrt{e}\,n}{\delta}} \le \frac{2nL}{\varepsilon_0}\sqrt{2\ln\frac{3.3n}{\delta}} = \sigma, \tag{13.30}$$

   where the first inequality uses $\varepsilon_0 \le 1 = 2\ln\sqrt{e}$, and the last equality is the σ of Definition 13.5. Noise of variance $\sigma^2 \ge \sigma_0^2$ is noise of variance σ_0^2 plus an independent Gaussian of variance $\sigma^2 - \sigma_0^2$, a post-processing, so the step is $(\varepsilon_0, \delta_0)$-DP with respect to z_i.
2. *One step, random index.* By Lemma 13.4, each step is $(\varepsilon_1, \delta_0/n)$-DP with respect to the dataset, where $\varepsilon_1 \le (e^{\varepsilon_0} - 1)/n \le 2\varepsilon_0/n$.
3. *All steps.* Each step is chosen adaptively from the previous iterates. Advanced composition over $T = n^2$ steps with $\delta' = \delta/2$ gives total failure probability $n^2 \cdot \delta/(2n^2) + \delta/2 = \delta$, and privacy parameter

$$\varepsilon_1\sqrt{2n^2\ln(2/\delta)} + \frac{n^2\varepsilon_1^2}{2} \le 2\varepsilon_0\sqrt{2\ln(2/\delta)} + 2\varepsilon_0^2 = \frac{\varepsilon}{2} + 2\varepsilon_0^2. \tag{13.31}$$

4. Finally, the second term is small because $\varepsilon \le 1$ and $\ln(2/\delta) > \ln 2 > 1/8$:

$$2\varepsilon_0^2 = \frac{\varepsilon^2}{16\ln(2/\delta)} \le \frac{\varepsilon}{2}. \tag{13.32}$$

   The average $\bar{\theta}$ is a post-processing of the iterates.

□

The privacy cost per step is about ε/n, and composition over n^2 steps multiplies it by only $\sqrt{n^2} = n$. Each piece of the argument is generic, so the same recipe extends to models with no convergence theorem.

13.6. Summary

Table 13.1.: The chapter's results.

Result	Statement
Proposition 13.1	the exact median is not (ε, δ)-DP for any $\delta < 1$
Theorem 13.1	the minimizer of loss plus λ·(1-strongly convex) has sensitivity $2L/\lambda$
Theorem 13.2	adding Gaussian noise to that minimizer is (ε, δ)-DP
Theorem 13.3	the exact minimizer of a Gaussian-perturbed objective is (ε, δ)-DP

Result	Statement
Theorem 13.4	projected SGD has excess risk $\|\Theta\|_2 G/\sqrt{T}$
Lemma 13.4	sampling one record of n turns ε_0 into $\ln(1+(e^{\varepsilon_0}-1)/n)$
Theorem 13.6	private SGD with n^2 steps is (ε,δ)-DP
Theorem 13.5	its excess risk is $O(\|\Theta\|_2 L\sqrt{d\ln(1/\delta)}/\varepsilon)$ up to logarithmic factors

13.7. References

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14. Private Deep Learning

The private stochastic gradient descent (SGD) of the previous chapter needs a convex, Lipschitz loss, and its privacy analysis loses large constants. The losses of neural networks are neither convex nor Lipschitz, and in practice the constants decide whether a guarantee is meaningful. This chapter adapts noisy SGD to neural networks under differential privacy (DP) by clipping each gradient. Then it replaces composition with a tighter accounting, built on the moments of the privacy loss. Last, it turns any non-private learner into a private one by a vote among models trained on disjoint data.

14.1. Neural networks

The setting is that of the previous chapter. Each record is a pair of a feature vector and a label, and training minimizes the total loss of the records over a parameter vector θ,

$$\mathcal{L}(\theta; D) = \sum_{i=1}^{n} \ell(\theta; x_i, y_i). \tag{14.1}$$

Here a feature vector x lies in $\mathbb{R}^{m_0}$, and a label is one of K classes, $y \in \{1, \dots, K\}$. A neural network maps a feature vector to one score per class, and training adjusts its parameters θ. The simplest network is a stack of layers, each a linear map followed by a fixed nonlinear function.

Definition 14.1 (Multilayer perceptron)**.** A network with J layers sets $v^{(0)} = x$ and, for $j = 1, \dots, J$,

$$v^{(j)} = \phi(W^{(j)} v^{(j-1)} + c^{(j)}), \tag{14.2}$$

where $W^{(j)}$ is an $m_j \times m_{j-1}$ matrix of weights, $c^{(j)} \in \mathbb{R}^{m_j}$ is a vector of offsets, and the **activation function** $\phi \colon \mathbb{R} \to \mathbb{R}$ is applied to each coordinate. The last layer has no activation and outputs $o \in \mathbb{R}^K$, one score per class. The parameter vector θ collects all weights and offsets.

Without the activation, the network is a composition of linear maps, and so a single linear map. Common activations are the rectified linear unit $\phi(u) = \max(0, u)$, the usual choice without privacy, and the bounded $\tanh(u) = (e^u - e^{-u})/(e^u + e^{-u})$.

The **softmax** function turns the scores into a probability vector over the classes, $\hat{y}_k = e^{o_k} / \sum_{j=1}^{K} e^{o_j}$, which gives the confidence of the network in each class. The prediction is the class with the largest $\hat{y}_k$. The loss is the **cross-entropy**, the negative log of the probability of the true class,

$$\ell(\theta; x, y) = -\ln \hat{y}_y = -\sum_{k=1}^{K} \mathbf{1}[k = y] \ln \hat{y}_k, \tag{14.3}$$

where $\hat{y}$ depends on θ through every layer. The analysis of the previous chapter does not apply to this loss. It is not convex in θ, so no convergence theorem holds, although SGD works well in practice. It is not Lipschitz either, because in a two-layer network the second-layer weights multiply the gradient of the first-layer weights, and they can be arbitrarily large. So a single gradient has unbounded sensitivity.

14.2. Differentially private SGD

Two changes make noisy SGD practical. First, each gradient is **clipped** to a fixed norm, which bounds sensitivity directly. Clipping shrinks a vector that is too long and leaves a short one alone. Second, each step uses a random batch of records, called a **lot**, chosen by Poisson sampling.

Definition 14.2 (Clipping)**.** For a clipping norm C, a positive number, the clipping of a vector g is

$$\text{clip}_C(g) = g \cdot \min\left(1, \frac{C}{\|g\|_2}\right). \tag{14.4}$$

So every clipped vector has norm at most C.

DP-SGD combines the two changes. Each step clips the gradient of every record in the lot, adds Gaussian noise to the sum, and divides by the expected lot size qn. The **noise multiplier** σ sets the noise, as the ratio of its standard deviation to the clipping norm. So, unlike the σ of the previous chapter, it is not the standard deviation itself.

Definition 14.3 (DP-SGD)**.** Let the parameter vector θ have d coordinates. Fix a sampling rate q in $(0, 1]$, a noise multiplier $\sigma > 0$, a clipping norm C, a number of steps T, learning rates η_t and a start θ_0. For $t = 1, \dots, T$:

1. Form the lot B_t by including each index $i \in \{1, \dots, n\}$ independently with probability q.
2. Compute the noisy average of clipped gradients,

$$\tilde{g}_t = \frac{1}{qn}\left(\sum_{i \in B_t} \text{clip}_C(\nabla \ell(\theta_{t-1}; z_i)) + b_t\right), \qquad b_t \sim \mathcal{N}(0, \sigma^2 C^2 I_d). \tag{14.5}$$

3. Step: $\theta_t = \theta_{t-1} - \eta_t \tilde{g}_t$.

Output θ_T.

Unlike the private SGD of the previous chapter, a step uses a lot of expected size qn instead of one record. And clipping, not an assumption, bounds the gradient. The noise is calibrated to C, the most that one record can contribute to the sum.

Lemma 14.1. *Adding or removing one record changes the clipped sum* $\sum_{i \in B} \operatorname{clip}_C(\nabla \ell(\theta; z_i))$ *by a vector of norm at most* C*. Replacing one record changes it by at most* $2C$*.*

Proof. Adding or removing a record adds or removes one clipped vector, of norm at most C. A replacement does both, and the triangle inequality gives $2C$. □

Clipping has a cost, because the clipped average is a biased estimate of the gradient. So even for convex losses the convergence theorem no longer applies, but clipped SGD works well in practice.

The privacy analysis depends on how the lot is drawn. **Poisson sampling** includes each record independently with probability q, so the lot size is random. Even at $q = 1/n$, this is not the same as the choice of one record, because the lot may be empty or hold several records. Other schemes, such as **sampling without replacement** of a lot of fixed size, need other analyses. The analyses below are for Poisson sampling. They use the add/remove relation, under which neighbours $D \sim D'$ differ by one record, so Lemma 14.1 gives sensitivity C.

Implementations often shuffle the data once and cut it into consecutive batches of fixed size. Each batch is then a uniform sample, but the batches are dependent, so the Poisson analysis does not apply to them.

14.3. Privacy accounting

The analysis of the previous chapter applies to DP-SGD too, because without sampling a single step is the Gaussian mechanism with noise multiplier σ. If that step is $(\varepsilon_0, \delta_0)$-DP, then amplification by Poisson sampling makes it about $(2q\varepsilon_0, q\delta_0)$-DP, and advanced composition over T steps gives

$$\Big(O(q\varepsilon_0\sqrt{T\ln(1/\delta)}),\ Tq\delta_0 + \delta\Big)\text{-DP}. \tag{14.6}$$

Abadi and coauthors proved a better bound for the same algorithm, which needs less noise for the same guarantee.

Theorem 14.1 (Moments accountant, asymptotic form)**.** *There are constants* c_1 *and* c_2 *such that, for every sampling rate* q *and number of steps* T*, every* $\varepsilon < c_1 q^2 T$ *and every* $\delta > 0$*, DP-SGD is* (ε, δ)*-DP under add/remove neighbours if*

$$\sigma \geq c_2 \, \frac{q\sqrt{T\ln(1/\delta)}}{\varepsilon}. \tag{14.7}$$

The proof is Theorem 1 of Abadi and coauthors, 2016, and the rest of this section proves its ingredients. Compared with composition, the gain is a factor of order $\sqrt{\ln(1/\delta_0)}$ in σ and no failure probability $Tq\delta_0$, as Corollary 14.1 shows below. Such factors in ε separate a meaningful guarantee from an empty one.

The accountant starts from the privacy loss. For a mechanism M whose output has density p_D on dataset D, the **privacy loss** at an output y is $\ln\left(p_D(y)/p_{D'}(y)\right)$. It is a random variable when

y is drawn from $M(D)$. By the privacy loss criterion of the approximate DP chapter, (ε, δ)-DP follows if, for every ordered pair of neighbours, the loss exceeds ε with probability at most δ. But tail probabilities do not compose simply, and moments do, by addition. So the accountant takes, for an order $\lambda > 0$, the logarithm of the average λ-th power of the ratio $p_D(y)/p_{D'}(y)$, for the worst pair of neighbours.

Definition 14.4 (Log-moments of the privacy loss)**.** Let p_D be the density of the output $M(D)$. For $\lambda > 0$, the λ-th log-moment of M is

$$\kappa_M(\lambda) = \sup_{D \sim D'} \ln \mathbb{E}_{y \sim M(D)} \left[\left(\frac{p_D(y)}{p_{D'}(y)} \right)^{\lambda} \right]. \tag{14.8}$$

Up to a factor, it is the Rényi divergence of order $\alpha = \lambda + 1$ between the two output distributions. A bound on it for every pair of neighbours is Rényi differential privacy (RDP). By Markov's inequality, a bound on one log-moment already gives approximate DP.

Theorem 14.2 (From moments to approximate DP)**.** *For every $\lambda > 0$ and $\varepsilon \geq 0$, the mechanism M is (ε, δ)-DP with*

$$\delta = \exp\left(\kappa_M(\lambda) - \lambda\varepsilon\right). \tag{14.9}$$

Proof.

1. Fix neighbours $D \sim D'$ and write $L = \ln(p_D(Y)/p_{D'}(Y))$ for the privacy loss random variable, with $Y \sim M(D)$. The map $u \mapsto e^{\lambda u}$ is increasing, so $\Pr[L > \varepsilon] \leq \Pr[e^{\lambda L} \geq e^{\lambda\varepsilon}]$.
2. Markov's inequality bounds the right side by $\mathbb{E}[e^{\lambda L}]\, e^{-\lambda\varepsilon}$, and $\mathbb{E}[e^{\lambda L}] \leq e^{\kappa_M(\lambda)}$ by definition.
3. So the privacy loss exceeds ε with probability at most δ, and the privacy loss criterion gives (ε, δ)-DP.

□

The conversion is useful because log-moments are easy to track over many steps. The next lemma shows that they add up when mechanisms run in sequence, even when the outputs of the previous mechanisms choose the next one.

Lemma 14.2 (Composition of moments)**.** *Let M_1 have output y_1, and let $M_2(\cdot\,; y_1)$ be a mechanism chosen as a function of y_1 with $\kappa_{M_2(\cdot; y_1)}(\lambda) \leq \kappa_2$ for every y_1. The mechanism M that outputs (y_1, y_2) satisfies $\kappa_M(\lambda) \leq \kappa_{M_1}(\lambda) + \kappa_2$. By induction, T adaptive steps have log-moments at most the sum of theirs.*

Proof.

1. The density of (y_1, y_2) on D is $p_D(y_1)\, p_D(y_2 \mid y_1)$, so the ratio for neighbours factors as $r_1(y_1)\, r_2(y_2 \mid y_1)$. Here r_1 is the density ratio of y_1, and r_2 is the density ratio of y_2 given y_1.

2. The expectation over y_2 first, given y_1, gives

$$\mathbb{E}[r_1^\lambda r_2^\lambda] = \mathbb{E}_{y_1}\Big[r_1(y_1)^\lambda \, \mathbb{E}_{y_2}[r_2(y_2 \mid y_1)^\lambda \mid y_1]\Big] \le e^{\kappa_2} \, \mathbb{E}_{y_1}[r_1(y_1)^\lambda] \le e^{\kappa_2 + \kappa_{M_1}(\lambda)}. \tag{14.10}$$

3. The logarithm and the supremum over neighbours give the claim.

□

For the Gaussian mechanism the log-moments have a closed form.

Lemma 14.3. *Let $M(D) = f(D) + b$ with $b \sim \mathcal{N}(0, \sigma^2 I_d)$, where f has ℓ_2 sensitivity $\Delta_2 f$. Then for every $\lambda > 0$,*

$$\kappa_M(\lambda) = \frac{\lambda(\lambda+1)(\Delta_2 f)^2}{2\sigma^2}. \tag{14.11}$$

Proof.

1. Fix neighbours, write $\mu = f(D)$ and $\mu' = f(D')$, and let $r = \|\mu - \mu'\|_2 \le \Delta_2 f$. For $y = \mu + \sigma\xi$ with ξ standard normal, the privacy loss is

$$\frac{\|y-\mu'\|^2 - \|y-\mu\|^2}{2\sigma^2} = \frac{r^2}{2\sigma^2} + \frac{\langle \xi, \mu - \mu' \rangle}{\sigma}. \tag{14.12}$$

2. So the privacy loss is normal with mean $m = r^2/(2\sigma^2)$ and variance $r^2/\sigma^2 = 2m$.
3. A normal variable Z with mean m and variance s^2 has $\mathbb{E}[e^{\lambda Z}] = e^{\lambda m + \lambda^2 s^2/2}$. Here the exponent is $\lambda m + \lambda^2 m = \lambda(\lambda+1) r^2/(2\sigma^2)$.
4. This increases with r, and the supremum is reached at $r = \Delta_2 f$.

□

The three results give a guarantee for many Gaussian steps without sampling. The log-moments of the steps add up, and the conversion turns their sum into (ε, δ)-DP at the best order λ.

Corollary 14.1. *Let $\rho = (\Delta_2 f)^2/(2\sigma^2)$. Any adaptive sequence of T Gaussian mechanisms, each with sensitivity $\Delta_2 f$ and noise $\mathcal{N}(0, \sigma^2 I_d)$, is (ε, δ)-DP for every $\delta \in (0,1)$, with*

$$\varepsilon = T\rho + 2\sqrt{T\rho \ln(1/\delta)}. \tag{14.13}$$

Proof.

1. By Lemma 14.3 and Lemma 14.2, the sequence has $\kappa(\lambda) \le \lambda(\lambda+1)T\rho$.
2. By Theorem 14.2, it is (ε, δ)-DP whenever $\lambda(\lambda+1)T\rho - \lambda\varepsilon \le \ln \delta$, that is, whenever $\varepsilon \ge (\lambda+1)T\rho + \ln(1/\delta)/\lambda$.
3. The right side is smallest at $\lambda = \sqrt{\ln(1/\delta)/(T\rho)}$, where it equals $T\rho + 2\sqrt{T\rho \ln(1/\delta)}$.

□

For comparison, take the same T Gaussian steps with $\Delta_2 f = 1$, so that $T\rho = T/(2\sigma^2)$. Advanced composition first needs a guarantee per step. The Gaussian mechanism of the approximate DP chapter gives ε_0 with $\sigma = \sqrt{2\ln(1/\delta_0) + \varepsilon_0}/\varepsilon_0$, so ε_0 is about $\sqrt{2\ln(1/\delta_0)}/\sigma$. Then it gives $\varepsilon \approx \varepsilon_0\sqrt{2T\ln(1/\delta)}$, which is $2\sqrt{T\ln(1/\delta_0)\ln(1/\delta)}/\sigma$. The corollary gives $\sqrt{2T\ln(1/\delta)}/\sigma$ plus a lower-order term. That is smaller by a factor of about $\sqrt{2\ln(1/\delta_0)}$, with no failure probability $T\delta_0$ to pay.

DP-SGD adds Poisson sampling to each step, and its log-moments come from a one-dimensional pair of distributions. By Lemma 14.1, after a division by C, a step compares the noise distribution $\mathcal{N}(0, \sigma^2)$ with the mixture $(1-q)\,\mathcal{N}(0, \sigma^2) + q\,\mathcal{N}(1, \sigma^2)$. In the mixture, the extra record is sampled with probability q.

Theorem 14.3 (Log-moments of the sampled Gaussian)**.** *For every integer $\lambda \geq 1$, one step of DP-SGD with sampling rate q and noise multiplier σ has, under add/remove neighbours,*

$$\kappa(\lambda) \leq \ln A_{\lambda+1}, \qquad A_a = \sum_{k=0}^{a} \binom{a}{k} (1-q)^{a-k} q^k \exp\left(\frac{k^2 - k}{2\sigma^2}\right). \tag{14.14}$$

The proof is Section 3.3 of Mironov, Talwar and Zhang (2019). It expands the a-th power of the density ratio between the mixture and $\mathcal{N}(0, \sigma^2)$ with the binomial theorem, which gives A_a. Each term is then a Gaussian integral, $\mathbb{E}[e^{k\xi/\sigma^2}] = e^{k^2/(2\sigma^2)}$ for $\xi \sim \mathcal{N}(0, \sigma^2)$. The hard part of the proof is to show that this pair, in this order, is the worst case over all neighbours and over the reverse order. For $\sigma \geq 1$, small q and moderate λ, the bound is $q^2\lambda(\lambda+1)/((1-q)\sigma^2) + O(q^3\lambda^3/\sigma^3)$, which gives Theorem 14.1.

These pieces together give the **moments accountant**. It computes the log-moment of one step for every λ, multiplies by T by Lemma 14.2, and converts with Theorem 14.2 at the best λ:

$$\varepsilon(\delta) = \min_{\lambda \geq 1} \frac{T \ln A_{\lambda+1} + \ln(1/\delta)}{\lambda}. \tag{14.15}$$

The computation depends only on q, σ, T and δ, never on the data. So you can fix the privacy cost of a training run before you touch the data, and set the noise or the number of steps to meet a target. Figure 14.1 compares the accountant with the composition route on a typical setting.

Example 14.1. With $q = 0.01$, $\sigma = 4$ and $\delta = 10^{-5}$, Figure 14.1 shows the two accountings of the same algorithm. After 100 epochs, or $T = 10{,}000$ steps, the composition route certifies $\varepsilon = 22.08$ and the moments accountant certifies $\varepsilon = 1.26$. After 400 epochs the values are 61.41 and 2.58. The first is no guarantee at all, and the second is a reasonable one.

These analyses use add/remove neighbours, while this book uses replace-one. A replacement is a removal followed by an addition, so a guarantee for one relation converts to the other at the cost of about twice the ε.

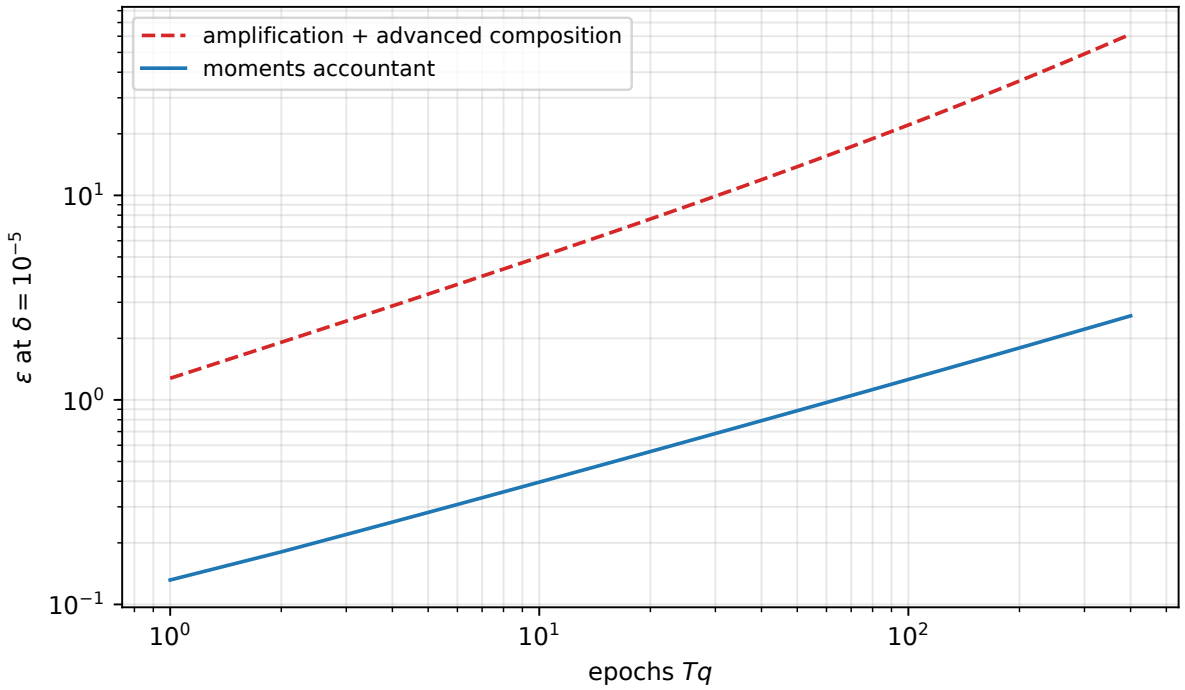


Figure 14.1.: Privacy parameter ε at $\delta = 10^{-5}$ of DP-SGD with sampling rate $q = 0.01$ and noise multiplier $\sigma = 4$, against the number of epochs of $1/q$ steps. Dashed: each step is $(\varepsilon_0, \delta_0)$-DP by Corollary 14.1 with $T = 1$, Poisson sampling makes it $(\ln(1 + q(e^{\varepsilon_0} - 1)), q\delta_0)$-DP, and advanced composition combines the steps, with δ split evenly. Solid: the moments accountant Equation 14.15.

14.4. DP-SGD in practice

DP-SGD replaces SGD directly, but its behaviour differs in three ways.

Accuracy. On the handwritten-digit benchmark MNIST, a well-tuned DP-SGD reached 98.1% accuracy at $\varepsilon = 2.93$ and $\delta = 10^{-5}$, against about 99.8% without privacy (Papernot and coauthors, 2020). On the harder image benchmark CIFAR-10, it reached only 66.2% at $\varepsilon = 7.53$, which is weak in both accuracy and privacy.

Architecture. Bounded activations such as tanh work better than the rectified linear unit under DP-SGD, and the size of the network matters too. In each step, the noise vector b_t has expected squared norm $d\sigma^2 C^2$, while the sum of the gradients clipped to norm C has norm at most qnC. So the ratio of signal to noise is at most about $qn/(\sigma\sqrt{d})$, which falls as the number of parameters d grows. So a network must be large enough to fit the task and no larger.

Hyperparameters. Each training run on the private data that tunes the learning rate, lot size, clipping norm or number of epochs spends privacy budget. Results that tune on the private data and do not count this cost overstate the privacy of the final model. Private selection picks the best of many private runs at a small constant factor in ε (Liu and Talwar). Public data helps too, for tuning or for pretraining a network whose last layers alone are then trained privately, which makes d smaller.

14.5. Private aggregation of teacher ensembles

DP-SGD changes the inside of one training algorithm. Private aggregation of teacher ensembles (PATE) instead treats the learner as a black box, so any improvement in non-private learning carries over. The price is a different model of the data. In addition to a sensitive labelled training set, PATE needs a supply of **public unlabelled** feature vectors.

PATE is an instance of **sample and aggregate**. It splits the data into disjoint parts, runs a non-private algorithm on each part, and combines the answers by a private aggregation. One record affects only one part, so the aggregation has low sensitivity. In PATE, the parts train m classifiers, the **teachers**, and the aggregation is a vote.

The simplest task is one private prediction for a public feature vector x. Each teacher votes for a label, and c_k is the number of votes for label k. The **mode** a is the label with the most votes, the smallest one in case of a tie. The **gap** $\Gamma = c_a - \max_{k \neq a} c_k$ measures how far the vote is from a change of winner. The private mode releases the mode only when a noisy gap is above a threshold, because a large gap means that one changed record cannot change the winner.

Definition 14.5 (Private mode of the teachers)**.** Let $\varepsilon > 0$ and $\delta \in (0, 1/2)$. Split the dataset into m fixed disjoint parts by position, train one teacher on each, and collect their votes on x. Let a be the mode of the votes and Γ its gap, as above. Draw $Z \sim \mathrm{Lap}(2/\varepsilon)$, Laplace noise with scale $2/\varepsilon$. If

$$\Gamma + Z > \tau, \qquad \tau = 2 + \frac{2}{\varepsilon} \ln \frac{1}{2\delta}, \tag{14.16}$$

output the mode a. Otherwise output a symbol $\perp$ for "no answer", or a uniformly random label.

When the teachers agree, the mode is stable, and its exact release costs little privacy. This is the propose-test-release idea of the chapter on beyond global sensitivity. It tests privately that the data is far from a dataset with another answer, and if so, releases the answer without noise. The private mode is the stable mode of that chapter applied to the votes, with a lower threshold so that the failure probability is δ, not $\delta/2$.

Theorem 14.4. *The private mode is (ε, δ)-DP. The teachers' training algorithm may be randomized, with coins independent of everything else.*

Proof.

1. Fix the teachers' coins; a guarantee for every fixed choice of coins implies one for random coins. Neighbours differ in one record, which lies in one part, so at most one teacher's vote changes. That moves one vote from one label to another, and changes each count by at most 1.
2. *The gap has sensitivity* 2. If the mode stays the same, both terms of Γ change by at most 1. If the mode changes from a on D to $a' \neq a$ on D', then $c_{a'} + 1 \geq c'_{a'} \geq c'_a \geq c_a - 1$, so $\Gamma \leq c_a - c_{a'} \leq 2$, and likewise $\Gamma' \leq 2$. In both cases $|\Gamma - \Gamma'| \leq 2$. So, by the Laplace mechanism, the indicator that the test passes is ε-DP.
3. *Same mode.* The output is a function of the test indicator and of the common mode, so it is ε-DP by post-processing.
4. *Different modes.* By step 2, $\Gamma \leq 2$. The test then passes on D with probability at most $\Pr[Z > \tau - 2] = \frac{1}{2} e^{-\varepsilon(\tau-2)/2} = \delta$. Outside that event the output is $\perp$, so for every set S of outputs,

$$\Pr[M(D) \in S] \leq \Pr[M(D) = \perp]\, \mathbf{1}[\perp \in S] + \delta \leq e^{\varepsilon} \Pr[M(D') = \perp]\, \mathbf{1}[\perp \in S] + \delta \leq e^{\varepsilon} \Pr[M(D') \in S] + \delta, \tag{14.17}$$

where the middle step is the ε-DP of the test from step 2. Replacing $\perp$ by a random label is post-processing.

□

The accuracy depends on how good the teachers are. If the records are independent draws from a distribution, then teachers trained on disjoint parts vote independently on a fixed x.

Proposition 14.1. *Let $\beta \in (0, 1)$ be a failure probability, and let τ be the threshold of Definition 14.5. Suppose each teacher, independently, votes for the correct label y of x with probability at least $3/4$. If*

$$m \geq \max\left(4\tau + \frac{8}{\varepsilon} \ln \frac{1}{\beta},\ 32 \ln \frac{2}{\beta}\right), \tag{14.18}$$

then the private mode outputs y with probability at least $1 - \beta$.

Proof.

1. Let V be the number of correct votes. Its mean is at least $3m/4$, and Hoeffding's inequality for independent indicators gives $\Pr[V \leq 5m/8] \leq e^{-2m(1/8)^2} = e^{-m/32} \leq \beta/2$.

2. If $V > 5m/8$, then y has a majority, so it is the mode, and every other label has at most $m - V$ votes. So $\Gamma \geq 2V - m > m/4$.
3. The test then fails with probability $\Pr[Z \leq \tau - \Gamma] \leq \frac{1}{2} e^{-\varepsilon(m/4-\tau)/2}$, because $m/4 > \tau$. The first lower bound on m makes this at most $\beta/2$.
4. A union bound over the two failure events gives the claim.

□

Example 14.2. With $\varepsilon = 1$, $\delta = 10^{-5}$ and $\beta = 0.01$, the threshold is $\tau = 2 + 2\ln(50{,}000) = 23.64$. The two lower bounds are $94.56 + 8 \ln 100 = 131.4$ and $32 \ln 200 = 169.5$, so 170 teachers suffice. If a teacher needs n_0 records to be right three times in four, the private prediction needs $170\, n_0$.

The number of teachers grows like $\ln(1/\delta)/\varepsilon$. This is the general price of sample and aggregate, about $1/\varepsilon$ times more data than the non-private algorithm. That is often more than the best private algorithm for a specific task needs.

To answer k queries with k runs of the private mode costs about $\varepsilon\sqrt{2k \ln(1/\delta')}$, by advanced composition. The sparse vector technique of the sparse vector chapter does better when most queries are stable, because it pays only for the queries whose answer is above a threshold. Here each query asks whether the gap on the next point is small. So the technique pays only for the unstable points, where it outputs a random label, and it answers the stable ones with the exact mode. Its privacy cost grows like the square root of the number of unstable queries, not of all k (Bassily, Thakkar and Thakurta).

The answers $(x_1, \hat{y}_1), \ldots, (x_k, \hat{y}_k)$, with public features and private labels, form a labelled dataset that is safe to release. PATE trains a non-private learner, the **student**, on it. If the labelling is (ε, δ)-DP and the public features do not depend on the private data, the student is (ε, δ)-DP too, by post-processing. The student then answers any number of queries at no further cost. So PATE turns a bounded number of private predictions into a private model.

14.6. Summary

Table 14.1.: The chapter's results.

Result	Statement
Lemma 14.1	one record changes the clipped sum by at most C, or $2C$ under replacement
Theorem 14.1	DP-SGD is (ε, δ)-DP with $\sigma = O(q\sqrt{T \ln(1/\delta)}/\varepsilon)$
Theorem 14.2	a log-moment bound $\kappa(\lambda)$ gives $(\varepsilon, e^{\kappa(\lambda)-\lambda\varepsilon})$-DP
Lemma 14.2	log-moments add up under adaptive composition

Result	Statement
Lemma 14.3	the Gaussian mechanism has $\kappa(\lambda) = \lambda(\lambda+1)(\Delta_2 f)^2/(2\sigma^2)$
Corollary 14.1	T Gaussian steps give $\varepsilon = T\rho + 2\sqrt{T\rho\ln(1/\delta)}$
Theorem 14.3	a sampled Gaussian step has $\kappa(\lambda) \le \ln A_{\lambda+1}$
Equation 14.15	the moments accountant for T steps of DP-SGD
Theorem 14.4	releasing the teachers' mode when its noisy gap exceeds τ is (ε, δ)-DP
Proposition 14.1	$O(\ln(1/\delta)/\varepsilon + \ln(1/\beta)/\varepsilon)$ accurate teachers suffice

14.7. References

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15. Private Mean Estimation

Earlier chapters estimated properties of a fixed dataset. In statistics, the dataset is a sample, and the goal is a property of the distribution that generated it. This chapter asks how many samples an algorithm with differential privacy (DP) needs to estimate the mean of a distribution to within α. It also asks how much more that is than without privacy.

For bounded data, privacy costs a lower-order term. Unbounded data needs prior knowledge, at least a bound R on the mean. Clipping to $[-R, R]$ costs samples linear in R, but an algorithm that first locates the data costs only samples logarithmic in R. For heavy-tailed data, the clipping must introduce bias, and the price of privacy grows.

15.1. The mean of a dataset and of a distribution

The chapter uses two facts from earlier chapters. A Laplace variable with scale b, written $Z \sim \mathrm{Lap}(b)$, has density proportional to $e^{-|z|/b}$, so its tail is

$$\Pr[|Z| > t] = e^{-t/b} \qquad (t \geq 0). \tag{15.1}$$

The Laplace mechanism privatizes a function f of the dataset with values in $\mathbb{R}^d$. Its **sensitivity** $\Delta_1 f$ is the largest change that the replacement of one record can cause,

$$\Delta_1 f = \max_{D \sim D'} \|f(D) - f(D')\|_1. \tag{15.2}$$

If $\Delta_1 f$ is finite, the release of $f(D)$ plus independent $\mathrm{Lap}(\Delta_1 f/\varepsilon)$ noise in each coordinate is ε-DP.

Start with bits. Each of the n records is a bit $X_i \in \{0, 1\}$, and their empirical mean is $\tilde{p} = \frac{1}{n}\sum_i X_i$. A replaced bit moves $\tilde{p}$ by at most $1/n$, so the Laplace mechanism releases

$$\hat{p} = \tilde{p} + \mathrm{Lap}\left(\frac{1}{\varepsilon n}\right). \tag{15.3}$$

Now suppose the bits are a sample, $X_1, \ldots, X_n$ independent draws from $\mathrm{Bern}(p)$, and the target is the unknown p.

Theorem 15.1 (Private Bernoulli mean)**.** *The estimator Equation 15.3 is ε-DP. If $X_1, \ldots, X_n$ are independent* $\mathrm{Bern}(p)$ *variables, then for every $\beta \in (0, 1)$, with probability at least $1 - \beta$,*

$$|\hat{p} - p| \leq \sqrt{\frac{\ln(4/\beta)}{2n}} + \frac{\ln(2/\beta)}{\varepsilon n}. \tag{15.4}$$

Proof.

1. Privacy is the Laplace mechanism with sensitivity $1/n$.
2. By the triangle inequality, $|\hat{p} - p| \le |\tilde{p} - p| + |\hat{p} - \tilde{p}|$.
3. The bits are independent, take values in $[0, 1]$, and have mean p. Hoeffding's inequality gives $\Pr[|\tilde{p} - p| > t] \le 2e^{-2nt^2}$, which equals $\beta/2$ at $t = \sqrt{\ln(4/\beta)/(2n)}$.
4. The noise $\hat{p} - \tilde{p}$ is $\mathrm{Lap}(1/(\varepsilon n))$, so by the Laplace tail it exceeds $\ln(2/\beta)/(\varepsilon n)$ with probability $\beta/2$.
5. A union bound over the two failures gives the claim.

□

Each term is at most $\alpha/2$ once

$$n \ge \frac{2\ln(4/\beta)}{\alpha^2} + \frac{2\ln(2/\beta)}{\alpha\varepsilon}, \tag{15.5}$$

so $n = O(1/\alpha^2 + 1/(\alpha\varepsilon))$ samples are enough for a fixed β. The first term is the sample complexity without privacy, and the second is the **cost of privacy**. For a constant ε and small α, the cost of privacy is smaller by a factor of α, so here privacy is nearly free. The proof split the error into the **sampling error** $\tilde{p} - p$, present without privacy, and the **noise error** $\hat{p} - \tilde{p}$ that privacy adds. That split recurs in the chapter.

15.2. Unbounded data

The Bernoulli case was easy because each record lies in $[0, 1]$. Take instead real records, such as heights in centimetres. If one height of 160 is replaced by a recorded 10^5, the empirical mean moves by about $10^5/n$, and nothing bounds the move. So the empirical mean has infinite sensitivity. The next result shows that no private algorithm can fix that. It uses **group privacy**. If a mechanism M is ε-DP and two datasets D, D' differ in k entries, then for every event S,

$$\Pr[M(D) \in S] \le e^{k\varepsilon} \Pr[M(D') \in S], \tag{15.6}$$

by the definition applied along a path of k neighbours. So every ε-DP algorithm is far from the empirical mean on some real dataset, however large the allowed error α.

Proposition 15.1. *Let $M : \mathbb{R}^n \to \mathbb{R}$ be ε-DP and $\alpha > 0$. Some dataset $D \in \mathbb{R}^n$ has $\Pr[|M(D) - \bar{X}| \le \alpha] < 1/2$, where $\bar{X}$ is its empirical mean.*

Proof.

1. Suppose instead that $\Pr[|M(D) - \bar{X}| \le \alpha] \ge 1/2$ for every D. Take an integer $m > 2e^{\varepsilon n}$ and, for $j = 0, \dots, m-1$, the dataset $D^{(j)}$ whose n entries all equal $3\alpha j$, so that its empirical mean is $3\alpha j$.
2. The intervals $I_j = [3\alpha j - \alpha, 3\alpha j + \alpha]$ are disjoint, and $\Pr[M(D^{(j)}) \in I_j] \ge 1/2$ by assumption.

3. The datasets $D^{(0)}$ and $D^{(j)}$ differ in n entries, so group privacy gives $\Pr[M(D^{(0)}) \in I_j] \geq e^{-\varepsilon n}/2$.
4. The sum over the disjoint intervals gives $1 \geq me^{-\varepsilon n}/2 > 1$, a contradiction.

□

The way out is prior knowledge that does not come from the data. No human is taller than 300 cm, so the algorithm can force each height into $[0, 300]$ before the average.

Definition 15.1 (Clipping)**.** For $u \leq v$, the **clipping** map onto $[u, v]$ is $g_{[u,v]}(x) = \min(\max(x, u), v)$. It leaves points of $[u, v]$ unchanged and moves every other point to the nearer endpoint.

Clipping to a range fixed in advance bounds how far one record can move the mean, so the Laplace mechanism applies to the clipped mean.

Lemma 15.1. *Let $u < v$ be fixed before seeing the data, and let $g_{[u,v]}$ be the clipping map of Definition 15.1. The estimator*

$$M(D) = \frac{1}{n}\sum_{i=1}^{n} g_{[u,v]}(X_i) + \mathrm{Lap}\Big(\frac{v-u}{\varepsilon n}\Big) \tag{15.7}$$

is ε-DP. If every X_i lies in $[u, v]$, then $|M(D) - \bar{X}| \leq (v-u)\ln(1/\beta)/(\varepsilon n)$ with probability at least $1 - \beta$.

Proof.

1. Each clipped value lies in $[u, v]$, so a replaced record moves the clipped mean by at most $(v-u)/n$. The Laplace mechanism gives privacy.
2. If every X_i lies in $[u, v]$, clipping changes nothing, and $M(D) - \bar{X}$ is the Laplace noise, whose tail gives the bound.

□

With heights in $[0, 300]$, $n = 1{,}000$ and $\varepsilon = 1$, the noise scale is $300/1{,}000 = 0.3$ cm. Fix the range before you see the data. A range chosen after a look, for example to exclude one absurd record, depends on that record, and the guarantee no longer holds.

15.3. Estimating a parameter of a distribution

From here on the data is a sample from an unknown distribution, and the target is the mean of that distribution. An estimator must be private on every dataset, and accurate on samples from the distribution.

Definition 15.2 (Private estimation)**.** Let $\mathcal{P}$ be a class of distributions on $\mathbb{R}$, each with a mean $\mu(P)$. An algorithm $M \colon \mathbb{R}^n \to \mathbb{R}$ is an ε**-DP** (α, β)**-estimator** for $\mathcal{P}$ if

1. M is ε-DP, as a map on all datasets in $\mathbb{R}^n$, and
2. for every $P \in \mathcal{P}$, if $X_1, \dots, X_n$ are independent draws from P, then $\Pr[|M(D) - \mu(P)| \le \alpha] \ge 1 - \beta$.

The smallest such n is the **sample complexity** of the class.

Privacy is a worst-case guarantee. It holds even for a dataset that has probability zero under every $P \in \mathcal{P}$. So the people in the dataset do not pay if the model of the analyst is wrong. Accuracy, in contrast, is an average over samples from P, and it needs assumptions on P even without privacy. For example, a Cauchy distribution has no mean, and the empirical mean of its samples does not converge.

The main class is the Gaussians with unit variance and a bounded mean,

$$\mathcal{G}_R = \{\, \mathcal{N}(\mu, 1) : |\mu| \le R \,\}. \tag{15.8}$$

A bound on the mean plays the role of the range of heights. The argument of Proposition 15.1 shows that some bound is necessary.

Proposition 15.2. *Let $\alpha > 0$, $\beta \in (0, 1)$, and $m = \lfloor 2R/(3\alpha) \rfloor + 1$. Every ε-DP (α, β)-estimator for $\mathcal{G}_R$ uses*

$$n \ge \frac{\ln\big((1-\beta)m\big)}{\varepsilon} \tag{15.9}$$

samples. Without a bound on $|\mu|$ ($R = \infty$), no finite n suffices.

Proof.

1. The means $\mu_j = -R + 3\alpha j$, for $j = 0, \dots, m-1$, lie in $[-R, R]$, and the intervals $I_j = [\mu_j - \alpha, \mu_j + \alpha]$ are disjoint.
2. Let D be a sample from $\mathcal{N}(\mu_0, 1)$ and set $D^{(j)} = D + (\mu_j - \mu_0)$, which shifts every entry. Then $D^{(j)}$ is a sample from $\mathcal{N}(\mu_j, 1)$, and it differs from D in all n entries.
3. For each fixed D, group privacy gives $\Pr_M[M(D) \in I_j] \ge e^{-\varepsilon n} \Pr_M[M(D^{(j)}) \in I_j]$. The average over D, with accuracy for $\mathcal{N}(\mu_j, 1)$, gives $\Pr[M(D) \in I_j] \ge e^{-\varepsilon n}(1 - \beta)$.
4. The intervals are disjoint, so the sum over j gives $1 \ge m e^{-\varepsilon n}(1 - \beta)$. This inequality, solved for n, gives the bound, and $m = \infty$ when $R = \infty$.

□

So a term of order $\ln(R/\alpha)/\varepsilon$ is unavoidable, and the rest of the chapter finds algorithms that pay little more than that.

15.4. Clipping the Gaussian

A bound on the mean is not a bound on the records. But with the light tails of the Gaussian, it gives one with high probability. The lemma below adapts part 3 of Lemma 5.3 in the approximate DP chapter, which says that a standard normal W has $\Pr[W > u] \le e^{-u^2/2}$ for every $u \ge 0$, to both tails.

Lemma 15.2 (Gaussian tail)**.** *If $Y \sim \mathcal{N}(0,1)$ and $t \ge 0$, then $\Pr[|Y| \ge t] \le 2e^{-t^2/2}$.*

Proof.

1. Part 3 of Lemma 5.3 with $u = t$ gives $\Pr[Y \ge t] = \Pr[Y > t] \le e^{-t^2/2}$, because Y has a density.
2. The lower tail is the same by symmetry, and the two add.

□

Fix $\beta \in (0,1)$ and write

$$\tau = \sqrt{2\ln(2n/\beta)}. \tag{15.10}$$

For a sample from $\mathcal{N}(\mu,1)$, each point leaves $[\mu-\tau, \mu+\tau]$ with probability at most $2e^{-\tau^2/2} = \beta/n$. So by a union bound all n points stay inside with probability at least $1-\beta$, and the width τ grows only like $\sqrt{\ln n}$.

The first estimator clips to an interval known to contain the mean, widened by τ. It takes any public centre c and radius r, because the later algorithms call it with a centre that they computed privately.

Theorem 15.2 (Clipped Laplace estimator)**.** *Let $c \in \mathbb{R}$ and $r \ge 0$ be public, $\beta \in (0,1)$, and τ as in Equation 15.10. With $g_{[u,v]}$ the clipping map of Definition 15.1, the estimator*

$$M_{c,r}(D) = \frac{1}{n}\sum_{i=1}^{n} g_{[c-r-\tau,\, c+r+\tau]}(X_i) + Z, \qquad Z \sim \mathrm{Lap}\left(\frac{2(r+\tau)}{\varepsilon n}\right), \tag{15.11}$$

is ε-DP. Let $X_1, \dots, X_n$ be independent $\mathcal{N}(\mu,1)$ draws. With probability at least $1-2\beta$ over the sample, an event E holds on which every $M_{c,r}$ with $|\mu - c| \le r$ satisfies

$$|M_{c,r}(D) - \mu| \le \sqrt{\frac{2\ln(2/\beta)}{n}} + |Z|. \tag{15.12}$$

In particular, if $|\mu - c| \le r$, then with probability at least $1-3\beta$,

$$|M_{c,r}(D) - \mu| \le s_\varepsilon(r) = \sqrt{\frac{2\ln(2/\beta)}{n}} + \frac{2(r+\tau)\ln(1/\beta)}{\varepsilon n}. \tag{15.13}$$

Proof.

1. *Privacy.* The clipping interval has length $2(r+\tau)$, so the clipped mean has sensitivity $2(r+\tau)/n$, and the Laplace mechanism applies.
2. *The event.* Let G be the event that every X_i lies in $[\mu-\tau, \mu+\tau]$, with probability at least $1-\beta$ by the union bound above. Let S be the event that the empirical mean $\tilde{\mu} = \frac{1}{n}\sum_i X_i$ satisfies $|\tilde{\mu}-\mu| \le \sqrt{2\ln(2/\beta)/n}$. Since $\tilde{\mu}-\mu \sim \mathcal{N}(0, 1/n)$, Lemma 15.2 with $t = \sqrt{2\ln(2/\beta)}$ gives $\Pr[S] \ge 1-\beta$. Set $E = G \cap S$.
3. *No clipping on E.* If $|\mu - c| \le r$, the interval $[\mu-\tau, \mu+\tau]$ lies inside $[c-r-\tau, c+r+\tau]$. So on G clipping changes no point, and $M_{c,r}(D) = \tilde{\mu} + Z$.
4. *Error on E.* By the triangle inequality, $|M_{c,r}(D) - \mu| \le |\tilde{\mu} - \mu| + |Z|$, and S bounds the first term.
5. *Noise.* The Laplace tail gives $|Z| \le 2(r+\tau)\ln(1/\beta)/(\varepsilon n)$ with probability $1-\beta$. A union bound with E gives Equation 15.13.

□

The proof splits the error in three. Take any clipping map g, write $\bar{g} = \frac{1}{n}\sum_i g(X_i)$ for the clipped mean and $\tilde{\mu}$ for the empirical mean, and let the estimate be $\hat{\mu} = \bar{g} + Z$ with noise Z. Then

$$\hat{\mu} - \mu = \underbrace{(\tilde{\mu} - \mu)}_{\text{sampling}} + \underbrace{(\bar{g} - \tilde{\mu})}_{\text{clipping bias}} + \underbrace{Z}_{\text{noise}}. \tag{15.14}$$

Here the clipping interval is wide enough that the bias term is zero with high probability. Until Section 15.8, where the bias must be traded against the noise, the chapter keeps that choice.

With $c = 0$ and $r = R$, each term of Equation 15.13 is at most $\alpha/2$ when

$$n \ge \frac{8\ln(2/\beta)}{\alpha^2} + \frac{4(R+\tau)\ln(1/\beta)}{\alpha\varepsilon}, \tag{15.15}$$

that is, $n = \tilde{O}(1/\alpha^2 + R/(\alpha\varepsilon))$, where $\tilde{O}$ hides factors logarithmic in n and $1/\beta$. When R is a small constant, this matches the Bernoulli case. But R measures how little the analyst knows, and for an unfamiliar quantity R may have to be 10^6, so that the noise swamps the signal. The next two sections replace R by $\ln R$.

15.5. A coarse estimate from a histogram

The first improvement locates the data before it takes the average. Cut the range into bins of width 2τ. The whole sample fits in an interval of that width, so one or two bins hold all the points. A private histogram can find one of them at a cost logarithmic in the number of bins.

Lemma 15.3 (Coarse estimate)**.** *Let τ be as in Equation 15.10, $m = \lfloor R/\tau \rfloor + 2$, and $a_j = -R - \tau + 2\tau j$ for $j = 0, \dots, m-1$. Let h_j count the records in the bin $B_j = [a_j, a_j + 2\tau)$. The algorithm outputs $c = a_{\hat{\jmath}} + \tau$, where $\hat{\jmath}$ maximizes $h_j + Z_j$ over j, with independent $Z_j \sim \mathrm{Lap}(2/\varepsilon)$. It is ε-DP. If $X_1, \dots, X_n$ are independent $\mathcal{N}(\mu, 1)$ draws with $|\mu| \le R$ and*

$$n > \frac{8}{\varepsilon} \ln \frac{m}{\beta}, \tag{15.16}$$

then $|\mu - c| \le 2\tau$ with probability at least $1 - 2\beta$.

Proof.

1. *Privacy.* A replaced record moves out of at most one bin and into at most one bin, so the vector of counts has sensitivity 2. Adding $\mathrm{Lap}(2/\varepsilon)$ to each count is ε-DP by the Laplace mechanism, and the argmax is post-processing.
2. *The bins catch the sample.* On the event G of Theorem 15.2, every record lies in $[\mu - \tau, \mu + \tau] \subseteq [-R - \tau, R + \tau]$. The bins cover $[-R - \tau, -R - \tau + 2\tau m)$, and $2\tau m > 2R + 2\tau$, so every record is counted.
3. *One or two bins hold everything.* The interval $[\mu - \tau, \mu + \tau]$ has length 2τ, so it meets at most two bins. Hence some bin has $h_j \ge n/2$, and every bin that misses the interval has $h_j = 0$.
4. *The noise is small.* Set $t = (2/\varepsilon) \ln(m/\beta)$. Each $|Z_j| > t$ with probability β/m, so with probability at least $1 - \beta$ all $|Z_j| \le t$. Then the fullest bin has noisy count at least $n/2 - t$, and an empty bin at most t. Since $n > 4t$, the winning bin $\hat{\jmath}$ is not empty.
5. *The winner is near the mean.* On G, the bin $B_{\hat{\jmath}}$ holds a record x with $|x - \mu| \le \tau$ and $a_{\hat{\jmath}} \le x < a_{\hat{\jmath}} + 2\tau$. So $\mu \in [a_{\hat{\jmath}} - \tau, a_{\hat{\jmath}} + 3\tau] = [c - 2\tau, c + 2\tau]$. The two failures have total probability at most 2β.

□

After the coarse step, the mean is known to within $2\tau = O(\sqrt{\ln(n/\beta)})$ instead of R. The fine step is Theorem 15.2 on that short interval. Basic composition, from the Laplace mechanism chapter, says that an ε_1-DP algorithm followed by an ε_2-DP algorithm that may depend on the first output is $(\varepsilon_1 + \varepsilon_2)$-DP. So each step gets half the budget.

Theorem 15.3 (Histogram, then clipping)**.** *Let m be the number of bins in Lemma 15.3. Run that algorithm with $\varepsilon/2$ (noise $\mathrm{Lap}(4/\varepsilon)$) to get c, then output $\hat{\mu} = M_{c,2\tau}(D)$ from Theorem 15.2 with $\varepsilon/2$. The result is ε-DP. If $X_1, \dots, X_n$ are independent $\mathcal{N}(\mu, 1)$ draws with $|\mu| \le R$ and $n > (16/\varepsilon) \ln(m/\beta)$, then with probability at least $1 - 4\beta$,*

$$|\hat{\mu} - \mu| \le \sqrt{\frac{2 \ln(2/\beta)}{n}} + \frac{12\,\tau \ln(1/\beta)}{\varepsilon n}. \tag{15.17}$$

Proof.

1. *Privacy.* The two steps are $\varepsilon/2$-DP each, and basic composition adds them.

2. *Coarse step.* Lemma 15.3 with $\varepsilon/2$ needs $n > (16/\varepsilon)\ln(m/\beta)$. Its two failure events are G and the event that some $|Z_j|$ is large, of total probability at most 2β. Outside them, $|\mu - c| \le 2\tau$.
3. *Fine step.* On the event $E = G \cap S$ of Theorem 15.2, the fine estimate satisfies $|\hat{\mu} - \mu| \le \sqrt{2\ln(2/\beta)/n} + |Z|$, since $|\mu - c| \le 2\tau$. The event S adds probability β of failure.
4. The noise Z is $\mathrm{Lap}(2(2\tau + \tau)/((\varepsilon/2)n)) = \mathrm{Lap}(12\tau/(\varepsilon n))$, whatever c is, and it exceeds $12\tau\ln(1/\beta)/(\varepsilon n)$ with probability β. The union bound over G, the histogram noise, S and Z gives $1 - 4\beta$.

□

The error is at most α once $n \ge 8\ln(2/\beta)/\alpha^2 + 24\tau\ln(1/\beta)/(\alpha\varepsilon)$, and the coarse step needs $n > (16/\varepsilon)\ln(m/\beta)$ with $m \le R/\tau + 2$. So

$$n = \tilde{O}\Big(\frac{1}{\alpha^2} + \frac{1}{\alpha\varepsilon} + \frac{\ln R}{\varepsilon}\Big). \tag{15.18}$$

The dependence on R fell from linear to logarithmic. The first term is needed without privacy, Proposition 15.2 shows that the last is needed, and the middle one is needed too, up to logarithmic factors.

15.6. Shrinking confidence intervals

The histogram finds the data in one shot, but a histogram over a region of $\mathbb{R}^d$ needs a number of bins exponential in d. A second route applies the clipped Laplace estimator repeatedly. Each call returns a confidence interval instead of a point, a constant factor shorter than the one it started from, which with high probability still contains μ.

One step takes an interval $[c - r, c + r]$ and a budget ε', computes the estimate $c' = M_{c,r}(D)$ with ε', and returns the interval $[c' - s, c' + s]$ with $s = s_{\varepsilon'}(r)$ from Equation 15.13. The radius s depends only on r and on public parameters, not on the data.

Lemma 15.4 (One step)**.** *Let τ be as in Equation 15.10, $n \ge 16$ and $n \ge 16\ln(1/\beta)/\varepsilon'$. Consider one step, as described above, from the interval $[c - r, c + r]$ to the interval $[c' - s, c' + s]$. Then*

$$s_{\varepsilon'}(r) \le \tfrac{1}{2}\max(r, \tau). \tag{15.19}$$

Moreover, on the event E of Theorem 15.2, if $|\mu - c| \le r$, then $\mu \in [c' - s, c' + s]$ with probability at least $1 - \beta$ over the step's noise.

Proof.

1. Since $2\ln(2/\beta) \le 2\ln(2n/\beta) = \tau^2$ and $n \ge 16$, the first term of $s_{\varepsilon'}(r)$ is at most $\tau/\sqrt{n} \le \tau/4$.
2. Since $n \ge 16\ln(1/\beta)/\varepsilon'$, the second term is at most $2(r + \tau)/16 = (r + \tau)/8 \le \max(r, \tau)/4$.
3. Adding the two terms gives the first claim, because $\tau/4 \le \max(r, \tau)/4$.
4. On E, Theorem 15.2 gives $|c' - \mu| \le \sqrt{2\ln(2/\beta)/n} + |Z|$, where Z is the step's Laplace noise. That noise is at most $2(r + \tau)\ln(1/\beta)/(\varepsilon' n)$ with probability $1 - \beta$, and then $|c' - \mu| \le s$.

□

The first claim is deterministic, and the second needs the data to be Gaussian. Each step halves the radius until it reaches the floor τ, where the tails of the data stop the clipping from doing better.

Theorem 15.4 (Shrinking intervals, then clipping)**.** *Let $T = \lceil \log_2(R/\tau) \rceil$ (or $T = 0$ if $R \le \tau$), $\varepsilon' = \varepsilon/(2T)$, $r_0 = R$, $c_0 = 0$, and $r_t = s_{\varepsilon'}(r_{t-1})$. For $t = 1, \dots, T$, let $c_t = M_{c_{t-1}, r_{t-1}}(D)$ computed with ε'. Output $\hat{\mu} = M_{c_T, r_T}(D)$ computed with $\varepsilon/2$. The result is ε-DP. Suppose $X_1, \dots, X_n$ are independent $\mathcal{N}(\mu, 1)$ draws with $|\mu| \le R$,*

$$n \ge 16 \qquad \text{and} \qquad n \ge \frac{32\,T \ln(1/\beta)}{\varepsilon}. \tag{15.20}$$

Then with probability at least $1 - (T+3)\beta$,

$$|\hat{\mu} - \mu| \le \sqrt{\frac{2\ln(2/\beta)}{n}} + \frac{4(r_T + \tau)\ln(1/\beta)}{\varepsilon n} \le \sqrt{\frac{2\ln(2/\beta)}{n}} + \frac{8\,\tau \ln(1/\beta)}{\varepsilon n}. \tag{15.21}$$

Proof.

1. *Privacy.* The radii r_t depend only on n, ε, β and R, so each step is an instance of Theorem 15.2 with public parameters. There are T steps of $\varepsilon/(2T)$ and one of $\varepsilon/2$, each using the outputs before it, and basic composition gives ε.
2. *The radii shrink.* The condition on n is the one of Lemma 15.4 with $\varepsilon' = \varepsilon/(2T)$, so $r_t \le \max(r_{t-1}, \tau)/2$. By induction, $r_t \le \max(R/2^t, \tau/2)$. Since $2^T \ge R/\tau$, this gives $r_T \le \tau$.
3. *The intervals cover μ.* The event E holds with probability at least $1 - 2\beta$, and it concerns only the sample, so it serves every step. On E, by induction on t, Lemma 15.4 keeps $|\mu - c_t| \le r_t$ unless the noise of step t fails, which has probability β.
4. *Final step.* On E and with $|\mu - c_T| \le r_T$, Theorem 15.2 with $\varepsilon/2$ bounds the error by the first expression, unless its noise fails, with probability β. Step 2 gives $r_T \le \tau$, hence the second expression.
5. The failures are E (2β), the T steps ($T\beta$) and the final noise (β).

□

The steps need $n = O(\ln(R)\ln(1/\beta)/\varepsilon)$, and the final step needs $n = \tilde{O}(1/\alpha^2 + 1/(\alpha\varepsilon))$ for error α. So shrinking intervals reach the same rate Equation 15.18 as the histogram, with only the Laplace mechanism. The example compares the three bounds when the prior bound R is large.

Example 15.1. Take $n = 4{,}000$, $\varepsilon = 1$, $R = 10^6$ and $\beta = 0.002$, so that $\tau = 5.51$. Then $T = \lceil \log_2(10^6/5.51) \rceil = 18$ steps, which need $n \ge 32 \cdot 18 \ln 500 = 3{,}580$, and all $T + 3 = 21$ events hold with probability at least $1 - 21 \cdot 0.002 = 0.958$. The radius reached by the recursion gives the error bound 0.098. The coarse step of Theorem 15.3 uses $m = 181{,}360$ bins, needs $n > 293$, and gives the bound 0.162. The clipped estimator on $[-R, R]$ alone has the bound 3,107.

The bounds are conservative, and Figure 15.1 compares the simulated errors.

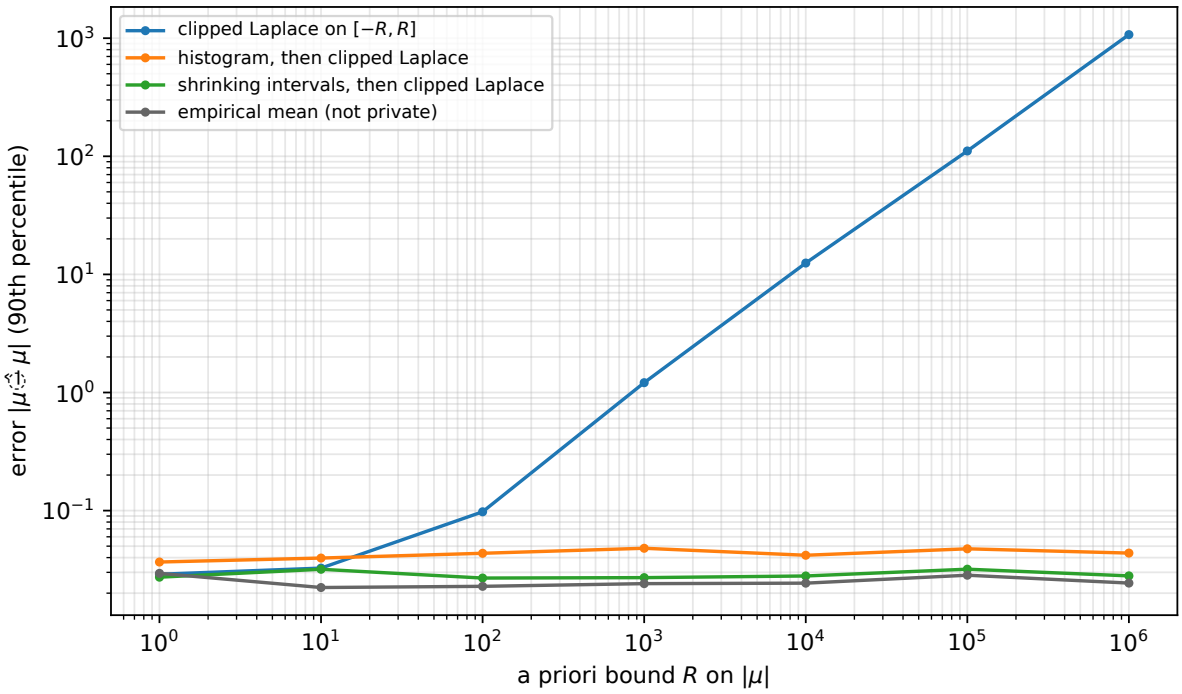


Figure 15.1.: Error of three ε-DP estimators of the mean of $\mathcal{N}(\mu, 1)$ with $\mu = R/3$, as the a priori bound R grows, for $n = 4{,}000$, $\varepsilon = 1$ and $\beta = 0.002$. Each point is the 90th percentile of $|\hat{\mu} - \mu|$ over 100 runs. The clipped estimator on $[-R, R]$ degrades linearly in R. The two estimators that first locate the data stay within a factor of about two of the non-private empirical mean for every R up to 10^6. At $R = 10^6$, over 300 runs, the 90th percentiles are 0.0250 without privacy, 0.0461 for the histogram and 0.0295 for shrinking intervals.

15.7. Many dimensions

The shrinking argument has an analogue in many dimensions. It keeps a confidence set for the parameter, starting from the prior knowledge, and repeats three moves.

1. Clip the data to the confidence set, widened by a tail bound for the class of distributions.
2. Compute the empirical estimate of the parameter, and add noise scaled to its sensitivity, which the clipping bounds.
3. Replace the confidence set by one centred at the noisy estimate, whose size accounts for the sampling error and the noise.

For a Gaussian $\mathcal{N}(\mu, I_d)$ in $\mathbb{R}^d$ with $\|\mu\|_2 \le R$, the confidence set is a Euclidean ball, and the noise is Gaussian, because clipping to a ball bounds the ℓ_2 sensitivity. Two facts replace those used above. First, the Gaussian mechanism adds $\mathcal{N}(0, \sigma^2 I_d)$ to a function f whose ℓ_2 sensitivity is $\Delta_2 f$. For every $\varepsilon > 0$ and $\delta \in (0,1)$ it is (ε, δ)-DP with $\sigma = \Delta_2 f \sqrt{2\ln(1/\delta) + \varepsilon}/\varepsilon$. Approximate DP also composes by adding the ε's and the δ's. Second, a standard Gaussian vector $Y \sim \mathcal{N}(0, I_d)$ has norm about $\sqrt{d}$, with

$$\Pr\left[\|Y\|_2 \ge \sqrt{d} + t\right] \le e^{-t^2/2} \qquad (t \ge 0). \tag{15.22}$$

This follows from the Gaussian concentration inequality for 1-Lipschitz functions (Boucheron, Lugosi and Massart, 2013, Theorem 5.6), applied to the norm, with $\mathbb{E}\|Y\|_2 \le \sqrt{\mathbb{E}\|Y\|_2^2} = \sqrt{d}$.

Clipping to the ball of centre c and radius ρ maps x to $c + (x-c)\min(1, \rho/\|x-c\|_2)$, which moves no point already inside the ball. Two radii play the role of τ in one dimension,

$$\tau_d = \sqrt{d} + \sqrt{2\ln(n/\beta)}, \qquad L = \sqrt{d} + \sqrt{2\ln(1/\beta)}. \tag{15.23}$$

The first widens each ball before clipping. By Equation 15.22 and a union bound, every record lies within τ_d of μ with probability at least $1-\beta$. The second bounds the norm of a single standard Gaussian vector with probability at least $1-\beta$, and so it bounds both the sampling error and the noise.

Theorem 15.5 (Multivariate Gaussian mean)**.** *Let $\varepsilon > 0$ and $\delta \in (0,1)$. A step on the ball $B(c, r)$ with budget (ε', δ') clips every record to $B(c, r + \tau_d)$, averages, and adds $\mathcal{N}(0, \sigma^2 I_d)$ with $\sigma = 2(r + \tau_d)\sqrt{2\ln(1/\delta') + \varepsilon'}/(\varepsilon' n)$. Its new radius is*

$$s(r) = \frac{L}{\sqrt{n}} + \sigma L. \tag{15.24}$$

Run $T = \lceil \log_2(R/\tau_d) \rceil$ steps with $(\varepsilon/(2T), \delta/(2T))$ from $B(0, R)$, then one more with $(\varepsilon/2, \delta/2)$, and output its noisy mean $\hat{\mu}$. This is (ε, δ)-DP. If $X_1, \dots, X_n$ are independent $\mathcal{N}(\mu, I_d)$ draws with $\|\mu\|_2 \le R$, and

$$n \ge 16, \qquad n \ge \frac{32\, T L \sqrt{2\ln(2T/\delta) + \varepsilon}}{\varepsilon}, \tag{15.25}$$

then with probability at least $1-(T+3)\beta$,

$$\|\hat{\mu}-\mu\|_2 \le \frac{L}{\sqrt{n}} + \frac{8\,\tau_d L\sqrt{2\ln(2/\delta)+\varepsilon}}{\varepsilon n}. \tag{15.26}$$

Proof. The proof is that of Theorem 15.4, with these changes.

1. *Privacy.* The clipped mean has ℓ_2 sensitivity $2(r+\tau_d)/n$, so each step is the Gaussian mechanism. The radii are public, and composition adds the budgets to (ε,δ).
2. *No clipping.* By Equation 15.22 with $t=\sqrt{2\ln(n/\beta)}$ and a union bound, every $\|X_i-\mu\|_2 \le \tau_d$ with probability at least $1-\beta$. If $\|\mu-c\|_2 \le r$, the ball $B(c, r+\tau_d)$ then contains every record.
3. *Sampling.* The empirical mean satisfies $\tilde{\mu}-\mu = Y/\sqrt{n}$ with $Y\sim\mathcal{N}(0,I_d)$, so Equation 15.22 gives $\|\tilde{\mu}-\mu\|_2 \le L/\sqrt{n}$ with probability $1-\beta$.
4. *Noise.* The noise is $\sigma Y'$ with $Y'\sim\mathcal{N}(0,I_d)$, so its norm is at most σL with probability $1-\beta$. With steps 2 and 3, the new ball has radius $s(r)$ and contains μ.
5. *Halving.* Since $L\le\tau_d$ and $n\ge 16$, the first term of $s(r)$ is at most $\tau_d/4$. With $\varepsilon'=\varepsilon/(2T)$ and $\delta'=\delta/(2T)$, $2\ln(1/\delta')+\varepsilon' \le 2\ln(2T/\delta)+\varepsilon$, so $\sigma L \le 4T(r+\tau_d)L\sqrt{2\ln(2T/\delta)+\varepsilon}/(\varepsilon n)$, and the condition on n makes this at most $(r+\tau_d)/8$. So $s(r)\le \max(r,\tau_d)/2$, and after T steps the radius is at most τ_d.
6. *Final step.* With $r_T \le \tau_d$, the final noise, with budget $(\varepsilon/2,\delta/2)$, has $\sigma = 4(r_T+\tau_d)\sqrt{2\ln(2/\delta)+\varepsilon/2}/(\varepsilon n) \le 8\tau_d\sqrt{2\ln(2/\delta)+\varepsilon}/(\varepsilon n)$, and steps 3 and 4 give the bound. The failures add up as before.

□

Since L and τ_d are $\sqrt{d}$ up to logarithmic terms, error α with $\varepsilon\le 1$ needs

$$n = \tilde{O}\left(\frac{d}{\alpha^2} + \frac{d\sqrt{\ln(1/\delta)}}{\alpha\varepsilon} + \frac{\sqrt{d}\,\ln R\,\sqrt{\ln(1/\delta)}}{\varepsilon}\right). \tag{15.27}$$

The first two terms are d times the one-dimensional ones. The cost of locating the data grows only like $\sqrt{d}$, because the noise of each step is compared with a radius that already includes the $\sqrt{d}$ spread of the data.

15.8. Heavy-tailed data

So far the arguments used only that the data has light tails. So they hold for **sub-Gaussian** distributions, whose tails decay at least as fast as those of a Gaussian. Then τ is multiplied by the scale of the tails. Real data often has heavier tails, and then only a few moments may be bounded. Take the class of distributions on $\mathbb{R}$ with

$$\mu = \mathbb{E}[X]\in[-1,1], \qquad \mathrm{Var}(X)\le 1. \tag{15.28}$$

The problem is to estimate μ to a precision α well below 1.

The previous strategy clips only where no point lands. Chebyshev's inequality gives $\Pr[|X-\mu| \geq \sqrt{n/\beta}] \leq \beta/n$. So the interval $[-1-C, 1+C]$, with the clipping norm C set to $\sqrt{n/\beta}$, clips no point with probability $1-\beta$. But the noise scale is then $2(1+C)/(\varepsilon n)$, which is more than $2/(\varepsilon\sqrt{\beta n})$. The tail bound on the noise error is this scale times $\ln(1/\beta)$, and it is at most α only if

$$n \geq \frac{4\ln^2(1/\beta)}{\beta\,\alpha^2\varepsilon^2}, \tag{15.29}$$

which is worse than the Gaussian rate in both α and ε. The remedy is to clip more aggressively and accept a bias, the clipping bias of Equation 15.14, which the next lemma bounds.

Lemma 15.5 (Clipping bias)**.** *Let X have mean μ and variance at most σ_X^2, and let $[u,v] \supseteq [\mu-C, \mu+C]$ with $C>0$. Then $g = g_{[u,v]}$ satisfies*

$$\big|\mathbb{E}[g(X)] - \mu\big| \leq \frac{\sigma_X^2}{C}, \qquad \mathrm{Var}(g(X)) \leq \sigma_X^2. \tag{15.30}$$

Proof.

1. Since $\mu = \mathbb{E}[X]$, $|\mathbb{E}[g(X)] - \mu| = |\mathbb{E}[g(X) - X]| \leq \mathbb{E}|g(X) - X|$.
2. If $|X-\mu| \leq C$, then $X \in [u,v]$ and $g(X) = X$. Otherwise X moves towards μ and does not pass it, because $\mu \in [u,v]$, so $|g(X)-X| \leq |X-\mu|$.
3. So $|g(X)-X| \leq |X-\mu|\,\mathbf{1}[|X-\mu|>C] \leq (X-\mu)^2/C$, and the expectation gives the first bound.
4. Clipping is 1-Lipschitz. With X' an independent copy of X, $\mathrm{Var}(g(X)) = \frac{1}{2}\mathbb{E}[(g(X)-g(X'))^2] \leq \frac{1}{2}\mathbb{E}[(X-X')^2] = \mathrm{Var}(X)$.

□

With the bias bounded, the clipping interval no longer has to catch every point. The next estimator clips at a distance C from the range of possible means, and the three terms of Equation 15.14 decide how large C must be.

Theorem 15.6 (Heavy-tailed mean)**.** *For $C>0$, the estimator*

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} g_{[-1-C,\,1+C]}(X_i) + \mathrm{Lap}\!\left(\frac{2(1+C)}{\varepsilon n}\right) \tag{15.31}$$

is ε-DP. If $X_1, \dots, X_n$ are independent draws from a distribution with mean $\mu \in [-1,1]$ and variance at most 1, then with probability at least $1-2\beta$,

$$|\hat{\mu} - \mu| \leq \frac{1}{C} + \frac{1}{\sqrt{\beta n}} + \frac{2(1+C)\ln(1/\beta)}{\varepsilon n}. \tag{15.32}$$

Proof.

1. Privacy is Lemma 15.1 with an interval of length $2(1 + C)$.
2. Since $|\mu| \le 1$, the interval $[-1 - C, 1 + C]$ contains $[\mu - C, \mu + C]$. Write $\nu = \mathbb{E}[g(X_1)]$ and $\bar{g}$ for the clipped mean. Then $\hat{\mu} - \mu = (\nu - \mu) + (\bar{g} - \nu) + Z$, with Z the noise.
3. *Bias.* Lemma 15.5 with $\sigma_X^2 = 1$ gives $|\nu - \mu| \le 1/C$.
4. *Sampling.* By Lemma 15.5, $\mathrm{Var}(\bar{g}) \le 1/n$, so Chebyshev's inequality gives $|\bar{g} - \nu| \le 1/\sqrt{\beta n}$ with probability at least $1 - \beta$.
5. *Noise.* The Laplace tail gives $|Z| \le 2(1 + C)\ln(1/\beta)/(\varepsilon n)$ with probability $1 - \beta$. A union bound finishes the proof.

□

The bias falls as C grows and the noise rises, so there is a best clipping norm. If each of the three terms of Equation 15.32 is at most $\alpha/3$, the sample complexity is simple.

Corollary 15.1. *Let $\alpha \in (0, 1]$, and let $\hat{\mu}$ be the estimator of Theorem 15.6 with $C = 3/\alpha$. If*

$$n \ge \frac{9}{\beta\alpha^2} \qquad \text{and} \qquad n \ge \frac{24\ln(1/\beta)}{\alpha^2\varepsilon}, \tag{15.33}$$

then $|\hat{\mu} - \mu| \le \alpha$ with probability at least $1 - 2\beta$.

Proof.

1. The bias term is $1/C = \alpha/3$, and the first condition makes the sampling term at most $\alpha/3$.
2. Since $\alpha \le 1$, $1 + C \le 4/\alpha$, so the noise term is at most $8\ln(1/\beta)/(\alpha\varepsilon n)$, which the second condition makes at most $\alpha/3$.

□

So the clipping norm must be about $1/\alpha$, and $n = O(1/(\alpha^2\varepsilon))$ samples are enough for a fixed β and $\varepsilon \le 1$. This is more than the $O(1/\alpha^2 + 1/(\alpha\varepsilon))$ of the Gaussian case, because the two terms now multiply instead of adding. The increase is necessary. For distributions with only a bounded variance, no ε-DP estimator has error below order $1/\sqrt{\varepsilon n}$ (Barber and Duchi, 2014). In terms of n, the clipping norm that minimizes $1/C + 2C\ln(1/\beta)/(\varepsilon n)$ is

$$C^* = \sqrt{\frac{\varepsilon n}{2\ln(1/\beta)}}, \tag{15.34}$$

and with $n = 4{,}000$, $\varepsilon = 1$ and $\beta = 0.05$ it is $C^* = 25.8$, where Equation 15.32 equals 0.150. Figure 15.2 shows the trade-off.

15.9. Summary

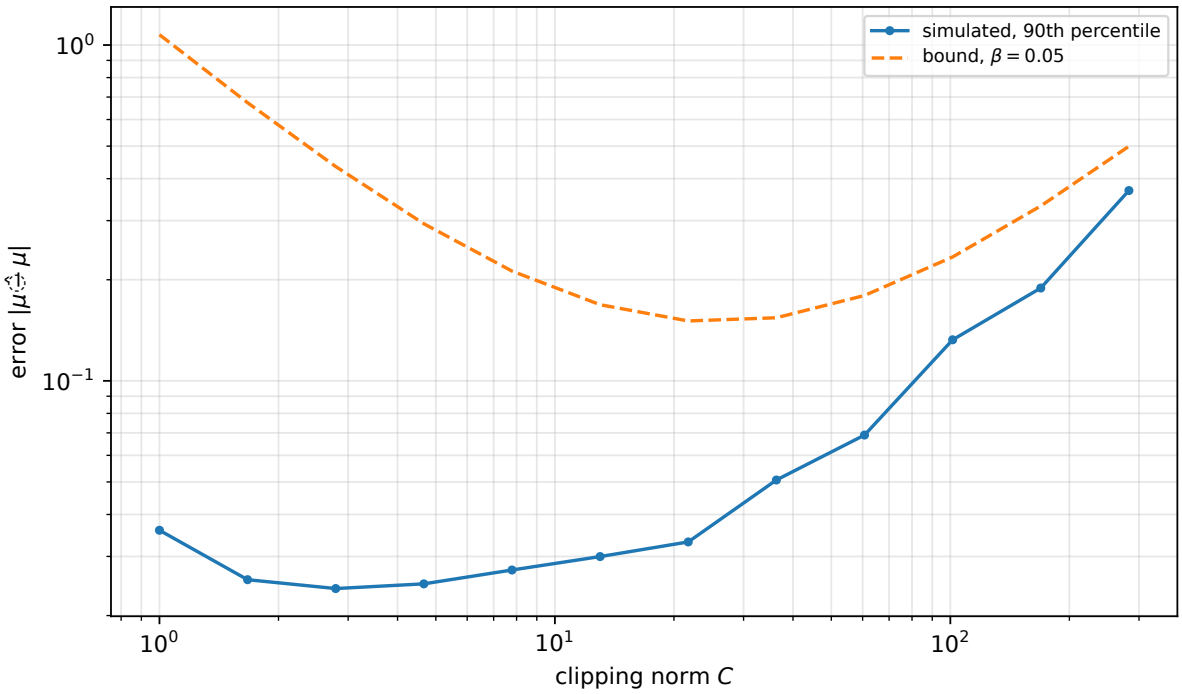


Figure 15.2.: Error of the clipped estimator of Theorem 15.6 on a Student t distribution with 3 degrees of freedom, scaled to variance 1, with $\mu = 0.5$, $n = 4{,}000$ and $\varepsilon = 1$, as the clipping norm C varies. The solid line is the 90th percentile of $|\hat{\mu} - \mu|$ over 300 runs, and the dashed line is the bound Equation 15.32 with $\beta = 0.05$. Both rise steeply at large C, where the noise dominates. The simulated error is smallest at a smaller C than the bound, because the clipping bias of this distribution is far below the worst case $1/C$.

Table 15.1.: The chapter's results.

Result	Statement
Theorem 15.1	Bernoulli mean with error $\sqrt{\ln(4/\beta)/(2n)} + \ln(2/\beta)/(\varepsilon n)$
Proposition 15.1	no ε-DP algorithm estimates the mean of every real dataset to finite accuracy
Lemma 15.1	clipping to a public $[u, v]$ gives sensitivity $(v-u)/n$
Proposition 15.2	estimating a Gaussian mean with $\lvert\mu\rvert \le R$ needs $n \ge \ln((1-\beta)m)/\varepsilon$, with m about $2R/(3\alpha)$
Theorem 15.2	clipping to $[-R-\tau, R+\tau]$ needs $n = \tilde{O}(1/\alpha^2 + R/(\alpha\varepsilon))$
Lemma 15.3	a noisy histogram locates μ within 2τ once $n > (8/\varepsilon)\ln(m/\beta)$
Theorem 15.3	histogram, then clipping, needs $n = \tilde{O}(1/\alpha^2 + 1/(\alpha\varepsilon) + \ln R/\varepsilon)$
Lemma 15.4	one clipped step halves the confidence interval down to the floor τ
Theorem 15.4	$T = \lceil \log_2(R/\tau) \rceil$ shrinking steps, then clipping, reach the same rate
Theorem 15.5	in $\mathbb{R}^d$, $n = \tilde{O}(d/\alpha^2 + d/(\alpha\varepsilon) + \sqrt{d}\ln R/\varepsilon)$ under (ε, δ)-DP, up to factors in $\ln(1/\delta)$
Lemma 15.5	clipping at distance C from the mean biases it by at most σ_X^2/C
Corollary 15.1	with variance 1 and $C = 3/\alpha$, $n = O(1/(\alpha^2\varepsilon))$ samples suffice

15.10. References

- Cynthia Dwork, Frank McSherry, Kobbi Nissim and Adam Smith, "Calibrating Noise to Sensitivity in Private Data Analysis", 2006, which introduces the Laplace mechanism and the sensitivity of the mean of bounded data.
- Vishesh Karwa and Salil Vadhan, "Finite Sample Differentially Private Confidence Intervals", 2018, on the histogram approach to the Gaussian mean, with lower bounds that make the rate Equation 15.18 optimal up to logarithmic factors.
- Cynthia Dwork and Aaron Roth, *The Algorithmic Foundations of Differential Privacy*, 2014, on the Laplace and Gaussian mechanisms and the composition theorems of this chapter.
- Stéphane Boucheron, Gábor Lugosi and Pascal Massart, *Concentration Inequalities*, 2013, whose Theorem 5.6 is the Gaussian concentration inequality behind Equation 15.22.

- Rina Foygel Barber and John Duchi, "Privacy and Statistical Risk: Formalisms and Minimax Bounds", 2014, whose Proposition 2 gives the $1/\sqrt{\varepsilon n}$ lower bound for bounded variance.

16. Adaptive Data Analysis

A lab tests hypothesis after hypothesis on one dataset, each suggested by the earlier results, until one is significant. Often, fresh data does not reproduce that result. Each test may be sound alone, but the sequence used the same data to choose the questions and to answer them. So the answers describe the sample instead of the population.

This chapter makes that failure precise for averages of predicates, called statistical queries, and shows how differential privacy (DP) prevents it. Queries fixed in advance need a sample size logarithmic in the number k of queries, and adaptively chosen queries can overfit unless it is of order k. A mechanism that answers through a DP algorithm needs only about $\sqrt{k}$, because a private algorithm cannot find a query on which the sample and the population disagree.

16.1. Statistical queries

The data is a sample from an unknown distribution $\mathcal{D}$ on a universe $\mathcal{X}$, and the analyst cares about the distribution, not the sample. Each question asks for the fraction of records with a yes-or-no property, so it has a value on the sample and a value on the distribution.

Definition 16.1 (Statistical query)**.** A **statistical query** is a function $q\colon \mathcal{X} \to \{0,1\}$. Its value on the distribution and its empirical value on a dataset $D = (X_1, \dots, X_n)$ are

$$q(\mathcal{D}) = \mathbb{E}_{x\sim\mathcal{D}}[q(x)], \qquad q(D) = \frac{1}{n}\sum_{j=1}^{n} q(X_j). \tag{16.1}$$

So $q(\mathcal{D})$ is the fraction of the population with the property q, such as red hair, and $q(D)$ is the fraction in the sample. These are the counting queries of earlier chapters, divided by n. If the records are independent draws from $\mathcal{D}$, the empirical value is an unbiased estimate of the population value. For a fixed list of queries, a sample whose size is logarithmic in the length of the list is enough.

Proposition 16.1. *Let $\alpha > 0$ be the allowed error and $\beta \in (0,1)$ the allowed failure probability. Let $q_1, \dots, q_k$ be statistical queries fixed before the sample is drawn, and let $X_1, \dots, X_n$ be independent draws from $\mathcal{D}$. If*

$$n \geq \frac{\ln(2k/\beta)}{2\alpha^2}, \tag{16.2}$$

then some query has error at least α with probability at most β,

$$\Pr\left[\exists i\colon |q_i(D) - q_i(\mathcal{D})| \geq \alpha\right] \leq \beta. \tag{16.3}$$

Proof.

1. For a fixed i, the values $q_i(X_1), \dots, q_i(X_n)$ are independent, lie in $[0,1]$, and have mean $q_i(\mathcal{D})$. Hoeffding's inequality gives $\Pr[|q_i(D) - q_i(\mathcal{D})| \geq \alpha] \leq 2e^{-2\alpha^2 n}$.
2. A union bound over the k queries gives failure probability at most $2ke^{-2\alpha^2 n}$, which the condition on n makes at most β.

□

So a sample of size n supports about $e^{2\alpha^2 n}$ queries. But step 1 needs the query to be independent of the sample, and adaptivity breaks that step.

16.2. Adaptivity breaks the bound

In an adaptive analysis, each query may depend on the answers to the earlier ones. The definition sets up this interaction, and says when the mechanism that answers is private whatever the analyst does.

Definition 16.2 (Adaptive analysis)**.** An **analyst** $\mathcal{A}$ and a **mechanism** M interact for k rounds. In round i, the analyst chooses a statistical query q_i as a function of the earlier answers $a_1, \dots, a_{i-1}$ and its own random bits, called its coins. The mechanism returns an answer a_i, computed from the dataset D, its own coins and the history. The mechanism is (ε, δ)**-DP against adaptive analysts** if for every analyst $\mathcal{A}$ the map from D to the transcript $(q_1, a_1, \dots, q_k, a_k)$ is (ε, δ)-DP.

The empirical mechanism $a_i = q_i(D)$ answers every query exactly on the sample, and an adaptive analyst makes it fail badly.

Proposition 16.2. *Let $\mathcal{X} = \{1, \dots, N\}$, let $\mathcal{D}$ be uniform on $\mathcal{X}$, and let the mechanism answer $a_i = q_i(D)$. Some analyst with $k = N + 1$ queries ends with a query q_k such that, for every sample,*

$$q_k(D) - q_k(\mathcal{D}) \geq 1 - \frac{n}{k-1}. \tag{16.4}$$

Proof.

1. For $i = 1, \dots, N$, the analyst asks $q_i(x) = \mathbf{1}[x = i]$. The answer a_i is the fraction of records equal to i, so after N rounds the analyst knows the set $S = \{X_1, \dots, X_n\}$ of values in the sample.
2. The last query is $q_k(x) = \mathbf{1}[x \in S]$. Every record lies in S, so $q_k(D) = 1$.
3. The set S has at most n elements, so $q_k(\mathcal{D}) = |S|/N \leq n/N = n/(k-1)$.

□

The first N queries are harmless one at a time, and each could be fixed in advance. The analyst builds the last one from their answers, and on it the sample and the population disagree by almost 1 unless n is of order k. So the number of queries that a sample supports drops from exponential in n to linear in n.

A simple defence is to give each query fresh data. The analyst can still choose adaptively, but each answer comes from records that no earlier answer has seen.

Proposition 16.3 (Sample splitting)**.** *Let k divide n. Split D into k disjoint parts $D^{(1)}, \dots, D^{(k)}$ of n/k records each, and answer $a_i = q_i(D^{(i)})$. If the records are independent draws from $\mathcal{D}$ and*

$$n \geq \frac{k \ln(2k/\beta)}{2\alpha^2}, \tag{16.5}$$

then for every analyst $\Pr\left[\exists i \colon |a_i - q_i(\mathcal{D})| \geq \alpha\right] \leq \beta$.

Proof.

1. The query q_i depends only on the coins of the analyst and on $a_1, \dots, a_{i-1}$, which depend only on the parts $D^{(1)}, \dots, D^{(i-1)}$. So q_i is independent of $D^{(i)}$.
2. Conditioned on q_i, the part $D^{(i)}$ is still n/k independent draws from $\mathcal{D}$. Step 1 of the proof of Proposition 16.1 gives $\Pr[|a_i - q_i(\mathcal{D})| \geq \alpha] \leq 2e^{-2\alpha^2 n/k}$.
3. A union bound over i gives at most $2ke^{-2\alpha^2 n/k}$, which the condition makes at most β.

□

Sample splitting is valid but expensive, because it needs k times the data of the non-adaptive case. A DP mechanism does much better. With Gaussian noise, a sample of size

$$n = O\left(\frac{\sqrt{k}\,\ln^{3/2}(k/(\alpha\beta))}{\alpha^2}\right) \tag{16.6}$$

is enough for error α (Corollary 16.1). Under a standard cryptographic assumption, no computationally efficient mechanism can do better than $\sqrt{k}$ in general.

16.3. Privacy implies generalization

To overfit, the analyst must learn too much about individual records, as the analyst of Proposition 16.2 did. If every answer passes through a DP mechanism, no answer reveals much about one record, so the analyst can learn only about the population. A **transfer theorem** makes this precise. It turns accuracy on the sample and privacy into accuracy on the population.

Theorem 16.1 (Transfer theorem)**.** *Let $\alpha, \beta \in (0, 0.1)$, and let M answer k adaptively chosen statistical queries. Suppose that*

1. *the mechanism is $(\alpha/64, \alpha\beta/32)$-DP against adaptive analysts, and*

2. *the mechanism is accurate on the sample, that is, for every analyst and every dataset D,* $\Pr\left[\exists i\colon |a_i - q_i(D)| > \alpha/8\right] \le \alpha\beta/16$, *with probability over the coins of M and of the analyst.*

Then for every analyst and every distribution $\mathcal{D}$, if the n records are independent draws from $\mathcal{D}$,

$$\Pr\left[\exists i\colon |a_i - q_i(\mathcal{D})| > \alpha\right] \le \beta. \tag{16.7}$$

This is Theorem 3.4 of Bassily, Nissim, Smith, Steinke, Stemmer and Ullman (2016), for queries of sensitivity $1/n$. The failure probability on the sample must be of order $\alpha\beta$, not β, for the conclusion to hold with failure probability β. The proof runs about $1/\beta$ independent copies of the interaction, each on its own sample. A monitor, which knows $\mathcal{D}$ and exists only in the proof, picks the copy and the query with the largest error on the population. Its choice post-processes DP outputs, so it is DP, and a version of Lemma 16.1 below for several samples shows that it cannot find a large population error on average.

Two results from earlier chapters apply the theorem. The Gaussian mechanism answers a query f with Gaussian noise, and the sensitivity Δf sets the standard deviation. With $\sigma = \Delta f\sqrt{2\ln(1/\delta_0) + \varepsilon_0}/\varepsilon_0$, the mechanism is $(\varepsilon_0, \delta_0)$-DP for every $\varepsilon_0 > 0$ and $\delta_0 \in (0, 1)$. A corollary of advanced composition says that, for $\varepsilon \in (0, 1]$ and $\delta' \in (0, 1/2]$, k adaptively chosen mechanisms that are each $(\varepsilon_0, \delta_0)$-DP with

$$\varepsilon_0 = \frac{\varepsilon}{2\sqrt{2k\ln(1/\delta')}} \tag{16.8}$$

are, together, $(\varepsilon, k\delta_0 + \delta')$-DP. A statistical query has sensitivity $1/n$, because a replaced record changes one term of the average by at most 1. The corollary adds Gaussian noise to each empirical answer, at a scale that meets both hypotheses of Theorem 16.1, and gives the sample size that this needs.

Corollary 16.1 (Gaussian answers generalize)**.** *Let $\alpha, \beta \in (0, 0.1)$. Answer each query by $a_i = q_i(D) + Z_i$ with independent $Z_i \sim \mathcal{N}(0, \sigma^2)$, where*

$$\varepsilon_0 = \frac{\alpha}{128\sqrt{2k\ln(64/(\alpha\beta))}}, \qquad \sigma = \frac{\sqrt{2\ln(64k/(\alpha\beta)) + \varepsilon_0}}{\varepsilon_0\, n}. \tag{16.9}$$

If

$$n \ge \frac{2048\sqrt{3}}{\alpha^2}\sqrt{k\ln\frac{64}{\alpha\beta}}\ \ln\frac{64k}{\alpha\beta}, \tag{16.10}$$

then for every analyst, $\Pr\left[\exists i\colon |a_i - q_i(\mathcal{D})| > \alpha\right] \le \beta$.

Proof.

1. *Each answer is private.* With $\delta_0 = \alpha\beta/(64k)$, the noise is the Gaussian mechanism's for sensitivity $1/n$ and budget $(\varepsilon_0, \delta_0)$. For a fixed analyst, the i-th answer is this mechanism applied to a query fixed by the earlier answers and the coins of the analyst.

2. *All answers are* $(\alpha/64, \alpha\beta/32)$*-DP.* The corollary of advanced composition applies with $\varepsilon = \alpha/64$ and $\delta' = \alpha\beta/64$. The total δ is $k\delta_0 + \delta' = \alpha\beta/32$. The queries in the transcript are a post-processing of the answers and the coins of the analyst, so the transcript has the same guarantee.
3. *Accuracy on the sample.* The error $a_i - q_i(D)$ is the noise Z_i, and a Gaussian variable exceeds t standard deviations in absolute value with probability at most $2e^{-t^2/2}$. With $t = \sqrt{2\ln(32k/(\alpha\beta))}$, each $|Z_i|$ exceeds $t\sigma$ with probability at most $\alpha\beta/(16k)$, so by a union bound all k answers are within $t\sigma$ of the sample values with probability at least $1 - \alpha\beta/16$.
4. *The threshold.* Write $\ell = \ln(64k/(\alpha\beta))$. Since $\varepsilon_0 < 1 < \ell$, we have $2\ell + \varepsilon_0 \le 3\ell$, and $t \le \sqrt{2\ell}$. So

$$t\sigma \le \frac{\sqrt{6}\,\ell}{\varepsilon_0 n} = \frac{256\sqrt{3}}{\alpha n}\sqrt{k \ln \frac{64}{\alpha\beta}}\;\ell, \tag{16.11}$$

which Equation 16.10 makes at most $\alpha/8$. Steps 2 to 4 are the hypotheses of Theorem 16.1, which gives the claim. □

The $\sqrt{k}$ in Equation 16.10 is the $\sqrt{k}$ of advanced composition. The constant is large because the constants of the transfer theorem are. But the number of queries that a sample supports grows from linear in n, for sample splitting, to quadratic in n. For a finite universe and a very large k, private multiplicative weights does better still. For the average error over the queries, it needs a sample only logarithmic in k (Bassily et al., 2016, Corollary 6.3).

16.4. An easier transfer theorem

A weaker transfer theorem, with accuracy in expectation instead of with high probability, has a short complete proof. It turns a bound on the expected worst error on the sample into one on the population.

Theorem 16.2 (Transfer in expectation)**.** *Let M be (ε, δ)-DP against adaptive analysts. Suppose that for every analyst, if the records are independent draws from $\mathcal{D}$, then*

$$\mathbb{E}\Big[\max_{i \le k} |a_i - q_i(D)|\Big] \le \alpha. \tag{16.12}$$

Then for every analyst,

$$\mathbb{E}\Big[\max_{i \le k} |a_i - q_i(\mathcal{D})|\Big] \le \alpha + e^{\varepsilon} - 1 + \delta. \tag{16.13}$$

For small ε, the excess is about $\varepsilon + \delta$. Most of the work is in a lemma about one query. It says that a query chosen by a DP algorithm has on average about the same value on the sample as on the population. So a DP algorithm cannot produce the last query of Proposition 16.2, on which the two disagree.

Lemma 16.1 (A private query generalizes)**.** *Let W be an (ε, δ)-DP algorithm that maps a dataset in $\mathcal{X}^n$ to a statistical query. Let $X_1, \dots, X_n$ be independent draws from $\mathcal{D}$ and $q = W(D)$. Then*

$$\left|\mathbb{E}[q(D)] - \mathbb{E}[q(\mathcal{D})]\right| \leq e^{\varepsilon} - 1 + \delta, \tag{16.14}$$

with expectations over the sample and the coins of W.

Proof.

1. *Linearity.* Since q takes values in $\{0, 1\}$,

$$\mathbb{E}[q(D)] = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[q(X_i)] = \frac{1}{n}\sum_{i=1}^{n} \Pr[q(X_i) = 1]. \tag{16.15}$$

2. *A fresh record.* Fix i, draw X_i' from $\mathcal{D}$ independently of everything else, and let $D^{(i)}$ be D with entry i replaced by X_i'. The datasets D and $D^{(i)}$ are neighbours, $D \sim D^{(i)}$, since they differ in one replaced record.
3. *Privacy.* Write $W(D)(X_i)$ for the value of the query $W(D)$ on the record X_i. For fixed values of D and X_i', the queries q' with $q'(X_i) = 1$ form an event about the output of W. DP for the neighbours D and $D^{(i)}$ gives

$$\Pr_W\left[W(D)(X_i) = 1\right] \leq e^{\varepsilon} \Pr_W\left[W(D^{(i)})(X_i) = 1\right] + \delta. \tag{16.16}$$

The average over D and X_i' keeps the inequality. 4. *Swapping.* The pair $(D^{(i)}, X_i)$ has the same distribution as the pair (D, X_i'), since each is n independent draws from $\mathcal{D}$ plus one more independent draw. Hence $\Pr[W(D^{(i)})(X_i) = 1] = \Pr[q(X_i') = 1]$. The record X_i' is independent of $q = W(D)$, so this probability is $\mathbb{E}[q(\mathcal{D})]$. 5. *One direction.* Steps 1, 3 and 4 give $\mathbb{E}[q(D)] \leq e^{\varepsilon}\,\mathbb{E}[q(\mathcal{D})] + \delta$. Since $\mathbb{E}[q(\mathcal{D})] \leq 1$,

$$\mathbb{E}[q(D)] - \mathbb{E}[q(\mathcal{D})] \leq (e^{\varepsilon} - 1)\,\mathbb{E}[q(\mathcal{D})] + \delta \leq e^{\varepsilon} - 1 + \delta. \tag{16.17}$$

6. *Other direction.* DP applied from $D^{(i)}$ to D gives $\Pr[W(D^{(i)})(X_i) = 1] \leq e^{\varepsilon} \Pr[W(D)(X_i) = 1] + \delta$. With steps 1 and 4, this is $\mathbb{E}[q(\mathcal{D})] \leq e^{\varepsilon}\,\mathbb{E}[q(D)] + \delta$, and the same argument as in step 5 bounds $\mathbb{E}[q(\mathcal{D})] - \mathbb{E}[q(D)]$ by $e^{\varepsilon} - 1 + \delta$.

□

The proof of Theorem 16.2 lets a private algorithm pick out the query with the worst error, and two tricks make that query fit the lemma.

Proof.

1. *Removing the absolute value.* Whenever the analyst asks q_i and receives a_i, add the query $1 - q_i$ with the answer $1 - a_i$. The extended transcript is a post-processing of the original, so it is still (ε, δ)-DP. The errors on the sample are unchanged, because $|(1 - a_i) - (1 - q_i)(D)| = |a_i - q_i(D)|$. Since $(1 - q_i)(\mathcal{D}) - (1 - a_i) = -(q_i(\mathcal{D}) - a_i)$, the $2k$ queries q_j satisfy

$$\max_{i \le k} |q_i(\mathcal{D}) - a_i| = \max_{j \le 2k} \left(q_j(\mathcal{D}) - a_j\right). \tag{16.18}$$

2. *The selector.* Let W run the interaction on D and output the query $q^* = q_{j^*}$, with answer $a^* = a_{j^*}$, where j^* maximizes $q_j(\mathcal{D}) - a_j$. The selector uses $\mathcal{D}$, which is fixed and is not the data, so W is a post-processing of the transcript and is (ε, δ)-DP. It is a device of the proof and is never run.
3. *Splitting the error.* By linearity of expectation,

$$\mathbb{E}\Big[\max_i |q_i(\mathcal{D}) - a_i|\Big] = \mathbb{E}\big[q^*(\mathcal{D}) - q^*(D)\big] + \mathbb{E}\big[q^*(D) - a^*\big]. \tag{16.19}$$

4. *The two terms.* Lemma 16.1, applied to W, bounds the first term by $e^{\varepsilon} - 1 + \delta$. The second is at most $\mathbb{E}[\max_j |q_j(D) - a_j|] \le \alpha$, by the accuracy hypothesis and step 1.

□

The Gaussian mechanism applies this theorem too, with a proof that cites nothing. The only change from Corollary 16.1 is that the error on the sample is bounded in expectation.

Corollary 16.2. *Let $\alpha \in (0, 1]$. Answer each query by $a_i = q_i(D) + Z_i$ with independent $Z_i \sim \mathcal{N}(0, \sigma^2)$, where*

$$\varepsilon_0 = \frac{\alpha}{2\sqrt{2k \ln(2/\alpha)}}, \qquad \sigma = \frac{\sqrt{2 \ln(2k/\alpha) + \varepsilon_0}}{\varepsilon_0 \, n}. \tag{16.20}$$

If

$$n \ge \frac{4\sqrt{3}}{\alpha^2} \sqrt{k \ln \frac{2}{\alpha}} \; \ln \frac{2k}{\alpha}, \tag{16.21}$$

then for every analyst, $\mathbb{E}[\max_i |a_i - q_i(\mathcal{D})|] \le 4\alpha$.

Proof.

1. *Privacy.* As in steps 1 and 2 of the proof of Corollary 16.1, each answer is the Gaussian mechanism with budget $(\varepsilon_0, \alpha/(2k))$, and the corollary of advanced composition with $\varepsilon = \alpha$ and $\delta' = \alpha/2$ makes the mechanism (α, α)-DP against adaptive analysts.
2. *Expected worst noise.* For $\lambda > 0$, Jensen's inequality and $e^{\lambda \max_i |Z_i|} \le \sum_i e^{\lambda |Z_i|}$ give

$$e^{\lambda\,\mathbb{E}[\max_i |Z_i|]} \le \sum_{i=1}^{k} \mathbb{E}\big[e^{\lambda |Z_i|}\big] \le 2k\, e^{\lambda^2\sigma^2/2}, \tag{16.22}$$

because $\mathbb{E}[e^{\lambda|Z|}] \le \mathbb{E}[e^{\lambda Z}] + \mathbb{E}[e^{-\lambda Z}] = 2e^{\lambda^2\sigma^2/2}$. The logarithm, with $\lambda = \sqrt{2\ln(2k)}/\sigma$, gives $\mathbb{E}[\max_i |Z_i|] \le \sigma\sqrt{2\ln(2k)}$. 3. *The threshold.* Write $\ell = \ln(2k/\alpha) \ge \ln 2$. Since $\varepsilon_0 \le 1/(2\sqrt{2\ln 2}) < \ln 2 \le \ell$, we have $2\ell + \varepsilon_0 \le 3\ell$, and $\ln(2k) \le \ell$. So

$$\sigma\sqrt{2\ln(2k)} \le \frac{\sqrt{6}\,\ell}{\varepsilon_0 n} = \frac{4\sqrt{3}}{\alpha n}\sqrt{k\ln\frac{2}{\alpha}}\;\ell, \tag{16.23}$$

which the condition on n makes at most α. Hence the expected worst error on the sample is at most α. 4. Theorem 16.2 with $\varepsilon = \delta = \alpha$ gives at most $\alpha + (e^\alpha - 1) + \alpha \le 4\alpha$, since $e^\alpha - 1 \le 2\alpha$ for $\alpha \le 1$. □

16.5. Summary

Table 16.1.: The chapter's results.

Result	Statement
Proposition 16.1	k fixed queries generalize once $n \ge \ln(2k/\beta)/(2\alpha^2)$
Proposition 16.2	exact answers to k adaptive queries can be off by $1 - n/(k-1)$ on the population
Proposition 16.3	sample splitting generalizes once $n \ge k\ln(2k/\beta)/(2\alpha^2)$
Theorem 16.1	an $(\alpha/64, \alpha\beta/32)$-DP mechanism with error $\alpha/8$ on the sample, except with probability $\alpha\beta/16$, has error α on the population, except with probability β
Corollary 16.1	Gaussian answers generalize with n of order $\sqrt{k}$ up to logarithmic factors
Lemma 16.1	a query chosen by an (ε, δ)-DP algorithm has $\lvert\mathbb{E}[q(D)] - \mathbb{E}[q(\mathcal{D})]\rvert \le e^\varepsilon - 1 + \delta$
Theorem 16.2	expected worst error on the population is at most $\alpha + e^\varepsilon - 1 + \delta$
Corollary 16.2	Gaussian answers give expected worst error 4α with n of order $\sqrt{k}$

16.6. References

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- Raef Bassily, Kobbi Nissim, Adam Smith, Thomas Steinke, Uri Stemmer and Jonathan Ullman, "Algorithmic Stability for Adaptive Data Analysis", 2016, with the transfer theorem (Theorem 3.4) and the bound of Lemma 16.1 (Lemma 3.1).
- Christopher Jung, Katrina Ligett, Seth Neel, Aaron Roth, Saeed Sharifi-Malvajerdi and Moshe Shenfeld, "A New Analysis of Differential Privacy's Generalization Guarantees", 2020, with a simpler and stronger transfer theorem.
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- Cynthia Dwork, Guy N. Rothblum and Salil Vadhan, "Boosting and Differential Privacy", 2010, which proves advanced composition.
- Cynthia Dwork and Aaron Roth, *The Algorithmic Foundations of Differential Privacy*, 2014, on the Gaussian mechanism and advanced composition (Theorem 3.20).

Part V.

Deployments

The earlier parts built differential privacy (DP) with a trusted curator who holds the data and releases noisy answers. This part asks what changes when the definition meets real systems with millions of users, daily data collection and outside analysts. The first chapter removes the curator, so that each person randomizes their own data, and measures the price in error. The second finds a middle ground with a shuffler, then returns to the central model to see how large deployments select, compose and budget.

Table 16.2.: The chapters of this part.

Chapter	Question it answers
Local differential privacy	What does it cost to trust no one, and how do large deployments live with that cost?
The shuffle model and central deployments	How much does an anonymizing shuffler recover, and how do central deployments select, compose and budget?

17. Local Differential Privacy

Almost every mechanism so far ran on a trusted curator who saw the raw data. This chapter drops that trust, so that each person randomizes their own record before it leaves their device. We define this local model and show that, with n people, it costs a factor of about $\sqrt{n}$ in accuracy on the simplest task. No noninteractive protocol can escape that cost.

The rest of the chapter studies three large deployments of the local model, each of which solves a problem the textbook protocol ignores. Google's protects people who answer the same question every day, Apple's finds values that no one knew to ask about, and Microsoft's sends numbers in a single bit and copes with values that drift.

17.1. The local model

In the central model of the first chapter on the definition, people send their data to a curator, and the trust barrier sits at the output of the curator. In the local model the barrier moves to each person. An adversary may see anything that leaves the device of a person, and the curator is now just an analyst who receives messages.

Definition 17.1 (Local randomizer)**.** Let $\varepsilon \geq 0$. A randomized algorithm $R\colon \mathcal{X} \to \mathcal{Y}$ is an **ε-local randomizer** if for all inputs $x, x' \in \mathcal{X}$ and all events $S \subseteq \mathcal{Y}$,

$$\Pr[R(x) \in S] \leq e^{\varepsilon} \Pr[R(x') \in S]. \tag{17.1}$$

This is the definition of differential privacy (DP) applied to a dataset with one record. In the simplest protocol, everyone speaks once and at the same time.

Definition 17.2 (Noninteractive local protocol)**.** A **noninteractive ε-local protocol** gives each person i an ε-local randomizer R_i. Person i sends $Y_i = R_i(X_i)$, using randomness independent of everyone else's, and the analyst outputs a function of the messages $(Y_1, \dots, Y_n)$.

Local privacy implies central privacy, because the view of the analyst is a vector of messages and only one of them depends on any one record.

Proposition 17.1. *The messages $(Y_1, \dots, Y_n)$ of a noninteractive ε-local protocol, viewed as the output of a mechanism on D, are ε-DP. So is any function of them that the analyst computes.*

Proof.

1. Let $D \sim D'$ be neighbours, which differ by replacing entry j, and take an event S. Write y_{-j} for a value of all the messages except the j-th, and $S_{y_{-j}} = \{y_j : (y_j, y_{-j}) \in S\}$ for the slice of S through it.
2. The other messages have the same distribution under D and D', because they depend only on the other records and on independent randomness. Conditioning on them gives

$$\Pr_{D}[Y \in S] = \mathbb{E}_{Y_{-j}}\left[\Pr[R_j(X_j) \in S_{Y_{-j}}]\right] \le e^{\varepsilon}\, \mathbb{E}_{Y_{-j}}\left[\Pr[R_j(X'_j) \in S_{Y_{-j}}]\right] = e^{\varepsilon} \Pr_{D'}[Y \in S], \qquad (17.2)$$

where the inequality applies Definition 17.1 to each slice. 3. The analyst's output is a post-processing of the messages, so it is ε-DP by the post-processing property of the Laplace mechanism chapter. □

Randomized response, with each person reporting their true bit with probability $e^{\varepsilon}/(1+e^{\varepsilon})$, is an ε-local randomizer. So the first private algorithm of this book was a local protocol.

Protocols can also interact. In a **sequentially interactive** protocol the analyst addresses people one at a time, and the randomizer of person i may depend on the messages before theirs, but each person speaks only once. In a **fully interactive** protocol the analyst may return to a person many times, and all the messages that person ever sends must together form an ε-local randomizer of their record. This chapter proves its bounds for noninteractive protocols only.

17.2. Counting bits, central and local

The cost of moving the barrier shows on the simplest task, in which each person holds a bit $X_i \in \{0,1\}$, and the analyst wants the mean $\mu = \frac{1}{n}\sum_{i=1}^{n} X_i$. Write $\mathrm{Lap}(b)$ for the Laplace distribution with density $\frac{1}{2b}e^{-|z|/b}$. The Laplace mechanism adds $\mathrm{Lap}(\Delta f/\varepsilon)$ noise to a query f whose value changes by at most Δf between neighbours, and it is ε-DP. So a central curator, who sees the bits, can add Laplace noise to their mean.

Proposition 17.2. *For every $\varepsilon > 0$ and $\beta \in (0,1)$, the central estimate $\hat{\mu} = \mu + \mathrm{Lap}(1/(\varepsilon n))$ is ε-DP, and with probability at least $1-\beta$,*

$$|\hat{\mu} - \mu| \le \frac{\ln(1/\beta)}{\varepsilon n}. \qquad (17.3)$$

Proof.

1. Changing one bit changes μ by at most $1/n$, so the Laplace mechanism with $\Delta f = 1/n$ is ε-DP.
2. The Laplace tail is $\Pr[|\mathrm{Lap}(b)| > t] = 2\int_t^{\infty} \frac{1}{2b}e^{-z/b}\,dz = e^{-t/b}$. Setting $b = 1/(\varepsilon n)$ and $t = b\ln(1/\beta)$ makes the tail equal to β.

□

In the local model each person runs randomized response and sends Y_i, equal to X_i with probability $e^\varepsilon/(1+e^\varepsilon)$ and to $1-X_i$ otherwise. A report equals 1 with probability $(1+(e^\varepsilon-1)X_i)/(e^\varepsilon+1)$, which is linear in X_i. Solving for X_i gives the unbiased estimate of the analyst,

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} \frac{(e^\varepsilon+1)\,Y_i - 1}{e^\varepsilon - 1}. \tag{17.4}$$

This is the debiased estimator of the randomized response chapter with bias $2\kappa = (e^\varepsilon-1)/(e^\varepsilon+1)$, written in terms of ε. That chapter proved that, with probability at least $1-\beta$,

$$|\hat{\mu}-\mu| \le \coth\Big(\frac{\varepsilon}{2}\Big)\sqrt{\frac{\ln(2/\beta)}{2n}}, \qquad \coth\Big(\frac{\varepsilon}{2}\Big) \ge \frac{2}{\varepsilon}. \tag{17.5}$$

The two bounds reach an error of α at different sizes. The central bound does so once $n \ge \ln(1/\beta)/(\varepsilon\alpha)$. The local bound does so once $n \ge \coth^2(\varepsilon/2)\ln(2/\beta)/(2\alpha^2)$, a threshold of at least $2\ln(2/\beta)/(\varepsilon^2\alpha^2)$. So the local estimate needs quadratically more data in $1/\alpha$ and in $1/\varepsilon$, as Figure 17.1 shows.

Example 17.1. Take $\varepsilon = 1$, $\alpha = 0.01$ and $\beta = 0.05$. The central estimate needs $n \ge \ln 20/0.01$, that is, 300 people. The local one needs $n \ge \coth^2(0.5)\ln 40/(2\times 10^{-4})$, that is, 86,370 people, about 288 times as many.

This gap is why only companies with hundreds of millions of users have deployed local privacy. With a few thousand users, a local protocol returns mostly noise.

17.3. The price is unavoidable

The local estimator is not merely a poor choice, because no noninteractive local protocol can do better than error of order $1/(\varepsilon\sqrt{n})$. The reason is that a local randomizer shrinks the difference between two input distributions by a factor that depends only on ε, and n independent messages cannot recover more than n such shrunken differences.

Two distances between distributions P and Q on a countable set measure this. The **total variation distance** is $\mathrm{TV}(P,Q) = \frac{1}{2}\sum_y |P(y)-Q(y)|$, the largest gap $P(S)-Q(S)$ over events S. The **chi-squared divergence** is

$$\chi^2(P\,\|\,Q) = \sum_y \frac{(P(y)-Q(y))^2}{Q(y)}, \tag{17.6}$$

where a term in which both $P(y)$ and $Q(y)$ are zero is dropped, and a term in which only $Q(y)$ is zero makes the divergence infinite. The lower bound needs two facts about these two distances. Total variation is at most a constant times the square root of the chi-squared divergence, and the chi-squared divergence behaves simply on products, which is where the n independent messages enter.

Lemma 17.1. *For distributions on a countable set,*

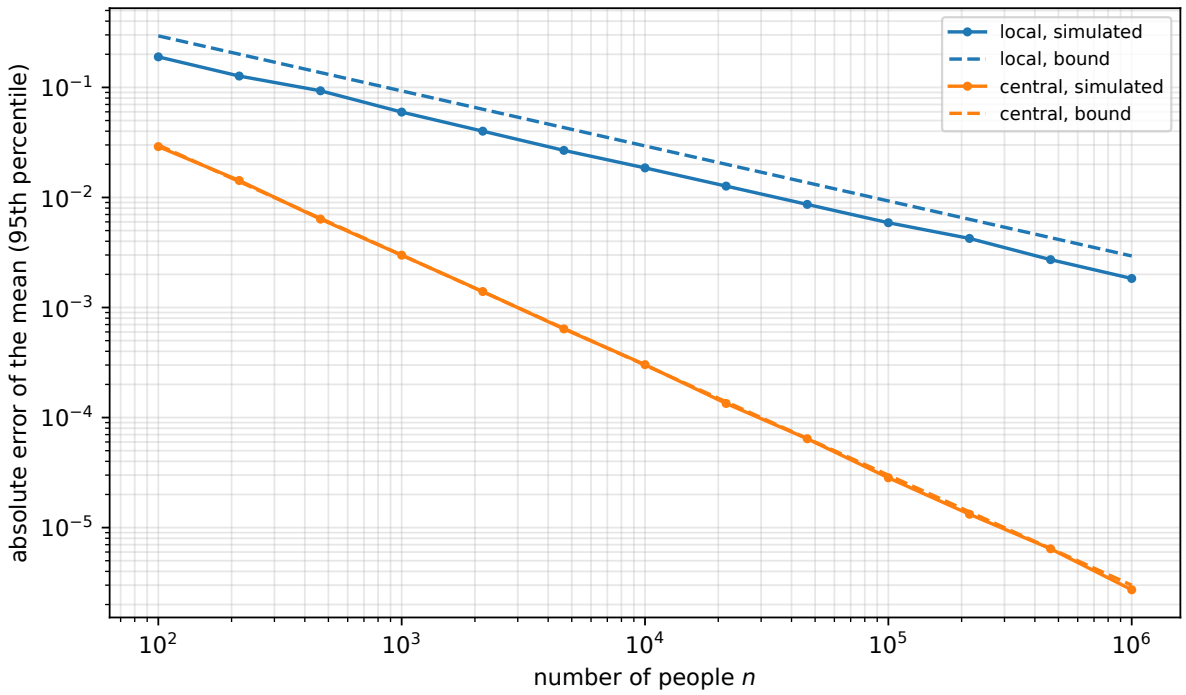


Figure 17.1.: Error of the mean of n bits at $\varepsilon = 1$. Solid lines are the 95th percentile of the error over 2,000 simulated runs with $\mu = 0.3$, dashed lines the bounds of Proposition 17.2 and Equation 17.5 with $\beta = 0.05$. The central error falls as $1/n$ and the local one as $1/\sqrt{n}$, so the gap widens with n.

1. $\mathrm{TV}(P,Q) \le \frac{1}{2}\sqrt{\chi^2(P \,\|\, Q)}$;
2. *for product distributions,* $1+\chi^2(P_1 \times P_2 \,\|\, Q_1 \times Q_2) = (1+\chi^2(P_1 \,\|\, Q_1))(1+\chi^2(P_2 \,\|\, Q_2))$.

Proof.

1. Write $|P(y)-Q(y)| = \frac{|P(y)-Q(y)|}{\sqrt{Q(y)}}\sqrt{Q(y)}$ and apply the Cauchy–Schwarz inequality to the sum. Since $\sum_y Q(y) = 1$, this gives $\sum_y |P(y)-Q(y)| \le \sqrt{\chi^2(P \,\|\, Q)}$.
2. Expanding the square shows $1+\chi^2(P \,\|\, Q) = \sum_y P(y)^2/Q(y)$. For a product, this sum factorizes as $\sum_{y_1} P_1(y_1)^2/Q_1(y_1) \cdot \sum_{y_2} P_2(y_2)^2/Q_2(y_2)$.

□

The key step is a contraction. Feed a local randomizer an input drawn from P_0 or from P_1, and compare the two resulting distributions of its message.

Lemma 17.2. *Let R be an ε-local randomizer with a countable message set, and let P_0, P_1 be distributions on $\mathcal{X}$. Write W_0 and W_1 for the distributions of the message $R(x)$ when x is drawn from P_0 and from P_1. Then*

$$\chi^2(W_0 \,\|\, W_1) \le (e^\varepsilon - 1)^2 \,\mathrm{TV}(P_0, P_1)^2. \tag{17.7}$$

Proof.

1. Fix a message y and write $Q(y \mid x) = \Pr[R(x) = y]$ and $c_y = \inf_x Q(y \mid x)$. By Definition 17.1, every $Q(y \mid x)$ lies in $[c_y, e^\varepsilon c_y]$. If $c_y = 0$, then $Q(y \mid x) = 0$ for every x, and y contributes nothing.
2. Since $\sum_x (P_0(x) - P_1(x)) = 0$, subtracting the midpoint $m_y = (1+e^\varepsilon)c_y/2$ changes nothing:

$$W_0(y) - W_1(y) = \sum_x (Q(y \mid x) - m_y)(P_0(x) - P_1(x)). \tag{17.8}$$

3. Each factor $|Q(y \mid x) - m_y|$ is at most $(e^\varepsilon - 1)c_y/2$, and $\sum_x |P_0(x) - P_1(x)| = 2\,\mathrm{TV}(P_0, P_1)$. So $|W_0(y) - W_1(y)| \le (e^\varepsilon - 1)c_y \,\mathrm{TV}(P_0, P_1)$.
4. The denominator satisfies $W_1(y) \ge c_y$. Hence each term of χ^2 is at most $(e^\varepsilon - 1)^2 c_y \,\mathrm{TV}(P_0, P_1)^2$.
5. Summing over y and using $\sum_y c_y \le \sum_y Q(y \mid x) = 1$ for any fixed x gives the bound.

□

Now let the data be random. If each X_i is drawn independently from $\mathrm{Bern}(p)$, the Bernoulli distribution with mean p, then an accurate estimate of the mean must tell $p = \frac{1}{2} - \alpha$ from $p = \frac{1}{2} + \alpha$.

Theorem 17.1 (Lower bound for local counting)**.** *Let $\alpha \in (0, \frac{1}{2}]$, and let $X_1, \dots, X_n$ be independent draws from* $\mathrm{Bern}(\frac{1}{2}+\alpha)$ *or from* $\mathrm{Bern}(\frac{1}{2}-\alpha)$*. Write P_+ and P_- for the two distributions of the messages of a noninteractive ε-local protocol with countable message sets. Then*

$$\mathrm{TV}(P_+, P_-) \le \frac{1}{2}\sqrt{\exp(4n(e^\varepsilon - 1)^2\alpha^2) - 1}. \tag{17.9}$$

Consequently, if an estimate $\tilde{\mu}$ computed from the messages satisfies $|\tilde{\mu} - p| < \alpha$ with probability at least $3/4$ in both cases, then

$$n \ge \frac{\ln 2}{4(e^\varepsilon - 1)^2\alpha^2}, \qquad \text{and for } \varepsilon \le 1, \quad n \ge \frac{1}{18\,\varepsilon^2\alpha^2}. \tag{17.10}$$

Proof.

1. The two input distributions have $\mathrm{TV}(\mathrm{Bern}(\frac{1}{2}+\alpha), \mathrm{Bern}(\frac{1}{2}-\alpha)) = 2\alpha$. By Lemma 17.2, each person's message satisfies $\chi^2 \le 4(e^\varepsilon - 1)^2\alpha^2$ between the two cases.
2. The messages are independent, because the records and the coins are. By the product rule of Lemma 17.1 and $1 + z \le e^z$, we get $1 + \chi^2(P_+ \,\|\, P_-) \le \exp(4n(e^\varepsilon - 1)^2\alpha^2)$. The first part of the same lemma gives the bound on TV.
3. Take the test that answers "$p = \frac{1}{2} + \alpha$" when $\tilde{\mu} \ge \frac{1}{2}$. In each case it errs only if $|\tilde{\mu} - p| \ge \alpha$, so each error probability is at most $\frac{1}{4}$ and the two add up to at most $\frac{1}{2}$. For any test, the error probabilities add up to at least $1 - \mathrm{TV}(P_+, P_-)$, because the chance of answering "+" differs between the cases by at most TV. So $\mathrm{TV}(P_+, P_-) \ge \frac{1}{2}$.
4. Combining steps 2 and 3 gives $\exp(4n(e^\varepsilon - 1)^2\alpha^2) \ge 2$, which is the first bound on n.
5. For $\varepsilon \le 1$, convexity of e^ε gives $e^\varepsilon - 1 \le (e-1)\varepsilon$, and $\ln 2/(4(e-1)^2) = 0.0587 > 1/18$.

□

The theorem is about data drawn at random, but it also constrains estimates of the empirical mean on fixed data. Suppose a protocol is within $\alpha/2$ of the empirical mean with probability at least $7/8$ on every dataset. When $n \ge 2\ln(16)/\alpha^2$, Hoeffding's inequality puts the empirical mean strictly within $\alpha/2$ of p with probability at least $7/8$. So the protocol is strictly within α of p with probability at least $3/4$, and the theorem applies to it. This reduction says nothing about smaller n, and Chan, Shi and Song (2012) give a direct argument for fixed data.

For $\varepsilon \le 1$, comparing the theorem with Equation 17.5 shows that randomized response is optimal up to a constant factor. The central estimate, which needs only about $1/(\varepsilon\alpha)$ people, is out of reach of every noninteractive local protocol.

17.4. Repeated questions: RAPPOR

Google's first large deployment, RAPPOR, collected statistics from the Chrome browser. It met a problem that a one-off survey never has. The same question goes to the same person every day, and fresh randomness each day leaks the answer.

Example 17.2 (Averaging attack)**.** Take $\varepsilon = \ln 4$, so that randomized response tells the truth with probability $4/5$. A single "yes" from a person whose answer is "no" happens with probability $1/5$, which leaves good deniability. Now ask the same question on 50 days with fresh coins, and suppose 40 reports say "yes". If the answer is "no", this happens with probability $\binom{50}{40}(1/5)^{40}(4/5)^{10} = 1.21 \times 10^{-19}$. If it is "yes", it happens with probability 0.140. The likelihood ratio is $4^{30} \approx 1.15 \times 10^{18}$, and the deniability is gone.

The attack is composition at work. Basic composition, from the Laplace mechanism chapter, says that k answers with fresh coins are $k\varepsilon$-DP, which here is $50 \ln 4 = 69.3$. The attack meets this bound exactly, because the all-"yes" outcome has likelihood ratio 4^{50}. So the fix is to randomize once and repeat.

Definition 17.3 (Memoization)**.** A person who is asked a question for the first time runs their randomizer and stores the result. When asked the same question again, they send the stored result.

Memoization keeps the privacy cost of a single answer, however often the question comes.

Proposition 17.3. *If the stored result comes from an ε-local randomizer, then any number of memoized answers to the same question is ε-locally private.*

Proof. The sequence of answers is a function of the single stored result, so the claim follows from post-processing. □

Memoization protects honest repetition, not repetition by an adversarial analyst. For example, an analyst can ask for "your PIN" and then for "your PIN reversed", two questions with the same information that each get a fresh randomization.

RAPPOR handles values from any finite domain, not just bits, so it first encodes a value as a bit vector.

Definition 17.4 (Bloom filter)**.** Fix a length K and h hash functions $H_1, \dots, H_h \colon \mathcal{X} \to \{1, \dots, K\}$. The Bloom filter of a value v is the vector $B \in \{0,1\}^K$ with $B_j = 1$ exactly when $H_l(v) = j$ for some l.

With $h = 1$ and no collisions the Bloom filter is a one-hot encoding of v. In general, B has at most h ones. RAPPOR then applies two layers of randomized response, bit by bit.

Definition 17.5 (RAPPOR)**.** Fix $\nu \in (0,1)$, the noise level of the permanent layer, and probabilities $p_0 < p_1$ in $(0,1)$, the chances of reporting a one in the instantaneous layer. A person with value v and Bloom filter B of length K, from Definition 17.4, does the following.

1. **Permanent randomized response.** Once, they draw $B' \in \{0,1\}^K$ with independent coordinates, where B'_j is 1 with probability $\nu/2$, 0 with probability $\nu/2$, and B_j otherwise. They memoize B'.
2. **Instantaneous randomized response.** Each time they report, they send a fresh $Y \in \{0,1\}^K$ with independent coordinates, where $Y_j = 1$ with probability p_1 if $B'_j = 1$ and with probability p_0 if $B'_j = 0$.

The permanent layer carries the long-term guarantee. It holds against an analyst who sees every report the person ever sends, because all of them come from the one memoized vector.

Theorem 17.2. *Run RAPPOR with h hash functions and permanent noise level ν. As a function of the value v, the memoized vector B' is an ε_∞-local randomizer with*

$$\varepsilon_\infty = 2h \ln \frac{1-\nu/2}{\nu/2}. \tag{17.11}$$

So is the whole sequence of reports Y, however long.

Proof.

1. Let v, v' have Bloom filters $B, \tilde{B}$. Each has at most h ones, so they differ in at most $2h$ coordinates.
2. Fix an outcome $b \in \{0,1\}^K$. Coordinates are independent, so $\Pr[B' = b]$ is a product over j. The factors at coordinates where $B_j = \tilde{B}_j$ are equal for v and v' and cancel in the ratio.
3. Each remaining factor is $1-\nu/2$ or $\nu/2$, so each ratio is at most $(1-\nu/2)/(\nu/2)$, and there are at most $2h$ of them. By the pointwise lemma of the randomized response chapter, this bound on single outcomes gives ε_∞-privacy.
4. The reports depend on v only through B', so they are a post-processing of it.

□

The instantaneous layer gives a stronger guarantee for an analyst who sees one report. With both layers, a reported bit is 1 with probability $p_1^* = (1-\frac{\nu}{2})p_1 + \frac{\nu}{2}p_0$ when $B_j = 1$ and $p_0^* = \frac{\nu}{2}p_1 + (1-\frac{\nu}{2})p_0$ when $B_j = 0$. Then $p_1^* > p_0^*$, because $p_1^* - p_0^* = (1-\nu)(p_1 - p_0)$.

Theorem 17.3. *Run RAPPOR with h hash functions, and let p_1^* and p_0^* be the combined probabilities above. A single report Y, as a function of v, is an ε_1-local randomizer with*

$$\varepsilon_1 = h \ln \frac{p_1^*(1-p_0^*)}{p_0^*(1-p_1^*)}. \tag{17.12}$$

Proof.

1. Given B, the coordinates of Y are independent, with $\Pr[Y_j = 1] = p_1^*$ or p_0^* according to B_j. As before, coordinates where $B_j = \tilde{B}_j$ cancel.
2. Let A be the set of coordinates with $B_j = 1$ and $\tilde{B}_j = 0$, and A' the set with the reverse. Both have at most h elements, since each lies inside the ones of one filter.
3. On A the ratio is $p_1^*/p_0^* > 1$ if $y_j = 1$ and $(1-p_1^*)/(1-p_0^*) < 1$ if $y_j = 0$, so it is at most p_1^*/p_0^*. On A' it is at most $(1-p_0^*)/(1-p_1^*)$ by the same reasoning.
4. Multiplying, the ratio is at most $\left(p_1^*(1-p_0^*)/(p_0^*(1-p_1^*))\right)^h$, and the pointwise lemma finishes the proof.

□

Typical parameters show how far apart the two guarantees can be.

Example 17.3. With $h = 2$, $\nu = 1/2$, $p_0 = 1/2$ and $p_1 = 3/4$, the long-term guarantee is $\varepsilon_\infty = 4 \ln 3 = 4.39$. The combined probabilities are $p_1^* = 0.6875$ and $p_0^* = 0.5625$, so a single report has $\varepsilon_1 = 1.07$.

The second layer has one more purpose, which local privacy does not capture. A memoized B' of a few hundred bits is almost surely unique, so two reports that both reveal it link to the same person, as a cookie would. Fresh noise on each report breaks that link for an analyst who sees few reports. An analyst who sees many reports can average the noise away and recover B', but then still faces the ε_∞ guarantee of Theorem 17.2.

The analyst decodes the reports by debiasing the bit counts and fitting candidate values against them, in the spirit of Equation 17.4. Memoization fails when the value drifts. For example, a person's age in days changes daily, each new value needs a new B', and k distinct values cost $k\,\varepsilon_\infty$ by composition.

17.5. Values no one knew: Apple

Apple deployed local privacy on its phones and computers. Around the randomizers it put measures outside the model of DP, for example it discards network addresses and limits the number of reports per day. Independent researchers later measured from the software of macOS that each report had ε of 1 or 2, but that a day could spend up to 16.

Most of Apple's protocols build on a private count mean sketch, a hashed table that estimates frequencies. A device hashes its value into a short table with one of several hash functions, and randomizes the table before it sends it. A variant with the Hadamard transform lets each device send a single bit. That matters on a phone, where a report must cost a few bytes at most.

The new problem Apple solved is **discovery**. A hash maps a value to its encoding, but the encoding cannot be inverted. So to estimate how often a word is typed, the analyst must already have the word in its dictionary and test its encoding against the reports. Slang and new words are exactly the values that no dictionary holds, and a search of all strings is hopeless, since there are $26^8 \approx 2.1 \times 10^{11}$ strings of eight letters.

The **sequence fragment puzzle** breaks a word into pieces short enough to enumerate. Take, for illustration, eight-letter words and fragments of two letters, and fix an eight-bit hash function H from words to $\{0, \dots, 255\}$. A person with an eight-letter word w picks one of the starting positions $j \in \{1, 3, 5, 7\}$ uniformly at random. They send j together with a private report of the *puzzle piece* $(w_j w_{j+1}, H(w))$, the two letters at position j and the hash of the whole word. The pieces come from a set of $26^2 \times 256 = 173{,}056$ candidates, small enough for the analyst to estimate the frequency of every one at each position.

The position needs no protection, because the person chooses it at random, independently of the word, as the next result shows.

Proposition 17.4. *If the private report of the pair is an ε-local randomizer, then the whole message (j, report) is ε-locally private.*

Proof. The position j is drawn independently of w. Conditional on j, the message is an ε-local randomizer of w. For an event S, write S_j for the set of reports r with $(j, r) \in S$. Then $\Pr[(j, \text{report}) \in S \mid w] = \sum_j \frac{1}{4} \Pr[\text{report} \in S_j \mid w, j]$ is at most e^ε times the same sum for any other word. □

An example shows how the analyst puts the pieces back together.

Example 17.4. Suppose many people type "snowfall", and $H(\text{snowfall}) = 117$. Their messages are spread over $(1, \text{sn}, 117)$, $(3, \text{ow}, 117)$, $(5, \text{fa}, 117)$ and $(7, \text{ll}, 117)$. Each of the four pairs is frequent at its position. Pieces that share the hash 117 are joined in order of position, giving the candidate "snowfall". A second round of ordinary frequency estimation over the discovered words then counts it, and it removes candidates that a hash collision assembled from different words.

With this method Apple found abbreviations and slang that no dictionary held, and measured the frequency of each emoji, which differed between English and French keyboards.

17.6. Numbers in one bit: Microsoft

Microsoft collects usage telemetry from Windows, such as how long people spend in each application. These values are numbers in a range $[0, m]$, and the analyst wants their mean μ. The Laplace mechanism works locally. Person i sends $Y_i = X_i + \text{Lap}(m/\varepsilon)$, which is ε-local because the identity map on $[0, m]$ changes by at most m. The mean of the reports then has standard deviation exactly $\sqrt{2}\, m/(\varepsilon\sqrt{n})$, which is optimal for local protocols by Theorem 17.1 applied to values in $\{0, m\}$. But each report is a real number of size about m/ε, so it costs many bits, while a single bit sent with a probability that grows linearly with the value suffices.

Definition 17.6 (One-bit mean protocol)**.** Person i, with $X_i \in [0, m]$, sends a bit Y_i with

$$\Pr[Y_i = 1] = \frac{1}{e^\varepsilon + 1} + \frac{X_i}{m} \cdot \frac{e^\varepsilon - 1}{e^\varepsilon + 1}. \tag{17.13}$$

The analyst outputs

$$\hat{\mu} = \frac{m}{n} \sum_{i=1}^{n} \frac{(e^\varepsilon + 1)\, Y_i - 1}{e^\varepsilon - 1}. \tag{17.14}$$

At $X_i = 0$ and $X_i = m$ this is randomized response, and between them it interpolates linearly. The estimate of the analyst inverts that linear relation, as Equation 17.4 did for bits, and it keeps the privacy and the accuracy of randomized response.

Theorem 17.4. *The one-bit protocol is ε-locally private, and $\hat{\mu}$ is unbiased. With probability at least $1 - \beta$,*

$$|\hat{\mu} - \mu| \le m \coth\Big(\frac{\varepsilon}{2}\Big) \sqrt{\frac{\ln(2/\beta)}{2n}}, \tag{17.15}$$

and the standard deviation of $\hat{\mu}$ is at most $m \coth(\varepsilon/2)/(2\sqrt{n})$.

Proof.

1. *Privacy.* For every $x \in [0, m]$, $\Pr[Y = 1]$ lies between $1/(e^\varepsilon + 1)$ and $e^\varepsilon/(e^\varepsilon + 1)$, and so does $\Pr[Y = 0] = 1 - \Pr[Y = 1]$. The ratio of two numbers in that interval is at most e^ε, and the pointwise lemma applies.
2. *Unbiasedness.* Substituting $\mathbb{E}[Y_i] = \Pr[Y_i = 1]$ gives $(e^\varepsilon + 1)\mathbb{E}[Y_i] - 1 = (X_i/m)(e^\varepsilon - 1)$, so each term of the sum has mean X_i/m, and $\mathbb{E}[\hat{\mu}] = \mu$.
3. *Error.* Each term $Z_i = m((e^\varepsilon + 1)Y_i - 1)/(e^\varepsilon - 1)$ takes the two values $-m/(e^\varepsilon - 1)$ and $me^\varepsilon/(e^\varepsilon - 1)$, an interval of length $m(e^\varepsilon + 1)/(e^\varepsilon - 1) = m \coth(\varepsilon/2)$. The Z_i are independent, and Hoeffding's inequality for their average gives the bound, as in the randomized response chapter.
4. *Variance.* A bit has variance at most $1/4$, so $\mathrm{Var}(Z_i) \le m^2 \coth^2(\varepsilon/2)/4$, and the average of n independent terms has variance at most $m^2 \coth^2(\varepsilon/2)/(4n)$.

□

Since $\coth(\varepsilon/2)/2 \approx 1/\varepsilon$ for small ε, one bit loses nothing against the $\sqrt{2}/\varepsilon$ of the Laplace protocol. At $\varepsilon = 1$ the one-bit standard deviation is at most 0.765 times the Laplace one.

17.6.1. Memoization with drifting values

Daily usage is never exactly the same two days running, so the memoization of RAPPOR does not apply. And rounding to bins before memoizing poses a dilemma. Narrow bins change often and force new randomizations, while wide bins make each report wrong by up to half a bin. Randomized rounding solves the second problem. It rounds each value to 0 or to m at random, with probabilities chosen so that the rounded value is right on average, so two bins suffice without bias.

Definition 17.7 (Memoized one-bit protocol)**.** Before any data arrive, person i does three things, independently of their data and of each other.

1. They run the one-bit protocol on the input 0 and store the bit $A_i(0)$.
2. They run it on the input m and store the bit $A_i(m)$.
3. They draw an offset u_i uniformly from $[0, m)$.

On a day when their value is $x \in [0, m]$, they round it to $r_i(x) = m$ if $x + u_i \ge m$ and to $r_i(x) = 0$ otherwise, and they send $A_i(r_i(x))$.

The random offset is what makes the rounding right on average.

Lemma 17.3. *For every $x \in [0, m]$, $\Pr[r_i(x) = m] = x/m$, so $\mathbb{E}[r_i(x)] = x$.*

Proof. The offset is uniform on $[0, m)$, so $\Pr[u_i \ge m - x] = (m - (m - x))/m = x/m$. □

Unbiased rounding, together with stored bits that are themselves outputs of the one-bit protocol, makes the bit of each day look like a fresh run of that protocol.

Theorem 17.5. *On each day, the bit sent by the memoized protocol has exactly the distribution of the one-bit protocol run on that day's value x. So each day's bit is ε-locally private, and the estimator of Definition 17.6, applied to one day's bits, is unbiased with the error bound of Theorem 17.4.*

Proof. The offset and the two stored bits are independent. By Lemma 17.3,

$$\Pr[\text{bit} = 1] = \frac{x}{m} \cdot \frac{e^{\varepsilon}}{e^{\varepsilon}+1} + \left(1 - \frac{x}{m}\right)\frac{1}{e^{\varepsilon}+1} = \frac{1}{e^{\varepsilon}+1} + \frac{x}{m} \cdot \frac{e^{\varepsilon}-1}{e^{\varepsilon}+1}, \tag{17.16}$$

which is the probability in Definition 17.6. Different people use independent coins, so the bits of one day are independent, as the error bound requires. □

Over many days, the bits depend on the data only through the sequence of rounded values. While the value of a person stays on one side of their threshold $m - u_i$, they repeat one stored bit, and the analyst learns nothing new. A crossing shows in the reports only if the two stored bits differ, which happens with probability $(1 + e^{2\varepsilon})/(1 + e^{\varepsilon})^2$, or 0.607 at $\varepsilon = 1$. Then the analyst learns the day on which the value crossed a threshold drawn uniformly at random. No ε-local guarantee for arbitrary sequences of values covers this leak, so the deployed protocol adds a further layer of randomness to the sent bits to blur the crossing points.

17.7. Summary

Table 17.1.: The chapter's results.

Result	Statement
Proposition 17.1	a noninteractive ε-local protocol is ε-DP in the central sense
Proposition 17.2	central mean of bits: error $\ln(1/\beta)/(\varepsilon n)$
Equation 17.5	local mean of bits by randomized response: error $\coth(\varepsilon/2)\sqrt{\ln(2/\beta)/(2n)}$
Lemma 17.2	an ε-local randomizer shrinks χ^2 to at most $(e^{\varepsilon}-1)^2\,\mathrm{TV}^2$
Theorem 17.1	every noninteractive local protocol needs $n \geq \ln 2/(4(e^{\varepsilon}-1)^2\alpha^2)$
Proposition 17.3	memoized answers to one question cost ε once
Theorem 17.2	RAPPOR's memoized vector is $2h\ln((1-\nu/2)/(\nu/2))$-locally private
Theorem 17.3	one RAPPOR report is $h\ln\left(p_1^*(1-p_0^*)/(p_0^*(1-p_1^*))\right)$-locally private
Proposition 17.4	a fragment report with a public random position keeps its ε
Theorem 17.4	one bit per person: unbiased, ε-local, error $m\coth(\varepsilon/2)\sqrt{\ln(2/\beta)/(2n)}$

Result	Statement
Theorem 17.5	memoizing two bits with a random offset reproduces the one-bit protocol each day

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18. The Shuffle Model and Central Deployments

The local model of the previous chapter trusts no one, and pays with an error $\sqrt{n}$ times larger than the central model's. This chapter first studies a middle ground for differential privacy (DP), the shuffle model, in which a party that only mixes messages lets each person add far less noise. It then turns to the central model, where a company holds the data, and studies a query service for outside analysts and two public data releases.

18.1. Encode, shuffle, analyze

Google's Prochlo system started from a failure of local privacy. Its designers wanted to count which of a thousand applications call which of a hundred system interfaces, 10^5 pairs in all. By their estimate, a local protocol such as RAPPOR would need reports from about a trillion people to count them usefully, more people than exist.

Their architecture has three stages, **encode, shuffle, analyze**. Each person *encodes* their data on their own device, for example with a local randomizer. A *shuffler* collects the encoded messages, removes their metadata such as arrival times and network addresses, and sends them on in a uniformly random order. The *analyst* sees only the shuffled batch and decodes statistics from it. So the shuffler has one job, which is to break the link between a message and its sender.

The users must trust the shuffler not to reveal the order. Prochlo runs it inside trusted hardware, an Intel SGX enclave, so trust rests on the hardware vendor. Side-channel attacks such as Spectre have shown that such hardware can leak. Another choice is a mix network of several servers, each of which shuffles and re-encrypts the batch. Then the order stays hidden unless all of them collude.

18.2. The shuffle model

Prochlo led to a formal model between the local and central ones. In it, the adversary sees the shuffled messages but not who sent which.

Definition 18.1 (Shuffle model)**.** A protocol in the **shuffle model** gives each person the same randomizer $R\colon \mathcal{X} \to \mathcal{Y}$. Person i sends $Y_i = R(X_i)$, using randomness independent of everyone else's, and the **shuffler** outputs $(Y_{\pi(1)}, \dots, Y_{\pi(n)})$ for a permutation π drawn uniformly at random. The protocol is (ε, δ)**-DP in the shuffle model** if this shuffled output, as a mechanism on the dataset D, is (ε, δ)-DP in the central sense.

The uniform order adds no information, so the shuffled output carries the same information as the multiset of messages. A local protocol is also a shuffle protocol with the same guarantee, because the shuffled output is a post-processing of the messages. The interesting protocols are those whose messages are *not* private one by one but become private once mixed.

The benchmark task is again a sum of bits. In the central model, a count plus Laplace noise of scale $1/\varepsilon$ has error of order $1/\varepsilon$. In the local model each person adds noise of that size, so the n independent noises add up to a standard deviation of order $\sqrt{n}/\varepsilon$, and by the previous chapter's lower bound no local protocol does better. The question is how close the shuffle model can get to $1/\varepsilon$.

18.3. Amplification by shuffling

The first answer holds for any local randomizer, because shuffling makes it more private, just as subsampling did in an earlier chapter. In the form first proved, a message that is ε_0-private on its own gives a shuffled batch whose parameter is smaller by a factor of order $\sqrt{n}$.

Theorem 18.1 (Amplification by shuffling)**.** *Let $n \geq 1000$, $\delta \in (0, \frac{1}{100})$, and let R be an ε_0-local randomizer with $0 < \varepsilon_0 < \frac{1}{2}$. Then the shuffled output of $R(X_1), \ldots, R(X_n)$ is (ε, δ)-DP with*

$$\varepsilon = 12\,\varepsilon_0 \sqrt{\frac{\ln(1/\delta)}{n}}. \tag{18.1}$$

The proof is in Erlingsson and coauthors (2019, Theorem 7 and Corollary 9). When everyone uses the same randomizer, shuffling the records before randomizing them gives the same output distribution, and the proof works in that form. It moves the one differing record to the front, swaps it into a uniformly random position, and then bounds how much each later message reveals about that position.

The theorem applies only to randomizers that are already very private, but few deployments want $\varepsilon_0 \leq \frac{1}{2}$, because local accuracy is worst there. Feldman, McMillan and Talwar (2021) cover the useful regime, ε_0 of order $\ln n$, where ε is still a constant. For every $\varepsilon_0 \leq \ln\left(n/(16\ln(2/\delta))\right)$, they give $\varepsilon = O((1 - e^{-\varepsilon_0})\sqrt{e^{\varepsilon_0}\ln(1/\delta)/n})$. In their proof, the message of each other person is, with probability $e^{-\varepsilon_0}$, a "clone" of what the differing person might have sent, and the count of clones hides that person. The next section carries out this idea in full for one simple randomizer.

18.4. Counting bits in the shuffle model

For a sum of bits, a direct protocol reaches error of order $\sqrt{\ln(1/\delta)}/\varepsilon$, which does not depend on n. Its randomizer is randomized response in which almost everyone tells the truth.

Definition 18.2 (Shuffled randomized response)**.** Fix a coin probability $\nu \in (0, 1]$. Person i, with bit X_i, sends $Y_i = X_i$ with probability $1 - \nu$, and otherwise sends a fresh fair coin. The analyst sees the shuffled reports, counts their ones as $Z = \sum_{i=1}^{n} Y_i$, and estimates the count $c = \sum_{i=1}^{n} X_i$ by

$$\tilde{c} = \frac{Z - n\nu/2}{1-\nu}. \tag{18.2}$$

In the local model ν would be a constant. Here it is of order $\ln(1/\delta)/(\varepsilon^2 n)$, so only about $\ln(1/\delta)/\varepsilon^2$ people in total send coins. These coins hide each single person, as the clones did above. Two lemmas prepare the proof, and both use $\tau = \tanh(\varepsilon/2)$, the number for which $(1+\tau)/(1-\tau) = e^{\varepsilon}$. The first says that, when s is large, the number of ones among s fair coins hides whether one more bit, of any distribution, is added to it.

Lemma 18.1. *Let $\varepsilon > 0$, let $\tau = \tanh(\varepsilon/2)$, and let $s \geq 2/\tau$ be an integer. Let N be binomial with s trials and success probability $\frac{1}{2}$, and let B and B' be Bernoulli variables independent of N, with any parameters. Then for every set S of integers,*

$$\Pr[N + B \in S] \leq e^{\varepsilon} \Pr[N + B' \in S] + 2e^{-s\tau^2/8}. \tag{18.3}$$

Proof.

1. Write $P(k) = \Pr[N = k] = \binom{s}{k} 2^{-s}$. For $1 \leq k \leq s$, the ratio of consecutive probabilities is $P(k-1)/P(k) = k/(s+1-k)$.
2. Let K be the set of integers k with $|k - \frac{s+1}{2}| \leq \frac{s+1}{2}\tau$. Writing $k = \frac{s+1}{2}(1+v)$ with $|v| \leq \tau$, the ratio becomes $(1+v)/(1-v)$, which lies in $[e^{-\varepsilon}, e^{\varepsilon}]$. Since $\tau < 1$, every $k \in K$ satisfies $1 \leq k \leq s$.
3. For $k \in K$ and any Bernoulli parameter b, $\Pr[N + B = k] = (1-b)P(k) + bP(k-1)$ is at most $\max(P(k), P(k-1))$. By step 2 this is at most $e^{\varepsilon} \min(P(k), P(k-1))$, which is at most $e^{\varepsilon} \Pr[N + B' = k]$.
4. Outside K, use $|N + B - \frac{s+1}{2}| \leq |N - \frac{s}{2}| + \frac{1}{2}$. So $N + B \notin K$ requires $|N - \frac{s}{2}| > \frac{(s+1)\tau - 1}{2}$, and this threshold is at least $s\tau/4$ because $s\tau \geq 2$. Hoeffding's inequality for s fair coins gives $\Pr[N + B \notin K] \leq 2\exp(-2(s\tau/4)^2/s) = 2e^{-s\tau^2/8}$.
5. Split S into $S \cap K$, bounded by step 3, and the rest, bounded by step 4.

□

The second lemma makes sure that enough people send coins. It is a lower tail bound for the binomial distribution, sharper than Hoeffding's inequality when the mean is small.

Lemma 18.2. *Let N be binomial with mean μ. Then*

$$\Pr\left[N \leq \frac{\mu}{2}\right] \leq \exp\left(-\frac{(1 - \ln 2)\,\mu}{2}\right). \tag{18.4}$$

Proof.

1. Let N have m trials with success probability p, so $\mu = mp$. For $\lambda > 0$, Markov's inequality applied to $e^{-\lambda N}$ gives $\Pr[N \leq a] \leq e^{\lambda a}\, \mathbb{E}[e^{-\lambda N}]$.
2. The trials are independent, so $\mathbb{E}[e^{-\lambda N}] = (1 - p + pe^{-\lambda})^m$. Using $1 + z \leq e^z$, this is at most $\exp\left(\mu(e^{-\lambda} - 1)\right)$.

3. Take $a = \mu/2$ and $\lambda = \ln 2$. The exponent becomes $\mu \ln 2/2 - \mu/2$.

□

The privacy theorem chooses ν so that, by Lemma 18.2, enough people send coins with high probability. By Lemma 18.1, those coins then hide the one person in question.

Theorem 18.2 (Privacy of shuffled randomized response)**.** *Let $\varepsilon > 0$, $\delta \in (0, 1)$, $\tau = \tanh(\varepsilon/2)$ and $n \geq 1 + 16 \ln(4/\delta)/\tau^2$, and set*

$$\nu = \frac{16 \ln(4/\delta)}{\tau^2 (n-1)}. \tag{18.5}$$

Then shuffled randomized response is (ε, δ)-DP in the shuffle model.

Proof.

1. *Reduce to the count.* Given the count Z, every arrangement of the shuffled bits with Z ones is equally likely. So the shuffled output and Z are functions of each other, up to independent randomness, and it suffices to prove that Z is (ε, δ)-DP.
2. *Isolate one person.* The count does not depend on the order of the people, so we may take neighbours $D \sim D'$ that differ by replacing the record of person 1. Let J be the set of persons $i \geq 2$ who send coins. Its law does not depend on the data, and given J,

$$Z = C_J + N + Y_1, \qquad C_J = \sum_{i \geq 2,\ i \notin J} X_i, \tag{18.6}$$

where N counts the ones among the $|J|$ fair coins. Both C_J and the law of N are the same under D and D', and only the Bernoulli variable Y_1 changes. 3. *Enough coins.* Let $s_0 = 8 \ln(4/\delta)/\tau^2$. The size $|J|$ is binomial with mean $(n-1)\nu = 2s_0$. By Lemma 18.2, $\Pr[|J| < s_0] \leq \exp(-(1 - \ln 2)\, 8 \ln(4/\delta)/\tau^2)$. Since $8(1 - \ln 2) > 2$ and $\tau < 1$, this is at most $(\delta/4)^2 \leq \delta/4$. 4. *Each large J is private.* If $|J| \geq s_0$, then $|J|\tau \geq 8 \ln 4/\tau > 2$, so Lemma 18.1 applies. Its error term is $2e^{-|J|\tau^2/8} \leq 2e^{-\ln(4/\delta)} = \delta/2$. Shifting by the constant C_J preserves the inequality, so for every event S, $\Pr_D[Z \in S \mid J] \leq e^{\varepsilon} \Pr_{D'}[Z \in S \mid J] + \delta/2$. 5. *Average over J.* Summing over sets with $|J| \geq s_0$ and adding the probability of the others,

$$\Pr_D[Z \in S] \leq e^{\varepsilon} \Pr_{D'}[Z \in S] + \frac{\delta}{2} + \frac{\delta}{4} \leq e^{\varepsilon} \Pr_{D'}[Z \in S] + \delta. \tag{18.7}$$

□

The coins also add error to the count. But only about $n\nu$ people send them, so the error depends on $n\nu$, which Equation 18.5 keeps independent of n.

Theorem 18.3 (Accuracy of shuffled randomized response)**.** *For every dataset, $\tilde{c}$ is an unbiased estimate of c with variance at most $n\nu/(2(1-\nu)^2)$. With ν as in Equation 18.5 and $\beta \in (0,1)$, with probability at least $1-\beta$,*

$$|\tilde{c} - c| \le \frac{1}{1-\nu}\sqrt{\frac{n\nu}{2\beta}} \le \frac{4}{(1-\nu)\,\tau}\sqrt{\frac{\ln(4/\delta)}{\beta}}. \tag{18.8}$$

For $\varepsilon \le 1$ and $\nu \le \frac{1}{2}$, the right-hand side is at most $18\sqrt{\ln(4/\delta)/\beta}\,/\varepsilon$.

Proof.

1. Each report is 1 with probability $(1-\nu)X_i + \nu/2$, so $\mathbb{E}[Z] = (1-\nu)c + n\nu/2$, and $\tilde{c}$ is unbiased.
2. Each report is a Bernoulli variable with parameter $\nu/2$ or $1-\nu/2$, so its variance is $\frac{\nu}{2}(1-\frac{\nu}{2}) \le \frac{\nu}{2}$. The reports are independent, so $\mathrm{Var}(Z) \le n\nu/2$ and $\mathrm{Var}(\tilde{c}) \le n\nu/(2(1-\nu)^2)$.
3. Chebyshev's inequality gives the first bound. For the second, $n\nu = 16\ln(4/\delta)\,n/(\tau^2(n-1)) \le 32\ln(4/\delta)/\tau^2$ when $n \ge 2$.
4. The function tanh is concave on $[0,\infty)$, so $\tau \ge \tanh(\frac{1}{2})\,\varepsilon = 0.462\,\varepsilon$ for $\varepsilon \le 1$. With $1/(1-\nu) \le 2$, the constant is $8/0.462 < 18$.

□

So the shuffled count has error $O(\sqrt{\ln(1/\delta)}/\varepsilon)$, which does not grow with n. That is within a factor $\sqrt{\ln(1/\delta)}$ of the central model and far from the local model's $\sqrt{n}/\varepsilon$, as Figure 18.1 shows over n and the example below shows at one setting.

Example 18.1. Take $\varepsilon = 1$, $\delta = 10^{-6}$ and $n = 10^6$. Then $\tau = 0.4621$, and Equation 18.5 gives $\nu = 0.001139$, so about 1,139 people send coins. The standard deviation of $\tilde{c}$ is at most 23.9. Local randomized response at the same ε has standard deviation $\sqrt{n}\,e^{1/2}/(e-1) = 959.5$, and central Laplace noise has $\sqrt{2} = 1.414$.

The price is trust in the shuffler. Each message on its own is randomized response that keeps the true bit with probability $1-\nu/2$, so it is only ε_0-locally private with $\varepsilon_0 = \ln((2-\nu)/\nu) = 7.47$ in the example. So a shuffler that leaked the order would expose almost every bit. This ε_0 is below $\ln\left(n/(16\ln(2/\delta))\right) = 8.37$, so the protocol lies in the range that the clone analysis of the previous section covers.

18.5. A query service: LinkedIn

LinkedIn's audience engagements service lets advertisers ask how members in a target audience engage with content, such as which articles or topics they click on most. It runs in the central model, so LinkedIn holds the data and answers each query with a private mechanism. This section covers two results of the research for it, a sharper analysis of the exponential mechanism and a fast way to run it.

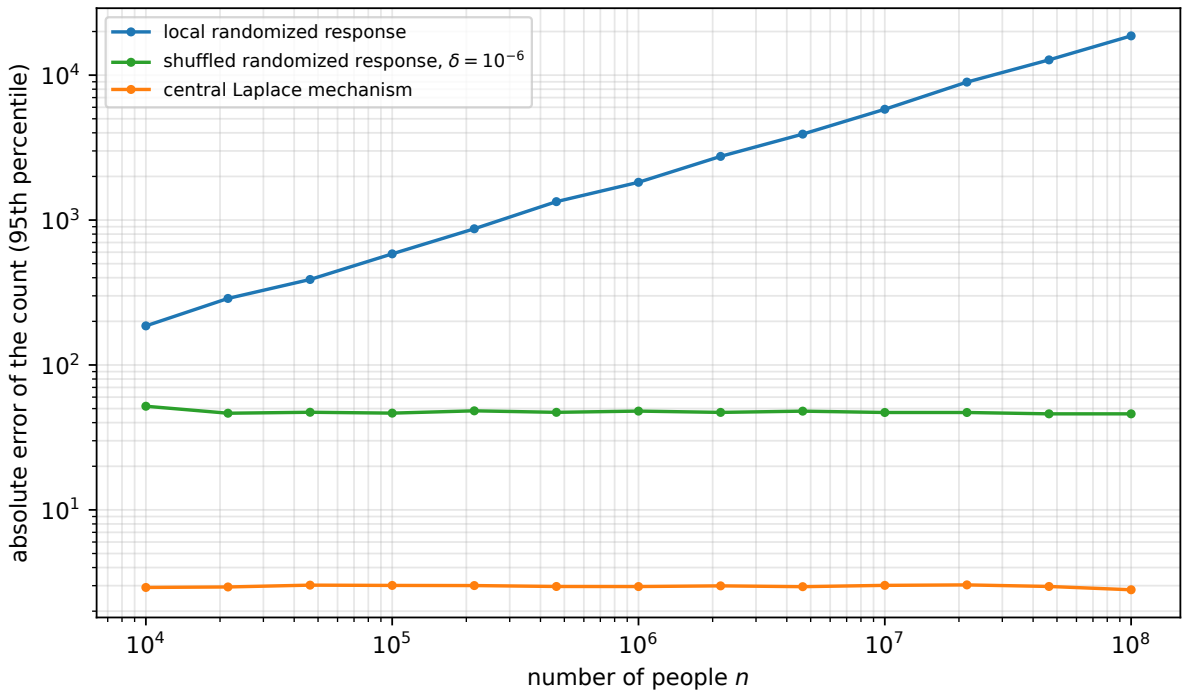


Figure 18.1.: Error of three private counts of n bits at $\varepsilon = 1$, as the 95th percentile of the absolute error over 2,000 simulated runs on data with half the bits equal to one. The local error grows as $\sqrt{n}$. The shuffled error, with ν from Equation 18.5 and $\delta = 10^{-6}$, stays near 47, and the central Laplace error stays near 3.

18.5.1. Bounded range

Recall the exponential mechanism. It takes a score $u(D, r)$ for each candidate r in a countable set, whose value changes by at most Δu between neighbours. It outputs r with probability proportional to $\exp(\varepsilon u(D, r)/(2\Delta u))$, and it is ε-DP. Write

$$L(y) = \ln \frac{\Pr[M(D) = y]}{\Pr[M(D') = y]} \tag{18.9}$$

for the privacy loss of output y between neighbours D and D'. Pure ε-DP says that $|L(y)| \le \varepsilon$, so the losses lie in an interval of width 2ε. The exponential mechanism satisfies a stronger condition.

Definition 18.3 (Bounded range)**.** A mechanism M with a countable output set is **ε-bounded range** if for all neighbours $D \sim D'$ and all outputs y, y',

$$\Pr[M(D) = y] \Pr[M(D') = y'] \le e^{\varepsilon} \Pr[M(D') = y] \Pr[M(D) = y']. \tag{18.10}$$

When the outputs have positive probability, the definition says that $L(y) - L(y') \le \varepsilon$. So the privacy losses lie in an interval of width ε, which need not be centred at zero. The next result shows that the two notions are within a factor of two of each other.

Proposition 18.1. *An ε-bounded range mechanism is ε-DP. An ε-DP mechanism with a countable output set is 2ε-bounded range.*

Proof.

1. *Same support.* Let M be ε-bounded range. Suppose some output y has $\Pr[M(D) = y] > 0$ but $\Pr[M(D') = y] = 0$. With this y, the right-hand side of the definition is zero for every y', and the left-hand side is $\Pr[M(D) = y]$ times $\Pr[M(D') = y']$. So the definition forces $\Pr[M(D') = y'] = 0$ for every y', which is impossible. So $M(D)$ and $M(D')$ give positive probability to the same outputs, and L is finite on them.
2. *Both signs occur.* Both distributions sum to one over the common support, so some output y_+ has $L(y_+) \ge 0$ and some output y_- has $L(y_-) \le 0$.
3. *Width ε around zero.* For every y, bounded range gives $L(y) \le L(y_-) + \varepsilon \le \varepsilon$ and $L(y) \ge L(y_+) - \varepsilon \ge -\varepsilon$. The pointwise lemma of the randomized response chapter turns $|L| \le \varepsilon$ into ε-DP.
4. *Converse.* If $|L| \le \varepsilon$ everywhere, then $L(y) - L(y') \le 2\varepsilon$.

□

The general conversion loses a factor of two, but the exponential mechanism does not, because its privacy losses already lie in an interval of width ε.

Proposition 18.2. *The exponential mechanism with parameter ε is ε-bounded range.*

Proof.

1. Write $Z_D = \sum_r \exp(\varepsilon u(D,r)/(2\Delta u))$ for the normalizing constant. Then $L(r) = \frac{\varepsilon}{2\Delta u}(u(D,r) - u(D',r)) + \ln(Z_{D'}/Z_D)$.
2. The constant cancels in a difference: $L(r) - L(r') = \frac{\varepsilon}{2\Delta u}[(u(D,r) - u(D',r)) - (u(D,r') - u(D',r'))]$.
3. Each bracketed difference is at most Δu in absolute value, so $L(r) - L(r') \le \varepsilon$.

□

The factor of two matters when many selections are composed, and two lemmas make the composition precise. The first turns a tail bound on the privacy loss into a privacy guarantee. It holds for any output space, with L the logarithm of the ratio of mass functions or of densities.

Lemma 18.3. *Let $\varepsilon' \ge 0$ and $\delta \in [0,1]$. If for all neighbours $D \sim D'$ the loss satisfies $\Pr_{y\sim M(D)}[L(y) > \varepsilon'] \le \delta$, then M is (ε', δ)-DP.*

Proof. Fix an event S and let $G = \{y : L(y) \le \varepsilon'\}$. On $S \cap G$ the probability under D is at most $e^{\varepsilon'}$ times that under D', so $\Pr[M(D) \in S \cap G] \le e^{\varepsilon'} \Pr[M(D') \in S]$. The rest of S has probability at most $\Pr[M(D) \notin G] \le \delta$ under D. □

The second lemma bounds the mean loss. It uses Hoeffding's lemma, the key step in the proof of Hoeffding's inequality: a random variable W in an interval of length ℓ satisfies $\mathbb{E}[e^{\lambda(W - \mathbb{E}W)}] \le e^{\lambda^2 \ell^2/8}$ for every real λ.

Lemma 18.4. *If M is ε-bounded range and $D \sim D'$, then $\mathbb{E}_{y\sim M(D)}[L(y)] \le \varepsilon^2/8$.*

Proof.

1. Under $y \sim M(D)$, the variable $W = -L(y)$ lies in an interval of length ε, and $\mathbb{E}[e^W] = \sum_y \Pr[M(D) = y] \frac{\Pr[M(D')=y]}{\Pr[M(D)=y]} = 1$.
2. Hoeffding's lemma with $\lambda = 1$ gives $1 = \mathbb{E}[e^W] \le e^{\mathbb{E}W + \varepsilon^2/8}$, so $\mathbb{E}[L] = -\mathbb{E}[W] \le \varepsilon^2/8$.

□

The two lemmas combine into a composition theorem. The total privacy loss of k steps is a sum of k losses, each with mean at most $\varepsilon^2/8$ and a range of width ε. So the sum concentrates around $k\varepsilon^2/8$, and Lemma 18.3 turns that concentration into a guarantee.

Theorem 18.4 (Composition of bounded-range mechanisms)**.** *Run k mechanisms in sequence on the same dataset, where the j-th may be chosen as a function of the outputs of the first $j-1$, and each is ε-bounded range whatever those outputs are. For every $\delta \in (0,1)$, the sequence of outputs is (ε', δ)-DP with*

$$\varepsilon' = \frac{k\varepsilon^2}{8} + \varepsilon\sqrt{\frac{k \ln(1/\delta)}{2}}. \tag{18.11}$$

Proof.

1. The probability of an output sequence is the product of the conditional probabilities of its terms, so the loss is a sum $L = \sum_{j=1}^{k} L_j$, where L_j is the loss of the j-th output given the earlier ones.
2. Given the earlier outputs, L_j lies in an interval of length ε, by bounded range, and its conditional mean is at most $\varepsilon^2/8$, by Lemma 18.4. Let V_j be L_j minus its conditional mean, so $L \le \sum_j V_j + k\varepsilon^2/8$.
3. For $\lambda > 0$, Hoeffding's lemma applied conditionally gives $\mathbb{E}[e^{\lambda V_j} \mid \text{earlier outputs}] \le e^{\lambda^2\varepsilon^2/8}$. Taking conditional expectations from the last term to the first gives $\mathbb{E}[e^{\lambda \sum_j V_j}] \le e^{k\lambda^2\varepsilon^2/8}$.
4. By Markov's inequality, $\Pr[\sum_j V_j \ge t] \le e^{-\lambda t + k\lambda^2\varepsilon^2/8}$. Choosing $\lambda = 4t/(k\varepsilon^2)$ makes this $e^{-2t^2/(k\varepsilon^2)}$, which equals δ at $t = \varepsilon\sqrt{k\ln(1/\delta)/2}$.
5. So $\Pr[L > \varepsilon'] \le \delta$ for every pair of neighbours, and Lemma 18.3 finishes the proof.

□

Advanced composition gives $\varepsilon\sqrt{2k\ln(1/\delta)} + k\varepsilon\tanh(\varepsilon/2)$ for k mechanisms that are known only to be ε-DP. The square-root term of Theorem 18.4 is half as large. Its linear term $k\varepsilon^2/8$ is also about a quarter of $k\varepsilon\tanh(\varepsilon/2) \approx k\varepsilon^2/2$ for small ε.

Example 18.2. Compose $k = 100$ exponential mechanisms, each with $\varepsilon = 0.1$, at $\delta = 10^{-6}$. Basic composition gives 10.00, advanced composition 5.76, and Theorem 18.4 gives 2.75.

18.5.2. Top-k selection with Gumbel noise

An advertiser usually wants the few most popular items in a histogram, such as the top topics among an audience, not every count. LinkedIn's algorithms depend on two things. The **domain** of items is known in advance or not. And a person's **contribution** is **m-restricted** when each person adds one to at most m counts, and unrestricted otherwise. For example, hires are 1-restricted, because a hire adds one to one employer, but skills are not, because one person can list many.

With m-restricted contributions, replacing one person changes the vector of counts by at most $2m$ in ℓ_1 norm. So Laplace noise of scale $2m/\varepsilon$ on every count, followed by taking the top k, is ε-DP, but its noise grows with m. For unrestricted contributions it is better to select with the exponential mechanism, with the count of an item as its score. A count changes by at most one between neighbours, whatever the number of counts that a person touches. **Peeling** selects the top item, removes it, and repeats k times. Each round is the exponential mechanism, so by Proposition 18.2 and Theorem 18.4 its cost depends on k and not on m.

Sampling from the exponential mechanism needs its normalizing constant, and peeling needs it k times. Adding noise of the right shape and taking a maximum avoids both, and the right shape is the Gumbel distribution.

Definition 18.4 (Gumbel distribution)**.** The **Gumbel distribution** with scale $b > 0$ has cumulative distribution function $\Pr[G \le g] = \exp(-e^{-g/b})$ for real g.

Adding independent Gumbel noise to each score and taking the largest gives exactly the distribution of the exponential mechanism, and taking the k largest gives exactly peeling.

Theorem 18.5 (Gumbel-max trick)**.** *Let $u_1, \dots, u_d$ be scores, let $G_1, \dots, G_d$ be independent Gumbel variables with scale b, and write $w_i = e^{u_i/b}$. Then*

1. *the index of the largest noisy score $u_i + G_i$ is i with probability $w_i / \sum_j w_j$;*
2. *the indices of the k largest noisy scores, in decreasing order, have the distribution of k rounds of peeling, where each round picks a remaining item i with probability proportional to w_i.*

Proof.

1. *Exponential race.* Let $E_i = e^{-(u_i+G_i)/b}$. Then $\Pr[E_i > t] = \Pr[G_i < -b \ln t - u_i] = \exp(-w_i t)$ for $t > 0$. So the E_i are independent exponential variables with rates w_i, and the noisy scores in decreasing order are the E_i in increasing order.
2. *The first finisher.* With $W = \sum_j w_j$, the chance that E_i is the smallest is

$$\int_0^\infty w_i e^{-w_i t} \prod_{j \neq i} e^{-w_j t} \, dt = \frac{w_i}{W}. \tag{18.12}$$

This proves the first claim. 3. *Memorylessness.* Condition on E_i being smallest, with value t. This is the event that $E_j > t$ for every $j \neq i$, and given it, the excesses $E_j - t$ are independent exponential variables with the original rates w_j, because $\Pr[E_j > t + s \mid E_j > t] = e^{-w_j s}$. 4. *Induction.* By step 3, the order of the remaining E_j is a fresh exponential race among the remaining items. Step 2 applied to it shows that the next item is chosen with probability proportional to its w_j, and repeating gives the second claim. □

Counts change by at most one between neighbours, so take $b = 2/\varepsilon_0$. Then $w_i = e^{\varepsilon_0 u_i/2}$, and the first claim is the exponential mechanism with parameter ε_0. By the second claim, adding Gumbel noise once and reporting the k largest noisy counts *is* peeling, so it has the guarantee of peeling from Theorem 18.4. It draws d noise values and selects the top k once, in time of order d plus the time of that selection. Peeling takes time of order kd.

18.5.3. Queries from outside

Here, unlike in the previous chapter, outside analysts choose the queries, and they can be careless or malicious. Each advertiser has a privacy budget for its own queries. Two advertisers with budget ε each can combine their answers, and then basic composition gives only 2ε about the members. Many advertisers together would erase the guarantee. The mechanisms cannot rule this out, so LinkedIn relies on vetting and on agreements not to share results. It also chooses ε and the monthly number of queries so that the attacks it models fail.

18.6. Data releases

The last two deployments do not answer queries. Each publishes a fixed dataset once, computed in the central model.

18.6.1. Mobility reports

During the Covid-19 pandemic, Google published community mobility reports from the location history of users who opted in. For each region and day, they gave the change in visits to places such as parks, workplaces and residences, relative to a baseline from before the pandemic. The counts of visits get Laplace noise after a bound on what each person adds. A person counts at most once per category and place, and in at most four such pairs per day, kept at random. So the contributions are m-restricted, as in the counts of LinkedIn. The reports drop metrics whose noisy count of people is below 100, and changes that the noise makes too uncertain.

18.6.2. A URL dataset with zero-concentrated privacy

Facebook and the academic organization Social Science One released a dataset on how users interact with web links shared on Facebook. It covers more than 38 million URLs, and researchers get access through a proposal process. The raw data are events, such as "a woman aged 35 to 44 in Germany shared this URL on this date". The release aggregates them by URL, month, country, age and gender, and publishes noisy counts of views, clicks, shares, likes and other reactions.

The noise is Gaussian, and the guarantee is zero-concentrated differential privacy, a variant of DP that suits Gaussian noise. It uses the Rényi divergence. For an order $\alpha > 1$ and two distributions P and Q, write p and q for their densities, or their mass functions on a countable set. The **Rényi divergence** of order α is

$$D_\alpha(P \,\|\, Q) = \frac{1}{\alpha - 1} \ln \int p(y)^\alpha q(y)^{1-\alpha} \, dy, \tag{18.13}$$

with a sum in place of the integral on a countable set. It is zero when $P = Q$ and grows as they move apart. Zero-concentrated privacy asks that the divergence between the outputs on neighbours grow at most linearly in the order α.

Definition 18.5 (Zero-concentrated differential privacy)**.** Let $\rho \geq 0$. A mechanism M is **ρ-zero-concentrated differentially private** if for all neighbours $D \sim D'$ and all $\alpha > 1$,

$$D_\alpha(M(D) \,\|\, M(D')) \leq \rho \, \alpha. \tag{18.14}$$

It has one parameter, like pure DP, but composes like advanced composition, as the next three results show. It lies between the two, because an ε-DP mechanism is $(\varepsilon^2/2)$-zero-concentrated, and the converse holds with a δ.

The first two results follow from private deep learning, where Definition 14.4 defines the log-moment $\kappa_M(\lambda)$. It is the worst case over neighbours of the logarithm of the average λ-th power of the density ratio p/q under p. For one pair of neighbours and $\alpha = \lambda + 1$, that logarithm is a multiple of the Rényi divergence of order α, by its definition:

$$\ln \int p^\alpha q^{1-\alpha} \, dy = (\alpha - 1) \, D_\alpha(M(D) \,\|\, M(D')). \tag{18.15}$$

So a mechanism is ρ-zero-concentrated exactly when $\kappa_M(\lambda) \leq \rho\lambda(\lambda + 1)$ for every $\lambda > 0$.

Lemma 18.5. *Let M_1 be ρ_1-zero-concentrated, and let $M_2(\cdot\,; y_1)$ be a mechanism chosen as a function of the output y_1 of M_1, which is ρ_2-zero-concentrated for every y_1. Then the mechanism that outputs (y_1, y_2) is $(\rho_1 + \rho_2)$-zero-concentrated. In particular, this holds for two mechanisms run with independent coins.*

Proof. By the equivalence above, $\kappa_{M_1}(\lambda) \le \rho_1\lambda(\lambda+1)$ and $\kappa_{M_2(\cdot;y_1)}(\lambda) \le \rho_2\lambda(\lambda+1)$ for every y_1 and every $\lambda > 0$. Lemma 14.2 adds the two bounds, so $\kappa_M(\lambda) \le (\rho_1+\rho_2)\lambda(\lambda+1)$, which is the claim. □

The second result gives the guarantee of the Gaussian mechanism, which is why the release uses it.

Proposition 18.3 (Gaussian mechanism)**.** *Let $f\colon \mathcal{X}^n \to \mathbb{R}^d$ change by at most $\Delta_2 f$ in Euclidean norm between neighbours. Then $M(D) = f(D) + Z$, with Z a vector of d independent normal variables of mean 0 and variance σ^2, is $(\Delta_2 f)^2/(2\sigma^2)$-zero-concentrated.*

Proof. Lemma 14.3 gives $\kappa_M(\lambda) = \lambda(\lambda+1)(\Delta_2 f)^2/(2\sigma^2)$ for every $\lambda > 0$, which by the equivalence above is ρ-zero-concentrated privacy with $\rho = (\Delta_2 f)^2/(2\sigma^2)$. □

The third result converts a zero-concentrated guarantee into an (ε, δ) guarantee, the form in which a release usually states its budget.

Proposition 18.4. *A ρ-zero-concentrated mechanism is $(\rho + 2\sqrt{\rho\ln(1/\delta)}, \delta)$-DP for every $\delta \in (0,1)$.*

Proof.

1. If $\rho = 0$, every Rényi divergence is 0, so $M(D)$ and $M(D')$ have the same distribution and the claim holds. Otherwise let $y \sim M(D)$, with density p, and let $L = \ln(p(y)/q(y))$, where q is the density of $M(D')$. For $\alpha > 1$, $\mathbb{E}[e^{(\alpha-1)L}] = \int p^\alpha q^{1-\alpha} = e^{(\alpha-1)D_\alpha} \le e^{(\alpha-1)\alpha\rho}$.
2. Markov's inequality gives $\Pr[L > t] \le e^{(\alpha-1)(\alpha\rho - t)}$.
3. Take $t = \rho + 2\sqrt{\rho\ln(1/\delta)}$ and $\alpha = 1 + \sqrt{\ln(1/\delta)/\rho}$. Writing $\ell = \ln(1/\delta)$, the exponent is $\sqrt{\ell/\rho}\,(\rho + \sqrt{\rho\ell} - \rho - 2\sqrt{\rho\ell}) = -\ell$, so $\Pr[L > t] \le \delta$.
4. Lemma 18.3 finishes the proof.

□

The budget of the release raises a question that recurs in deployments: what is a neighbour? **User-level** privacy hides everything one person did, and **action-level** privacy hides one action, such as one click. User-level privacy is the goal, but a few users act thousands of times, and hiding them would drown every count in noise. So the release protects each action strongly enough that a user with at most a high percentile of the activity distribution gets a stated user-level guarantee. Heavier users still have the action-level one.

Example 18.3. Suppose each action adds one to a single cell, and the actions of one user fall in different cells. Replacing one action changes two counts by one each, a Euclidean change of $\sqrt{2}$. Replacing all k actions of a user changes at most $2k$ counts, a change of at most $\sqrt{2k}$. Then by Proposition 18.3, k actions cost k times the action-level ρ. To give users with at most 20 actions $(0.5, 10^{-6})$-DP, solve $\rho + 2\sqrt{\rho \ln 10^6} = 0.5$ for $\rho = 0.00444$ at the user level. The action level is then $\rho = 0.000222$, which Gaussian noise of standard deviation $\sigma = \sqrt{2}/\sqrt{2\rho} = 67.1$ achieves. A single action alone is protected at $(0.111, 10^{-6})$-DP.

The largest deployment, the US Census Bureau's use of DP for the 2020 census, is complex enough to need a separate treatment.

18.7. Summary

Table 18.1.: The chapter's results.

Result	Statement
Definition 18.1	the adversary sees the messages in a uniformly random order, not who sent which
Theorem 18.1	shuffling turns ε_0-local into $(12\varepsilon_0\sqrt{\ln(1/\delta)/n}, \delta)$-DP, for $\varepsilon_0 < \frac{1}{2}$ and $n \geq 1000$
Theorem 18.2	randomized response with $\nu = 16\ln(4/\delta)/(\tau^2(n-1))$ is (ε, δ)-DP once shuffled
Theorem 18.3	its count has error $O(\sqrt{\ln(1/\delta)/\beta}/\varepsilon)$, independent of n
Proposition 18.1	ε-bounded range implies ε-DP, which implies 2ε-bounded range
Proposition 18.2	the exponential mechanism is ε-bounded range
Theorem 18.4	k bounded-range steps cost $k\varepsilon^2/8 + \varepsilon\sqrt{k\ln(1/\delta)/2}$
Theorem 18.5	Gumbel noise and a maximum sample the exponential mechanism, and the top k is peeling
Proposition 18.3	Gaussian noise with sensitivity $\Delta_2 f$ is $(\Delta_2 f)^2/(2\sigma^2)$-zero-concentrated
Proposition 18.4	ρ-zero-concentrated implies $(\rho + 2\sqrt{\rho\ln(1/\delta)}, \delta)$-DP

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A. Glossary

Acronyms used in the book and the pages that use them.

API Application programming interface. Used in The Shuffle Model and Central Deployments.
DNA Deoxyribonucleic acid. Used in Attempts at Data Privacy.
DP Differential privacy. Used in The Basics of Differential Privacy, Randomized Response and the Definition, The Laplace Mechanism, Approximate Differential Privacy, Advanced Composition, The Exponential Mechanism, Query Release and Its Limits, Private Multiplicative Weights, Sparse Vector, Beyond Global Sensitivity, Packing Lower Bounds, Private Learning and Statistics, What Differential Privacy Protects, Private Empirical Risk Minimization, Private Deep Learning, Private Mean Estimation, Adaptive Data Analysis, Deployments, Local Differential Privacy, The Shuffle Model and Central Deployments.
ERM Empirical risk minimization. Used in Private Empirical Risk Minimization.
LP Linear programming. Used in Reconstruction Attacks.
PAC Probably approximately correct. Used in The Exponential Mechanism.
PATE Private aggregation of teacher ensembles. Used in Private Deep Learning.
PIN Personal identification number. Used in Local Differential Privacy.
RDP Rényi differential privacy. Used in Private Deep Learning.
SGD Stochastic gradient descent. Used in Private Empirical Risk Minimization, Private Deep Learning.
SVM Support vector machine. Used in Private Empirical Risk Minimization.
ZIP Zone Improvement Plan code, the US postal code. Used in Attempts at Data Privacy.

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